Learning to Lens

An Introduction to General Relativity and Strong Gravitational Lensing

with Mathematica

Based on the published works of

P. Schneider, J. Ehlers & E. E. Falco, *Gravitational Lenses* (1992)
R. Narayan & M. Bartelmann, *Lectures on Gravitational Lensing* (1997)
A. O. Petters, H. Levine & J. Wambsganss, *Singularity Theory and Gravitational Lensing* (2001)
S. Carroll, *Spacetime and Geometry* (2004)
A. B. Congdon & C. R. Keeton, *Principles of Gravitational Lensing* (2018)
M. Meneghetti, *Introduction to Gravitational Lensing* (2021)
P. Saha, D. Sluse, J. Wagner & L. L. R. Williams, *Essentials of Strong Gravitational Lensing* (2024)
P. Schneider, C. S. Kochanek & J. Wambsganss, *Gravitational Lensing: Strong, Weak and Micro* (2006)

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with

Claude (Anthropic)
via Claude Code

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About This Document

This tutorial suite was produced by **Rodrigo Córdova Rosado** (Harvard–Smithsonian Center for Astrophysics) with the assistance of **Claude** (Anthropic), an AI language model, via **Claude Code** — Anthropic's command-line tool for collaborative software engineering and technical writing.

The physics content in these notes draws from the following published textbooks and review articles:

- S. Carroll, *Spacetime and Geometry: An Introduction to General Relativity* (Addison Wesley, 2004)
- A. B. Congdon & C. R. Keeton, *Principles of Gravitational Lensing* (Springer, 2018)
- R. Narayan & M. Bartelmann, *Lectures on Gravitational Lensing* (1997; arXiv:astro-ph/9606001)
- P. Saha, D. Sluse, J. Wagner & L. L. R. Williams, *Essentials of Strong Gravitational Lensing*, Space Science Reviews **220**, 12 (2024)
- M. Meneghetti, *Introduction to Gravitational Lensing: With Python Examples* (Springer, 2021)
- P. Schneider, J. Ehlers & E. E. Falco, *Gravitational Lenses* (Springer, 1992)
- A. O. Petters, H. Levine & J. Wambsganss, *Singularity Theory and Gravitational Lensing* (Birkhäuser, 2001)
- P. Schneider, C. S. Kochanek & J. Wambsganss, *Gravitational Lensing: Strong, Weak and Micro*, Saas-Fee Advanced Course 33 (Springer, 2006)

Full bibliographic details for these and all works cited in the text appear in the **References** at the end of this document.

The workflow was as follows: the author directed Claude to co-write the LaTeX lecture notes, Wolfram Mathematica scripts, and interactive notebooks, drawing on the reference texts above. Claude drafted the text and code; the author reviewed, revised, and approved all content.

Importantly, all mathematical derivations and symbolic results in these notes have been verified deterministically using Wolfram Mathematica's computer algebra system — not generated probabilistically by an LLM. Every equation, identity, and worked solution was checked by running the companion `.wl` scripts (via `wolframscript`), which perform exact symbolic computation. The Mathematica verification outputs are reproducible and accompany each module in the repository.

The AI assistant was used for:

- Drafting expository prose and pedagogical explanations
- Structuring the curriculum and organizing modules
- Writing Mathematica code (subsequently executed and verified)
- Formatting LaTeX and managing cross-references

The AI assistant was **not** used as a source of mathematical truth. All physics content derives from the published works listed above, and all symbolic computations are verified by Mathematica.

This is a **living document**. Additional details, examples, and lessons will be added over time as the author continues to learn and teach this material. The most current version is always available at: https://github.com/rodelcr/Learning_to_Lens

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Notation Guide

Symbol	Meaning	Units / Notes
Coordinates and Angles		
$\boldsymbol{\theta}$	Image position (angular)	radians or arcsec
$\boldsymbol{\beta}$	Source position (angular)	radians or arcsec
$\boldsymbol{\alpha}$	Reduced deflection angle	radians
$\hat{\alpha}$	Physical deflection angle	radians
$\boldsymbol{\xi}$	Position in lens plane	physical (kpc, Mpc)
Distances		
$D_{\rm d}$	Observer–lens angular diameter distance	Mpc
$D_{\rm s}$	Observer–source angular diameter distance	Mpc
$D_{\rm ds}$	Lens–source angular diameter distance	Mpc
Lensing Quantities		
$\theta_{\rm E}$	Einstein radius	arcsec
$\Sigma_{\rm cr}$	Critical surface mass density	$\mathrm{g\,cm^{-2}}$ or $M_\odot\,\mathrm{pc^{-2}}$
κ	Convergence $= \Sigma/\Sigma_{\rm cr}$	dimensionless
γ	Shear magnitude	dimensionless
ψ	Effective lensing potential	dimensionless
μ	Magnification	dimensionless
$\mathcal{A}$	Jacobian (inverse magnification) matrix	2×2
Cosmology		
H_0	Hubble constant	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
$\Omega_{\rm m}$	Matter density parameter	dimensionless
Ω_Λ	Dark energy density parameter	dimensionless
GR / Metric		
$g_{\mu\nu}$	Metric tensor	
$\Gamma^{\alpha}_{\ \mu\nu}$	Christoffel symbols	
$R^{\alpha}_{\ \beta\mu\nu}$	Riemann curvature tensor	
$R_{\mu\nu}$	Ricci tensor	
R	Ricci scalar	

Mathematica files are cross-linked throughout these notes. Links formatted as `filename.nb` or `filename.wl` point to companion Mathematica notebooks and scripts in the `Mathematica/`

directory of this repository.

Part I

General Relativity Foundations

Chapter 1

Special Relativity and Tensor Basics

> *General relativity (GR) is Einstein's theory of space, time, and gravitation. At heart it is a very simple subject . . . The essential idea is perfectly straightforward: while most forces of nature are represented by fields defined on spacetime, gravity is inherent in spacetime itself.* — Sean Carroll, *Spacetime and Geometry*

Before we can understand how gravity bends light, we need the language of special relativity and tensors. This chapter reviews the foundations: Minkowski spacetime, Lorentz transformations, four-vectors, and the index notation that will carry us through the rest of these notes. The emphasis is on building fluency with the notation and the key physical ideas — time dilation, length contraction, the invariant interval — so that the transition to curved spacetime in later modules feels natural.

Companion Mathematica files:

- `lorentz_transforms.wl` — symbolic construction and verification of Lorentz boost matrices
- `four_vectors.nb` — interactive visualizations of worldlines, light cones, and time dilation

1.1 From Newton to Einstein

In Newtonian mechanics, space and time are independent. The gravitational force between two masses M and m separated by a distance r is (cf. Carroll eq. 1.1)

$$\mathbf{F} = -\frac{GMm}{r^2}\,\hat{\mathbf{r}}\,, \tag{1.1}$$

and the gravitational potential Φ satisfies Poisson's equation,

$$\nabla^2\Phi = 4\pi G\rho\,. \tag{1.2}$$

In GR, both of these are replaced by statements about the curvature of spacetime. The key equation we will eventually arrive at is Einstein's field equation (Carroll eq. 1.5, in units with $c = 1$):

$$R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} = 8\pi G\,T_{\mu\nu}\,. \tag{1.3}$$

But we are getting ahead of ourselves. For now, we focus on *flat* spacetime — the arena of special relativity — and learn the tensor language needed to write down equations like (1.3).

1.2 Spacetime and the Invariant Interval

Special relativity (SR) unifies space and time into a single four-dimensional continuum called **Minkowski space**. An individual point in spacetime is an **event**, and the path of a particle through spacetime is its **worldline**.

Unlike Newtonian mechanics, SR has no absolute notion of simultaneity. Instead, the fundamental invariant is the **spacetime interval** between two events (Carroll eq. 1.10):

$$\boxed{(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \tag{1.4}$$

where we work in **natural units** with $c = 1$ throughout this chapter (we will restore factors of c when needed for astrophysical applications).

Remark 1.2.1. We adopt the **mostly-plus** sign convention $(-,+,+,+)$, following Carroll. Some references (especially in particle physics) use the opposite convention $(+,-,-,-)$. Be careful when comparing equations across textbooks.

The interval classifies the separation between events:

- $(\Delta s)^2 < 0$: **timelike** separation (events can be causally connected)
- $(\Delta s)^2 = 0$: **null** or **lightlike** separation (connected by a light ray)
- $(\Delta s)^2 > 0$: **spacelike** separation (no causal connection possible)

The **proper time** $\Delta\tau$ along a timelike path is defined as (Carroll eq. 1.17)

$$(\Delta\tau)^2 = -(\Delta s)^2 = (\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2 \,. \tag{1.5}$$

This is the time measured by a clock traveling along the worldline. For an infinitesimal displacement, we write the **line element** (Carroll eq. 1.20):

$$ds^2 = \eta_{\mu\nu}\, dx^\mu\, dx^\nu \,, \tag{1.6}$$

where we have introduced the **Minkowski metric** and the **Einstein summation convention** (repeated indices are summed over).

1.3 Index Notation and the Minkowski Metric

Spacetime coordinates are labeled by Greek letters $(\mu, \nu, \rho, \sigma, \ldots)$ running from 0 to 3 (Carroll eq. 1.12):

$$x^\mu : \quad x^0 = t, \quad x^1 = x, \quad x^2 = y, \quad x^3 = z \,. \tag{1.7}$$

Latin indices $i, j, k, \ldots$ run over the spatial components $1, 2, 3$ only.

The **Minkowski metric** is the 4×4 matrix (Carroll eq. 1.15):

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{1.8}$$

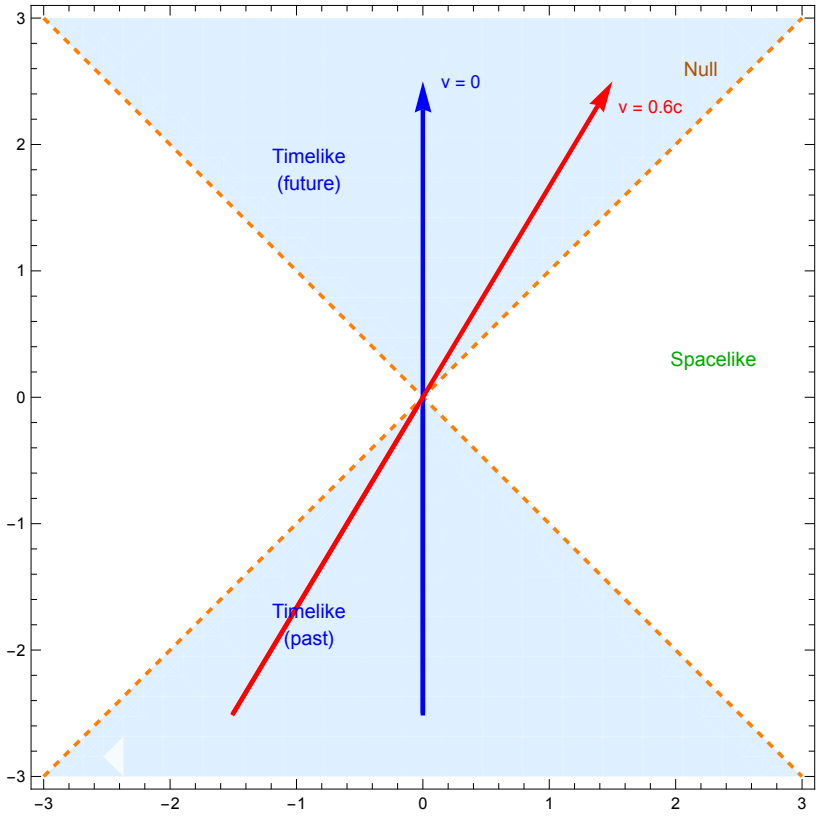


Figure 1.1: Spacetime diagram in 1+1 dimensions showing the light cone structure. The shaded (blue) region is timelike; the dashed orange lines are the light cone (null); the unshaded region is spacelike. The blue arrow is the worldline of a stationary particle; the red arrow is a particle moving at $v = 0.6c$. (cf. Carroll Fig. 1.5)

Its inverse, $\eta^{\mu\nu}$, has the same components. The metric is used to **raise and lower indices**:

$$V_\mu = \eta_{\mu\nu}\, V^\nu, \qquad \omega^\mu = \eta^{\mu\nu}\, \omega_\nu\,. \tag{1.9}$$

Note that in Minkowski space this simply flips the sign of the time component: $V_0 = -V^0$, while $V_i = V^i$.

1.4 Lorentz Transformations

The coordinate transformations that leave the interval (1.6) invariant are called **Lorentz transformations**. They act on the coordinates as (Carroll eq. 1.25)

$$x^{\mu'} = \Lambda^{\mu'}{}_{\nu}\, x^\nu\,, \tag{1.10}$$

where the matrix Λ satisfies the defining condition (Carroll eq. 1.28):

$$\boxed{\eta = \Lambda^{\mathrm{T}}\, \eta\, \Lambda} \tag{1.11}$$

or equivalently $\eta_{\rho\sigma} = \Lambda^{\mu'}{}_{\rho}\, \Lambda^{\nu'}{}_{\sigma}\, \eta_{\mu'\nu'}$.

1.4.1 Rotations

A spatial rotation in the x–y plane by angle θ is (Carroll eq. 1.31):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{1.12}$$

1.4.2 Boosts

Rotations mix two spatial directions using sin and cos (circular functions that preserve the positive-definite sum $x^2 + y^2$). A **Lorentz boost** mixes a space and a time direction, and must preserve the *indefinite* form $-t^2 + x^2$ (the Minkowski interval). The functions that do this are the hyperbolic functions sinh and cosh, which satisfy the identity $\cosh^2\phi - \sinh^2\phi = 1$ — exactly mirroring the Minkowski signature (Carroll Sec. 1.3, pp. 12–14).

The boost along the x-axis with **rapidity** parameter ϕ is therefore (Carroll eq. 1.32):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} \cosh\phi & -\sinh\phi & 0 & 0 \\ -\sinh\phi & \cosh\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{1.13}$$

The rapidity ϕ parameterizes the boost in a way that is analogous to the rotation angle θ: just as two successive rotations by θ_1 and θ_2 give a rotation by $\theta_1 + \theta_2$, two successive boosts with rapidities ϕ_1 and ϕ_2 compose to a single boost with rapidity $\phi_1 + \phi_2$ (Exercise 1.11.2).

To connect to the more familiar velocity v of the boosted frame (Carroll eqs. 1.33–1.35), note that a particle at rest in S' has $x' = 0$, so from the boost matrix: $0 = -\sinh\phi\cdot t + \cosh\phi\cdot x$, giving $x/t = \tanh\phi$. Since x/t is the velocity v of the S' frame as seen from S, we identify:

$$v = \tanh\phi\,. \tag{1.14}$$

The Lorentz factor γ then follows from the hyperbolic identity $\cosh^2\phi - \sinh^2\phi = 1$:

$$\gamma \equiv \frac{1}{\sqrt{1-v^2}} = \cosh\phi\,, \qquad \gamma v = \sinh\phi\,. \tag{1.15}$$

Substituting into the boost matrix, we recover the familiar form (Carroll eq. 1.35):

$$\begin{aligned} t' &= \gamma(t - vx)\,, \\ x' &= \gamma(x - vt)\,. \end{aligned} \tag{1.16}$$

Exercise 1.4.1. Verify that the boost matrix (1.13) satisfies the Lorentz condition (1.11) by explicit matrix multiplication. Then confirm this symbolically using `lorentz_transforms.wl`.

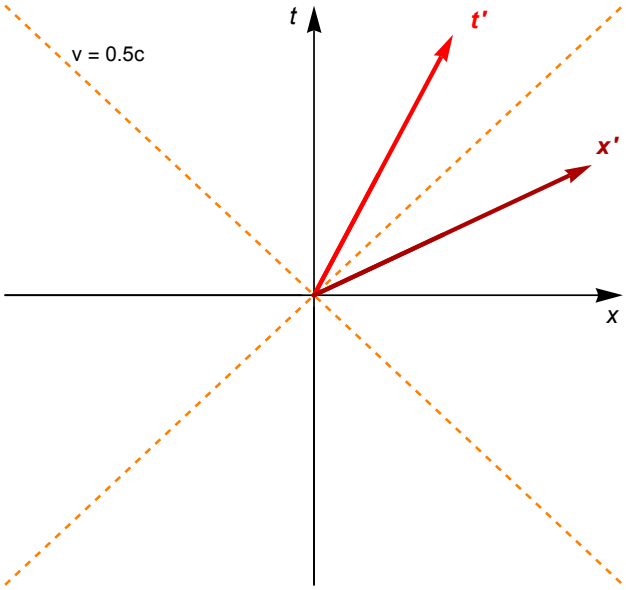


Figure 1.2: A Lorentz boost with $v = 0.5c$ tilts both the time axis (t', red) and the space axis (x', dark red) toward the light cone. The light cone itself (orange dashed) is invariant under boosts. (cf. Carroll Fig. 1.7)

1.4.3 Physical Consequences

The boost equations (1.16) immediately lead to the classic relativistic effects (Carroll Sec. 1.3; Congdon & Keeton Sec. 3.1.2). In each case the derivation is a single substitution:

- **Time dilation:** A clock at rest in S' has $\Delta x' = 0$. The inverse boost gives $\Delta t = \gamma(\Delta t' + v \cdot 0) = \gamma\,\Delta t'$, so the moving clock appears to run *slow* by a factor $\gamma > 1$.

- **Length contraction:** A rod at rest in S' has proper length $L_0 = \Delta x'$. We measure it in S by noting the positions of both ends *simultaneously* ($\Delta t = 0$). From $\Delta x' = \gamma(\Delta x - v \cdot 0)$ we get $L_0 = \gamma\,\Delta x$, hence $L = \Delta x = L_0/\gamma$ — the rod is shorter in the frame where it moves.

- **Relativity of simultaneity:** Events simultaneous in S' ($\Delta t' = 0$) satisfy $0 = \gamma(\Delta t - v\,\Delta x)$, giving $\Delta t = v\,\Delta x$: events separated in space are *not* simultaneous in S.

Exercise 1.4.2. (Twin paradox.) An observer travels from event $A = (0, 0)$ to $B' = (\Delta t/2,\, v\Delta t/2)$ and back to $C = (\Delta t, 0)$ at constant speed v, while a stationary twin travels from A directly to C. Show that the traveling twin ages by $\Delta\tau_{AB'C} = \sqrt{1 - v^2}\,\Delta t$ while the stationary twin ages by $\Delta\tau_{ABC} = \Delta t$. (cf. Carroll eqs. 1.18–1.19. Visualize with `four_vectors.nb`.)

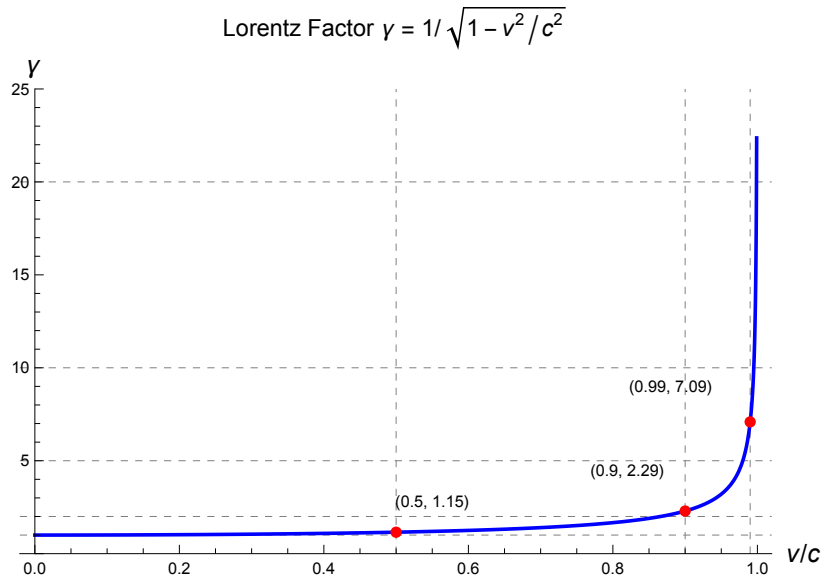


Figure 1.3: The Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$ as a function of velocity. Time dilation and length contraction both scale with γ. Note the dramatic increase as $v \to c$.

1.5 Four-Vectors

A **four-vector** V^μ is a set of four components that transforms under Lorentz transformations as (Carroll eq. 1.39):

$$V^{\mu'} = \Lambda^{\mu'}{}_{\nu} V^\nu \,. \tag{1.17}$$

Key examples:

Four-position: $x^\mu = (t, x, y, z)$.

Four-velocity: For a massive particle with worldline $x^\mu(\tau)$, the four-velocity is defined as the derivative of the position with respect to *proper time* τ (the time measured by a clock traveling with the particle):

$$U^\mu = \frac{dx^\mu}{d\tau} \,. \tag{1.18}$$

To express this in terms of coordinate time t, use $d\tau = dt/\gamma$ (from time dilation, eq. (1.5); see also Carroll eq. 1.17), giving $U^\mu = \gamma\, dx^\mu/dt = \gamma(1, v^x, v^y, v^z)$ where $v^i = dx^i/dt$ is the ordinary three-velocity.

The norm of the four-velocity is always $\eta_{\mu\nu} U^\mu U^\nu = -1$ (in units with $c = 1$; Carroll eq. 1.100). This follows directly from the definition: $U^\mu U_\mu = (dx^\mu/d\tau)(dx_\mu/d\tau) = ds^2/d\tau^2 = -d\tau^2/d\tau^2 = -1$, since $ds^2 = -d\tau^2$ along a timelike worldline.

Four-momentum: The four-momentum is defined as $p^\mu = m\, U^\mu$, where m is the rest mass (Carroll eq. 1.101). Expanding: $p^\mu = m\gamma(1, v^x, v^y, v^z) = (E, p^x, p^y, p^z)$, where $E = \gamma m$ is the relativistic energy (which reduces to m at rest, i.e., $E = mc^2$ in conventional units) and

$\mathbf{p} = \gamma m\,\mathbf{v}$ is the relativistic three-momentum (which reduces to $m\mathbf{v}$ for $v \ll c$; Congdon & Keeton eqs. 3.18–3.23). The invariant norm gives the **energy–momentum relation**:

$$\boxed{\eta_{\mu\nu}\,p^\mu p^\nu = -E^2 + |\mathbf{p}|^2 = -m^2} \tag{1.19}$$

or, restoring c: $E^2 = (pc)^2 + (mc^2)^2$.

For a **photon** ($m = 0$), $E = |\mathbf{p}|$ and the four-momentum is null: $p^\mu p_\mu = 0$. This will be crucial when we trace light rays through curved spacetime in later modules.

1.6 Dual Vectors (One-Forms)

Associated with every vector space is a **dual vector space** (or cotangent space T_p^*). A dual vector (or **one-form**) ω_μ transforms under the *inverse* Lorentz transformation (Carroll eq. 1.50):

$$\omega_{\mu'} = \Lambda^{\nu}{}_{\mu'}\,\omega_\nu\,. \tag{1.20}$$

The prototypical one-form is the **gradient** of a scalar field ϕ (Carroll eq. 1.54):

$$(\mathrm{d}\phi)_\mu = \partial_\mu \phi \equiv \frac{\partial \phi}{\partial x^\mu}\,. \tag{1.21}$$

The contraction of a vector with a dual vector is a Lorentz scalar: $\omega_\mu V^\mu \in \mathbf{R}$.

1.7 Tensors

A **tensor** of type (k, l) is a multilinear map that takes k one-forms and l vectors and produces a real number (Carroll eq. 1.56). In components, the transformation law is (Carroll eq. 1.63):

$$T^{\mu'_1\cdots\mu'_k}{}_{\nu'_1\cdots\nu'_l} = \Lambda^{\mu'_1}{}_{\alpha_1}\cdots\Lambda^{\mu'_k}{}_{\alpha_k}\,\Lambda^{\beta_1}{}_{\nu'_1}\cdots\Lambda^{\beta_l}{}_{\nu'_l}\,T^{\alpha_1\cdots\alpha_k}{}_{\beta_1\cdots\beta_l}\,. \tag{1.22}$$

1.7.1 Important Operations

- **Contraction:** Summing over one upper and one lower index reduces the rank by $(1, 1)$ (Carroll eq. 1.70): $S^{\mu\rho}{}_{\sigma} = T^{\mu\nu\rho}{}_{\sigma\nu}$.
- **Raising/lowering:** Use the metric to move indices (Carroll eq. 1.72): $T^{\alpha\beta\mu}{}_{\delta} = \eta^{\mu\gamma}\,T^{\alpha\beta}{}_{\gamma\delta}$.
- **Symmetrization/antisymmetrization:** Round brackets denote symmetrization; square brackets denote antisymmetrization (Carroll eqs. 1.79–1.80).

1.7.2 Key Tensors in Minkowski Space

- **Metric:** $\eta_{\mu\nu}$ — a symmetric $(0, 2)$ tensor.
- **Kronecker delta:** $\delta^\mu{}_\nu$ — the identity map.

- **Levi-Civita symbol:** $\tilde{\epsilon}_{\mu\nu\rho\sigma}$ (Carroll eq. 1.68) — totally antisymmetric, with $\tilde{\epsilon}_{0123} = +1$.
- **Electromagnetic field strength:** $F_{\mu\nu} = -F_{\nu\mu}$ (Carroll eq. 1.69) — a $(0, 2)$ antisymmetric tensor encoding **E** and **B**.

1.8 The Metric as the Central Object

The key insight that connects SR to GR is that the metric is the fundamental object describing the geometry of spacetime. In SR, the metric is the constant $\eta_{\mu\nu}$. In GR, the metric becomes a field $g_{\mu\nu}(x)$ that varies from point to point, encoding the curvature of spacetime due to matter and energy.

Everything we need for gravitational lensing — the bending of light, the distortion of images, the time delays between multiple images — follows from understanding how the metric determines the paths of light rays. That story begins in the next module, where we introduce curved spacetime and the geodesic equation.

1.9 Energy and Momentum in SR

The **stress-energy tensor** $T^{\mu\nu}$ encodes the energy and momentum content of matter and fields. For a perfect fluid with energy density ρ, pressure p, and four-velocity U^μ:

$$T^{\mu\nu} = (\rho + p)\, U^\mu U^\nu + p\, \eta^{\mu\nu}\,. \tag{1.23}$$

Conservation of energy and momentum is expressed as

$$\partial_\mu T^{\mu\nu} = 0\,. \tag{1.24}$$

This will generalize to $\nabla_\mu T^{\mu\nu} = 0$ in curved spacetime, and the stress-energy tensor will appear as the source term on the right-hand side of Einstein's field equation (1.3).

1.10 Summary

- Spacetime is a four-dimensional continuum; the invariant quantity is the interval $ds^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu$.
- Lorentz transformations ($x^{\mu'} = \Lambda^{\mu'}{}_\nu x^\nu$) leave the interval invariant. Boosts mix space and time.
- Four-vectors and tensors are classified by how they transform under Lorentz transformations.
- The metric $\eta_{\mu\nu}$ is the central object — it defines distances, raises/lowers indices, and generalizes to $g_{\mu\nu}$ in GR.
- Photons travel on null paths ($ds^2 = 0$), which will be the starting point for understanding gravitational lensing.

1.11 Problems

Exercise 1.11.1. Classify the following pairs of events as timelike, spacelike, or null separated:

(a) $A = (0, 0, 0, 0)$, $B = (2, 1, 1, 0)$

(b) $A = (0, 0, 0, 0)$, $B = (1, 1, 0, 0)$

(c) $A = (0, 0, 0, 0)$, $B = (3, 1, 2, 2)$

Check your answers with Mathematica.

Exercise 1.11.2. Show that two successive boosts along the x-axis with rapidities ϕ_1 and ϕ_2 are equivalent to a single boost with rapidity $\phi = \phi_1 + \phi_2$. Relate this to the velocity addition formula $v = (v_1 + v_2)/(1 + v_1 v_2)$.

Exercise 1.11.3. Prove that the four-velocity of any massive particle satisfies $U^\mu U_\mu = -1$ (in units with $c = 1$).

Exercise 1.11.4. A photon with energy E in frame S is observed in frame S' moving with velocity v along the x-axis.

(a) Show that the photon energy in S' is $E' = \gamma E(1 - v\cos\theta)$, where θ is the angle between the photon's direction and the x-axis in frame S.

(b) Specialize to $\theta = 0$ (head-on) and $\theta = \pi/2$ (transverse) and interpret physically.

Exercise 1.11.5. The electromagnetic field strength tensor has components (Carroll eq. 1.69):

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

(a) Show that $F_{\mu\nu}$ is antisymmetric.

(b) Compute $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$ and verify that the signs of the electric field components change.

(c) Compute the Lorentz scalar $F_{\mu\nu}F^{\mu\nu}$ and express it in terms of $\mathbf{E}$ and $\mathbf{B}$.

Chapter 2

Differential Geometry and the Metric Tensor

> *The appropriate mathematical structure used to describe curvature is that of a differentiable manifold: essentially, a kind of set that looks locally like flat space, but might have a very different global geometry.* — Sean Carroll, *Spacetime and Geometry*

In Module 1a we worked exclusively in flat Minkowski spacetime. To describe gravity, we must allow spacetime to be *curved*. This chapter introduces the mathematical machinery of differential geometry — manifolds, the general metric tensor, the covariant derivative, Christoffel symbols, and the curvature tensors — that forms the foundation of general relativity and, ultimately, gravitational lensing.

Companion Mathematica files:

- `christoffel_symbols.wl` — general-purpose symbolic computation of Christoffel symbols and curvature tensors from any input metric
- `curved_surfaces.nb` — interactive visualizations of parallel transport, geodesics, and curvature on 2D surfaces

2.1 Manifolds and Coordinate Charts

A **manifold** is a space that may be curved globally but looks like $\mathbb{R}^n$ in every sufficiently small neighborhood. More precisely, an n-dimensional manifold M is a topological space equipped with a collection of **coordinate charts** — invertible maps $\phi : U \to \mathbb{R}^n$ from open subsets $U \subset M$ — that cover all of M and are smoothly compatible on overlaps (Carroll Sec. 2.2).

Examples of manifolds include:

- $\mathbb{R}^n$ (trivially)
- The 2-sphere S^2 (the surface of a ball)
- The 2-torus T^2 (the surface of a doughnut)
- Spacetime itself (a 4-dimensional Lorentzian manifold)

A key fact: in general, *no single coordinate system can cover an entire manifold*. The sphere S^2, for instance, requires at least two charts (e.g., stereographic projections from the north and south poles; Carroll eqs. 2.9–2.10). We will always work locally in a single chart, keeping in mind that our results are coordinate-independent.

2.2 The General Metric Tensor

On a manifold, we need a way to measure distances and angles. This is provided by the **metric tensor** $g_{\mu\nu}(x)$, a symmetric, nondegenerate $(0,2)$ tensor field. The line element is (cf. Carroll eq. 1.20, now generalized):

$$\boxed{ds^2 = g_{\mu\nu}(x)\, dx^\mu\, dx^\nu} \tag{2.1}$$

In flat Minkowski space, $g_{\mu\nu} = \eta_{\mu\nu}$ everywhere. In curved spacetime, $g_{\mu\nu}$ varies from point to point and encodes the gravitational field.

The **inverse metric** $g^{\mu\nu}$ is defined by

$$g^{\mu\lambda} g_{\lambda\nu} = \delta^\mu{}_\nu \,. \tag{2.2}$$

As before, the metric raises and lowers indices: $V_\mu = g_{\mu\nu} V^\nu$, $V^\mu = g^{\mu\nu} V_\nu$.

Example 2.2.1. The metric on the 2-sphere of radius R in spherical coordinates (θ, ϕ) is

$$ds^2 = R^2\big(d\theta^2 + \sin^2\theta\, d\phi^2\big)\,, \tag{2.3}$$

so $g_{\theta\theta} = R^2$, $g_{\phi\phi} = R^2 \sin^2\theta$, and all other components vanish.

Example 2.2.2. The flat plane in polar coordinates (r, ϕ) has metric

$$ds^2 = dr^2 + r^2\, d\phi^2 \,. \tag{2.4}$$

Despite the coordinate-dependent appearance, this space is flat — the curvature tensors vanish. We will verify this in Mathematica.

2.3 Vectors and Tensors on Manifolds

On a general manifold, vectors live in the **tangent space** T_p at each point p. A vector V at p is identified with a directional derivative operator along a curve $x^\mu(\lambda)$ through p (Carroll Sec. 2.3):

$$V = V^\mu \partial_\mu \,, \tag{2.5}$$

where $\partial_\mu \equiv \partial/\partial x^\mu$ form a **coordinate basis** for T_p.

Under a coordinate transformation $x^\mu \to x^{\mu'}$, vector components transform as (Carroll eq. 2.19):

$$V^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\nu}\, V^\nu \,. \tag{2.6}$$

This generalizes the Lorentz transformation law from Module 1a to arbitrary coordinate changes.

Dual vectors (one-forms) and general tensors transform analogously, with one factor of $\partial x^{\mu'}/\partial x^\nu$ for each upper index and $\partial x^\nu/\partial x^{\mu'}$ for each lower index.

2.4 The Covariant Derivative

Ordinary partial derivatives of tensors are *not* tensors on a curved manifold: the transformation of $\partial_\mu V^\nu$ picks up an extra, non-tensorial term from the coordinate dependence of the basis vectors (Carroll Sec. 3.2).

To fix this, we introduce the **covariant derivative** ∇_μ, which adds a correction term involving the **connection coefficients** $\Gamma^\rho{}_{\mu\sigma}$ (Carroll eq. 3.5):

$$\boxed{\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu{}_{\mu\lambda} V^\lambda} \tag{2.7}$$

For a one-form (Carroll eq. 3.16):

$$\nabla_\mu \omega_\nu = \partial_\mu \omega_\nu - \Gamma^\lambda{}_{\mu\nu} \omega_\lambda \,. \tag{2.8}$$

The general rule for a tensor of arbitrary rank: $+\Gamma$ for each upper index, $-\Gamma$ for each lower index (Carroll eq. 3.17).

The connection coefficients are *not* tensors — they transform inhomogeneously (Carroll eq. 3.10) — but the covariant derivative $\nabla_\mu V^\nu$ *is* a tensor.

2.5 Christoffel Symbols

In GR we demand two additional properties of the connection:

1. **Torsion-free:** $\Gamma^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\nu\mu}$
2. **Metric compatibility:** $\nabla_\rho g_{\mu\nu} = 0$

These uniquely determine the **Christoffel symbols** (also called the Levi-Civita connection; Carroll eq. 3.27):

$$\boxed{\Gamma^\sigma{}_{\mu\nu} = \frac{1}{2} g^{\sigma\rho}\big(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big)} \tag{2.9}$$

This is one of the most important formulas in GR. Given any metric, we can compute the Christoffel symbols and from them all of the geometry.

Example 2.5.1. For the plane in polar coordinates, $ds^2 = dr^2 + r^2 d\phi^2$ (Example 2.2.2), the only nonvanishing Christoffel symbols are (Carroll eqs. 3.29–3.31):

$$\Gamma^r{}_{\phi\phi} = -r \,, \qquad \Gamma^\phi{}_{r\phi} = \Gamma^\phi{}_{\phi r} = \frac{1}{r} \,. \tag{2.10}$$

We derive these and more general cases in `christoffel_symbols.wl`.

2.6 Parallel Transport

In flat space, we can freely slide a vector from one point to another while keeping it "constant." On a curved manifold, this is no longer unambiguous — the result depends on the *path taken*.

The notion of keeping a vector constant along a curve $x^\mu(\lambda)$ is formalized as **parallel transport**: a vector V^μ is parallel-transported if (Carroll eq. 3.40)

$$\frac{dV^\mu}{d\lambda} + \Gamma^\mu{}_{\rho\sigma}\frac{dx^\rho}{d\lambda}V^\sigma = 0\,. \tag{2.11}$$

An important consequence of metric compatibility is that parallel transport preserves the inner product: if V^μ and W^μ are both parallel-transported along a curve, then $g_{\mu\nu}V^\mu W^\nu = \text{const}$ (Carroll eq. 3.42).

On the 2-sphere, parallel transporting a vector around a closed loop results in a rotation of the vector — the angle of rotation is proportional to the enclosed area and the curvature (Figure 2.1).

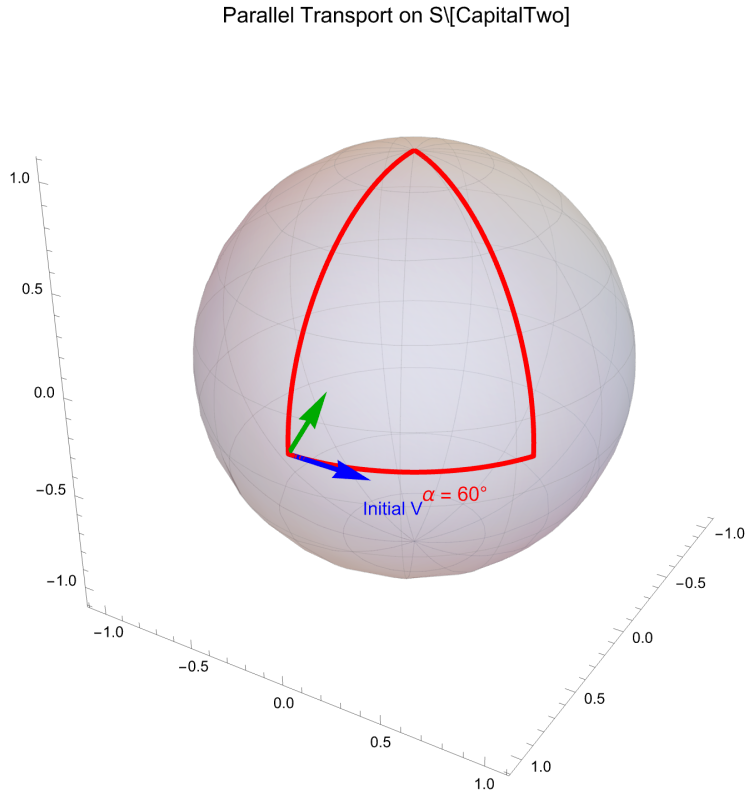


Figure 2.1: Parallel transport on the 2-sphere. A vector (blue) initially pointing east at the equator is transported north to the pole along $\phi = 0$, then south along $\phi = 60$ back to the equator. The final vector (green) has rotated by $\alpha = 60$ — equal to the solid angle enclosed by the path. This rotation is a direct manifestation of curvature. (cf. Carroll Fig. 3.2)

This is also visualized interactively in `curved_surfaces.nb`.

2.7 The Geodesic Equation

A **geodesic** is the curved-space generalization of a straight line: it is the path that parallel-transports its own tangent vector. Setting $V^\mu = dx^\mu/d\lambda$ in (2.11), we obtain the **geodesic**

equation (Carroll eq. 3.44):

$$\boxed{\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu{}_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0} \tag{2.12}$$

This can equivalently be derived by extremizing the proper time/interval functional (Carroll Sec. 3.3, eqs. 3.45–3.55; Congdon & Keeton eq. 3.60).

Key properties:

- **Timelike geodesics** ($ds^2 < 0$): paths of freely falling massive particles; parameterized by proper time τ.
- **Null geodesics** ($ds^2 = 0$): paths of light rays; parameterized by an affine parameter λ.
- **Spacelike geodesics** ($ds^2 > 0$): shortest spatial distances.
- The character (timelike/null/spacelike) is preserved along the geodesic (Carroll Sec. 3.4).

In flat space with Cartesian coordinates, $\Gamma^\mu{}_{\rho\sigma} = 0$, so the geodesic equation reduces to $d^2x^\mu/d\lambda^2 = 0$ — a straight line, as expected.

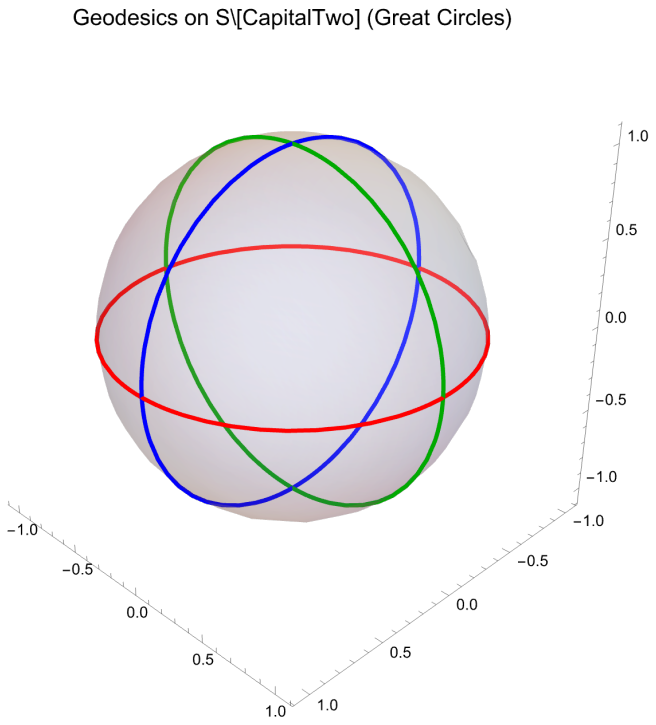


Figure 2.2: Geodesics on the 2-sphere are great circles. Three orthogonal great circles are shown, illustrating that the shortest path between any two points on a sphere follows a great circle arc.

2.8 Curvature Tensors

The **Riemann curvature tensor** $R^{\rho}{}_{\sigma\mu\nu}$ measures how much parallel transport around a closed loop rotates a vector. It is defined by (Carroll eq. 3.4):

$$\boxed{R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}{}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}{}_{\mu\sigma} + \Gamma^{\rho}{}_{\mu\lambda}\,\Gamma^{\lambda}{}_{\nu\sigma} - \Gamma^{\rho}{}_{\nu\lambda}\,\Gamma^{\lambda}{}_{\mu\sigma}} \tag{2.13}$$

A manifold is flat if and only if $R^{\rho}{}_{\sigma\mu\nu} = 0$ everywhere. This is the mathematical statement that curvature is characterized entirely by the Riemann tensor.

From the Riemann tensor we construct:

- **Ricci tensor** (contraction on 1st and 3rd indices):

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu} \tag{2.14}$$

- **Ricci scalar** (trace of the Ricci tensor):

$$R = g^{\mu\nu} R_{\mu\nu} \tag{2.15}$$

These are the quantities that appear in Einstein's field equation

$$R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = 8\pi G\, T_{\mu\nu}\,, \tag{2.16}$$

relating the curvature of spacetime (left side) to its matter and energy content (right side). Here $c = 1$; with factors of c restored the right-hand side reads $8\pi G\, T_{\mu\nu}/c^4$, the form used from Module 1c on.

Remark 2.8.1. The Riemann tensor in n dimensions has $n^2(n^2-1)/12$ independent components. In 4D spacetime, that is 20. In 2D, there is only 1 — the Gaussian curvature determines everything.

2.9 The Metric Determines Everything

Let us collect the chain of logic:

Metric $g_{\mu\nu}$
$\downarrow$ eq. (2.9)
Christoffel symbols $\Gamma^{\sigma}{}_{\mu\nu}$
$\downarrow$ eq. (2.12) $\quad$ $\downarrow$ eq. (2.13)
Geodesics (particle paths) $\quad$ **Curvature** $R^{\rho}{}_{\sigma\mu\nu}$

This is the central message: *the metric is the fundamental object in GR.* Once you have $g_{\mu\nu}$, you can compute everything — how particles move, how light bends, and how spacetime curves.

For gravitational lensing, the key application is: given the metric around a massive object (e.g., the Schwarzschild metric for a point mass), solve the null geodesic equation to trace light rays and determine how images are formed.

2.10 Computation Cookbook: A Step-by-Step Guide

Computing the geometric quantities of GR by hand is a rite of passage. This section gives explicit, algorithmic recipes for going from a metric to Christoffel symbols, to the Riemann tensor, to the Ricci tensor and scalar. Work through the 2-sphere example in full, then use `christoffel_symbols.wl` to check every result.

2.10.1 Algorithm: Christoffel Symbols from a Metric

Given a metric $g_{\mu\nu}$ in coordinates $\{x^\mu\}$, compute all Christoffel symbols $\Gamma^\sigma{}_{\mu\nu}$ as follows.

1. **Write down the metric components.** List every nonvanishing $g_{\mu\nu}$ and note which coordinates each component depends on. For a diagonal metric, the only nonvanishing components are $g_{11}, g_{22}, \ldots$

2. **Compute the inverse metric $g^{\mu\nu}$.** For a diagonal metric this is trivial: $g^{\mu\mu} = 1/g_{\mu\mu}$ (no sum), and all off-diagonal components vanish.

3. **Compute the required partial derivatives.** For each target symbol $\Gamma^\sigma{}_{\mu\nu}$, you need the three derivatives
$$\partial_\mu g_{\nu\rho}, \quad \partial_\nu g_{\rho\mu}, \quad \partial_\rho g_{\mu\nu}$$
appearing in the Christoffel formula (2.9). Most of these vanish for diagonal metrics with few coordinate-dependent components.

4. **Apply the formula.**
$$\Gamma^\sigma{}_{\mu\nu} = \frac{1}{2}\, g^{\sigma\rho}\big(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big)$$
For a diagonal metric, the sum over ρ collapses to the single term $\rho = \sigma$ (since $g^{\sigma\rho} = 0$ for $\rho \neq \sigma$).

5. **Exploit symmetry.** $\Gamma^\sigma{}_{\mu\nu} = \Gamma^\sigma{}_{\nu\mu}$, so you only need to compute symbols with $\mu \leq \nu$.

Remark 2.10.1 (Diagonal metric shortcut). For a diagonal metric $g_{\mu\nu} = \mathrm{diag}(g_{11}, g_{22}, \ldots)$, the nonvanishing Christoffel symbols come in only three types (no sum on repeated indices here):

$$\Gamma^\mu{}_{\mu\mu} = \frac{1}{2g_{\mu\mu}}\frac{\partial g_{\mu\mu}}{\partial x^\mu}\,, \tag{2.17}$$

$$\Gamma^\mu{}_{\mu\nu} = \Gamma^\mu{}_{\nu\mu} = \frac{1}{2g_{\mu\mu}}\frac{\partial g_{\mu\mu}}{\partial x^\nu} \qquad (\mu \neq \nu)\,, \tag{2.18}$$

$$\Gamma^\mu{}_{\nu\nu} = -\frac{1}{2g_{\mu\mu}}\frac{\partial g_{\nu\nu}}{\partial x^\mu} \qquad (\mu \neq \nu)\,. \tag{2.19}$$

All other components vanish. These formulas dramatically reduce the work.

2.10.2 Algorithm: The Riemann Tensor from Christoffel Symbols

Given the Christoffel symbols, compute each component of ${R^\rho}_{\sigma\mu\nu}$ using (2.13):

$$ {R^\rho}_{\sigma\mu\nu} = \partial_\mu {\Gamma^\rho}_{\nu\sigma} - \partial_\nu {\Gamma^\rho}_{\mu\sigma} + {\Gamma^\rho}_{\mu\lambda}\,{\Gamma^\lambda}_{\nu\sigma} - {\Gamma^\rho}_{\nu\lambda}\,{\Gamma^\lambda}_{\mu\sigma} $$

1. **Index structure.** The Riemann tensor is antisymmetric in its last two indices: ${R^\rho}_{\sigma\mu\nu} = -{R^\rho}_{\sigma\nu\mu}$. So only compute components with $\mu < \nu$.

2. **Partial derivatives of Christoffels.** For each component, differentiate the relevant Christoffel symbols with respect to the coordinates x^μ and x^ν.

3. **Quadratic terms.** For each component, sum over the dummy index λ in the two $\Gamma\Gamma$ products. Only terms where *both* Christoffel symbols in the product are nonzero contribute.

4. **Independent components.** In n dimensions, the Riemann tensor has $n^2(n^2-1)/12$ independent components: 1 in 2D, 6 in 3D, 20 in 4D. In 2D, the single independent component determines everything.

2.10.3 Algorithm: Ricci Tensor and Ricci Scalar

1. **Ricci tensor.** Contract the first and third indices of the Riemann tensor:

$$ R_{\sigma\nu} = {R^\lambda}_{\sigma\lambda\nu} = \sum_\lambda {R^\lambda}_{\sigma\lambda\nu}\,. $$

   In 2D this is a single sum; in 4D it is a sum of four terms.

2. **Ricci scalar.** Trace with the inverse metric:

$$ R = g^{\sigma\nu} R_{\sigma\nu} = \sum_\sigma g^{\sigma\sigma} R_{\sigma\sigma} $$

   (the last equality holding for a diagonal metric).

2.10.4 Worked Example: The 2-Sphere

We carry out the *complete* calculation for the 2-sphere metric

$$ ds^2 = R^2\,d\theta^2 + R^2\sin^2\theta\,d\phi^2\,, \tag{2.20} $$

with coordinates $x^1 = \theta$, $x^2 = \phi$.

Step 1: Metric components.

$$ g_{\theta\theta} = R^2\,, \qquad g_{\phi\phi} = R^2\sin^2\theta\,, \qquad g_{\theta\phi} = g_{\phi\theta} = 0\,. $$

Only $g_{\phi\phi}$ depends on a coordinate (θ).

Step 2: Inverse metric.

$$g^{\theta\theta} = \frac{1}{R^2}\,, \qquad g^{\phi\phi} = \frac{1}{R^2 \sin^2\theta}\,.$$

Step 3: Partial derivatives of the metric. The only nonvanishing derivative is

$$\partial_\theta g_{\phi\phi} = R^2 \cdot 2\sin\theta\cos\theta = R^2 \sin 2\theta\,.$$

All other $\partial_\mu g_{\nu\rho} = 0$.

Step 4: Christoffel symbols. Using the diagonal shortcut formulas (2.17)–(2.19):

- Type (2.17) ($\Gamma^\mu{}_{\mu\mu}$): Both $\partial_\theta g_{\theta\theta} = 0$ and $\partial_\phi g_{\phi\phi} = 0$, so $\Gamma^\theta{}_{\theta\theta} = 0$ and $\Gamma^\phi{}_{\phi\phi} = 0$.
- Type (2.18) ($\Gamma^\mu{}_{\mu\nu}$ with $\mu \neq \nu$):

$$\Gamma^\phi{}_{\phi\theta} = \Gamma^\phi{}_{\theta\phi} = \frac{1}{2g_{\phi\phi}}\,\partial_\theta g_{\phi\phi} = \frac{R^2 \sin 2\theta}{2R^2 \sin^2\theta} = \frac{\cos\theta}{\sin\theta} = \cot\theta\,.$$

  Also $\Gamma^\theta{}_{\theta\phi} = \frac{1}{2g_{\theta\theta}}\partial_\phi g_{\theta\theta} = 0$.
- Type (2.19) ($\Gamma^\mu{}_{\nu\nu}$ with $\mu \neq \nu$):

$$\Gamma^\theta{}_{\phi\phi} = -\frac{1}{2g_{\theta\theta}}\,\partial_\theta g_{\phi\phi} = -\frac{R^2 \sin 2\theta}{2R^2} = -\sin\theta\cos\theta\,.$$

  Also $\Gamma^\phi{}_{\theta\theta} = -\frac{1}{2g_{\phi\phi}}\partial_\phi g_{\theta\theta} = 0$.

Result: The nonvanishing Christoffel symbols are

$$\boxed{\Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta\,, \qquad \Gamma^\phi{}_{\theta\phi} = \Gamma^\phi{}_{\phi\theta} = \cot\theta} \tag{2.21}$$

Verify this with `christoffel_symbols.wl`.

Step 5: Riemann tensor. In 2D there is only one independent component. We compute $R^\theta{}_{\phi\theta\phi}$ (i.e., $\rho = \theta$, $\sigma = \phi$, $\mu = \theta$, $\nu = \phi$):

$$\begin{aligned}
R^\theta{}_{\phi\theta\phi} &= \partial_\theta \Gamma^\theta{}_{\phi\phi} - \partial_\phi \Gamma^\theta{}_{\theta\phi} + \Gamma^\theta{}_{\theta\lambda}\,\Gamma^\lambda{}_{\phi\phi} - \Gamma^\theta{}_{\phi\lambda}\,\Gamma^\lambda{}_{\theta\phi} \\
&= \partial_\theta(-\sin\theta\cos\theta) - 0 + \Gamma^\theta{}_{\theta\theta}\,\Gamma^\theta{}_{\phi\phi} + \Gamma^\theta{}_{\theta\phi}\,\Gamma^\phi{}_{\phi\phi} \\
&\quad - \Gamma^\theta{}_{\phi\theta}\,\Gamma^\theta{}_{\theta\phi} - \Gamma^\theta{}_{\phi\phi}\,\Gamma^\phi{}_{\theta\phi} \\
&= (-\cos^2\theta + \sin^2\theta) + 0 + 0 - 0 - (-\sin\theta\cos\theta)(\cot\theta) \\
&= -\cos^2\theta + \sin^2\theta + \cos^2\theta \\
&= \sin^2\theta\,.
\end{aligned} \tag{2.22}$$

Step 6: Ricci tensor.

$$R_{\phi\phi} = R^{\lambda}{}_{\phi\lambda\phi} = R^{\theta}{}_{\phi\theta\phi} + R^{\phi}{}_{\phi\phi\phi} = \sin^2\theta + 0 = \sin^2\theta\,. \tag{2.23}$$

For $R_{\theta\theta}$, we need $R^{\phi}{}_{\theta\phi\theta}$. Lower the index and use the antisymmetry of the fully lowered tensor in each index pair, $R_{\phi\theta\phi\theta} = -R_{\theta\phi\phi\theta} = +R_{\theta\phi\theta\phi}$:

$$R^{\phi}{}_{\theta\phi\theta} = g^{\phi\phi}\, R_{\phi\theta\phi\theta} = g^{\phi\phi}\, R_{\theta\phi\theta\phi} = g^{\phi\phi}\, g_{\theta\theta}\, R^{\theta}{}_{\phi\theta\phi} = \frac{R^2\sin^2\theta}{R^2\sin^2\theta} = 1\,,$$

so

$$R_{\theta\theta} = R^{\lambda}{}_{\theta\lambda\theta} = R^{\theta}{}_{\theta\theta\theta} + R^{\phi}{}_{\theta\phi\theta} = 0 + 1 = 1\,. \tag{2.24}$$

Step 7: Ricci scalar.

$$R = g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = \frac{1}{R^2}\cdot 1 + \frac{1}{R^2\sin^2\theta}\cdot\sin^2\theta = \frac{1}{R^2} + \frac{1}{R^2} = \boxed{\frac{2}{R^2}}\,. \tag{2.25}$$

This is the standard result: a sphere of radius R has constant positive Gaussian curvature $K = 1/R^2$, and in 2D the Ricci scalar equals $2K$.

2.10.5 Tips and Tricks

1. **Start with the diagonal shortcut.** Most metrics encountered in GR courses are diagonal. Use eqs. (2.17)–(2.19) rather than the full Christoffel formula.

2. **Spot vanishing components early.** A Christoffel symbol $\Gamma^{\sigma}{}_{\mu\nu}$ vanishes whenever the relevant partial derivatives of the metric are all zero. Before computing, ask: does $g_{\mu\nu}$ depend on x^{σ}? Does $g_{\sigma\mu}$ depend on x^{ν}? If not, the symbol is likely zero.

3. **Count before you compute.** In n dimensions there are $n^2(n+1)/2$ potentially independent Christoffel symbols (accounting for the $\mu\nu$ symmetry). In 2D that is 6; in 4D it is 40. Most are zero for typical metrics — identify the nonvanishing ones first.

4. **Use Mathematica to verify.** Always check hand calculations with `christoffel_symbols.wl`. Even experts make sign and index errors in these computations.

5. **Look for symmetry.** If the metric is independent of a coordinate x^{α} (a Killing symmetry), many Christoffel symbols simplify or vanish. In the 2-sphere example, the metric is ϕ-independent, which immediately kills several terms.

2.11 Standard Computational Tools

The hand-rolled functions defined in `christoffel_symbols.wl` are pedagogically valuable — they make the connection between the textbook formula and the computation explicit. However, Mathematica also provides built-in tools for GR calculations that you should know about.

Wolfram Function Repository. The `ResourceFunction` interface provides one-liner access to several GR computations:

- `ResourceFunction["ChristoffelSymbol"][g, coords]` — returns the Christoffel symbols as a plain $n \times n \times n$ array, identical in format to our custom function. We have verified that this matches our code exactly for both the 2-sphere and Schwarzschild metrics (see `verify_against_textbooks.wl`).
- `ResourceFunction["RicciScalar"][g, coords]` — returns the Ricci scalar. Works for simple metrics (e.g., the 2-sphere) but may not fully evaluate for more complex ones.
- `ResourceFunction["RiemannTensor"]` and `ResourceFunction["RicciTensor"]` — these exist but return opaque tensor objects rather than plain arrays, making them difficult to use for further computation.

Remark 2.11.1. For this reason, we use `ResourceFunction["ChristoffelSymbol"]` as a cross-check throughout the tutorial, but maintain our custom functions for the full Christoffel → Riemann → Ricci → scalar pipeline, since the built-in ResourceFunction versions of the higher-order tensors do not return plain arrays. All results have been cross-validated against textbook equations; see `Mathematica/verify_against_textbooks.wl` for the complete test suite.

The xAct Package. For professional-level tensor algebra in Mathematica, the **xAct** suite (available at http://www.xact.es/) is the gold standard. Its component packages include:

- **xTensor**: abstract tensor algebra (index manipulation, symmetry, canonicalization)
- **xCoba**: component computations in coordinate bases
- **xPert**: perturbation theory (useful for linearized gravity, Module 1e)

xAct excels at abstract manipulations — e.g., proving tensor identities, computing perturbation expansions, and working with complicated index symmetries. It is less convenient for the simple "plug in a metric, get numbers out" workflow used in this tutorial, which is why we do not use it as our primary tool. However, if your research involves heavy tensor algebra (e.g., higher-order perturbation theory, non-trivial spacetimes, or abstract index calculations), xAct is the right tool.

Installation:

```
(* Download from http://www.xact.es/download.html *)
(* Unpack into ~/Library/Wolfram/Applications/ (macOS) *)
(* Then: *)
Needs["xAct`xTensor`"]
Needs["xAct`xCoba`"]
```

2.12 Summary

- A manifold is locally flat but may have global curvature; the metric $g_{\mu\nu}(x)$ encodes the geometry.
- The covariant derivative ∇_μ generalizes the partial derivative to curved spacetime, using the connection $\Gamma^\sigma{}_{\mu\nu}$.

- The Christoffel symbols are uniquely determined by the metric via (2.9) (torsion-free, metric-compatible).
- Parallel transport depends on the path; this path-dependence is measured by the Riemann curvature tensor.
- Geodesics — paths that parallel-transport their own tangent vector — are the trajectories of freely falling particles and light rays.
- Einstein's equation relates curvature ($R_{\mu\nu}$) to matter content ($T_{\mu\nu}$).

2.13 Problems

Exercise 2.13.1. Compute all nonvanishing Christoffel symbols for the 2-sphere metric $ds^2 = R^2(d\theta^2 + \sin^2\theta\, d\phi^2)$ by hand, then verify with `christoffel_symbols.wl`.

Exercise 2.13.2. For the flat plane in polar coordinates, $ds^2 = dr^2 + r^2 d\phi^2$:

(a) Compute the Christoffel symbols.

(b) Compute the Riemann tensor and verify that the space is flat ($R^{\rho}{}_{\sigma\mu\nu} = 0$).

(c) Write down the geodesic equations and show that straight lines in Cartesian coordinates ($x = a + b\lambda$, $y = c + d\lambda$) satisfy them.

Exercise 2.13.3. For the 2-sphere metric, compute:

(a) The Riemann tensor $R^{\theta}{}_{\phi\theta\phi}$.

(b) The Ricci tensor $R_{\theta\theta}$ and $R_{\phi\phi}$.

(c) The Ricci scalar R. Show that $R = 2/R^2$ (constant positive curvature, as expected for a sphere of radius R).

Exercise 2.13.4. A vector initially pointing east is parallel-transported from the equator of a unit sphere ($R = 1$) along the following path: north along the meridian $\phi = 0$ to the pole, then south along $\phi = \alpha$ back to the equator. Show that the vector has rotated by angle α. (Hint: solve the parallel transport equation (2.11) along each segment, or use the geometric argument that the rotation equals the solid angle enclosed.)

Exercise 2.13.5. Show that great circles are geodesics of the 2-sphere. Specifically, show that the equator $\theta = \pi/2$, $\phi = \lambda$ satisfies the geodesic equation (2.12) with the Christoffel symbols from Exercise 2.13.1.

Chapter 3

The Schwarzschild Solution

> *In GR, the unique spherically symmetric vacuum solution is the Schwarzschild metric; it is second only to Minkowski space in the list of important spacetimes.* — Sean Carroll, *Spacetime and Geometry*

Having developed the general machinery of differential geometry in Module 1b, we now apply it to the most important exact solution in GR: the **Schwarzschild metric**, which describes the spacetime geometry outside any spherically symmetric, non-rotating mass. This metric governs the gravitational field of stars, planets, and non-rotating black holes, and is the starting point for understanding gravitational lensing.

Companion Mathematica files:

- `schwarzschild_metric.wl` — derive and verify the Schwarzschild metric: compute Christoffel symbols, Riemann tensor, verify $R_{\mu\nu} = 0$, and compute the Kretschmann scalar
- `gravitational_redshift.nb` — interactive plots of gravitational redshift, time dilation, and Flamm's paraboloid embedding diagram

3.1 The Equivalence Principle

Before diving into the Schwarzschild solution, it is worth pausing to ask a foundational question: *why* should gravity be described by the geometry of spacetime at all? The answer lies in the **equivalence principle**, arguably the single most important physical insight behind general relativity.

3.1.1 The Weak Equivalence Principle

The oldest form of the equivalence principle dates back to Galileo: *all objects fall at the same rate, regardless of their composition or internal structure.* More precisely, the **weak equivalence principle** (WEP) states that the **inertial mass** m_i (which appears in Newton's second law, $F = m_i\, a$) is equal to the **gravitational mass** m_g (which appears in Newton's law of gravitation, $F = m_g\, g$):

$$m_i = m_g\,. \tag{3.1}$$

This equality has no explanation within Newtonian mechanics — it is simply an empirical fact. Yet it has been tested to extraordinary precision: Eötvös-type torsion-balance experiments (Dicke, Braginsky, and the Eöt-Wash group) verified this to better than one part in 10^{13}, and the MICROSCOPE satellite mission (2022) extended the limit by two orders of magnitude:

$$\frac{|m_i - m_g|}{m_g} < 10^{-15}\,. \tag{3.2}$$

The universality of free fall is not an accident — Einstein elevated it to a *principle*.

3.1.2 Einstein's Elevator

Einstein's great insight was to take the WEP seriously as a fundamental principle, not merely an empirical coincidence. He expressed this through the famous **elevator thought experiment**:

> *An observer in a closed, windowless elevator cannot distinguish between the following two situations:*
>
> (a) *The elevator is at rest on the surface of the Earth, where the gravitational acceleration is g.*
> (b) *The elevator is in empty space, accelerating upward at g.*

Conversely, an observer in free fall in a gravitational field ("falling elevator") experiences exactly the same conditions as an observer floating in empty space, far from any gravitating body. Dropped objects float alongside the observer; there is no local experiment that can detect the gravitational field.

This equivalence between gravity and acceleration is *local*: it holds in a sufficiently small region of spacetime where tidal effects (variations of the gravitational field across the elevator) can be neglected.

3.1.3 The Strong Equivalence Principle

Einstein extended the elevator argument to its fullest form as the **strong equivalence principle** (SEP):

> *The results of any local experiment (gravitational or not) in a freely falling reference frame are indistinguishable from those of the same experiment in special relativity.*

This has a profound consequence. If a freely falling observer experiences special relativity locally, then gravity cannot be a "force" acting in Minkowski spacetime — it must instead be encoded in the *geometry* of spacetime itself. Different freely falling observers at different locations each see flat (Minkowski) spacetime locally, but the global structure connecting these local frames is curved. Gravity *is* the curvature of spacetime.

3.1.4 From the Equivalence Principle to Geodesics

The equivalence principle connects directly to the mathematical framework developed in Module 1b. A freely falling object experiences no forces in its local (freely falling) frame — it follows a "straight line" through its local Minkowski patch of spacetime. But a straight line in a curved manifold is precisely a **geodesic** (Module 1b, Section 2.7).

Therefore, the equivalence principle implies that *freely falling particles follow geodesics of the spacetime metric*. This is the physical content of the geodesic equation:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu{}_{\alpha\beta}\,\frac{dx^\alpha}{d\tau}\,\frac{dx^\beta}{d\tau} = 0\,. \tag{3.3}$$

In Newtonian language, the Christoffel symbols $\Gamma^\mu{}_{\alpha\beta}$ play the role of gravitational "forces," but in GR they are geometric quantities — they describe how the coordinate system (not the particle) is "accelerating."

Remark 3.1.1. The equivalence principle tells us that gravity *curves* spacetime, but it does not tell us *how* the curvature is related to the matter content. That information requires Einstein's field equations $G_{\mu\nu} = 8\pi G\,T_{\mu\nu}/c^4$, which were stated in Module 1b and whose simplest vacuum solution ($T_{\mu\nu} = 0$) we derive in this module. The field equations will be used throughout the remaining modules as the foundation for gravitational lensing.

3.2 The Most General Spherically Symmetric Metric

We seek the metric outside a static, spherically symmetric mass distribution. The most general line element consistent with these symmetries can be written (Carroll eq. 5.11; cf. Congdon & Keeton eq. 3.64):

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2\,d\Omega^2\,, \tag{3.4}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\,d\phi^2$ is the metric on the unit 2-sphere, and $\alpha(r)$ and $\beta(r)$ are functions to be determined from Einstein's equation.

Remark 3.2.1. The "static" assumption means the metric is independent of t and has no $dt\,dr$ cross terms. "Spherically symmetric" means the angular part takes the form $r^2d\Omega^2$, with r chosen so that spheres at constant r have area $4\pi r^2$.

3.3 Deriving the Schwarzschild Metric

We solve Einstein's vacuum equation $R_{\mu\nu} = 0$ for the metric (3.4). The strategy follows Carroll Sec. 5.1:

Step 1: Christoffel symbols. The nonvanishing Christoffel symbols for (3.4) are (Carroll eq. 5.12):

$$\begin{aligned}
&\Gamma^t{}_{tr} = \alpha' \,, \quad \Gamma^r{}_{tt} = e^{2(\alpha-\beta)}\alpha' \,, \quad \Gamma^r{}_{rr} = \beta' \,, \\
&\Gamma^r{}_{\theta\theta} = -r\, e^{-2\beta} \,, \quad \Gamma^r{}_{\phi\phi} = -r\, e^{-2\beta}\sin^2\theta \,, \\
&\Gamma^\theta{}_{r\theta} = \Gamma^\phi{}_{r\phi} = \frac{1}{r} \,, \quad \Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta \,, \quad \Gamma^\phi{}_{\theta\phi} = \cot\theta \,,
\end{aligned} \tag{3.5}$$

where primes denote d/dr.

Step 2: Ricci tensor. Computing the Ricci tensor components (Carroll eq. 5.14):

$$R_{tt} = e^{2(\alpha-\beta)}\left[\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{2}{r}\alpha'\right] , \tag{3.6}$$

$$R_{rr} = -\alpha'' - (\alpha')^2 + \alpha'\beta' + \frac{2}{r}\beta' \,, \tag{3.7}$$

$$R_{\theta\theta} = e^{-2\beta}\big[r(\beta' - \alpha') - 1\big] + 1 \,, \tag{3.8}$$

$$R_{\phi\phi} = R_{\theta\theta}\sin^2\theta \,. \tag{3.9}$$

Step 3: Solve $R_{\mu\nu} = 0$. From $e^{-2(\alpha-\beta)}R_{tt} + R_{rr} = 0$, we get $\frac{2}{r}(\alpha' + \beta') = 0$, hence (Carroll eq. 5.17):

$$\alpha = -\beta + \text{const} \,. \tag{3.10}$$

By rescaling the time coordinate, we set the constant to zero: $\alpha = -\beta$.

Setting $R_{\theta\theta} = 0$ (Carroll eq. 5.18):

$$e^{2\alpha}(1 + 2r\alpha') = 1 \quad \Longrightarrow \quad \partial_r(r\, e^{2\alpha}) = 1 \,. \tag{3.11}$$

Integrating:

$$e^{2\alpha} = 1 - \frac{R_S}{r} \,, \tag{3.12}$$

where R_S is an integration constant.

Step 4: Identify R_S. In the weak-field limit ($r \to \infty$), the metric must reduce to $g_{tt} \approx -(1 + 2\Phi/c^2)$ with $\Phi = -GM/r$. Comparing with $g_{tt} = -(1 - R_S/r)$, we identify the **Schwarzschild radius** (Carroll eq. 5.23):

$$\boxed{R_S = \frac{2GM}{c^2}} \tag{3.13}$$

Result: The Schwarzschild Metric.

$$\boxed{ds^2 = -\left(1 - \frac{R_S}{r}\right)c^2\, dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2} \tag{3.14}$$

This is the unique vacuum solution with spherical symmetry (**Birkhoff's theorem**; Carroll Sec. 5.2). We verify $R_{\mu\nu} = 0$ symbolically in `schwarzschild_metric.wl`.

3.4 Birkhoff's Theorem

Birkhoff's theorem states that the Schwarzschild metric is the *unique* spherically symmetric vacuum solution to Einstein's equation. The remarkable consequence: even a *time-dependent* spherically symmetric source (e.g., a pulsating star) produces a *static* exterior metric (Carroll Sec. 5.2).

This is the GR analogue of Newton's shell theorem: the gravitational field outside a spherically symmetric mass depends only on the total enclosed mass M, not on how the mass is distributed.

3.5 Gravitational Time Dilation

A clock at radial coordinate r in the Schwarzschild spacetime runs at a rate different from a clock at infinity. For a stationary clock ($dr = d\theta = d\phi = 0$), the proper time is related to coordinate time by (Congdon & Keeton eq. 3.65):

$$d\tau = \sqrt{1 - \frac{R_S}{r}}\, dt\,. \tag{3.15}$$

Since $\sqrt{1 - R_S/r} < 1$ for $r > R_S$, clocks closer to the mass run *slower* than clocks farther away. At $r = R_S$, the time dilation becomes infinite — this is the event horizon.

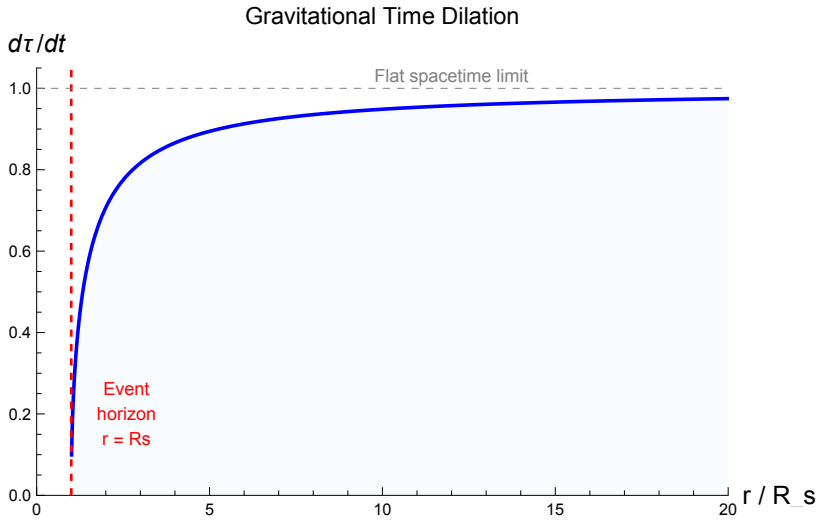


Figure 3.1: Gravitational time dilation factor $d\tau/dt = \sqrt{1 - R_S/r}$ as a function of r/R_S. Clocks near the horizon ($r \to R_S$) are dramatically slowed. Far from the mass ($r \gg R_S$), the factor approaches unity (flat spacetime).

3.6 Gravitational Redshift

A photon emitted at radius $r_{\rm em}$ with frequency $\nu_{\rm em}$ is observed at $r_{\rm obs} \to \infty$ with frequency:

$$\frac{\nu_{\rm obs}}{\nu_{\rm em}} = \frac{\sqrt{1 - R_S/r_{\rm em}}}{\sqrt{1 - R_S/r_{\rm obs}}} \xrightarrow{r_{\rm obs}\to\infty} \sqrt{1 - \frac{R_S}{r_{\rm em}}}\,. \tag{3.16}$$

Since $\nu_{\rm obs} < \nu_{\rm em}$, the photon is **redshifted** as it climbs out of the gravitational potential well. The gravitational redshift parameter is:

$$z_{\rm grav} = \frac{\nu_{\rm em}}{\nu_{\rm obs}} - 1 = \left(1 - \frac{R_S}{r_{\rm em}}\right)^{-1/2} - 1\,. \tag{3.17}$$

In the weak-field limit ($R_S/r \ll 1$): $z_{\rm grav} \approx GM/(rc^2)$ (Congdon & Keeton eq. 3.63).

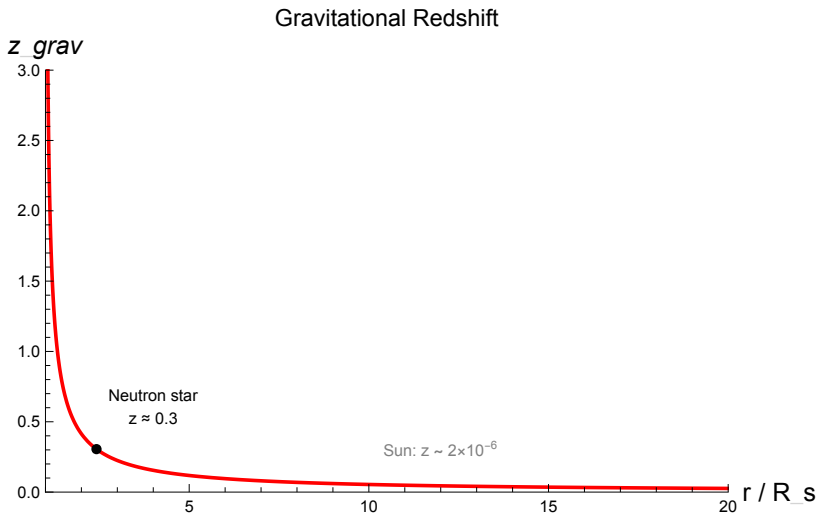


Figure 3.2: Gravitational redshift $z_{\rm grav}$ as a function of emission radius r/R_S. The redshift diverges at the horizon. The black dot marks a typical neutron star ($r/R_S \approx 2.4$, $z \approx 0.3$). For the Sun, $z \approx 2.1 \times 10^{-6}$.

Remark 3.6.1. The gravitational redshift was one of the three classical tests of GR proposed by Einstein, along with the perihelion precession of Mercury and the bending of light. It has been measured to high precision using atomic clocks at different altitudes (e.g., Pound–Rebka experiment, 1959; GPS satellite corrections).

3.7 The Event Horizon and Singularities

The Schwarzschild metric has two apparent singularities: at $r = 0$ and at $r = R_S$. Their nature is quite different (Carroll Sec. 5.3):

$r = R_S$ **(coordinate singularity).** The metric components diverge, but this is a *coordinate artifact*. The curvature remains finite, and the geometry is perfectly smooth. In appropriate coordinates (e.g., Eddington–Finkelstein or Kruskal–Szekeres), the metric is regular at $r = R_S$. However, $r = R_S$ is the **event horizon**: once a particle crosses it, it can never return.

$r = 0$ **(true singularity).** This is a genuine physical singularity where the curvature becomes infinite. We can confirm this with a curvature invariant: the **Kretschmann scalar**:

$$K = R_{\mu\nu\rho\sigma}\, R^{\mu\nu\rho\sigma} = \frac{12\, R_S^2}{r^6} = \frac{48\, G^2 M^2}{c^4\, r^6}\,. \tag{3.18}$$

This diverges as $r \to 0$ but is finite at $r = R_S$, confirming that only $r = 0$ is a true singularity. We verify eq. (3.18) in `schwarzschild_metric.wl`.

3.8 Inside the Event Horizon: The Coordinate Swap

The Schwarzschild metric (3.14) describes spacetime outside a spherical mass, but it also has something startling to say about the region $r < R_S$. Understanding this interior, even briefly, sharpens our physical intuition about what the Schwarzschild radius really means.

3.8.1 The Signature Flip

Recall the Schwarzschild line element:

$$ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2\,.$$

Outside the horizon ($r > R_S$), the factor $f(r) \equiv 1 - R_S/r$ is positive, so $g_{tt} < 0$ (timelike) and $g_{rr} > 0$ (spacelike).

Inside the horizon ($r < R_S$), this factor becomes *negative*: $f(r) < 0$. Consequently:

- $g_{tt} = -f(r)\, c^2 > 0$ — the t direction is now **spacelike**.
- $g_{rr} = f(r)^{-1} < 0$ — the r direction is now **timelike**.

The roles of r and t have swapped. Inside the horizon, r is a time coordinate and t is a space coordinate.

3.8.2 Physical Meaning: The Singularity Is a Moment in Time

Since r is timelike inside the horizon, decreasing r is the *future direction*. Every future-directed worldline must have $dr < 0$; the singularity at $r = 0$ is not a place you can steer away from — it is a *moment in time* that all interior observers inevitably reach.

Remark 3.8.1. You can no more avoid the singularity at $r = 0$ (once inside the horizon) than you can avoid next Tuesday. This is the precise sense in which a black hole traps everything: it is not that the singularity pulls you toward a point in space, but that the future itself points toward $r = 0$.

3.8.3 Why Nothing Escapes: Tilting Light Cones

For radial null rays ($ds^2 = 0$, $d\Omega = 0$) in Schwarzschild coordinates:

$$\frac{dr}{dt} = \pm c\left(1 - \frac{R_S}{r}\right). \tag{3.19}$$

Outside the horizon, the two signs give outgoing (+) and ingoing (−) light rays. At $r = R_S$, both solutions give $dr/dt = 0$: light "stands still" in these coordinates (a coordinate effect, not a physical one).

For $r < R_S$, the factor $(1 - R_S/r)$ is negative, so dr/dt has the *same sign* for both solutions — both "outgoing" and "ingoing" light rays have $dr < 0$ when $dt > 0$. In other words, **all future-directed light cones point toward smaller** r. No signal — not even light — can move to larger r inside the horizon.

3.8.4 Eddington–Finkelstein Coordinates

The pathological behavior at $r = R_S$ in Schwarzschild coordinates is a coordinate artifact. We can remove it by switching to **ingoing Eddington–Finkelstein coordinates**. Define the advanced time coordinate:

$$v = t + \frac{r_*}{c}, \qquad r_* = r + R_S \ln\left|\frac{r}{R_S} - 1\right|, \tag{3.20}$$

where r_* (units of length) is the **tortoise coordinate** (Carroll eq. 5.108), and dividing by c makes r_*/c a time shift so that v is a well-defined time coordinate in SI units. In terms of (v, r, θ, ϕ), the Schwarzschild line element becomes (Carroll eq. 5.111):

$$\boxed{ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dv^2 + 2c\, dv\, dr + r^2\, d\Omega^2} \tag{3.21}$$

This metric is manifestly regular at $r = R_S$: every component is finite and the determinant is nonzero. The horizon is simply the surface where outgoing light rays have $dr/dv = 0$; ingoing rays cross it smoothly.

Remark 3.8.2. The Eddington–Finkelstein form makes it transparent that the event horizon is a *one-way membrane*: ingoing light ($dv > 0$, $dr < 0$) crosses the horizon without incident, while outgoing light at $r = R_S$ remains at $r = R_S$ forever.

3.8.5 Relevance to Gravitational Lensing

Remark 3.8.3. For gravitational lensing, we work exclusively in the region $r \gg R_S$ — far outside the event horizon. Galaxy-mass lenses have $R_S \sim 0.1$ pc, while lensing occurs at scales of kiloparsecs. The black hole interior is therefore not directly relevant to lensing calculations. Nevertheless, understanding the interior clarifies *why* the Schwarzschild radius is special (it is where light cones tip over) and deepens physical intuition about the boundary conditions of the exterior solution.

3.9 The Weak-Field Limit

For $r \gg R_S$, the Schwarzschild metric reduces to the **weak-field** form (Congdon & Keeton eq. 3.71):

$$ds^2 \approx -\left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 + \left(1 - \frac{2\Phi}{c^2}\right) \left[dr^2 + r^2\, d\Omega^2\right], \tag{3.22}$$

where $\Phi = -GM/r$ is the Newtonian gravitational potential. This form is valid for *any* weak gravitational field, not just the Schwarzschild case, and will be the starting point for deriving the deflection angle of light in Module 2.

Remark 3.9.1. The weak-field metric (3.22) has an important feature: the spatial part is modified by a factor $(1 - 2\Phi/c^2)$ as well as the time part. This means that gravity affects both the *rate* of clocks (time dilation) and the *curvature of space*. Both effects contribute equally to the bending of light — this is why GR gives twice the Newtonian deflection angle.

3.10 Isotropic Coordinates

The standard Schwarzschild coordinates (t, r, θ, ϕ) have an important asymmetry: the radial and angular parts of the spatial metric carry *different* factors. In eq. (3.14), the dr^2 term is multiplied by $(1 - R_S/r)^{-1}$ while the angular part $r^2\, d\Omega^2$ has no such factor. The spatial part of the Schwarzschild metric is therefore *not* conformally flat — it is not proportional to the flat-space metric $dr^2 + r^2\, d\Omega^2$.

We can remedy this by introducing **isotropic coordinates**, in which the spatial metric *is* conformally flat.

3.10.1 The Isotropic Radial Coordinate

Define a new radial coordinate ρ via:

$$r = \rho \left(1 + \frac{R_S}{4\rho}\right)^2. \tag{3.23}$$

This transformation is invertible for $r > R_S$ (i.e., outside the event horizon), with the inverse $\rho = \frac{1}{2}(r - R_S/2 + \sqrt{r^2 - r\,R_S})$.

3.10.2 The Schwarzschild Metric in Isotropic Form

In the coordinates (t, ρ, θ, ϕ), the Schwarzschild metric becomes:

$$\boxed{ds^2 = -\left(\frac{1 - R_S/(4\rho)}{1 + R_S/(4\rho)}\right)^2 c^2\, dt^2 + \left(1 + \frac{R_S}{4\rho}\right)^4 \left(d\rho^2 + \rho^2\, d\Omega^2\right)} \tag{3.24}$$

The crucial feature is that the spatial part is now proportional to the *flat-space metric* $d\rho^2 + \rho^2\, d\Omega^2$ — the spatial geometry is **conformally flat**. The single conformal factor $(1 + R_S/(4\rho))^4$ multiplies all three spatial directions equally, hence the name "isotropic."

3.10.3 Weak-Field Expansion

To first order in R_S/ρ (the weak-field regime $\rho \gg R_S$), the isotropic metric reduces to:

$$ds^2 \approx -\left(1 - \frac{R_S}{\rho}\right) c^2\, dt^2 + \left(1 + \frac{R_S}{\rho}\right) \left(d\rho^2 + \rho^2\, d\Omega^2\right). \tag{3.25}$$

Since $R_S = 2GM/c^2$, we have $R_S/\rho = 2GM/(\rho\, c^2) = -2\Phi/c^2$ where $\Phi = -GM/\rho$ is the Newtonian potential. Equation (3.25) is therefore:

$$ds^2 \approx -\left(1 + \frac{2\Phi}{c^2}\right) c^2\, dt^2 + \left(1 - \frac{2\Phi}{c^2}\right) \left(d\rho^2 + \rho^2\, d\Omega^2\right), \tag{3.26}$$

which is precisely the weak-field metric (3.22) with r replaced by ρ. (In the weak-field regime, r and ρ agree to leading order.)

3.10.4 Connection to the Linearized Gravity Formalism

This result provides the explicit **bridge** between the exact Schwarzschild solution (this module) and the linearized gravity approach (Module 1e). In Module 1e, the weak-field metric is derived from first principles as a perturbation of flat spacetime: $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $h_{00} = -2\Phi/c^2$ and $h_{ij} = -2\Phi/c^2\, \delta_{ij}$. The isotropic form (3.26) shows that the Schwarzschild solution, when expanded to first order, reproduces exactly this linearized metric. The two descriptions — exact solution in curved coordinates, and perturbation theory in nearly flat coordinates — describe the *same physics* in different coordinate systems.

Remark 3.10.1. The isotropic form makes the connection to the parameterized post-Newtonian (PPN) formalism and the factor-of-2 in light deflection transparent. In eq. (3.26), both $h_{00} = -2\Phi/c^2$ (time perturbation) and $h_{ij} = -2\Phi/c^2\, \delta_{ij}$ (space perturbation) are proportional to R_S/ρ, confirming that the time and space perturbations are *equal*. In the PPN formalism, this equality corresponds to $\gamma_{\text{PPN}} = 1$, which is the GR prediction. As we saw in the weak-field limit discussion, this equal contribution of time and space curvature is precisely why GR predicts *twice* the Newtonian deflection angle for light.

3.11 Schwarzschild Radius for Astrophysical Objects

Object	**Mass**	R_S	**Actual Radius**
Earth	5.97×10^{24} kg	8.9 mm	6371 km
Sun	2×10^{30} kg	3.0 km	7×10^5 km
Neutron star	$1.4\, M_\odot$	4.1 km	~ 10 km
SgrA*	$4 \times 10^6\, M_\odot$	1.2×10^7 km	— (black hole)
Galaxy ($10^{12}\, M_\odot$)	$10^{12}\, M_\odot$	3×10^{12} km	—

For a galaxy-mass lens ($M \sim 10^{12}\, M_\odot$), $R_S \sim 0.1$ pc — far smaller than the galaxy itself. This confirms that the weak-field limit is appropriate for gravitational lensing by galaxies.

3.12 Embedding Diagram: Flamm's Paraboloid

To visualize the spatial curvature of the Schwarzschild geometry, we can embed the equatorial ($\theta = \pi/2$) spatial slice in 3D Euclidean space. At constant time t, the spatial metric on the equatorial plane is:

$$dl^2 = \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\phi^2\,. \tag{3.27}$$

Embedding this as a surface of revolution $z(r)$ in cylindrical coordinates (r, ϕ, z) where $dl^2 = (1 + z'^2)dr^2 + r^2 d\phi^2$, we find:

$$z(r) = 2\sqrt{R_S}\,\sqrt{r - R_S}\,, \tag{3.28}$$

which is **Flamm's paraboloid** (Figure 3.3).

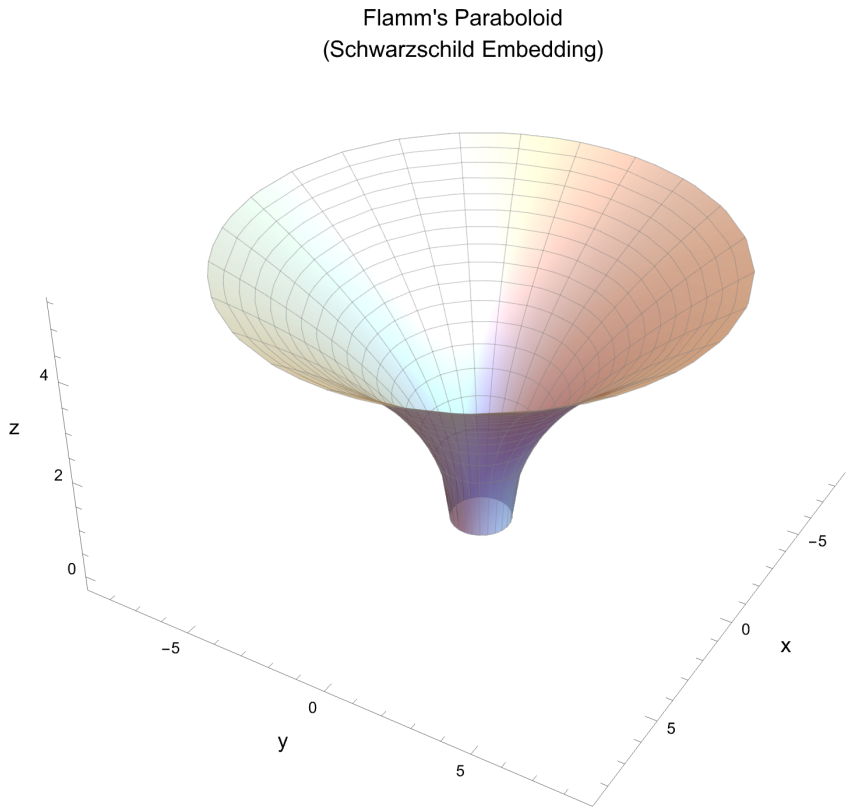


Figure 3.3: Flamm's paraboloid: the equatorial spatial geometry of the Schwarzschild spacetime embedded in 3D Euclidean space. The throat at $r = R_S$ represents the event horizon. The surface curves more steeply near the throat, visualizing the spatial curvature caused by gravity. An interactive version is in `gravitational_redshift.nb`.

3.13 Summary

- The Schwarzschild metric is the unique spherically symmetric vacuum solution (Birkhoff's theorem).

- It is parameterized by a single quantity: the mass M (or equivalently the Schwarzschild radius $R_S = 2GM/c^2$).
- Gravitational time dilation: clocks run slower in stronger gravitational fields; $d\tau = \sqrt{1 - R_S/r}\, dt$.
- Gravitational redshift: photons lose energy climbing out of a potential well.
- $r = R_S$ is a coordinate singularity (event horizon); $r = 0$ is a true curvature singularity.
- The weak-field limit ($ds^2 \approx -(1 + 2\Phi/c^2)c^2dt^2 + (1 - 2\Phi/c^2)d\ell^2$) will be the foundation for light deflection in Module 2.

3.14 Problems

Exercise 3.14.1. Using the reusable functions from Module 1b (`christoffel_symbols.wl`), verify that the Schwarzschild metric (3.14) satisfies $R_{\mu\nu} = 0$.

Exercise 3.14.2. Compute the Kretschmann scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ for the Schwarzschild metric and show that $K = 12\, R_S^2/r^6$. Verify that K is finite at $r = R_S$ but diverges at $r = 0$.

Exercise 3.14.3. Compute the Schwarzschild radius for:

(a) The Earth ($M = 5.97 \times 10^{24}$ kg)

(b) The Sun ($M = 1.99 \times 10^{30}$ kg)

(c) A typical galaxy lens ($M = 10^{12}\, M_\odot$)

In each case, compare R_S to the physical size of the object.

Exercise 3.14.4. A photon is emitted from the surface of a neutron star ($M = 1.4\, M_\odot$, $R = 10$ km).

(a) Compute the gravitational redshift z_{grav}.

(b) By what fraction is the photon's wavelength increased by the time it reaches a distant observer?

(c) Compare to the redshift from the surface of the Sun.

Exercise 3.14.5. Derive the Flamm's paraboloid embedding (3.28) by embedding the equatorial spatial slice of the Schwarzschild geometry in 3D Euclidean space with cylindrical coordinates (r, ϕ, z).

Chapter 4

Geodesics and Orbits in Schwarzschild Spacetime

> *Together the conserved quantities E and L provide a convenient way to understand the orbits of particles in the Schwarzschild geometry.* — Sean Carroll, *Spacetime and Geometry*

In Module 1c we derived the Schwarzschild metric. Now we ask: how do particles and light *move* in this spacetime? The answer comes from the geodesic equation, which we reduce to a one-dimensional effective potential problem using Killing vectors and conservation laws. This module derives the effective potential for massive and massless particles, identifies the key features (ISCO, photon sphere), and culminates in Mercury's perihelion precession — one of the three classical tests of GR.

Companion Mathematica files:

- `geodesic_equation.wl` — symbolic derivation of the geodesic equations, conserved quantities, effective potential, ISCO, photon sphere, and perihelion precession formula
- `effective_potential.nb` — interactive plots of effective potential with sliders for L and ϵ; orbit integration; precessing ellipses

4.1 Killing Vectors and Conservation Laws

The Schwarzschild metric is independent of both t and ϕ, which means it possesses two **Killing vectors** (Carroll Sec. 5.4):

$$K^\mu = (\partial_t)^\mu = (1, 0, 0, 0) \quad \text{(time translation)}, \tag{4.1}$$

$$R^\mu = (\partial_\phi)^\mu = (0, 0, 0, 1) \quad \text{(axial rotation)}. \tag{4.2}$$

For any geodesic $x^\mu(\lambda)$, the quantity $K_\mu \, dx^\mu/d\lambda$ is conserved along the path (Carroll eq. 5.54). Lowering the index with the metric gives the two conserved quantities (Carroll eqs. 5.61–5.62):

$$\boxed{E = \left(1 - \frac{R_S}{r}\right) \frac{dt}{d\lambda}} \tag{4.3}$$

$$\boxed{L = r^2 \frac{d\phi}{d\lambda}} \tag{4.4}$$

For massive particles ($\lambda = \tau$), E and L are the conserved energy and angular momentum *per unit mass*. For photons, they are the energy and angular momentum of the photon (up to normalization of the affine parameter).

Additionally, spherical symmetry allows us to restrict all orbits to the equatorial plane $\theta = \pi/2$ without loss of generality (Carroll eq. 5.56).

4.2 The Effective Potential

The normalization condition for the four-velocity gives (Carroll eq. 5.63):

$$-\left(1 - \frac{R_S}{r}\right)\left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{R_S}{r}\right)^{-1}\left(\frac{dr}{d\lambda}\right)^2 + r^2\left(\frac{d\phi}{d\lambda}\right)^2 = -\epsilon\,, \tag{4.5}$$

where $\epsilon = 1$ for massive particles (timelike geodesics) and $\epsilon = 0$ for photons (null geodesics).

Substituting the conserved quantities (4.3) and (4.4), we obtain a radial equation (Carroll eq. 5.65):

$$\frac{1}{2}\left(\frac{dr}{d\lambda}\right)^2 + V_{\text{eff}}(r) = \mathcal{E}\,, \tag{4.6}$$

where the **effective potential** is (Carroll eq. 5.66, with additive constant $+\frac{1}{2}\epsilon$ absorbed into $\mathcal{E}$; Congdon & Keeton eq. 3.80):

$$\boxed{V_{\text{eff}}(r) = -\epsilon\,\frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2 r^3}} \tag{4.7}$$

and $\mathcal{E} = \frac{1}{2}(E^2 - \epsilon c^2)$ is the effective "energy."

Remark 4.2.1. The three terms in V_{eff} have clear physical interpretations:

1. $-\epsilon\,GM/r$: the Newtonian gravitational potential (absent for photons since $\epsilon = 0$)
2. $L^2/(2r^2)$: the centrifugal barrier (identical to Newtonian mechanics)
3. $-GML^2/(c^2 r^3)$: the **GR correction** — this term has no Newtonian analogue and is responsible for all the distinctive features of relativistic orbits

The GR term is attractive at small r, which means the centrifugal barrier can be overcome — this is why particles can spiral into black holes, which is impossible in Newtonian gravity.

4.3 Circular Orbits and the ISCO

Circular orbits occur at extrema of the effective potential: $dV_{\text{eff}}/dr = 0$. For massive particles ($\epsilon = 1$), this gives (Carroll eq. 5.68):

$$GMr_c^2 - L^2 r_c + 3GML^2/c^2 = 0\,. \tag{4.8}$$

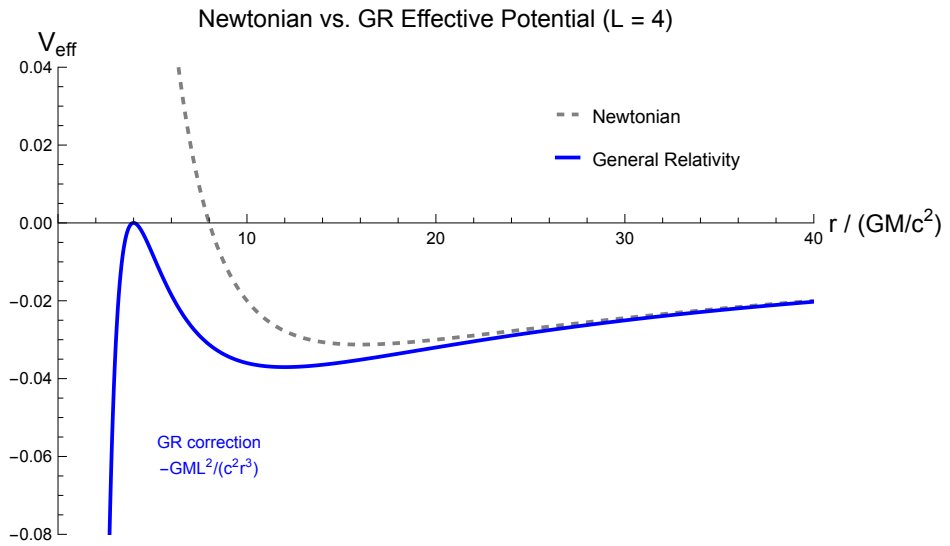


Figure 4.1: Comparison of Newtonian (dashed) and GR (solid) effective potentials for a massive particle with $L = 4\,GM/c$. The GR correction term $\propto 1/r^3$ pulls the potential down at small r, allowing particles to plunge past the centrifugal barrier. (cf. Carroll Figs. 5.4 and 5.5)

The two solutions are (Carroll eq. 5.71):

$$r_\pm = \frac{L^2}{2GM}\left(1 \pm \sqrt{1 - \frac{12G^2M^2}{L^2c^2}}\right). \tag{4.9}$$

- r_+ (outer): **stable** circular orbit (minimum of V_{eff})
- r_- (inner): **unstable** circular orbit (maximum of V_{eff})

Real solutions require $L^2 \geq 12\,G^2M^2/c^2$, i.e., $L \geq 2\sqrt{3}\,GM/c$ (Carroll eq. 5.73). The minimum stable circular orbit occurs when $r_+ = r_-$, giving the **innermost stable circular orbit (ISCO)** (Carroll eq. 5.74; Congdon & Keeton eq. 3.84):

$$\boxed{r_{\text{ISCO}} = \frac{6GM}{c^2} = 3\,R_S} \tag{4.10}$$

Remark 4.3.1. The ISCO is a fundamental scale in black hole astrophysics. Accreting matter in a thin disk spirals inward until it reaches the ISCO, then plunges into the black hole. The ISCO radius determines the efficiency of energy extraction from accretion and the inner edge of accretion disks observed in X-ray binaries and active galactic nuclei.

4.4 The Photon Sphere

For photons ($\epsilon = 0$), the effective potential simplifies to (Congdon & Keeton eq. 3.86):

$$V_{\text{eff}}^{\text{photon}}(r) = \frac{L^2}{2r^2} - \frac{GML^2}{c^2r^3}. \tag{4.11}$$

Setting $dV_{\text{eff}}/dr = 0$ gives a single circular orbit (Carroll eq. 5.70; Congdon & Keeton Sec. 3.3.3, prose after eq. 3.86):

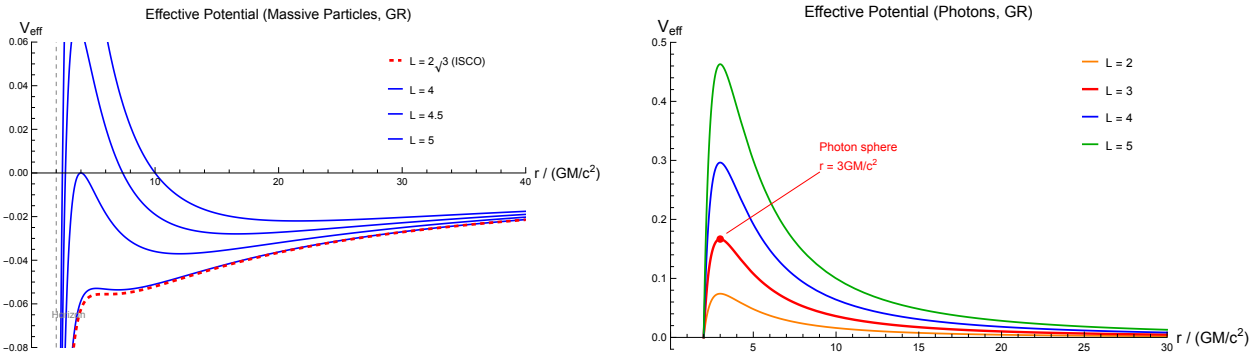
$$\boxed{r_{\text{photon}} = \frac{3GM}{c^2} = \frac{3}{2} R_S} \tag{4.12}$$

This is an *unstable* circular orbit — any perturbation causes the photon to either escape to infinity or spiral into the black hole. Nevertheless, the photon sphere plays a key role in strong gravitational lensing near black holes and is directly related to the "photon ring" observed by the Event Horizon Telescope.

4.5 Effective Potential: Newtonian vs. GR

The key qualitative differences between Newtonian and GR effective potentials are:

Feature	**Newtonian**	**GR**
Centrifugal barrier	Diverges as $r \to 0$	Overcome at small r
ISCO	None (stable at any r)	$r = 3R_S$
Photon orbits	None	Unstable at $r = 3R_S/2$
Plunge orbits	Impossible	Possible for $r < r_{\text{ISCO}}$



(a) Massive particles ($\epsilon = 1$)

(b) Photons ($\epsilon = 0$)

Figure 4.2: GR effective potentials for several values of angular momentum L (in GM/c units). **Left:** For massive particles, the red dashed curve shows $L = L_{\text{ISCO}} = 2\sqrt{3}\,GM/c$, where the stable and unstable circular orbits merge. **Right:** For photons, the potential always has a maximum at $r = 3GM/c^2$ (the photon sphere). Interactive versions in `effective_potential.nb`.

4.6 Perihelion Precession

In Newtonian gravity, bound orbits are perfect ellipses that close on themselves. In GR, the extra $-GML^2/(c^2r^3)$ term in the effective potential causes the orbit to *precess*: the perihelion (point of closest approach) advances by a small angle each orbit.

Following Carroll Sec. 5.5 and using the substitution $x = L^2/(GMr)$, the orbit equation becomes (Carroll eq. 5.79):

$$\frac{d^2x}{d\phi^2} + x = 1 + \frac{3G^2M^2}{L^2c^2}\,x^2\,. \tag{4.13}$$

The last term is the GR correction. Treating it as a perturbation ($x = x_0 + x_1$, with $x_0 = 1 + e\cos\phi$ the Keplerian solution), one finds the accumulated precession per orbit (Carroll eq. 5.92):

$$\boxed{\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)} = \frac{6\pi G^2M^2}{c^2L^2}} \tag{4.14}$$

where a is the semi-major axis and e the eccentricity.

For Mercury (Carroll eq. 5.96):

$$\Delta\phi_{\text{Mercury}} = 5.02 \times 10^{-7}\ \text{rad/orbit} = 0.104''/\text{orbit} = 43.0''/\text{century}\,. \tag{4.15}$$

This matched the previously unexplained 43 arcseconds/century discrepancy in Mercury's orbit and was one of Einstein's first confirmations of GR.

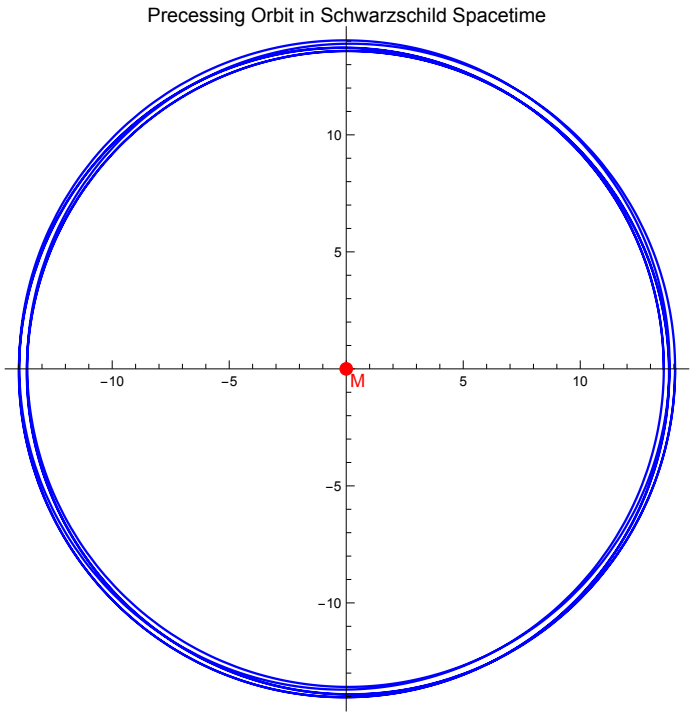


Figure 4.3: A precessing orbit in Schwarzschild spacetime, computed by numerically integrating the orbit equation. The perihelion advances by a small angle each orbit, tracing out a rosette pattern. (cf. Carroll Fig. 5.6)

4.7 Null Geodesics and the Deflection of Light (Preview)

For photons, the orbit equation (using $u = 1/r$) becomes (Congdon & Keeton eq. 3.87):

$$\frac{d\phi}{dr} = \pm\frac{L}{r^2}\left[\frac{E^2}{c^2} - \frac{L^2}{r^2}\left(1 - \frac{R_S}{r}\right)\right]^{-1/2} . \tag{4.16}$$

A photon approaching a mass M with impact parameter $b = cL/E$ (the perpendicular distance from the mass in the absence of deflection) will be deflected by an angle (Congdon & Keeton eq. 3.96):

$$\hat{\alpha} = \frac{4GM}{c^2 b} , \tag{4.17}$$

in the weak-field limit ($b \gg R_S$). This is *twice* the Newtonian prediction and was confirmed by Eddington's 1919 solar eclipse observations.

The full derivation of the deflection angle is the subject of Module 2. We mention it here because the effective potential framework makes it clear *why* light bends: photons follow null geodesics of the curved Schwarzschild spacetime, and the GR correction term in the effective potential deflects them more strongly than Newtonian gravity would predict.

4.8 Brief Overview: The Kerr Metric

Real astrophysical black holes are not perfectly described by the Schwarzschild solution because they *rotate*. The metric for a rotating, axially symmetric black hole is the **Kerr metric**, discovered by Roy Kerr in 1963. In Boyer–Lindquist coordinates, it takes the form:

$$ds^2 = -\left(1 - \frac{R_S r}{\Sigma}\right)c^2 dt^2 - \frac{2R_S r a \sin^2\theta}{\Sigma}\, c\, dt\, d\phi + \frac{\Sigma}{\Delta}\, dr^2 + \Sigma\, d\theta^2 + \frac{A \sin^2\theta}{\Sigma}\, d\phi^2 , \tag{4.18}$$

where

$$\Sigma = r^2 + a^2\cos^2\theta , \qquad \Delta = r^2 - R_S r + a^2 , \qquad A = (r^2 + a^2)^2 - a^2\Delta\sin^2\theta , \tag{4.19}$$

and $a = J/(Mc)$ is the **spin parameter** (angular momentum per unit mass, with dimensions of length). Note $0 \le a \le GM/c^2$ for a physical black hole.

Key features that differ from Schwarzschild:

- **Frame dragging:** The $dt\, d\phi$ cross-term forces particles near the black hole to co-rotate. This effect is absent in Schwarzschild.

- **Ergosphere:** The region $r_+ < r < (R_S + \sqrt{R_S^2 - 4a^2\cos^2\theta})/2$, where $r_+ = (R_S + \sqrt{R_S^2 - 4a^2})/2$ is the outer event horizon, is where even stationary observers must co-rotate with the black hole. (Note: at $a = 0$, $r_+ = R_S$, not $R_S/2$; the ergosphere collapses onto the Schwarzschild horizon. Only at maximum spin $a = GM/c^2$ does $r_+ = R_S/2$.)

- **Spin-dependent ISCO:** The ISCO radius depends on the spin parameter and the direction of the orbit (prograde vs. retrograde). For a maximally spinning black hole ($a = GM/c^2$): $r_{\text{ISCO}} = GM/c^2$ (prograde) vs. $r_{\text{ISCO}} = 9GM/c^2$ (retrograde), compared to $6GM/c^2$ for Schwarzschild.
- **Two horizons:** $r_\pm = GM/c^2 \pm \sqrt{(GM/c^2)^2 - a^2}$.

Remark 4.8.1. For gravitational lensing by galaxies and clusters, the Kerr metric is not needed: the lensing effect is dominated by the total enclosed mass, and the weak-field/thin-lens approximation (Module 1e and beyond) washes out spin effects at the angular scales of interest. However, Kerr lensing is an active research topic for *strong* lensing near individual black holes — for instance, the "photon ring" structure observed by the Event Horizon Telescope, where frame-dragging produces asymmetric brightness patterns.

4.9 Summary

- Killing vectors (∂_t, ∂_ϕ) yield conserved energy E and angular momentum L along geodesics.
- The radial motion reduces to a 1D effective potential problem: $V_{\text{eff}} = -\epsilon GM/r + L^2/(2r^2) - GML^2/(c^2r^3)$.
- The GR correction term $\propto 1/r^3$ is responsible for: ISCO at $r = 3R_S$, photon sphere at $r = 3R_S/2$, perihelion precession, and enhanced light deflection.
- Mercury's perihelion precesses by 43 arcsec/century — confirmed by observation and one of the first tests of GR.
- The Kerr metric generalizes Schwarzschild to rotating black holes (frame dragging, ergosphere, spin-dependent ISCO), but is not needed for galaxy/cluster-scale lensing.

4.10 Problems

Exercise 4.10.1. Starting from the normalization condition (4.5) and the conserved quantities E and L, derive the effective potential (4.7) for massive particles ($\epsilon = 1$). Show that in the Newtonian limit ($c \to \infty$, $R_S \to 0$), it reduces to $V_{\text{eff}}^{\text{Newton}} = -GM/r + L^2/(2r^2)$.

Exercise 4.10.2. Derive the ISCO radius by finding the value of L for which the two circular orbit solutions (4.9) coincide (i.e., the discriminant vanishes). Verify $r_{\text{ISCO}} = 6GM/c^2 = 3R_S$.

Exercise 4.10.3. Show that the only circular photon orbit in the Schwarzschild spacetime is at $r = 3GM/c^2 = 3R_S/2$, and that it is unstable (a maximum of $V_{\text{eff}}^{\text{photon}}$).

Exercise 4.10.4. Compute the perihelion precession of Mercury using eq. (4.14) with: $M = M_\odot$, $a = 5.79 \times 10^{10}$ m, $e = 0.2056$. Verify $\Delta\phi \approx 43''$/century (Mercury's orbital period is 87.97 days).

Exercise 4.10.5. Using the effective potential for massive particles with $L = 4GM/c$:

(a) Plot $V_{\text{eff}}(r)$ and identify the stable circular orbit radius, the unstable circular orbit radius, and the ISCO.

(b) For a particle with $\mathcal{E}$ slightly above $V_{\text{eff}}(r_+)$, describe qualitatively the type of orbit (bound precessing ellipse, scatter orbit, or plunge orbit).

Chapter 5

Linearized Gravity and the Weak-Field Metric

> *The weak-field limit described in this chapter is less restrictive, assuming that the field is still weak but it can vary with time, and without any restrictions on the motion of test particles. This will allow us to discuss phenomena that are absent or ambiguous in the Newtonian theory, such as gravitational radiation (where the field varies with time) and the deflection of light (which involves fast-moving particles).* — Sean Carroll, *Spacetime and Geometry*, Sect. 7.1

In Modules 1a–1d we developed the full machinery of general relativity and applied it to the exact Schwarzschild solution. That framework is necessary for understanding black holes and strong-field orbits, but *most gravitational lensing* — by galaxies, galaxy clusters, and large-scale structure — operates in a regime where gravity is weak: the Newtonian potential satisfies $|\Phi|/c^2 \ll 1$ and peculiar velocities are small compared to c. In this regime the metric can be treated as a small perturbation of flat Minkowski spacetime, and the resulting **linearized gravity** formalism is both simpler and more directly connected to the lens equation that we will develop in Module 4.

This module derives the weak-field metric from first principles, recovers Newtonian gravity as a limiting case, computes the deflection of light (reproducing the famous factor-of-two over the Newtonian prediction), and introduces the effective refractive index interpretation that provides the physical intuition behind Fermat's principle for lensing. The linearity of the perturbation equations also guarantees that deflection angles from multiple masses *superpose* — the foundation for treating extended mass distributions in later modules.

Companion Mathematica files:

- `weak_field_metric.wl` — symbolic verification of the Newtonian limit, weak-field deflection angle, factor-of-2, and figure generation
- `deflection_comparison.nb` — interactive comparison of exact Schwarzschild deflection vs. weak-field approximation; effective refractive index visualization

5.1 Introduction: Why Linearize?

Consider a photon passing the Sun at the solar limb. The relevant dimensionless quantity characterizing the strength of gravity is (Carroll eq. 4.21):

$$\frac{|\Phi|}{c^2} = \left.\frac{GM}{rc^2}\right|_{r=R_\odot} \approx \frac{6.674\times10^{-11}\times1.989\times10^{30}}{6.96\times10^{8}\times(3\times10^{8})^2} \approx 2.1\times10^{-6}\,. \tag{5.1}$$

For a typical galaxy lens ($M\sim10^{12}\,M_\odot$, $r\sim10$ kpc), the ratio is $|\Phi|/c^2\approx5\times10^{-6}$. Even for galaxy clusters ($M\sim10^{15}\,M_\odot$, $r\sim1$ Mpc), one finds $|\Phi|/c^2\sim5\times10^{-5}$; values of $\sim10^{-4}$ arise only near cluster cores at $r\lesssim100$ kpc.

In all of these astrophysical settings, gravity is *extremely* weak. This justifies expanding the metric perturbatively around flat spacetime and keeping only first-order terms — the **linearized gravity** approximation.

Remark 5.1.1. The only astrophysical setting where linearized gravity is *insufficient* for lensing is near black holes and neutron stars, where $|\Phi|/c^2$ approaches unity. Strong lensing by black holes (e.g., the photon ring observed by the Event Horizon Telescope) requires the full Schwarzschild or Kerr solution developed in Modules 1c and 1d.

5.2 The Weak-Field Expansion

We write the full spacetime metric as a perturbation of the Minkowski metric (Carroll eq. 7.1; Congdon & Keeton Sec. 3.3.2):

$$\boxed{g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}\,, \qquad |h_{\mu\nu}| \ll 1} \tag{5.2}$$

where $\eta_{\mu\nu}=\mathrm{diag}(-1,+1,+1,+1)$ is the flat Minkowski metric and $h_{\mu\nu}$ is the **metric perturbation**. The condition $|h_{\mu\nu}|\ll1$ means that spacetime is "almost flat" — deviations from Minkowski geometry are small.

The inverse metric to first order in h is (Carroll eq. 7.2):

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + \mathcal{O}\big(h^2\big)\,, \tag{5.3}$$

where indices on $h_{\mu\nu}$ are raised and lowered with η.

The Christoffel symbols to first order are (Carroll eq. 7.4):

$$\Gamma^{\alpha}_{\ \mu\nu} = \tfrac{1}{2}\eta^{\alpha\beta}\big(\partial_\mu h_{\nu\beta} + \partial_\nu h_{\mu\beta} - \partial_\beta h_{\mu\nu}\big) + \mathcal{O}\big(h^2\big)\,. \tag{5.4}$$

Remark 5.2.1. In linearized gravity we systematically discard all terms of order h^2 and higher. This means the Christoffel symbols are first order in h, the Riemann tensor is first order, and the Einstein equations become *linear* in $h_{\mu\nu}$. Linearity is the crucial simplification that allows superposition of gravitational effects.

5.3 The Newtonian Limit

We now show that the geodesic equation in the weak-field, slow-motion limit reduces to Newton's second law, thereby identifying the metric perturbation h_{00} with the Newtonian potential Φ.

5.3.1 Assumptions

The **Newtonian limit** requires three conditions (Carroll Sec. 4.1):

1. **Weak field:** $|h_{\mu\nu}| \ll 1$
2. **Slow motion:** $v \ll c$, so $dx^i/d\tau \ll dx^0/d\tau = c\,dt/d\tau$
3. **Static field:** $\partial_0 h_{\mu\nu} = 0$ (the gravitational field is not changing with time)

5.3.2 Derivation

The geodesic equation for a massive particle is (Carroll eq. 3.44):

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu{}_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0\,. \tag{5.5}$$

In the slow-motion limit, $dx^i/d\tau$ is negligible compared to $dx^0/d\tau = c\,dt/d\tau$, so the double sum over α, β is dominated by the $\alpha = \beta = 0$ term:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu{}_{00}\left(\frac{dx^0}{d\tau}\right)^2 \approx 0\,. \tag{5.6}$$

From eq. (5.4), using the static-field condition ($\partial_0 h_{\mu\nu} = 0$):

$$\Gamma^\mu{}_{00} = \tfrac{1}{2}\eta^{\mu\alpha}\big(\partial_0 h_{0\alpha} + \partial_0 h_{0\alpha} - \partial_\alpha h_{00}\big) = -\tfrac{1}{2}\eta^{\mu\alpha}\,\partial_\alpha h_{00}\,. \tag{5.7}$$

For the spatial components ($\mu = i$), this gives $\Gamma^i{}_{00} = -\frac{1}{2}\partial_i h_{00}$.

The spatial part of the geodesic equation then becomes:

$$\frac{d^2x^i}{d\tau^2} = \tfrac{1}{2}\partial_i h_{00}\left(\frac{dx^0}{d\tau}\right)^2 = \tfrac{1}{2}c^2\partial_i h_{00}\left(\frac{dt}{d\tau}\right)^2. \tag{5.8}$$

Using $d\tau \approx dt$ (valid for slow motion, since $\gamma \approx 1$), we get:

$$\frac{d^2x^i}{dt^2} = \tfrac{1}{2}c^2\,\partial_i h_{00}\,. \tag{5.9}$$

5.3.3 Identification with Newtonian Gravity

Newton's second law in a gravitational field is:

$$\frac{d^2x^i}{dt^2} = -\partial_i\Phi\,. \tag{5.10}$$

Comparing with eq. (5.9), we identify (Carroll eq. 4.19):

$$\boxed{h_{00} = -\frac{2\Phi}{c^2}} \tag{5.11}$$

The time-time component of the metric perturbation is determined by the Newtonian potential. For a point mass M, $\Phi = -GM/r$, so:

$$h_{00} = \frac{2GM}{c^2 r} = \frac{R_S}{r}\,, \tag{5.12}$$

where $R_S = 2GM/c^2$ is the Schwarzschild radius.

Exercise 5.3.1. Starting from the geodesic equation (5.5) with the perturbed metric $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, carry out the Newtonian limit derivation in full detail. Show that for slow-moving particles in a static, weak gravitational field:

(a) The only relevant Christoffel symbol is $\Gamma^i{}_{00} = -\frac{1}{2}\partial_i h_{00}$.

(b) The geodesic equation reduces to $d^2x^i/dt^2 = \frac{1}{2}c^2\,\partial_i h_{00}$.

(c) Comparing with Newton's law identifies $h_{00} = -2\Phi/c^2$.

Verify each step using `weak_field_metric.wl`.

5.4 The Weak-Field Metric

The Newtonian limit only determined h_{00}. To get the full weak-field metric — including the spatial components needed for light bending — we must go beyond the slow-motion assumption and use the linearized Einstein equations.

For a static, spherically symmetric source, the solution to the linearized Einstein equations in isotropic coordinates gives (Congdon & Keeton eq. 3.71; Carroll eq. 7.59):

$$\boxed{ds^2 = -\left(1 + \frac{2\Phi}{c^2}\right) c^2\,dt^2 + \left(1 - \frac{2\Phi}{c^2}\right)\left(dx^2 + dy^2 + dz^2\right)} \tag{5.13}$$

where $\Phi(\mathbf{x})$ is the Newtonian gravitational potential (with $\Phi < 0$ for attractive gravity). Equivalently:

$$h_{00} = -\frac{2\Phi}{c^2}\,, \qquad h_{ij} = -\frac{2\Phi}{c^2}\,\delta_{ij}\,. \tag{5.14}$$

Remark 5.4.1. The key point is that **both** the time and space components of the metric are perturbed, and by equal amounts. The Newtonian limit (which assumes slow motion) only probes h_{00}, because the spatial components of the four-velocity are negligible. But photons travel at c, so all components of the metric contribute equally to the propagation of light. This is why light bending in GR is *twice* the Newtonian prediction.

Exercise 5.4.1. The Schwarzschild metric in standard coordinates is:

$$ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2 dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\,d\Omega^2\,.$$

(a) Expand $(1 - R_S/r)^{-1} \approx 1 + R_S/r + \mathcal{O}((R_S/r)^2)$ and show that the metric becomes:

$$ds^2 \approx -\left(1 - \frac{R_S}{r}\right) c^2 dt^2 + \left(1 + \frac{R_S}{r}\right) dr^2 + r^2\,d\Omega^2\,.$$

(b) Transform to Cartesian-like coordinates and show that this matches the weak-field metric (5.13) with $\Phi = -GM/r$ to first order in R_S/r.

Verify with `weak_field_metric.wl`.

5.5 Null Geodesics in the Weak-Field Metric

We now derive the deflection of light in the weak-field metric. This is the central result of this module and the starting point for the entire gravitational lensing formalism.

5.5.1 Setup

Consider a photon propagating through the weak-field metric (5.13) generated by a point mass M at the origin, with $\Phi = -GM/r$. The photon travels along a null geodesic: $ds^2 = 0$.

For a photon moving primarily in the x-direction, passing the mass at impact parameter b (the closest approach distance in the absence of deflection), we can use the **Born approximation**: compute the deflection along the unperturbed (straight-line) path. This is valid when the deflection angle is small, which is guaranteed by the weak-field condition.

5.5.2 The Null Condition

Setting $ds^2 = 0$ in eq. (5.13):

$$\left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 = \left(1 - \frac{2\Phi}{c^2}\right) \left(dx^2 + dy^2 + dz^2\right). \tag{5.15}$$

The coordinate speed of light is therefore:

$$\frac{|\mathbf{dx}|}{dt} = c\sqrt{\frac{1 + 2\Phi/c^2}{1 - 2\Phi/c^2}} \approx c\left(1 + \frac{2\Phi}{c^2}\right) \tag{5.16}$$

to first order in Φ/c^2. Since $\Phi < 0$, the coordinate speed is *less* than c near a mass — gravity slows light.

5.5.3 Deflection Angle

The deflection due to the temporal perturbation h_{00} alone is computed by integrating the transverse gradient of $|\Phi|$ along the unperturbed (straight-line) path (Congdon & Keeton Sec. 3.4.1):

$$\hat{\alpha}_{\text{time}} = \frac{1}{c^2} \int_{-\infty}^{+\infty} \nabla_\perp |\Phi| \, dl \,, \tag{5.17}$$

where $\nabla_\perp$ denotes the gradient of $|\Phi|$ perpendicular to the photon's path and dl is the path length along the unperturbed trajectory. This gives the "Newtonian" contribution to the deflection.

The spatial perturbation h_{ij} contributes an *equal* amount, doubling the result. In the PPN (parametrized post-Newtonian) formalism, the full deflection angle is written as (Will 1993):

$$\boxed{\hat{\alpha} = \frac{1+\gamma_{\rm PPN}}{c^2}\int_{-\infty}^{+\infty} \nabla_\perp |\Phi|\, dl} \tag{5.18}$$

where $\gamma_{\rm PPN} = 1$ in GR. The Parametrized Post-Newtonian (PPN) formalism describes deviations from Newtonian gravity using a set of dimensionless parameters; $\gamma_{\rm PPN}$ measures how much *spatial* curvature is produced per unit rest mass. In GR, $\gamma_{\rm PPN} = 1$ (equal time and space perturbations to the metric, as derived in Section 5.4); in Brans–Dicke theory, $\gamma_{\rm PPN} = (1+\omega)/(2+\omega)$ where ω is the coupling constant. Solar system tests — most precisely the Cassini spacecraft's measurement of the Shapiro delay — constrain $|\gamma_{\rm PPN} - 1| < 2.3\times10^{-5}$, strongly supporting GR (Will 2014, *Living Reviews in Relativity*). The Newtonian prediction corresponds to $\gamma_{\rm PPN} = 0$ (no spatial curvature contribution), giving half the GR value.

5.5.4 Point Mass Deflection

For a point mass with $|\Phi| = GM/r$, the photon travels along the x-axis at perpendicular distance b (so $r = \sqrt{x^2+b^2}$ and $\nabla_\perp|\Phi| = (GM\,b)/(x^2+b^2)^{3/2}$). The line-of-sight integral evaluates to:

$$\int_{-\infty}^{+\infty} \nabla_\perp|\Phi|\, dl = GM\int_{-\infty}^{+\infty} \frac{b\,dx}{(x^2+b^2)^{3/2}} = GM\cdot\frac{2}{b} = \frac{2GM}{b}\,, \tag{5.19}$$

using the standard integral $\int_{-\infty}^{+\infty} b\,dx/(x^2+b^2)^{3/2} = 2/b$. With $\gamma_{\rm PPN} = 1$ (GR), the deflection angle is:

$$\boxed{\hat{\alpha} = \frac{4GM}{c^2 b}} \tag{5.20}$$

This is the **Einstein deflection angle** for a point mass — the single most important equation in gravitational lensing. We derived this here via the Born approximation in the weak-field metric; in Module 6 we will re-derive it from the exact null geodesic orbit equation in the full Schwarzschild metric, confirming that both approaches agree.

5.5.5 The Factor of Two

The factor of 4 in eq. (5.20) (rather than 2) is one of the most celebrated results in GR. Its origin is as follows:

- The temporal perturbation $h_{00} = -2\Phi/c^2$ contributes a deflection of $2GM/(c^2 b)$. This is the "Newtonian" part, which can also be derived by treating the photon as a Newtonian particle moving at speed c.
- The spatial perturbation $h_{ij} = -2\Phi/c^2\,\delta_{ij}$ contributes an *equal* additional deflection of $2GM/(c^2 b)$. This arises because photons, unlike massive particles, are equally sensitive to the spatial curvature.

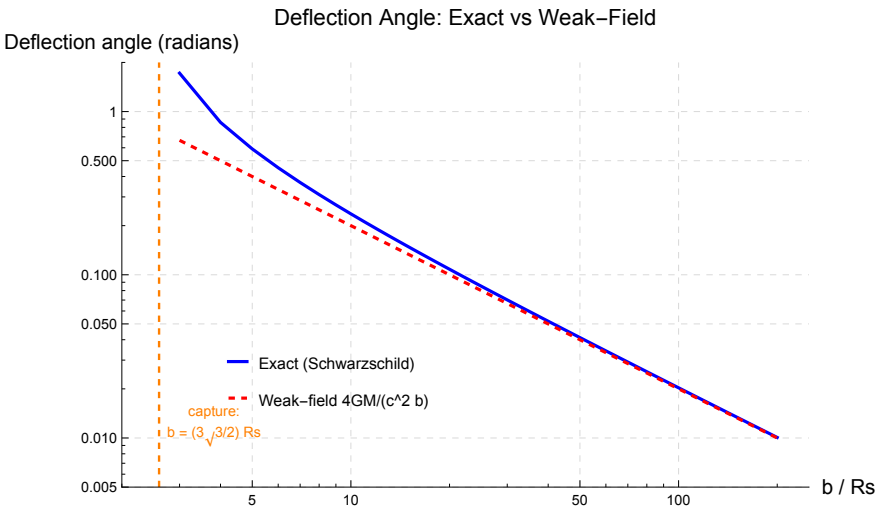


Figure 5.1: Comparison of the exact Schwarzschild deflection angle (solid blue) with the weak-field approximation $\hat{\alpha} = 4GM/(c^2 b)$ (dashed red) as a function of impact parameter b/R_S. The weak-field value falls short of the exact one by a fraction $\approx (15\pi/32)\, R_S/b$ (the second-order term): about 15% at $b = 10\, R_S$, 5% at $30\, R_S$ and 1.5% at $100\, R_S$. The orange dashed line marks the capture impact parameter $b = (3\sqrt{3}/2)\, R_S$. For the Sun, $R_S \approx 3$ km and the closest approach is $R_\odot \approx 7 \times 10^5$ km, so $b/R_S \approx 2.4 \times 10^5$ — deep in the weak-field regime. Interactive version: `deflection_comparison.nb`. (cf. Congdon & Keeton Fig. 3.7)

The total is $4GM/(c^2 b) = 2 \times (2GM/(c^2 b))$, i.e., **twice the Newtonian prediction**. This factor of 2 was confirmed by Eddington's 1919 solar eclipse expedition and is one of the classical tests of general relativity.

Remark 5.5.1. Historically, Einstein himself initially published the "Newtonian" value $2GM/(c^2 b)$ in 1911, before he had completed general relativity. The correct value $4GM/(c^2 b)$ appeared in 1915, after Einstein accounted for the spatial curvature. The Eddington expedition confirmed the GR value, not the 1911 prediction.

Exercise 5.5.1. Derive the Einstein deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ for a point mass by evaluating the integral (5.18) with $|\Phi| = GM/r$ along the unperturbed photon path $\mathbf{x}(l) = (l, b, 0)$.

(a) Show that $\nabla_\perp |\Phi| = (GM\, b)/(l^2 + b^2)^{3/2}$.

(b) Evaluate $\int_{-\infty}^{\infty} b\, dl/(l^2 + b^2)^{3/2} = 2/b$.

(c) Combine with $\gamma_{\mathrm{PPN}} = 1$ to obtain $\hat{\alpha} = 4GM/(c^2 b)$.

(d) Compute $\hat{\alpha}$ for the Sun at the solar limb ($b = R_\odot = 6.96 \times 10^8$ m) and verify $\hat{\alpha} \approx 1.75''$.

Check with `weak_field_metric.wl`.

5.6 The Effective Refractive Index

The weak-field metric leads to a beautiful analogy between gravitational lensing and optics. From eq. (5.16), the coordinate speed of light near a mass is:

$$v_{\text{coord}} = c\left(1 + \frac{2\Phi}{c^2}\right) . \tag{5.21}$$

We can define an **effective refractive index** by analogy with optics, where $v = c/n$:

$$\boxed{n(\mathbf{x}) = \frac{c}{v_{\text{coord}}} \approx 1 - \frac{2\Phi}{c^2} = 1 + \frac{2|\Phi|}{c^2}} \tag{5.22}$$

Since $\Phi < 0$ near a mass, $n > 1$: gravity acts as a medium with refractive index greater than unity, *slowing* light (in coordinate time) and bending it toward the mass.

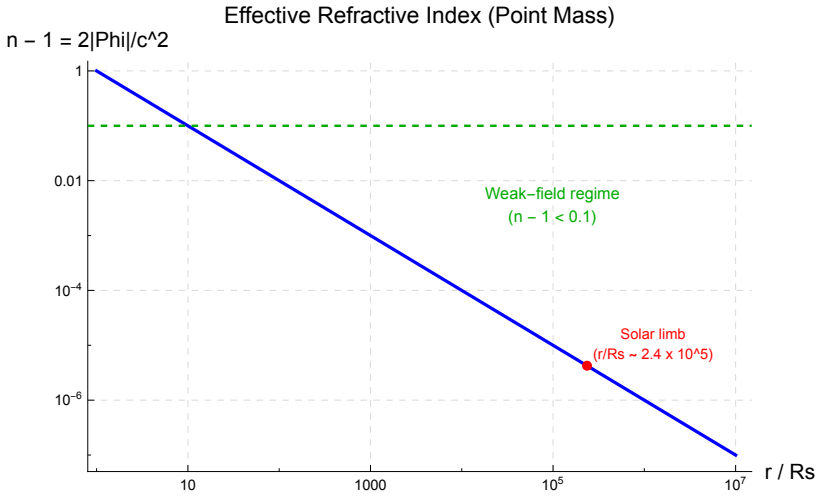


Figure 5.2: Effective refractive index $n = 1 + 2|\Phi|/c^2$ as a function of distance from a point mass, plotted for a solar-mass object. The refractive index is largest (light is slowest) at the surface of the mass and approaches unity far away.

This refractive index interpretation has two important consequences:

Fermat's Principle. In optics, light follows the path that extremizes the optical path length $\int n\, dl$. The same principle applies in gravitational lensing: photons follow paths that extremize the **time delay functional** (Carroll eq. 7.85; Congdon & Keeton eq. 4.25):

$$t[\mathbf{x}(l)] = \frac{1}{c}\int n\, dl = \frac{1}{c}\int \left(1 + \frac{2|\Phi|}{c^2}\right) dl\, . \tag{5.23}$$

This will be the foundation for the time delay formalism in Module 6.

Shapiro Time Delay. Light passing near a mass arrives later than it would in flat spacetime, by an amount proportional to $\int 2|\Phi|/c^2\, dl$ along the path. This **Shapiro delay** has been measured to high precision using radar signals bouncing off Mercury and Venus.

Exercise 5.6.1. Compute the effective refractive index n at the following distances from the Sun ($M = M_\odot$):

(a) The solar surface ($r = R_\odot = 6.96 \times 10^8$ m)

(b) The Earth's orbit ($r = 1$ AU $= 1.496 \times 10^{11}$ m)

(c) The orbit of Pluto ($r = 39.5$ AU)

For each, compute $n - 1 = 2GM/(c^2 r)$ and comment on the magnitude.

5.7 Superposition in Linearized Gravity

Because the metric perturbation $h_{\mu\nu}$ satisfies *linear* equations in the weak-field limit, we can superpose solutions. If we have two masses M_1 and M_2 at positions $\mathbf{x}_1$ and $\mathbf{x}_2$, the total Newtonian potential is:

$$\Phi(\mathbf{x}) = -\frac{GM_1}{|\mathbf{x} - \mathbf{x}_1|} - \frac{GM_2}{|\mathbf{x} - \mathbf{x}_2|}\,, \tag{5.24}$$

and the total deflection angle is simply the *vector sum* of the individual deflections:

$$\boxed{\hat{\boldsymbol{\alpha}} = \sum_i \frac{4GM_i}{c^2}\,\frac{\boldsymbol{\xi} - \boldsymbol{\xi}_i}{|\boldsymbol{\xi} - \boldsymbol{\xi}_i|^2}} \tag{5.25}$$

where $\boldsymbol{\xi}$ is the position in the lens plane and $\boldsymbol{\xi}_i$ is the projected position of mass M_i.

For a continuous mass distribution with surface mass density $\Sigma(\boldsymbol{\xi})$, this becomes (Congdon & Keeton eq. 4.9):

$$\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = \frac{4G}{c^2}\int \frac{(\boldsymbol{\xi} - \boldsymbol{\xi}')\,\Sigma(\boldsymbol{\xi}')}{|\boldsymbol{\xi} - \boldsymbol{\xi}'|^2}\, d^2\xi'\,. \tag{5.26}$$

This integral is the starting point for all lens models in Modules 7–10.

Remark 5.7.1. Superposition of deflection angles does *not* hold in the exact (nonlinear) theory. It is a consequence of linearization and is valid only when all masses satisfy $|\Phi|/c^2 \ll 1$. Fortunately, this condition is satisfied for essentially all astrophysical lensing systems.

Exercise 5.7.1. Two point masses $M_1 = M_2 = M$ are located at $\mathbf{x}_1 = (-d/2, 0)$ and $\mathbf{x}_2 = (+d/2, 0)$ in the lens plane. A photon passes through the midpoint at $\boldsymbol{\xi} = (0, b)$.

(a) Write down the deflection angle vector from each mass.

(b) Show that the x-components cancel by symmetry and that the total deflection is:

$$\hat{\alpha}_y = \frac{8GMb}{c^2(d^2/4 + b^2)}\,.$$

(c) Verify that in the limit $d \to 0$ (masses merge) this reduces to $\hat{\alpha} = 8GM/(c^2 b) = 4G(2M)/(c^2 b)$, i.e., the deflection by a single mass of $2M$.

Check with `problems_01e.wl`.

5.8 Preview: From Deflection to the Lens Equation

The deflection angle (5.20) is a *physical* angle, measured in the frame of the lens. To connect this to observed image positions on the sky, we need the **lens equation**, which relates the true source position $\boldsymbol{\beta}$ to the observed image position $\boldsymbol{\theta}$ through the deflection angle (Congdon & Keeton eq. 4.26):

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \frac{D_{\rm ds}}{D_{\rm s}}\,\boldsymbol{\alpha}(\boldsymbol{\theta})\,, \tag{5.27}$$

where $\boldsymbol{\alpha} = (D_{\rm d}/D_{\rm s})\,\hat{\boldsymbol{\alpha}}$ is the **reduced deflection angle** and $D_{\rm d}$, $D_{\rm s}$, $D_{\rm ds}$ are the angular diameter distances from observer to lens, observer to source, and lens to source.

The full derivation of this equation, along with the thin-lens approximation and the definition of angular diameter distances, will be developed in Modules 3 and 4. The key point for now is that the weak-field deflection angle computed in this module provides the essential input to the lens equation.

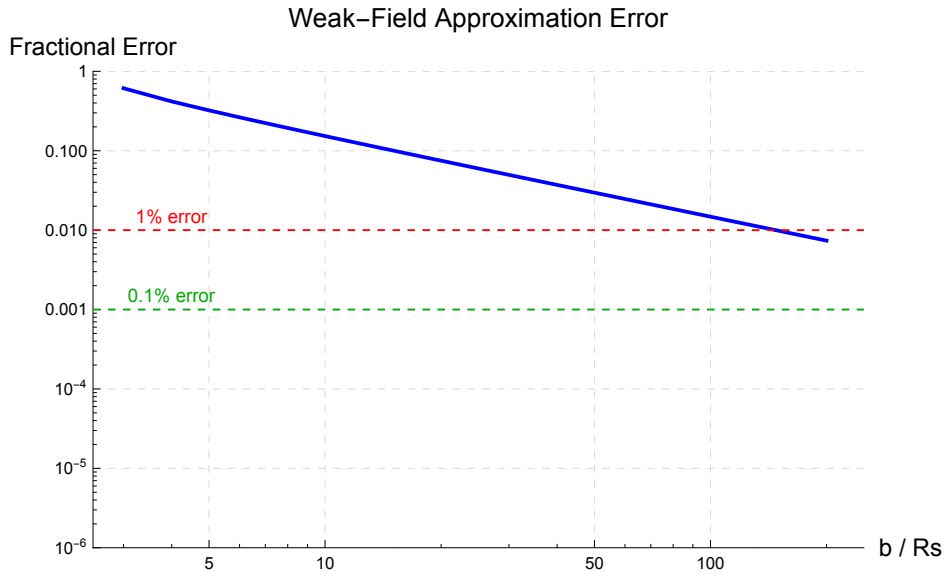


Figure 5.3: Fractional error $|\hat{\alpha}_{\rm exact} - \hat{\alpha}_{\rm weak}|/\hat{\alpha}_{\rm exact}$ of the weak-field approximation as a function of b/R_S. The error is $\approx (15\pi/32)\,R_S/b$: below 1% for $b/R_S \gtrsim 150$ and below 0.1% for $b/R_S \gtrsim 1500$. For typical astrophysical lenses ($b/R_S > 10^4$), the weak-field approximation is essentially exact.

5.9 Summary

- In the weak-field limit, the metric is $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$.
- The Newtonian limit of the geodesic equation identifies $h_{00} = -2\Phi/c^2$, where Φ is the Newtonian gravitational potential.

- The full weak-field metric has *equal* perturbations in time and space: $ds^2 = -(1 + 2\Phi/c^2)\, c^2 dt^2 + (1 - 2\Phi/c^2)\, d\mathbf{x}^2$.
- Null geodesics in this metric yield the Einstein deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ — **twice** the Newtonian value. The factor of 2 arises from the spatial curvature contribution.
- The weak-field metric defines an effective refractive index $n = 1 + 2|\Phi|/c^2 > 1$, connecting gravitational lensing to Fermat's principle and the Shapiro time delay.
- Linearized gravity allows **superposition**: deflection angles from multiple masses add vectorially. This is the foundation for treating galaxies, clusters, and continuous mass distributions as gravitational lenses.
- The deflection angle is the key input to the lens equation (Module 4), which maps source positions to image positions.

5.10 Problems

Exercise 5.10.1. (Newtonian limit from the geodesic equation.) Carry out the full Newtonian limit derivation described in Section 5.3. Starting from the geodesic equation with $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$:

(a) Compute the Christoffel symbol $\Gamma^i{}_{00}$ to first order in h.

(b) Apply the slow-motion and static-field approximations.

(c) Show that $d^2x^i/dt^2 = -\partial_i \Phi$ implies $h_{00} = -2\Phi/c^2$.

Exercise 5.10.2. (Weak-field Schwarzschild.) Expand the Schwarzschild metric to first order in R_S/r and verify that it matches the weak-field metric (5.13) with $\Phi = -GM/r$. Specifically:

(a) Expand $(1 - R_S/r)^{-1} \approx 1 + R_S/r$.

(b) Write the result in isotropic Cartesian coordinates.

(c) Identify $h_{00} = R_S/r$ and $h_{ij} = (R_S/r)\, \delta_{ij}$ and confirm that the time and spatial diagonal perturbations are equal (i.e., $h_{00} = h_{xx} = h_{yy} = h_{zz}$).

Exercise 5.10.3. (Effective refractive index.) Compute the effective refractive index $n = 1 + 2|\Phi|/c^2$ for a point mass $M = M_\odot$ at distances: (a) $r = R_\odot$, (b) $r = 1$ AU, (c) $r = 1$ kpc. Comment on whether the weak-field approximation is valid in each case.

Exercise 5.10.4. (Deflection angle from null geodesics.) Derive $\hat{\alpha} = 4GM/(c^2 b)$ by evaluating the integral (5.18) with $\gamma_{\rm PPN} = 1$ for a point mass along the unperturbed path. Compute the deflection angle for: (a) the Sun at the solar limb, (b) a $10^{12}\, M_\odot$ galaxy at $b = 10$ kpc. Express results in arcseconds.

Exercise 5.10.5. (Superposition of deflection angles.) Two equal point masses M are separated by distance d in the lens plane.

(a) Derive the total deflection angle for a photon passing at perpendicular distance b from the midpoint (along the symmetry axis perpendicular to the line joining the masses).

(b) Show that for $b \gg d$, the result approaches $\hat{\alpha} \to 4G(2M)/(c^2 b)$: at large distances the two masses act as a single mass $2M$.

(c) Plot $\hat{\alpha}(b)$ for $d = 1$ kpc and $M = 10^{11}\, M_\odot$ using Mathematica.

Part II

Gravitational Lensing Theory

Chapter 6

Light Deflection in Curved Spacetime

> *The great attraction of the theory is its logical consistency. If any deduction from it should prove untenable, it must be given up. A modification of it seems impossible without destruction of the whole.* — Albert Einstein, *The Times* (London), 28 November 1919
> (reprinted in *Science* **51**, 8, 1920)

In Module 1d we derived the effective potential for geodesic motion in Schwarzschild spacetime, and in Module 1e we obtained the deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ via the weak-field / Born approximation, showing how the factor of 2 over Newtonian gravity arises from the equal contributions of temporal and spatial curvature. Now we derive this same result rigorously from the *exact* null geodesic orbit equation in the full Schwarzschild metric. This is the single most important result in gravitational lensing theory — every lens equation, every magnification formula, and every time delay expression ultimately rests on it.

We also derive the Shapiro time delay, which provides the second observable consequence of light propagation through a gravitational field, and compute the deflection for several astrophysically relevant systems: the Sun (Eddington's 1919 test), Jupiter, and a typical galaxy lens.

Companion Mathematica files:

- `deflection_angle.wl` — symbolic derivation of the deflection angle integral, weak-field expansion, numerical evaluation for Sun/Jupiter/galaxy, Shapiro delay, exact vs. approximate comparison, and figure generation
- `deflection_geometry.nb` — interactive notebook: ray-bending geometry, sliders for M and b, exact vs. weak-field comparison with error percentage

6.1 Introduction: Null Geodesics and the Deflection Problem

Light follows **null geodesics** in curved spacetime: paths for which $ds^2 = 0$ at every point. In Module 1d we showed that for a photon in the equatorial plane of the Schwarzschild metric, the conserved energy E and angular momentum L reduce the equations of motion to a single

radial equation (cf. eqs. 4.3–4.4 and the null geodesic orbit equation (4.16) from Module 1d):

$$\frac{d\phi}{dr} = \pm\frac{L}{r^2}\left[\frac{E^2}{c^2} - \frac{L^2}{r^2}\left(1 - \frac{R_S}{r}\right)\right]^{-1/2}, \tag{6.1}$$

where $R_S = 2GM/c^2$ is the Schwarzschild radius (Congdon & Keeton eq. 3.87).

The physical setup is as follows. A photon approaches a mass M from infinity, reaches a distance of closest approach r_0, and then escapes back to infinity. The total change in the azimuthal angle ϕ is $\Delta\phi$; in flat spacetime this would be π (a straight line), so the **deflection angle** is

$$\hat{\alpha} = \Delta\phi - \pi\,. \tag{6.2}$$

Our goal in this chapter is to evaluate $\Delta\phi$ and show that, in the weak-field limit,

$$\boxed{\hat{\alpha} = \frac{4GM}{c^2\,b}} \tag{6.3}$$

where b is the impact parameter. This is the fundamental result of gravitational lensing.

6.2 Soldner's Newtonian Calculation (1801)

Before Einstein, Johann Georg von Soldner (1801) computed the deflection of light using Newtonian gravity, treating a photon as a corpuscle of mass m traveling at speed c.

Remark 6.2.1. Soldner's calculation predates general relativity by over a century. It is both historically important and pedagogically useful: it gives exactly *half* the correct GR answer, and understanding *why* illuminates the role of spatial curvature.

Consider a particle approaching a mass M on a hyperbolic trajectory with velocity $v_\infty = c$ at infinity. The Newtonian equations of motion in polar coordinates yield the orbit equation

$$r(\phi) = \frac{p}{1 + e\cos\phi}\,, \tag{6.4}$$

where $p = L^2/(GM)$ and $e = \sqrt{1 + 2\mathcal{E}L^2/(G^2M^2)}$ is the eccentricity (with $\mathcal{E} = \frac{1}{2}c^2$ for an unbound particle at speed c).

The asymptotic directions correspond to $r \to \infty$, i.e., $\cos\phi = -1/e$. The total angle swept is $\Delta\phi = \pi + 2\delta$, where $\sin\delta = 1/e$. For large impact parameter ($b \gg R_S$), the eccentricity is large and $\delta \approx 1/e \ll 1$, giving

$$\hat{\alpha}_{\rm Newton} = 2\delta \approx \frac{2}{e} = \frac{2GM}{c^2\,b}\,. \tag{6.5}$$

Exercise 6.2.1. Derive eq. (6.5) in detail. Start from the Newtonian orbit equation for a massless particle with $v_\infty = c$ and impact parameter b.

(a) Show that the semi-latus rectum is $p = b^2c^2/(GM)$ and the eccentricity is $e = \sqrt{1 + b^2c^4/(GM)^2}$.

(b) For $b \gg GM/c^2$, show that $\hat{\alpha}_{\rm Newton} = 2GM/(c^2 b)$.

Hint: The angular momentum is $L = bc$ and the energy is $\mathcal{E} = \frac{1}{2}c^2$.

6.3 The GR Deflection Angle — Full Derivation

We now follow Congdon & Keeton Sec. 3.4.1 (eqs. 3.87–3.96) to derive the deflection angle in general relativity. The result will be $\hat{\alpha} = 4GM/(c^2 b)$ — exactly **twice** the Newtonian prediction.

6.3.1 The Impact Parameter

The **impact parameter** b is defined as the perpendicular distance between the undeflected ray (the asymptotic straight line at $r \to \infty$) and the mass M. From the conserved quantities, it is related to E and L by (Congdon & Keeton eq. 3.90):

$$b = \frac{c\,L}{E}\,. \tag{6.6}$$

6.3.2 Distance of Closest Approach

At the turning point $r = r_0$, the radial velocity vanishes: $dr/d\lambda = 0$. From the radial equation of motion, this gives (Congdon & Keeton eq. 3.89):

$$\frac{E^2}{c^2} - \frac{L^2}{r_0^2}\left(1 - \frac{R_{\rm S}}{r_0}\right) = 0\,. \tag{6.7}$$

Substituting $b = cL/E$ and defining $m \equiv GM/c^2 = R_{\rm S}/2$, we obtain

$$\frac{r_0^2}{b^2} = 1 - \frac{2m}{r_0}\,. \tag{6.8}$$

In the weak-field limit $m/r_0 \ll 1$, we can solve perturbatively: $r_0 \approx b\,(1 - m/b + \cdots)$, so to leading order $r_0 \approx b$.

6.3.3 The Total Azimuthal Change

By symmetry, the total angle swept from $r = \infty$ (incoming) to $r = \infty$ (outgoing) is twice the angle from r_0 to infinity:

$$\Delta\phi = 2\int_{r_0}^{\infty} \frac{L}{r^2}\left[\frac{E^2}{c^2} - \frac{L^2}{r^2}\left(1 - \frac{R_{\rm S}}{r}\right)\right]^{-1/2} dr\,. \tag{6.9}$$

Substituting $b = cL/E$ and simplifying (Congdon & Keeton eq. 3.92):

$$\Delta\phi = 2\int_{r_0}^{\infty} \frac{dr}{r^2}\left[\frac{1}{b^2} - \frac{1}{r^2}\left(1 - \frac{2m}{r}\right)\right]^{-1/2}. \tag{6.10}$$

6.3.4 Change of Variables

Following Congdon & Keeton, we introduce the substitution $u = r_0/r$ (so $u = 0$ at $r = \infty$ and $u = 1$ at $r = r_0$):

$$\Delta\phi = 2\int_0^1 \frac{du}{\sqrt{\dfrac{r_0^2}{b^2} - u^2 + 2m\dfrac{u^3}{r_0}}}\,. \tag{6.11}$$

Using eq. (6.8) to replace $r_0^2/b^2 = 1 - 2m/r_0$ (Congdon & Keeton eq. 3.93):

$$\Delta\phi = 2\int_0^1 \frac{du}{\sqrt{(1-u^2) - \dfrac{2m}{r_0}(1-u^3)}}\,. \tag{6.12}$$

6.3.5 Weak-Field Expansion

We define the small parameter $\varepsilon \equiv m/r_0 \ll 1$ and expand the integrand to first order in ε. Writing $f(u) = (1-u^2) - 2\varepsilon(1-u^3)$, we use

$$\frac{1}{\sqrt{f}} \approx \frac{1}{\sqrt{1-u^2}} + \frac{\varepsilon(1-u^3)}{(1-u^2)^{3/2}} + \mathcal{O}\big(\varepsilon^2\big)\,. \tag{6.13}$$

Therefore (Congdon & Keeton eq. 3.94):

$$\Delta\phi \approx 2\int_0^1 \frac{du}{\sqrt{1-u^2}} + 2\varepsilon\int_0^1 \frac{1-u^3}{(1-u^2)^{3/2}}\,du + \mathcal{O}\big(\varepsilon^2\big)\,. \tag{6.14}$$

6.3.6 Evaluating the Integrals

Zeroth-order integral: With the substitution $u = \sin\theta$:

$$I_0 = 2\int_0^1 \frac{du}{\sqrt{1-u^2}} = 2\,[\arcsin u]_0^1 = 2\cdot\frac{\pi}{2} = \pi\,. \tag{6.15}$$

This is the flat-spacetime result: no deflection.

First-order integral: We need (the m/r_0 term of Congdon & Keeton eq. 3.94):

$$I_1 = 2\int_0^1 \frac{1-u^3}{(1-u^2)^{3/2}}\,du\,. \tag{6.16}$$

Factor the numerator: $1-u^3 = (1-u)(1+u+u^2)$ and $1-u^2 = (1-u)(1+u)$, so

$$\frac{1-u^3}{(1-u^2)^{3/2}} = \frac{(1-u)(1+u+u^2)}{(1-u)^{3/2}(1+u)^{3/2}} = \frac{1+u+u^2}{(1-u)^{1/2}(1+u)^{3/2}}\,. \tag{6.17}$$

The bare integral $\int_0^1 (1-u^3)(1-u^2)^{-3/2}\,du$ evaluates to 2 (as verified in Mathematica — see `deflection_angle.wl`), so

$$I_1 = 2\times 2 = 4\,. \tag{6.18}$$

Remark 6.3.1. The evaluation of $\int_0^1 (1-u^3)(1-u^2)^{-3/2}\,du = 2$ is a good exercise in integration techniques. One approach: split $1+u+u^2 = (1+u)^2 - u$ and evaluate each piece separately via trigonometric/hyperbolic substitution. The Mathematica script verifies this symbolically.

6.3.7 The Deflection Angle

Assembling the result (Congdon & Keeton eqs. 3.95–3.96): using $\Delta\phi = I_0 + \varepsilon\, I_1 + \mathcal{O}(\varepsilon^2)$,

$$\Delta\phi = \pi + 4\varepsilon + \mathcal{O}\big(\varepsilon^2\big) = \pi + \frac{4m}{r_0} + \mathcal{O}(m/r_0)^2 \ . \tag{6.19}$$

Since $r_0 \approx b$ to leading order and $m = GM/c^2$:

$$\boxed{\hat{\alpha} = \Delta\phi - \pi = \frac{4GM}{c^2\, b} + \mathcal{O}\big(GM/(c^2 b)\big)^2} \tag{6.20}$$

This is the **Einstein deflection angle** for a point mass in the weak-field limit.

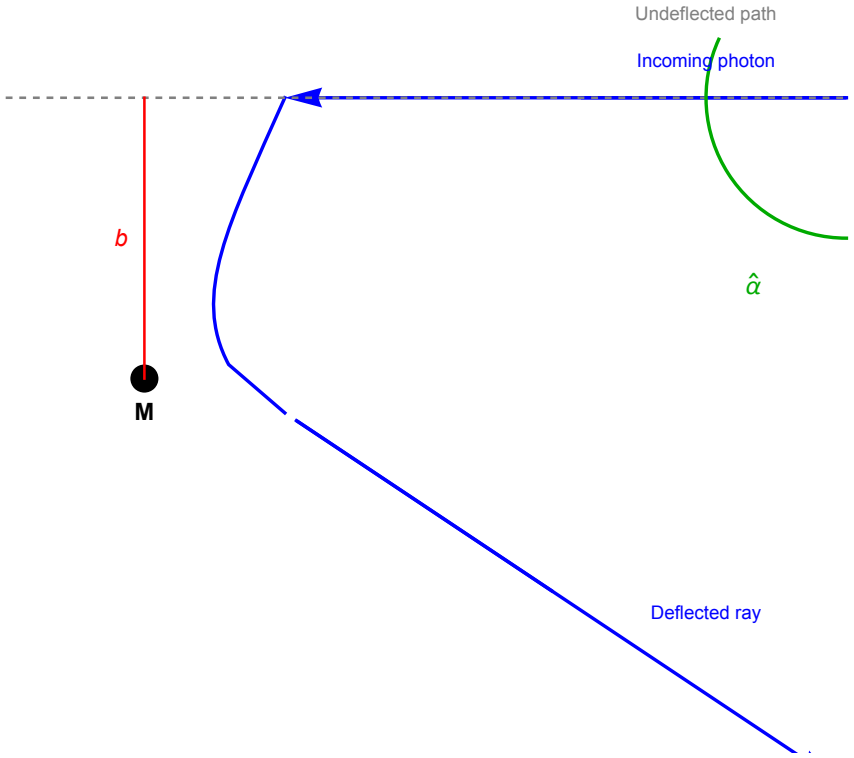


Figure 6.1: Geometry of gravitational light deflection by a point mass M. A photon approaches from the right with impact parameter b, reaches closest approach at $r_0 \approx b$, and is deflected by $\hat{\alpha} = 4GM/(c^2 b)$. The dashed line shows the undeflected path. Generated by `deflection_angle.wl`.

Exercise 6.3.1. Show that the GR deflection angle (6.20) is exactly twice the Newtonian prediction (6.5). Explain physically why:

(a) The Newtonian calculation accounts for the deflection due to the gravitational potential (temporal curvature of the metric, g_{tt}), but ignores spatial curvature (g_{rr}).

(b) In GR, both g_{tt} and g_{rr} deviate from flat space by the same amount (to leading order in m/r), so the spatial curvature doubles the deflection.

Hint: Expand the Schwarzschild line element in isotropic coordinates and identify the contributions from the time–time and space–space components of the metric.

6.4 The Shapiro Time Delay

In addition to deflecting light, a gravitational field *slows it down* — or more precisely, causes an excess travel time relative to the flat-spacetime path. This effect was predicted by Irwin Shapiro in 1964 and first measured using radar echoes from Mercury and Venus.

Following Congdon & Keeton Sec. 3.4.2 (eqs. 3.97–3.110), we derive the time delay for a photon passing near a point mass.

6.4.1 Coordinate Travel Time

For a photon on a null geodesic, we can write dt/dr using the conserved quantities (Congdon & Keeton eq. 3.97):

$$\frac{dt}{dr} = \frac{E/c^2}{(1 - R_{\rm S}/r)} \left[\frac{E^2}{c^2} - \frac{L^2}{r^2}\left(1 - \frac{R_{\rm S}}{r}\right) \right]^{-1/2} . \tag{6.21}$$

The coordinate time for the photon to travel from closest approach r_0 to a radial distance R is (Congdon & Keeton eq. 3.97):

$$T(R) = \int_{r_0}^{R} \frac{dt}{dr}\, dr \,. \tag{6.22}$$

In the weak-field limit ($m/r \ll 1$), expanding to first order gives (Congdon & Keeton eq. 3.100, second line; its expansion in r_0/R is their eq. 3.102):

$$T(R) \approx \frac{1}{c}\sqrt{R^2 - r_0^2} + \frac{2m}{c} \ln\left(\frac{R + \sqrt{R^2 - r_0^2}}{r_0} \right) + \frac{m}{c}\sqrt{\frac{R - r_0}{R + r_0}} \,. \tag{6.23}$$

6.4.2 The Two Components of Time Delay

The total time delay ΔT for a lensed photon (relative to the unlensed path) has two contributions (Congdon & Keeton eqs. 3.107–3.110):

$$\boxed{\Delta T = \underbrace{\frac{D_{\rm d} D_{\rm s}}{2c\, D_{\rm ds}} (\theta - \beta)^2}_{\text{geometric delay}} + \underbrace{\frac{4GM}{c^3} \ln \frac{\theta_{\rm ref}}{|\theta|}}_{\text{Shapiro (gravitational) delay}}} \tag{6.24}$$

where $D_{\rm d}$, $D_{\rm s}$, $D_{\rm ds}$ are the angular diameter distances from observer to lens, observer to source, and lens to source respectively; θ is the image position and β the (unlensed) source position; and $\theta_{\rm ref}$ is an arbitrary reference angle. Changing $\theta_{\rm ref}$ adds the same constant to ΔT for every image, so it drops out of the measurable delay *differences* between images.

Remark 6.4.1. **Geometric delay:** The lensed ray follows a longer path through space than the direct (unlensed) ray. This is a purely geometric effect and is always positive.

Shapiro (gravitational) delay: The gravitational potential slows the effective speed of light near the lens. Clocks near a mass run slower (gravitational redshift), so the signal takes longer to traverse the gravitational potential well.

Remark 6.4.2. The total time delay (6.24) is closely related to the **Fermat potential**, which we will study in detail in Module 6. The Fermat potential $\tau(\boldsymbol{\theta})$ encodes both the geometric and gravitational delays, and its gradient gives the lens equation. Images form at stationary points of the Fermat potential (Fermat's principle applied to gravitational lensing).

Exercise 6.4.1. Derive the Shapiro time delay for a radar signal passing near the Sun and bouncing off a planet at distance R from the Sun.

(a) Show that the excess round-trip travel time is approximately

$$\Delta T \approx \frac{4GM_\odot}{c^3} \ln\left(\frac{4R_\oplus R}{r_0^2}\right)$$

where $R_\oplus$ is the Earth–Sun distance and r_0 the closest approach distance.

(b) For a signal grazing the Sun ($r_0 = R_\odot$) bouncing off Mercury at inferior conjunction, compute ΔT numerically. You should find $\Delta T \approx 240\ \mu$s.

6.5 Deflection by the Sun

The first and most famous test of the GR deflection formula was **Eddington's 1919 solar eclipse expedition**. During a total eclipse, stars near the Sun become visible, and their apparent positions are shifted outward by the gravitational deflection.

For light grazing the limb of the Sun ($b = R_\odot$):

$$\hat{\alpha}_\odot = \frac{4GM_\odot}{c^2\, R_\odot} = \frac{4 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(2.998 \times 10^8)^2 \times 6.957 \times 10^8} \approx 8.49 \times 10^{-6}\ \text{rad}\,. \tag{6.25}$$

Converting to arcseconds:

$$\boxed{\hat{\alpha}_\odot = 1.75''} \tag{6.26}$$

This is the Newtonian value $0.875''$ *doubled* by general relativity. Eddington's measurements confirmed the GR prediction (within the substantial error bars of the time), making Einstein a global celebrity.

Example 6.5.1. Deflection by Jupiter. Jupiter has $M_J = 1.898 \times 10^{27}$ kg and $R_J = 7.149 \times 10^7$ m. The deflection for light grazing Jupiter's limb is:

$$\hat{\alpha}_J = \frac{4GM_J}{c^2\, R_J} \approx 0.016'' = 16\ \text{mas}\,. \tag{6.27}$$

This is small but measurable with modern astrometry (e.g., the Gaia satellite, which achieves $\sim 10\ \mu$as precision).

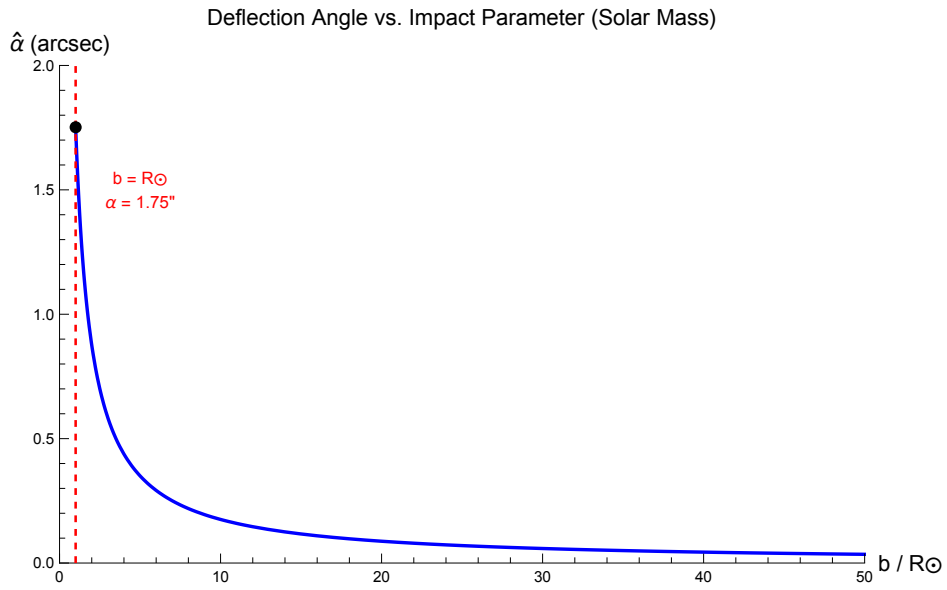


Figure 6.2: Deflection angle $\hat{\alpha}$ as a function of impact parameter b for a solar-mass lens, shown in arcseconds. The deflection falls as $1/b$. The vertical dashed line marks the solar radius ($b = R_\odot$) where $\hat{\alpha} = 1.75''$. Generated by `deflection_angle.wl`.

Exercise 6.5.1. Compute the deflection angle for light grazing the surface of:

(a) The Sun ($M = 1.989 \times 10^{30}$ kg, $R = 6.957 \times 10^8$ m): verify $\hat{\alpha} = 1.75''$.

(b) Jupiter ($M = 1.898 \times 10^{27}$ kg, $R = 7.149 \times 10^7$ m).

(c) A $10^{12}\, M_\odot$ galaxy with effective radius $R_e = 10$ kpc, using $b = R_e$. Express the result in arcseconds.

6.6 Deflection by Galaxies

The astrophysical power of gravitational lensing comes from the fact that the deflection angles produced by *galaxies* are of order arcseconds — large enough to produce multiple images, arcs, and rings that are observable with ground-based and space-based telescopes.

For a galaxy of mass $M = 10^{12}\, M_\odot$ with a typical scale of $b \sim 10$ kpc $\approx 3.09 \times 10^{20}$ m:

$$\hat{\alpha} \sim \frac{4GM}{c^2\, b} = \frac{4 \times 6.674 \times 10^{-11} \times 10^{12} \times 1.989 \times 10^{30}}{(2.998 \times 10^8)^2 \times 3.09 \times 10^{20}} \approx 3.9'' \tag{6.28}$$

The corresponding Einstein radius (which we will derive properly in Module 4) is of order $\sim 1''$, depending on the distance configuration.

Remark 6.6.1. The Einstein radius θ_{E} for a point mass lens at cosmological distances is:

$$\theta_{\mathrm{E}} = \sqrt{\frac{4GM}{c^2}\,\frac{D_{\mathrm{ds}}}{D_{\mathrm{d}}\, D_{\mathrm{s}}}}\,. \tag{6.29}$$

For a $10^{12}\,M_\odot$ galaxy at $z_d \sim 0.3$ with a source at $z_s \sim 1$, this gives $\theta_{\rm E} \sim 1''$. Multiple images form whenever the source lies within $\sim \theta_{\rm E}$ of the optical axis — this is the regime of **strong gravitational lensing**. We will develop this quantitatively in Modules 4–5.

6.7 Exact vs. Weak-Field Deflection

The weak-field formula $\hat{\alpha} = 4m/b$ is a first-order approximation valid when $b \gg R_{\rm S}$. For photons passing closer to the mass, higher-order corrections become important. The *exact* deflection angle is obtained by evaluating the integral (6.12) numerically (or using elliptic integrals), without the weak-field expansion.

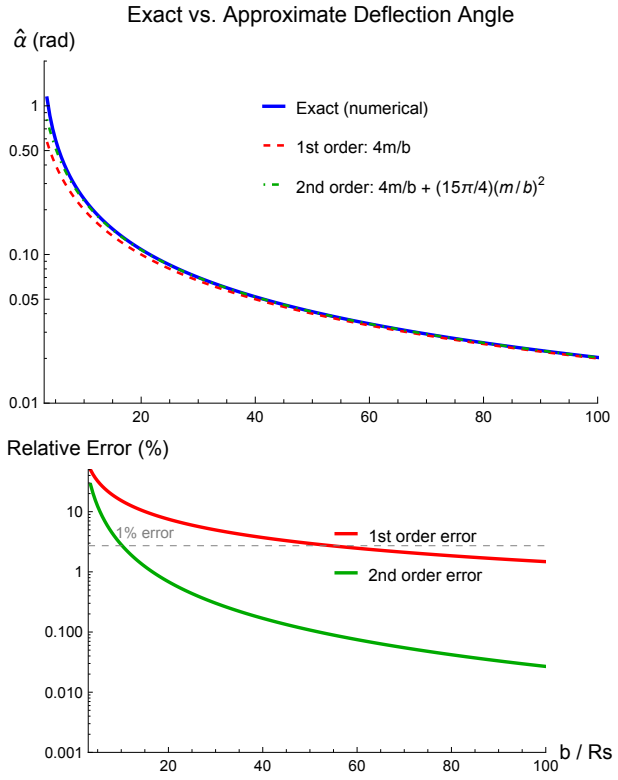


Figure 6.3: Comparison of the exact deflection angle (solid blue, from numerical integration) and the weak-field approximations $\hat{\alpha} = 4m/b$ (dashed red, first order) and including the second-order correction (dot-dashed green), as a function of $b/R_{\rm S}$. The lower panel shows the relative error. The first-order approximation has $\sim 15\%$ error at $b = 10\,R_{\rm S}$ and reaches 1% accuracy only for $b \gtrsim 150\,R_{\rm S}$. At the photon sphere $b = 3\sqrt{3}\,R_{\rm S}/2$, the exact deflection diverges (the photon orbits the mass). Generated by `deflection_angle.wl`.

The second-order correction has been computed by several authors; the result is (see, e.g., Keeton & Petters 2005):

$$\hat{\alpha} = \frac{4m}{b} + \frac{15\pi}{4}\left(\frac{m}{b}\right)^2 + \mathcal{O}(m/b)^3 \,. \tag{6.30}$$

The critical impact parameter below which the photon is captured (i.e., spirals into the mass) is $b_{\rm crit} = 3\sqrt{3}\,m = 3\sqrt{3}\,R_{\rm S}/2$. This corresponds to the photon sphere derived in Module 1d.

Exercise 6.7.1. For a photon passing a Schwarzschild black hole at impact parameter $b = 10\,R_{\rm S}$:

(a) Evaluate the deflection integral (6.12) numerically to obtain the exact deflection angle.

(b) Compare with the weak-field approximation $\hat{\alpha} = 4m/b = 0.2$ rad. What is the relative error?

(c) Repeat for several values of $b/R_{\rm S}$ from 5 to 100. At what $b/R_{\rm S}$ does the weak-field approximation first achieve 1% accuracy?

Use the Mathematica script `deflection_angle.wl` *to verify your results.*

6.8 Summary

- Light follows null geodesics in Schwarzschild spacetime. The orbit equation (6.1) reduces to an integral for the total azimuthal angle change $\Delta\phi$.
- **Newtonian prediction** (Soldner, 1801): $\hat{\alpha}_{\rm Newton} = 2GM/(c^2 b)$. This accounts for only the *temporal* curvature of spacetime.
- **GR prediction** (Einstein, 1915): $\hat{\alpha} = 4GM/(c^2 b)$. The factor of 2 enhancement comes from *spatial* curvature, which contributes equally to the deflection.
- **Shapiro time delay:** A gravitational field causes an excess travel time ΔT with geometric and gravitational components. This is connected to the Fermat potential (Module 6).
- **Solar deflection:** $\hat{\alpha}_\odot = 1.75''$, confirmed by Eddington (1919) and modern measurements to $< 0.01\%$.
- **Galaxy-scale lensing:** $\hat{\alpha} \sim 1''$–$4''$ for $\sim 10^{12}\,M_\odot$ galaxies at $b \sim 10$ kpc, producing multiple images, arcs, and Einstein rings in strong lensing.
- The weak-field approximation $\hat{\alpha} = 4m/b$ converges slowly; the error is $\sim 15\%$ at $b = 10\,R_{\rm S}$ and drops below 1% only for $b \gtrsim 150\,R_{\rm S}$. The exact deflection diverges at the photon sphere ($b_{\rm crit} = 3\sqrt{3}\,R_{\rm S}/2$).

6.9 Problems

Exercise 6.9.1. (Soldner's Newtonian deflection.) Derive the Newtonian deflection angle for a "corpuscle of light" traveling at speed c past a mass M at impact parameter b.

(a) Write the Newtonian orbit equation and identify the semi-latus rectum p and eccentricity e in terms of b, M, G, and c.

(b) Show that the deflection angle is $\hat{\alpha}_{\text{Newton}} = 2GM/(c^2 b)$ in the weak-field limit.

(c) Compute the Newtonian prediction for the solar deflection. You should get $0.875''$.

Exercise 6.9.2. (The factor of 2: GR vs. Newton.)

(a) Starting from the result $\Delta\phi = \pi + 4m/r_0$, show that the GR deflection angle is $\hat{\alpha} = 4GM/(c^2 b)$.

(b) The Newtonian result uses only $g_{tt} \approx -(1 - 2\Phi/c^2)$ where $\Phi = -GM/r$. In GR, the spatial part of the Schwarzschild metric also curves: $g_{rr} \approx 1 + 2GM/(c^2 r)$. Argue qualitatively (or using the isotropic form of the metric) that the spatial curvature contributes an *equal* amount of deflection, doubling the Newtonian result.

Exercise 6.9.3. (Deflection for Sun, Jupiter, and galaxy.) Compute $\hat{\alpha}$ for:

(a) Light grazing the Sun: $M = 1.989 \times 10^{30}$ kg, $b = R_\odot = 6.957 \times 10^8$ m.

(b) Light grazing Jupiter: $M = 1.898 \times 10^{27}$ kg, $b = R_J = 7.149 \times 10^7$ m.

(c) A $10^{12}\, M_\odot$ galaxy at $b = 10$ kpc.

Express all answers in arcseconds. Verify with `problems_02.wl`.

Exercise 6.9.4. (Shapiro time delay for radar echoes.) A radar signal is sent from Earth, passes near the Sun at closest approach r_0, bounces off Mercury, and returns.

(a) Using eq. (6.23), show that the excess round-trip time is

$$\Delta T \approx \frac{4GM_\odot}{c^3}\left[\ln\left(\frac{4\, R_\oplus\, R_{\text{Merc}}}{r_0^2}\right) + 1\right]$$

where $R_\oplus = 1.496 \times 10^{11}$ m and $R_{\text{Merc}} = 5.79 \times 10^{10}$ m are the Earth–Sun and Mercury–Sun distances.

(b) For $r_0 = R_\odot$, compute ΔT in microseconds.

(c) Shapiro's 1964 measurement using the Haystack radar gave $\Delta T = 200 \pm 20$ μs. Is this consistent?

Exercise 6.9.5. (Exact vs. weak-field deflection.) For a photon passing a Schwarzschild mass at impact parameter $b = 10\, R_{\text{S}}$:

(a) Evaluate the exact deflection angle by numerically integrating eq. (6.12).

(b) Compare with the first-order approximation $\hat{\alpha}_1 = 4m/b$ and the second-order approximation $\hat{\alpha}_2 = 4m/b + (15\pi/4)(m/b)^2$.

(c) Tabulate the relative error of $\hat{\alpha}_1$ and $\hat{\alpha}_2$ for $b/R_{\text{S}} = 5, 10, 20, 50, 100$.

(d) At what value of b/R_{S} does the first-order approximation first achieve 1% accuracy?

Use `deflection_angle.wl` *to generate and verify your results.*

Chapter 7

Cosmological Distances

> *Not only does God definitely play dice, but He sometimes confuses us by throwing them where they can't be seen.* — Stephen Hawking, *Does God Play Dice?* (public lecture, 1999)

Gravitational lensing is inherently a cosmological phenomenon: the lens and source are typically at redshifts $z \sim 0.1$–3, separated by distances of order gigaparsecs. To connect the physical deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ derived in Modules 1e and 2 to observable image positions on the sky, we need a precise definition of "distance" in an expanding universe. Unlike in Euclidean space, there is no single notion of distance in cosmology: the *angular diameter distance* (relevant for angles), the *luminosity distance* (relevant for fluxes), and the *comoving distance* (relevant for volumes) are all distinct.

This module develops the full framework of cosmological distances from first principles. In Part A we derive the Friedmann–Lemaître–Robertson–Walker (FLRW) metric from the cosmological principle, compute the Christoffel symbols and curvature tensors for this metric using the tools from Module 1b, and plug into Einstein's equations with a perfect-fluid stress-energy tensor to obtain the **Friedmann equations** — the dynamical equations governing the expansion of the universe. In Part B we define the various distance measures, derive the key distance ratios that enter the lens equation, and highlight the counterintuitive *non-monotonicity* of the angular diameter distance, which turns over at $z \sim 1.6$ for the concordance cosmology.

Companion Mathematica files:

- `friedmann_equations.wl` — symbolic computation of Christoffel symbols and curvature tensors for the FRW metric, derivation of the Friedmann equations, scale factor solutions, cosmological distance calculations, and figure generation
- `cosmological_distances.nb` — interactive notebook with sliders for Ω_m, Ω_Λ, H_0 showing how distance–redshift relations change; lensing geometry calculator

Part A: The FRW Metric and Friedmann Equations

7.1 The Cosmological Principle

Modern cosmology rests on the **cosmological principle**: on sufficiently large scales ($\gtrsim$ 100 Mpc), the universe is *spatially homogeneous* (the same at every point) and *spatially isotropic* (the same in every direction). These two symmetry assumptions are supported by:

- The near-perfect isotropy of the cosmic microwave background (CMB), which is uniform to one part in 10^5 across the sky.
- The large-scale distribution of galaxies, which becomes statistically uniform on scales $\gtrsim 100$ Mpc, as shown by surveys such as SDSS, 2dFGRS, and DESI.

Remark 7.1.1. Homogeneity and isotropy constrain only the *spatial* geometry of the universe at each instant. The universe is manifestly *not* homogeneous in time: it was denser and hotter in the past. The metric is therefore allowed to have time-dependent coefficients (the scale factor), and it is this time dependence that encodes the expansion history.

The mathematical consequence of the cosmological principle is powerful: the spatial part of the metric must be a maximally symmetric 3-dimensional Riemannian manifold. By a classic theorem in differential geometry (see Carroll Secs. 3.9 and 8.2), maximally symmetric n-dimensional spaces have $n(n+1)/2$ Killing vectors, and there are exactly three possibilities for $n = 3$: positive curvature (the 3-sphere S^3), zero curvature (flat $\mathbb{R}^3$), or negative curvature (the 3-hyperboloid H^3). These are characterized by a single parameter, the **spatial curvature constant** k:

$$k = \begin{cases} +1 & \text{positive curvature (closed, finite volume)} \\ 0 & \text{zero curvature (flat, infinite volume)} \\ -1 & \text{negative curvature (open, infinite volume)} \end{cases} \tag{7.1}$$

7.2 The Robertson–Walker Metric

Combining the cosmological principle with the requirement that spacetime be described by a Lorentzian 4-metric, the most general metric takes the **Robertson–Walker (RW)** form (Carroll eq. 8.43; Congdon & Keeton eq. 3.123a):

$$\boxed{ds^2 = -c^2\,dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2\big(d\theta^2 + \sin^2\theta\,d\phi^2\big)\right]} \tag{7.2}$$

where:

- t is **cosmic time** — the proper time measured by comoving observers (those at rest with respect to the Hubble flow).

- $a(t)$ is the **scale factor**, a dimensionless function of time that encodes how distances between comoving observers change. We adopt the convention $a(t_0) = 1$ at the present epoch.
- r, θ, ϕ are **comoving coordinates** — the coordinate labels assigned to comoving observers do not change as the universe expands. The *physical* distance between two comoving observers separated by coordinate distance Δr is $a(t)\,\Delta r$, which grows with time if the universe is expanding ($\dot{a} > 0$).
- $k = -1, 0, +1$ is the spatial curvature parameter.

Remark 7.2.1. The metric (7.2) is purely *kinematical*: it is the most general metric consistent with homogeneity and isotropy, but it tells us nothing about how $a(t)$ evolves. The dynamics of $a(t)$ come from plugging this metric into Einstein's field equations, which we do in the next sections to derive the Friedmann equations.

It is often useful to write the spatial part of the metric using the **comoving distance** χ defined by

$$d\chi = \frac{dr}{\sqrt{1 - kr^2}}\,, \tag{7.3}$$

so that

$$ds^2 = -c^2\,dt^2 + a^2(t)\left[d\chi^2 + S_k^2(\chi)\,d\Omega^2\right], \tag{7.4}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\,d\phi^2$ and the **transverse comoving distance** $S_k(\chi)$ is

$$S_k(\chi) = \begin{cases} \sin\chi & k = +1 \\ \chi & k = 0 \\ \sinh\chi & k = -1 \end{cases} \tag{7.5}$$

7.3 Christoffel Symbols for the FRW Metric

To derive the Friedmann equations, we need the Christoffel symbols, Ricci tensor, and Ricci scalar for the RW metric (7.2). We use the general functions developed in Module 1b (`christoffel_symbols.wl`) and verified symbolically in `friedmann_equations.wl`.

The coordinates are $x^\mu = (t, r, \theta, \phi)$, and the nonvanishing metric components are (reading off from eq. (7.2)):

$$\begin{aligned} g_{00} &= -c^2\,, & g_{11} &= \frac{a^2(t)}{1 - kr^2}\,, \\ g_{22} &= a^2(t)\,r^2\,, & g_{33} &= a^2(t)\,r^2\sin^2\theta\,. \end{aligned} \tag{7.6}$$

Since this is a diagonal metric, we use the formulas from Module 1b (Remark 2.10.1): for a diagonal metric with components g_{aa} (no sum),

$$\begin{aligned} \Gamma^a{}_{bb} &= -\frac{1}{2g_{aa}}\,\partial_a g_{bb} \qquad (a \neq b)\,, \\ \Gamma^a{}_{ab} &= \frac{1}{2g_{aa}}\,\partial_b g_{aa} \qquad (\text{no sum on } a)\,. \end{aligned} \tag{7.7}$$

The **nonvanishing Christoffel symbols** of the RW metric are (Carroll eq. 8.44; verified in `friedmann_equations.wl`):

Time-space mixed:

$$\Gamma^0{}_{11} = \frac{a\dot{a}}{c^2(1-kr^2)}\,, \quad \Gamma^0{}_{22} = \frac{a\dot{a}\,r^2}{c^2}\,, \quad \Gamma^0{}_{33} = \frac{a\dot{a}\,r^2\sin^2\theta}{c^2} \tag{7.8}$$

and their counterparts (mixed time–space, related by the symmetry $\Gamma^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\nu\mu}$):

$$\Gamma^1{}_{01} = \Gamma^2{}_{02} = \Gamma^3{}_{03} = \frac{\dot{a}}{a} \tag{7.9}$$

where the dot denotes d/dt.

Purely spatial:

$$\begin{aligned}
\Gamma^1{}_{11} &= \frac{kr}{1-kr^2}\,, & \Gamma^1{}_{22} &= -r(1-kr^2)\,,\\
\Gamma^1{}_{33} &= -r(1-kr^2)\sin^2\theta\,, & \Gamma^2{}_{12} &= \frac{1}{r}\,,\\
\Gamma^2{}_{33} &= -\sin\theta\cos\theta\,, & \Gamma^3{}_{13} &= \frac{1}{r}\,,\\
\Gamma^3{}_{23} &= \frac{\cos\theta}{\sin\theta}\,. & &
\end{aligned} \tag{7.10}$$

Exercise 7.3.1. Compute all nonvanishing Christoffel symbols of the RW metric by hand using the diagonal metric formulas (7.7). Verify your results with `friedmann_equations.wl`, which uses the general `computeChristoffel` function from Module 1b.

7.4 Ricci Tensor and Ricci Scalar

From the Christoffel symbols, we compute the Riemann tensor $R^\rho{}_{\sigma\mu\nu}$, contract to obtain the Ricci tensor $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$, and then contract again with the inverse metric to get the Ricci scalar $R = g^{\mu\nu}R_{\mu\nu}$. The computation is lengthy but straightforward; we quote the results here and verify them symbolically in `friedmann_equations.wl`.

The nonvanishing independent components of the **Ricci tensor** are (Carroll eq. 8.45):

$$R_{00} = -\frac{3\ddot{a}}{c^2\,a}\,, \tag{7.11}$$

$$R_{11} = \frac{1}{c^2(1-kr^2)}\left(a\ddot{a} + 2\dot{a}^2 + 2kc^2\right), \tag{7.12}$$

$$R_{22} = \frac{r^2}{c^2}\left(a\ddot{a} + 2\dot{a}^2 + 2kc^2\right), \tag{7.13}$$

$$R_{33} = \frac{r^2\sin^2\theta}{c^2}\left(a\ddot{a} + 2\dot{a}^2 + 2kc^2\right). \tag{7.14}$$

Notice that R_{11}, R_{22}, and R_{33} all share the same combination $a\ddot{a} + 2\dot{a}^2 + 2kc^2$, which is a consequence of spatial isotropy. We can write the spatial components compactly as:

$$R_{ij} = \frac{g_{ij}}{c^2\, a^2}\left(a\ddot{a} + 2\dot{a}^2 + 2kc^2\right). \tag{7.15}$$

The **Ricci scalar** is:

$$\boxed{R = g^{\mu\nu}R_{\mu\nu} = \frac{6}{c^2}\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right)} \tag{7.16}$$

7.5 Einstein's Equations with a Perfect Fluid

Einstein's field equations relate the geometry of spacetime (the Einstein tensor $G_{\mu\nu}$) to the matter content (the stress-energy tensor $T_{\mu\nu}$):

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu} \tag{7.17}$$

where Λ is the **cosmological constant**.

For a **perfect fluid** at rest in the comoving frame (the natural choice for a homogeneous, isotropic universe), the stress-energy tensor is (Carroll eq. 8.48):

$$T^{\mu\nu} = \left(\rho + \frac{p}{c^2}\right) U^\mu U^\nu + p\, g^{\mu\nu} \tag{7.18}$$

where ρ is the **mass-energy density** (energy density $/c^2$), p is the **pressure**, and $U^\mu = (1, 0, 0, 0)$ is the four-velocity of comoving observers [with $g_{00} = -c^2$, the standard normalisation $g_{\mu\nu}U^\mu U^\nu = -c^2$ gives $g_{00}(U^0)^2 = -c^2$, so $U^0 = 1$]. The components in the comoving frame are:

$$T^{00} = \rho\,, \qquad T^{ij} = p\, g^{ij}\,, \qquad T^{0i} = 0\,. \tag{7.19}$$

With indices lowered using $T_{\mu\nu} = g_{\mu\alpha}g_{\nu\beta}T^{\alpha\beta}$: $T_{00} = (g_{00})^2\, T^{00} = c^4\rho$ and $T_{ij} = p\, g_{ij}$.

Remark 7.5.1. In cosmology, the "fluid" is not a literal gas: ρ and p represent the averaged energy density and pressure of *all* matter and radiation in the universe, smoothed over scales $\gg 100$ Mpc. The cosmological principle guarantees that ρ and p depend only on time: $\rho = \rho(t)$, $p = p(t)$.

7.6 The Friedmann Equations

We now substitute the Ricci tensor (7.11)–(7.15) and the Ricci scalar (7.16) into Einstein's equations (7.17) with the perfect-fluid stress-energy tensor (7.19).

7.6.1 The First Friedmann Equation

The $(0,0)$ component of Einstein's equations gives:

$$R_{00} - \tfrac{1}{2}R\, g_{00} + \Lambda g_{00} = \frac{8\pi G}{c^4}\, T_{00}\,. \tag{7.20}$$

Substituting $R_{00} = -3\ddot{a}/(c^2 a)$, $g_{00} = -c^2$, $R = 6(\ddot{a}/a + \dot{a}^2/a^2 + kc^2/a^2)/c^2$, and $T_{00} = \rho c^4$:
Left side:

$$\begin{aligned} R_{00} - \tfrac{1}{2}R\, g_{00} + \Lambda g_{00} &= -\frac{3\ddot{a}}{c^2 a} + \frac{3}{c^2}\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right) - \Lambda c^2 \\ &= \frac{3}{c^2}\left(\frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right) - \Lambda c^2\,. \end{aligned} \tag{7.21}$$

Right side:

$$\frac{8\pi G}{c^4}\,\rho c^4 = 8\pi G\rho\,. \tag{7.22}$$

Setting these equal and rearranging:

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\rho}{3} - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}} \tag{7.23}$$

This is the **first Friedmann equation**. It relates the expansion rate $\dot{a}/a$ to the energy density ρ, spatial curvature k, and cosmological constant Λ.

7.6.2 The Second Friedmann Equation

The spatial (i,j) components of Einstein's equations, using the compact form (7.15) and $T_{ij} = p\, g_{ij}$, give (after dividing both sides by $g_{ij}/(c^2 a^2)$):

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}} \tag{7.24}$$

This is the **second Friedmann equation** (or **acceleration equation**). It shows that matter and radiation ($\rho > 0$, $p \geq 0$) cause the expansion to *decelerate* ($\ddot{a} < 0$), while a positive cosmological constant $\Lambda > 0$ causes *acceleration* ($\ddot{a} > 0$).

Remark 7.6.1. The combination $\rho + 3p/c^2$ in the second Friedmann equation is the key to understanding cosmic acceleration. Pressure contributes to gravity in GR (unlike in Newtonian gravity), and the factor of 3 comes from the three spatial dimensions. For a cosmological constant (with equation of state $p = -\rho c^2$), this combination is $\rho + 3(-\rho) = -2\rho < 0$, leading to acceleration.

7.6.3 The Fluid Conservation Equation

The Friedmann equations (7.23) and (7.24) are not independent: differentiating the first and substituting the second yields the **fluid conservation equation** (equivalently, this follows from $\nabla_\mu T^{\mu\nu} = 0$):

$$\boxed{\dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + \frac{p}{c^2}\right) = 0} \tag{7.25}$$

This is the relativistic analogue of the first law of thermodynamics: the energy density decreases as the universe expands, with the rate depending on both ρ and p. The $3\dot{a}/a$ factor reflects the three-dimensional expansion.

Exercise 7.6.1. Derive the Friedmann equations (7.23) and (7.24) from the Einstein equations (7.17) with the FRW Ricci tensor and the perfect-fluid stress-energy tensor. Verify each step with `friedmann_equations.wl`.

7.7 The Hubble Parameter and Density Parameters

7.7.1 The Hubble Parameter

The **Hubble parameter** is defined as the fractional expansion rate:

$$H(t) \equiv \frac{\dot{a}(t)}{a(t)}\,. \tag{7.26}$$

Its present-day value is the **Hubble constant**:

$$H_0 \equiv H(t_0) \approx 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}\,. \tag{7.27}$$

The associated timescale is $t_H \equiv 1/H_0 \approx 14$ Gyr, and the associated length scale is the **Hubble distance**:

$$d_H \equiv \frac{c}{H_0} \approx 4.28\ \mathrm{Gpc} \approx 1.32 \times 10^{26}\ \mathrm{m}\,. \tag{7.28}$$

7.7.2 Density Parameters

The first Friedmann equation (7.23) can be rewritten in terms of dimensionless **density parameters**. Define the **critical density** as the density required for a spatially flat ($k = 0$), $\Lambda = 0$ universe:

$$\rho_{\mathrm{cr}}(t) \equiv \frac{3H^2(t)}{8\pi G}\,, \qquad \rho_{\mathrm{cr},0} = \frac{3H_0^2}{8\pi G} \approx 9.2 \times 10^{-27}\ \mathrm{kg\,m^{-3}}\,. \tag{7.29}$$

The density parameters are:

$$\begin{aligned}
\Omega_{\rm m} &\equiv \frac{\rho_{m,0}}{\rho_{{\rm cr},0}}\,, && \text{(matter: non-relativistic)}\\
\Omega_r &\equiv \frac{\rho_{r,0}}{\rho_{{\rm cr},0}}\,, && \text{(radiation: relativistic)}\\
\Omega_\Lambda &\equiv \frac{\Lambda c^2}{3H_0^2}\,, && \text{(dark energy / cosmological constant)}\\
\Omega_k &\equiv -\frac{kc^2}{H_0^2}\,. && \text{(spatial curvature)}
\end{aligned} \tag{7.30}$$

With these definitions, the first Friedmann equation at the present epoch reduces to the elegant **cosmic sum rule**:

$$\boxed{\Omega_{\rm m} + \Omega_r + \Omega_\Lambda + \Omega_k = 1} \tag{7.31}$$

Current observations (Planck 2018) give $\Omega_{\rm m} \approx 0.31$, $\Omega_\Lambda \approx 0.69$, $\Omega_r \approx 9 \times 10^{-5}$, and $|\Omega_k| < 0.01$, consistent with a spatially flat ($k = 0$) universe. For most lensing calculations, we work with the **concordance cosmology**: $\Omega_{\rm m} = 0.3$, $\Omega_\Lambda = 0.7$, $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

7.8 Equations of State and Scale Factor Solutions

To solve the Friedmann equations, we need a relationship between p and ρ: the **equation of state**

$$p = w\rho c^2\,, \tag{7.32}$$

where w is a dimensionless constant for each component.

7.8.1 Matter, Radiation, and Dark Energy

The three primary components of the universe are:

Component	w	$\rho \propto$	$a(t) \propto$
Non-relativistic matter (dust)	0	a^{-3}	$t^{2/3}$
Radiation (photons, neutrinos)	1/3	a^{-4}	$t^{1/2}$
Cosmological constant (Λ)	-1	a^0 (constant)	e^{Ht}

These follow from the fluid conservation equation (7.25). With $p = w\rho c^2$:

$$\dot\rho + 3\frac{\dot a}{a}(1+w)\rho = 0 \qquad \Longrightarrow \qquad \rho \propto a^{-3(1+w)}\,. \tag{7.33}$$

For matter ($w = 0$): $\rho_m \propto a^{-3}$ (dilution by volume expansion). For radiation ($w = 1/3$): $\rho_r \propto a^{-4}$ (volume dilution plus cosmological redshift of each photon's energy). For Λ ($w = -1$): $\rho_\Lambda = $ constant.

7.8.2 Scale Factor Solutions

In a single-component universe with $k = 0$ (flat) and $\Lambda = 0$:

Matter-dominated: $H^2 = H_0^2\,\Omega_{\rm m}\,a^{-3}$ gives

$$a(t) = \left(\frac{3}{2}H_0\sqrt{\Omega_{\rm m}}\,t\right)^{2/3} \propto t^{2/3}\,. \tag{7.34}$$

Radiation-dominated: $H^2 = H_0^2\,\Omega_r\,a^{-4}$ gives

$$a(t) = \left(2H_0\sqrt{\Omega_r}\,t\right)^{1/2} \propto t^{1/2}\,. \tag{7.35}$$

Λ-dominated: $H^2 = H_0^2\,\Omega_\Lambda$ (constant) gives de Sitter expansion:

$$a(t) \propto e^{H_\Lambda t}\,, \qquad H_\Lambda = H_0\sqrt{\Omega_\Lambda}\,. \tag{7.36}$$

7.9 Redshift and the Scale Factor

As light propagates through the expanding universe, its wavelength stretches with the scale factor. A photon emitted at time $t_{\rm em}$ with wavelength $\lambda_{\rm em}$ is observed at t_0 with wavelength $\lambda_{\rm obs} = \lambda_{\rm em}\cdot a(t_0)/a(t_{\rm em})$. The **cosmological redshift** is therefore:

$$\boxed{1 + z = \frac{a(t_0)}{a(t_{\rm em})} = \frac{1}{a(t_{\rm em})}} \tag{7.37}$$

where we used $a(t_0) = 1$. High-redshift objects emitted their light when the universe was smaller: at $z = 1$, $a = 1/2$; at $z = 2$, $a = 1/3$; and so on.

Using eq. (7.37), the Friedmann equation can be rewritten entirely in terms of redshift. From the first Friedmann equation (7.23), $H^2 = 8\pi G\rho/3 - kc^2/a^2 + \Lambda c^2/3$. Each component's density scales as $\rho_i = \rho_{i,0}\,a^{-3(1+w_i)}$ (eq. 7.33), and $a = 1/(1+z)$ (eq. 7.37), so $\rho_m = \rho_{m,0}(1+z)^3$, $\rho_r = \rho_{r,0}(1+z)^4$, and $\rho_\Lambda = \rho_{\Lambda,0}$. Dividing through by H_0^2 and using the density parameter definitions (7.30): $H^2/H_0^2 = \Omega_{\rm m}(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda + \Omega_k(1+z)^2$. Taking the square root (cf. Congdon & Keeton eq. 3.143, which omits Ω_r; verified in `friedmann_equations.wl`):

$$\boxed{H(z) = H_0\sqrt{\Omega_{\rm m}(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda + \Omega_k(1+z)^2}} \tag{7.38}$$

For the concordance cosmology ($\Omega_{\rm m} = 0.3$, $\Omega_\Lambda = 0.7$, $\Omega_r \approx 0$, $\Omega_k = 0$), this simplifies to:

$$H(z) = H_0\sqrt{0.3\,(1+z)^3 + 0.7}\,. \tag{7.39}$$

Part B: Distance Measures for Lensing

7.10 Comoving Distance

A photon traveling radially ($d\theta = d\phi = 0$) through the RW metric satisfies $ds^2 = 0$, which gives $c\,dt = -a(t)\,d\chi$ (choosing the sign for incoming photons). The **comoving distance** to

an object at redshift z is therefore:

$$\chi(z) = \int_0^z \frac{c\,dz'}{H(z')}\,. \tag{7.40}$$

This integral must in general be evaluated numerically, using $H(z)$ from eq. (7.38). The comoving distance is the "distance at the present epoch" — the distance that would be measured by a chain of rulers laid end-to-end at time t_0.

For the concordance cosmology, some representative values:

z	$\chi(z)$ (Mpc)
0.1	420
0.5	1880
1.0	3300
2.0	5200
5.0	7775

7.11 Angular Diameter Distance

The **angular diameter distance** is defined so that an object of physical (proper) size ℓ at redshift z, subtending an angle $\delta\theta$ on the sky, satisfies the Euclidean-looking relation:

$$\delta\theta = \frac{\ell}{D_A(z)}\,. \tag{7.41}$$

From the RW metric (7.4), a physical transverse distance at redshift z (when $a = 1/(1+z)$) subtending angle $\delta\theta$ is $\ell = a(t)\,S_k(\chi)\,\delta\theta$, so

$$\boxed{D_A(z) = \frac{S_k(\chi(z))}{1+z}} \tag{7.42}$$

For a flat universe ($k = 0$), $S_k(\chi) = \chi$, and this simplifies to:

$$D_A(z) = \frac{\chi(z)}{1+z} = \frac{1}{1+z}\int_0^z \frac{c\,dz'}{H(z')}\,. \tag{7.43}$$

The angular diameter distance is the distance measure that enters the **lens equation**. The angular diameter distances $D_\mathrm{d} = D_A(z_d)$, $D_\mathrm{s} = D_A(z_s)$, and D_ds (defined below) determine the mapping from physical deflection angles to angular positions on the sky.

7.12 The Angular Diameter Distance Is Not Monotonic

One of the most counterintuitive results in cosmology is that the angular diameter distance $D_A(z)$ is **not a monotonically increasing function** of redshift. For the concordance cosmology, D_A reaches a maximum at $z \approx 1.6$ and then *decreases* at higher redshifts.

Physically, this means that objects at very high redshifts appear *larger* on the sky than one would naively expect. The turnover arises from two competing effects. For nearby objects, increasing redshift means greater comoving distance χ, so $D_A = \chi/(1+z)$ grows. But at high z, the factor $1/(1+z)$ dominates: the universe was much smaller when the light was emitted ($a = 1/(1+z) \ll 1$), so the *physical* transverse size of the object at emission — which determines the angle it subtends — was correspondingly smaller. A smaller physical size at fixed comoving size subtends a *larger* angle, making D_A decrease. The turnover at $z \approx 1.6$ (for the concordance cosmology) occurs where these two effects balance: $d\chi/dz = \chi/(1+z)$, i.e., where the fractional growth of comoving distance equals the fractional growth of the redshift factor (for D_A see Carroll Sec. 8.5, eq. 8.125; the turnover is computed numerically in `friedmann_equations.wl`).

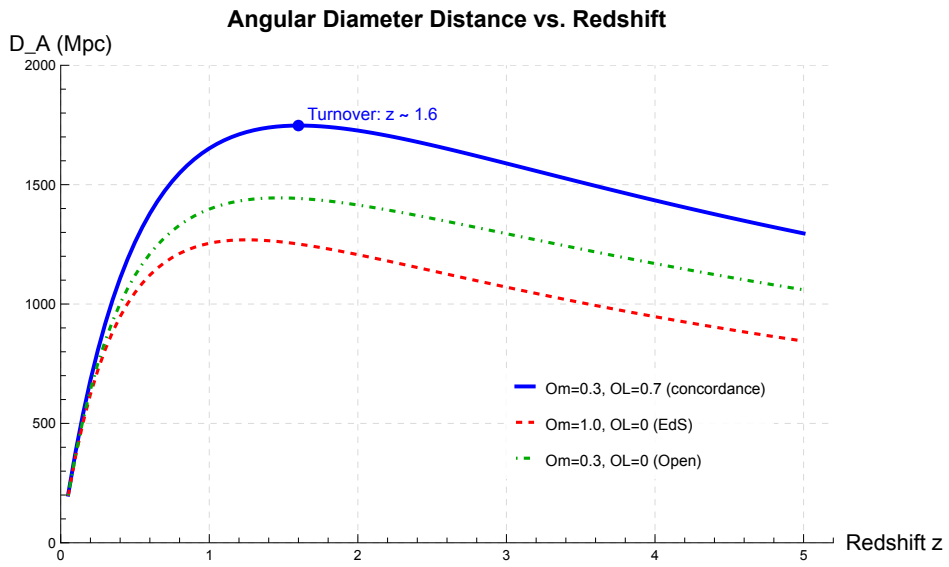


Figure 7.1: Angular diameter distance $D_A(z)$ for the concordance cosmology ($\Omega_{\mathrm{m}} = 0.3$, $\Omega_\Lambda = 0.7$, $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$). D_A reaches a maximum of approximately 1748 Mpc at $z \approx 1.6$ and then decreases. This turnover is a key consequence of the expanding universe and has no analogue in Euclidean geometry. Generated by `friedmann_equations.wl`.

Remark 7.12.1. The turnover of $D_A(z)$ has practical consequences for lensing: a lens at $z_d \sim 0.5$ and a source at $z_s \sim 2$ might have a surprisingly large distance ratio $D_{\mathrm{ds}}/D_{\mathrm{s}}$, and the "sweet spot" for lensing efficiency is typically $z_d \sim 0.2$–0.5 for sources at $z_s \sim 1$–3.

7.13 Luminosity Distance

The **luminosity distance** D_L is defined so that the observed flux F from a source of luminosity L satisfies the Euclidean inverse-square law:

$$F = \frac{L}{4\pi D_L^2}\,. \tag{7.44}$$

In an expanding universe, two effects reduce the observed flux beyond the $1/r^2$ geometric dilution: (1) photon energies are redshifted by a factor $(1+z)$, and (2) photon arrival rates are reduced by another factor $(1+z)$. This gives:

$$D_L(z) = (1+z)\, S_k(\chi(z)) = (1+z)^2\, D_A(z)\,. \tag{7.45}$$

The relation $D_L = (1+z)^2 D_A$ is the **Etherington reciprocity theorem** (or **distance-duality relation**). It holds for any metric theory of gravity in which photons travel on null geodesics and photon number is conserved.

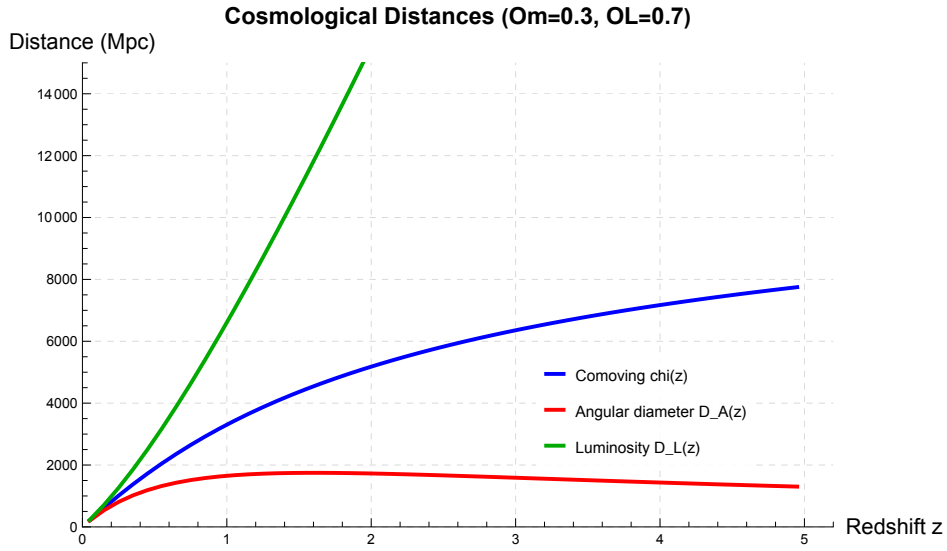


Figure 7.2: Comoving distance $\chi(z)$ (blue), angular diameter distance $D_A(z)$ (red), and luminosity distance $D_L(z)$ (green) for the concordance cosmology. At low redshift ($z \ll 1$) all three converge: $\chi \approx D_A \approx D_L \approx cz/H_0$. At high redshift they diverge dramatically. The turnover of D_A is clearly visible, while D_L grows monotonically. Generated by `friedmann_equations.wl`.

7.14 The Lens–Source Angular Diameter Distance

For gravitational lensing, we need not only $D_{\rm d} = D_A(z_d)$ and $D_{\rm s} = D_A(z_s)$, but also $D_{\rm ds}$: the angular diameter distance from the *lens* to the *source*. This is **not** simply $D_A(z_s) - D_A(z_d)$, because angular diameter distances do not add linearly.

In a flat universe ($k = 0$), the comoving distances *do* add: $\chi_{ds} = \chi_s - \chi_d$, and the lens–source angular diameter distance is:

$$\boxed{D_{\rm ds} = \frac{\chi_s - \chi_d}{1+z_s} = \frac{1}{1+z_s}\int_{z_d}^{z_s} \frac{c\,dz'}{H(z')}} \tag{7.46}$$

For a general (non-flat) universe one uses the Hogg (1999) form (eq. 19; equivalently Congdon & Keeton eq. 3.155):

$$D_{\rm ds} = \frac{1}{1+z_s}\left[D_M(z_s)\sqrt{1+\Omega_k\, D_M^2(z_d)/D_H^2} - D_M(z_d)\sqrt{1+\Omega_k\, D_M^2(z_s)/D_H^2}\right], \tag{7.47}$$

where $D_M(z) = (1+z)\,D_A(z)$ is the transverse comoving distance, $D_H = c/H_0$ is the Hubble distance, and $\Omega_k = -k\,(c/(H_0 R_0))^2$ is the curvature density parameter. For the spatially flat case ($\Omega_k = 0$), eq. (7.47) reduces to eq. (7.46), which is exact and sufficient for the rest of these notes.

Remark 7.14.1. A common mistake is to write $D_{\rm ds} = D_{\rm s} - D_{\rm d}$. This is **wrong** in general. Even in a flat universe, $D_{\rm ds} \neq D_{\rm s} - D_{\rm d}$ because the factors of $(1+z)$ are different. One must always compute $D_{\rm ds}$ using eq. (7.46) (or its curved-space generalization). To illustrate: for $z_d = 0.5$ and $z_s = 2.0$ in the concordance cosmology, $D_{\rm d} \approx 1259$ Mpc, $D_{\rm s} \approx 1727$ Mpc, but $D_{\rm ds} \approx 1097$ Mpc $\neq D_{\rm s} - D_{\rm d} \approx 468$ Mpc.

7.15 The Distance Ratio for Lensing

The **lens equation** (Module 4) and the **Einstein radius** both depend on a specific combination of distances. The Einstein radius of a point mass M is (Congdon & Keeton eq. 2.25):

$$\theta_{\rm E} = \sqrt{\frac{4GM}{c^2}\,\frac{D_{\rm ds}}{D_{\rm d}\,D_{\rm s}}} \tag{7.48}$$

and the critical surface mass density is:

$$\Sigma_{\rm cr} = \frac{c^2}{4\pi G}\,\frac{D_{\rm s}}{D_{\rm d}\,D_{\rm ds}}\,. \tag{7.49}$$

The key quantity is the **distance ratio**:

$$\mathcal{D} \equiv \frac{D_{\rm ds}}{D_{\rm d}\,D_{\rm s}}\,, \tag{7.50}$$

which determines the "lensing strength" of a given observer–lens–source configuration.

Note that $\mathcal{D}$ is *not* simply $1/D_{\rm d}$, because $D_{\rm ds}$ depends on both z_d and z_s in a non-trivial way. The distance ratio is maximized (lensing is most efficient) when the lens is roughly halfway between the observer and the source in terms of comoving distance.

7.16 Lensing Efficiency

For a fixed source redshift z_s, one can ask: at what lens redshift z_d is lensing most efficient? The answer is determined by the **lensing efficiency function**:

$$\mathcal{E}(z_d; z_s) \equiv \frac{D_{\rm d}\,D_{\rm ds}}{D_{\rm s}} \tag{7.51}$$

which is proportional to the square of the Einstein radius (at fixed mass). Note that $\theta_{\rm E}^2 \propto D_{\rm ds}/(D_{\rm d}\,D_{\rm s}) = \mathcal{D}$, while the physical Einstein ring area scales as $D_{\rm d}^2\theta_{\rm E}^2 \propto D_{\rm d}\,D_{\rm ds}/D_{\rm s} = \mathcal{E}$.

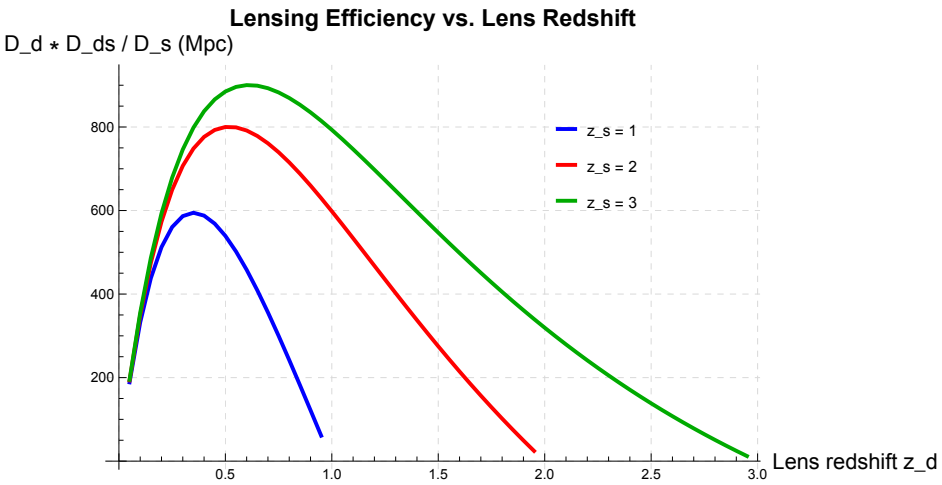


Figure 7.3: Lensing efficiency $D_{\mathrm{d}}\, D_{\mathrm{ds}}/D_{\mathrm{s}}$ as a function of lens redshift z_d for several source redshifts $z_s = 1, 2, 3$ in the concordance cosmology. For $z_s = 1$, lensing is most efficient at $z_d \approx 0.35$; for $z_s = 2$, at $z_d \approx 0.5$. The lens is typically most effective when placed roughly "halfway" (in comoving distance) between the observer and the source. Generated by `friedmann_equations.wl`.

7.17 Summary

- The **cosmological principle** (homogeneity + isotropy) constrains the spacetime metric to the Robertson–Walker form $ds^2 = -c^2dt^2 + a^2(t)\,[dr^2/(1-kr^2) + r^2 d\Omega^2]$.
- Plugging the RW metric into **Einstein's equations** with a perfect-fluid stress-energy tensor yields the **Friedmann equations**:

$$H^2 = \frac{8\pi G\rho}{3} - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}\,,$$
$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}\,.$$

- The density parameters Ω_{m}, Ω_Λ, Ω_k satisfy $\Omega_{\mathrm{m}} + \Omega_\Lambda + \Omega_k = 1$ (ignoring radiation at late times).
- The **angular diameter distance** $D_A(z) = \chi(z)/(1+z)$ is *not monotonic*: it turns over at $z \approx 1.6$ in the concordance cosmology.
- The **luminosity distance** is $D_L = (1+z)^2 D_A$ (Etherington reciprocity).
- The **lens–source distance** $D_{\mathrm{ds}} \neq D_{\mathrm{s}} - D_{\mathrm{d}}$; it must be computed as $D_{\mathrm{ds}} = (\chi_s - \chi_d)/(1+z_s)$ for a flat universe.
- The **Einstein radius** $\theta_{\mathrm{E}} \propto \sqrt{D_{\mathrm{ds}}/(D_{\mathrm{d}}\, D_{\mathrm{s}})}$ depends on the distance configuration, and lensing is most efficient when the lens is roughly halfway (in comoving distance) to the source.

7.18 Problems

Exercise 7.18.1. (Deriving the Friedmann equations from the RW metric.) Starting from the Robertson–Walker metric (7.2):

(a) Compute the Christoffel symbols for the FRW metric.

(b) Compute the Ricci tensor $R_{\mu\nu}$ and Ricci scalar R.

(c) Use Einstein's equations with the perfect-fluid stress-energy tensor to derive the first and second Friedmann equations.

(d) Show that differentiating the first Friedmann equation and using the second gives the fluid conservation equation.

Verify all steps with `friedmann_equations.wl`.

Exercise 7.18.2. (Age of the universe for different cosmologies.) The age of the universe is $t_0 = \int_0^\infty dz / [(1+z)\,H(z)]$.

(a) For an Einstein–de Sitter universe ($\Omega_\mathrm{m} = 1$, $\Omega_\Lambda = 0$, $k = 0$), show analytically that $t_0 = 2/(3H_0)$.

(b) Compute t_0 numerically for the concordance cosmology ($\Omega_\mathrm{m} = 0.3$, $\Omega_\Lambda = 0.7$). You should find $t_0 \approx 13.5$ Gyr for $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

(c) Compare with an open universe ($\Omega_\mathrm{m} = 0.3$, $\Omega_\Lambda = 0$, $\Omega_k = 0.7$).

Exercise 7.18.3. (Angular diameter distance turnover.)

(a) Plot $D_A(z)$ for the following cosmologies and identify the turnover redshift in each: (i) $\Omega_\mathrm{m} = 0.3$, $\Omega_\Lambda = 0.7$ (concordance); (ii) $\Omega_\mathrm{m} = 1.0$, $\Omega_\Lambda = 0$ (Einstein–de Sitter); (iii) $\Omega_\mathrm{m} = 0.3$, $\Omega_\Lambda = 0$ (open).

(b) Explain physically why D_A turns over: why do high-redshift objects appear *larger* than intermediate-redshift objects?

(c) For the concordance cosmology, find the redshift z_max at which D_A is maximum by numerically solving $dD_A/dz = 0$.

Exercise 7.18.4. (Distances for a real lens system: Q0957+561.) The first gravitational lens discovered (Walsh, Carswell & Weymann 1979) has lens redshift $z_d = 0.36$ and source redshift $z_s = 1.41$. Using the concordance cosmology:

(a) Compute D_d, D_s, and D_ds in Mpc.

(b) Verify that $D_\mathrm{ds} \neq D_\mathrm{s} - D_\mathrm{d}$.

(c) Compute the critical surface mass density Σ_cr in $M_\odot\,\mathrm{kpc}^{-2}$.

(d) Compute the Einstein radius θ_E for a lens mass $M = 10^{12}\,M_\odot$.

Exercise 7.18.5. (Einstein radius for a typical galaxy lens.) For a flat ΛCDM universe with $\Omega_{\rm m} = 0.3$, $\Omega_\Lambda = 0.7$, $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$:

(a) Compute $D_{\rm d}$, $D_{\rm s}$, $D_{\rm ds}$ for $z_d = 0.3$, $z_s = 1.0$.

(b) Compute the Einstein radius $\theta_{\rm E}$ for a lens mass $M = 10^{12}\,M_\odot$. You should find $\theta_{\rm E} \approx 2.4''$.

(c) How does $\theta_{\rm E}$ scale with M? What mass gives $\theta_{\rm E} = 10''$?

(d) Plot $\theta_{\rm E}$ as a function of z_d (with $z_s = 1$ and $M = 10^{12}\,M_\odot$ fixed). Does $\theta_{\rm E}$ have a maximum? At what z_d is the physical Einstein radius $R_{\rm E} = D_{\rm d}\theta_{\rm E}$ largest?

Verify with `problems_03.wl`.

Chapter 8

The Gravitational Lens Equation

> *Equation (14) is called the lens equation, or ray-tracing equation. It is nonlinear in the general case, and so it is possible to have multiple images $\vec{\theta}$ corresponding to a single source position $\vec{\beta}$.* — Narayan & Bartelmann (1997), Sect. 2.1.3

In Module 6 we derived the deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ for a point mass, and in Module 5 we showed that deflection angles from multiple masses superpose in the weak-field limit. In Module 7 we introduced the angular diameter distances D_d, D_s, and D_ds that describe the geometry of the observer–lens–source system at cosmological separations.

Now we assemble these ingredients into the **lens equation** — the fundamental mapping from source positions to image positions. This single equation is the starting point for everything that follows: image multiplicity, magnification, time delays, and lens modeling.

Companion Mathematica files:

- `lens_equation.wl` — symbolic derivation of the lens equation, Einstein radius, point mass image positions, magnification preview, critical surface mass density, and publication-quality figure generation
- `einstein_ring.nb` — interactive notebook: Manipulate for source position and image positions, Einstein ring visualization, interactive θ_E calculator

8.1 Introduction: From Deflection to Imaging

We now have all the pieces needed to build the gravitational lens equation:

1. **The deflection angle** (Modules 1e and 2): $\hat{\alpha} = 4GM/(c^2 b)$ for a point mass, and the superposition principle (5.25) for multiple masses and continuous distributions.

2. **The cosmological distances** (Module 3): the angular diameter distances D_d, D_s, D_ds relating angular separations on the sky to physical separations at the lens and source.

The task of this module is to combine these into a single equation that maps source positions $\boldsymbol{\beta}$ to image positions $\boldsymbol{\theta}$. Remarkably, this equation can produce *multiple solutions* — multiple images of a single source — which is the defining phenomenon of strong gravitational lensing.

8.2 The Thin-Screen Approximation

The physical extent of a gravitational lens along the line of sight is typically much smaller than the distances between observer, lens, and source. For a galaxy lens, the stellar and dark matter distribution extends over ~ 100 kpc, while $D_{\rm d} \sim 1$ Gpc. For a galaxy cluster, the virial radius is ~ 2 Mpc, compared to distances of several Gpc. In both cases the ratio

$$\frac{\text{lens extent}}{\text{cosmological distance}} \sim \frac{100 \text{ kpc}}{1 \text{ Gpc}} = 10^{-4} \tag{8.1}$$

is exceedingly small.

Remark 8.2.1. The thin-screen approximation is the gravitational analog of the **thin-lens** approximation in classical optics. Just as a glass lens can be treated as imparting a phase shift at a single plane (when its thickness is much less than the focal length), a gravitational lens can be treated as deflecting light at a single **lens plane** perpendicular to the line of sight at distance $D_{\rm d}$.

This justifies the following idealization: all the mass of the lens is projected onto a single plane (the **lens plane**) at angular diameter distance $D_{\rm d}$ from the observer. The deflection of a light ray occurs instantaneously as it crosses this plane. The three-dimensional mass distribution $\rho(\mathbf{r})$ is replaced by a two-dimensional **surface mass density** $\Sigma(\boldsymbol{\xi})$, obtained by integrating ρ along the line of sight:

$$\Sigma(\boldsymbol{\xi}) = \int_{-\infty}^{+\infty} \rho(\boldsymbol{\xi}, z)\, dz\,, \tag{8.2}$$

where $\boldsymbol{\xi}$ is the two-dimensional position in the lens plane and z is the coordinate along the line of sight.

8.3 Lensing Geometry

Figure 8.1 shows the geometry of a typical gravitational lensing configuration. The key elements are:

- **Observer (O):** located at the bottom of the diagram.
- **Lens plane:** a plane perpendicular to the optical axis at angular diameter distance $D_{\rm d}$ from the observer. The lens (galaxy, cluster, etc.) is projected onto this plane.
- **Source plane:** a plane perpendicular to the optical axis at angular diameter distance $D_{\rm s}$ from the observer. The background source (galaxy, quasar, etc.) lies in this plane.
- **Optical axis:** the line from the observer through the center of the lens, defining the origin of the angular coordinate system.

We define the following angles (all measured from the optical axis):

- $\boldsymbol{\theta}$: the **observed image position** on the sky,

- $\boldsymbol{\beta}$: the **true source position** (where the source would appear without lensing),
- $\hat{\alpha}$: the **physical deflection angle** at the lens plane,
- $\boldsymbol{\alpha}(\boldsymbol{\theta})$: the **reduced deflection angle**, defined below.

The physical positions in the lens and source planes are related to the angular positions by:

$$\boldsymbol{\xi} = D_{\mathrm{d}}\,\boldsymbol{\theta}\,, \tag{8.3}$$
$$\boldsymbol{\eta} = D_{\mathrm{s}}\,\boldsymbol{\beta}\,, \tag{8.4}$$

where $\boldsymbol{\xi}$ is the position in the lens plane and $\boldsymbol{\eta}$ is the position in the source plane (Narayan & Bartelmann eq. 15; Congdon & Keeton Sec. 2.2.2).

8.4 The Lens Equation

From the geometry of Figure 8.1, we can read off the fundamental relationship between source and image positions. A light ray that is observed at angle $\boldsymbol{\theta}$ from the optical axis was emitted from the source at position $\boldsymbol{\eta}$ in the source plane. Following the deflected ray from the lens plane to the source plane, the physical displacement in the source plane is (Narayan & Bartelmann Sec. 2.1.3; Congdon & Keeton eq. 2.16):

$$\boldsymbol{\eta} = \frac{D_{\mathrm{s}}}{D_{\mathrm{d}}}\,\boldsymbol{\xi} - D_{\mathrm{ds}}\,\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi})\,. \tag{8.5}$$

Substituting $\boldsymbol{\xi} = D_{\mathrm{d}}\,\boldsymbol{\theta}$ and $\boldsymbol{\eta} = D_{\mathrm{s}}\,\boldsymbol{\beta}$:

$$D_{\mathrm{s}}\,\boldsymbol{\beta} = D_{\mathrm{s}}\,\boldsymbol{\theta} - D_{\mathrm{ds}}\,\hat{\boldsymbol{\alpha}}(D_{\mathrm{d}}\,\boldsymbol{\theta})\,, \tag{8.6}$$

and dividing by D_{s}:

$$\boxed{\boldsymbol{\beta} = \boldsymbol{\theta} - \frac{D_{\mathrm{ds}}}{D_{\mathrm{s}}}\,\hat{\boldsymbol{\alpha}}(D_{\mathrm{d}}\,\boldsymbol{\theta})} \tag{8.7}$$

This is the **gravitational lens equation** in terms of the physical deflection angle $\hat{\boldsymbol{\alpha}}$. It is convenient to define the **reduced deflection angle**:

$$\boxed{\boldsymbol{\alpha}(\boldsymbol{\theta}) \equiv \frac{D_{\mathrm{ds}}}{D_{\mathrm{s}}}\,\hat{\boldsymbol{\alpha}}(D_{\mathrm{d}}\,\boldsymbol{\theta})} \tag{8.8}$$

so that the lens equation takes the elegant form:

$$\boxed{\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})} \tag{8.9}$$

Remark 8.4.1. Equation (8.9) is **THE** fundamental equation of gravitational lensing. Every result in strong lensing theory — image positions, magnifications, time delays, critical curves, caustics — derives from this single equation. The nonlinearity of the deflection angle $\boldsymbol{\alpha}(\boldsymbol{\theta})$ is what allows multiple solutions (multiple images) for a given source position $\boldsymbol{\beta}$.

Remark 8.4.2. The lens equation is a mapping from the image plane to the source plane: given an image position $\boldsymbol{\theta}$, it determines the corresponding source position $\boldsymbol{\beta}$. The *inverse* problem — given a source position $\boldsymbol{\beta}$, find all image positions $\boldsymbol{\theta}$ — is generally much harder, because the equation can have multiple solutions. For a point mass lens the inverse problem can be solved analytically (Section 8.7); for more complex lens models it typically requires numerical root-finding.

8.5 Surface Mass Density and the Deflection Integral

For a continuous mass distribution described by the surface mass density $\Sigma(\boldsymbol{\xi})$ (eq. 8.2), the physical deflection angle at position $\boldsymbol{\xi}$ in the lens plane is obtained by superposing the deflections from all mass elements (Congdon & Keeton eq. 4.9; cf. Module 1e eq. 5.26):

$$\boxed{\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = \frac{4G}{c^2} \int \frac{(\boldsymbol{\xi} - \boldsymbol{\xi}')\,\Sigma(\boldsymbol{\xi}')}{|\boldsymbol{\xi} - \boldsymbol{\xi}'|^2}\, d^2\xi'} \tag{8.10}$$

This is the two-dimensional analog of the Newtonian gravitational field of a surface mass distribution. The integral extends over the entire lens plane. In terms of the angular position $\boldsymbol{\theta}$ and using $\boldsymbol{\xi} = D_\mathrm{d}\,\boldsymbol{\theta}$, the reduced deflection angle becomes:

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \frac{D_\mathrm{ds}}{D_\mathrm{s}} \frac{4G}{c^2} D_\mathrm{d} \int \frac{(\boldsymbol{\theta} - \boldsymbol{\theta}')\,\Sigma(D_\mathrm{d}\,\boldsymbol{\theta}')}{|\boldsymbol{\theta} - \boldsymbol{\theta}'|^2}\, d^2\theta' \,. \tag{8.11}$$

Remark 8.5.1. The deflection integral (8.10) follows directly from the superposition principle derived in Module 5 (eq. 5.25). Each infinitesimal mass element $\Sigma(\boldsymbol{\xi}')\,d^2\xi'$ acts as a point mass, deflecting the light ray by $4G\Sigma(\boldsymbol{\xi}')\,d^2\xi'/(c^2\,|\boldsymbol{\xi} - \boldsymbol{\xi}'|)$ in the direction of $(\boldsymbol{\xi} - \boldsymbol{\xi}')/|\boldsymbol{\xi} - \boldsymbol{\xi}'|$. Superposition (valid because we are in the weak-field regime) gives the total deflection as the integral over the lens plane.

8.6 Critical Surface Mass Density

There is a natural surface mass density at which the lens is just strong enough to bend a ray from the source back to the optical axis. Consider a uniform disk of constant surface mass density Σ, and ask what Σ is required for the disk to act as a perfect focusing lens for a source at infinity, i.e., $\alpha(\theta) = \theta$ for every θ inside the disk.

For a uniform disk the two-dimensional enclosed mass within $\xi = D_\mathrm{d}\theta$ is $M_\mathrm{2D}(\xi) = \pi\xi^2\Sigma$, so the physical deflection $\hat{\alpha} = 4GM_\mathrm{2D}/(c^2\xi) = 4\pi G\Sigma\,\xi/c^2$. Converting to the reduced deflection via $\boldsymbol{\alpha} = (D_\mathrm{ds}/D_\mathrm{s})\,\hat{\alpha}$ and $\xi = D_\mathrm{d}\theta$:

$$\alpha(\theta) \;=\; \frac{D_\mathrm{ds}}{D_\mathrm{s}} \cdot \frac{4\pi G\Sigma\, D_\mathrm{d}\,\theta}{c^2}.$$

The focusing condition $\alpha(\theta) = \theta$ removes the factor of θ and gives $\Sigma = \Sigma_\mathrm{cr}$ with (Narayan & Bartelmann eq. 17; Congdon & Keeton eq. 2.68):

$$\boxed{\Sigma_\mathrm{cr} = \frac{c^2}{4\pi G} \frac{D_\mathrm{s}}{D_\mathrm{d}\, D_\mathrm{ds}}\,.} \tag{8.12}$$

Remark 8.6.1. The factor of 4π traces to the 2D Poisson equation $\nabla^2\psi = 2\kappa$ (Module 5): the 4π in eq. (8.12) is the product of $4G/c^2$ (from the deflection formula) and the π that converts the uniform-disk enclosed mass to the line-averaged surface density.

The **convergence** is then the dimensionless ratio of the actual surface mass density to the critical value:

$$\boxed{\kappa(\boldsymbol{\theta}) = \frac{\Sigma(D_\mathrm{d}\,\boldsymbol{\theta})}{\Sigma_\mathrm{cr}}} \tag{8.13}$$

The critical surface mass density sets the boundary between weak and strong lensing:

- If $\kappa < 1$ everywhere ($\Sigma < \Sigma_\mathrm{cr}$), the lens is **subcritical**: it can still deflect light, but it cannot produce multiple images of a generic source.
- If $\kappa \geq 1$ somewhere ($\Sigma \geq \Sigma_\mathrm{cr}$), the lens is **supercritical**: it can produce multiple images, arcs, and Einstein rings — this is the regime of **strong gravitational lensing**.

Example 8.6.1. Numerical value of Σ_cr. For a typical galaxy lens at $z_d = 0.3$ with a source at $z_s = 1$ in a flat ΛCDM cosmology ($H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\mathrm{m} = 0.3$, $\Omega_\Lambda = 0.7$), the angular diameter distances are $D_\mathrm{d} \approx 919$ Mpc, $D_\mathrm{s} \approx 1652$ Mpc, $D_\mathrm{ds} \approx 1055$ Mpc (computed in `friedmann_equations.wl`). Substituting into eq. (8.12):

$$\Sigma_\mathrm{cr} \approx \frac{(3\times10^8)^2}{4\pi \times 6.674\times10^{-11}} \times \frac{1652}{919\times1055}\ \mathrm{Mpc^{-1}} \approx 0.59\ \mathrm{g\,cm^{-2}} \approx 2840\ M_\odot\,\mathrm{pc^{-2}}\,. \tag{8.14}$$

This is a remarkably low surface mass density — only about $5.9\ \mathrm{kg\,m^{-2}}$. Nevertheless, typical galaxy and cluster lenses can achieve $\Sigma \geq \Sigma_\mathrm{cr}$ in their central regions, making strong lensing possible. The exact value is computed and verified in `lens_equation.wl`.

Exercise 8.6.1. Show that Σ_cr as defined in eq. (8.12) has units of mass per area. *Hint:* G has units of $\mathrm{m^3\,kg^{-1}\,s^{-2}}$, c has units of $\mathrm{m\,s^{-1}}$, and the distances have units of length.

8.7 The Einstein Radius

The lens equation simplifies dramatically for a **point mass lens**. For a mass M at the origin of the lens plane, the physical deflection angle is (from Module 2):

$$\hat{\alpha}(\xi) = \frac{4GM}{c^2\,\xi}\,, \tag{8.15}$$

where $\xi = |\boldsymbol{\xi}| = D_\mathrm{d}\,\theta$ is the physical distance from the mass in the lens plane. The reduced deflection angle is:

$$\alpha(\theta) = \frac{D_\mathrm{ds}}{D_\mathrm{s}}\,\frac{4GM}{c^2\,D_\mathrm{d}\,\theta} = \frac{\theta_\mathrm{E}^2}{\theta}\,, \tag{8.16}$$

where we have defined the **Einstein radius**:

$$\boxed{\theta_\mathrm{E} = \sqrt{\frac{4GM}{c^2}\,\frac{D_\mathrm{ds}}{D_\mathrm{d}\,D_\mathrm{s}}}} \tag{8.17}$$

This is **THE** characteristic angular scale of gravitational lensing. For a circularly symmetric lens, the Einstein radius is the angular radius of the ring that forms when the source lies exactly behind the lens ($\beta = 0$). The corresponding physical radius in the lens plane is:

$$R_{\rm E} = D_{\rm d}\,\theta_{\rm E} = \sqrt{\frac{4GM}{c^2}\,\frac{D_{\rm d}\,D_{\rm ds}}{D_{\rm s}}}\,. \tag{8.18}$$

8.7.1 The Point Mass Lens Equation

Substituting eq. (8.16) into the lens equation (8.9) (restricting to one angular dimension by the circular symmetry of the point mass):

$$\boxed{\beta = \theta - \frac{\theta_{\rm E}^2}{\theta}} \tag{8.19}$$

This is a quadratic equation in θ. Multiplying through by θ:

$$\theta^2 - \beta\,\theta - \theta_{\rm E}^2 = 0\,. \tag{8.20}$$

The two solutions are:

$$\boxed{\theta_\pm = \frac{1}{2}\left(\beta \pm \sqrt{\beta^2 + 4\,\theta_{\rm E}^2}\right)} \tag{8.21}$$

Since $\beta^2 + 4\,\theta_{\rm E}^2 > \beta^2$ for $\theta_{\rm E} > 0$, both solutions are always real. A point mass lens *always* produces two images, regardless of the source position.

8.7.2 The Einstein Ring

When the source is exactly on the optical axis ($\beta = 0$), the two image positions merge:

$$\theta_\pm = \pm\,\theta_{\rm E}\,. \tag{8.22}$$

By the circular symmetry of the lens, this corresponds to a complete **ring** of images at angular radius $\theta_{\rm E}$ — the **Einstein ring**. This is the most spectacular manifestation of gravitational lensing and has been observed in numerous galaxy-scale and cluster-scale lens systems.

8.7.3 Numerical Examples

Example 8.7.1. Einstein radii for typical lensing systems.

(a) Stellar microlensing ($M = 1\,M_\odot$, $D_{\rm d} \approx D_{\rm s}/2 \approx 5$ kpc, $D_{\rm ds} \approx 5$ kpc):

$$\theta_{\rm E} = \sqrt{\frac{4 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(3 \times 10^8)^2} \times \frac{1}{10 \times 3.086 \times 10^{19}}} \approx 0.9 \text{ mas}\,. \tag{8.23}$$

The Einstein radius for stellar microlensing is of order *milliarcseconds* — far too small to resolve the individual images, but the magnification is detectable photometrically.

(b) Galaxy-scale lensing ($M = 10^{12}\,M_\odot$, $z_d = 0.3$, $z_s = 1$): With $D_\mathrm{d} \approx 919$ Mpc, $D_\mathrm{s} \approx 1652$ Mpc, $D_\mathrm{ds} \approx 1055$ Mpc:

$$\theta_\mathrm{E} \approx 2.4'' \,. \tag{8.24}$$

Galaxy-scale Einstein radii are of order *arcseconds* — easily resolved with ground-based telescopes. This is the regime of strong lensing that produces multiple images and arcs.

(c) Cluster-scale lensing ($M_\mathrm{E} \sim 2 \times 10^{14}\,M_\odot$ inside the Einstein radius, $z_d = 0.3$, $z_s = 2$): With $D_\mathrm{d} \approx 919$ Mpc, $D_\mathrm{s} \approx 1727$ Mpc, $D_\mathrm{ds} \approx 1329$ Mpc, the point-mass formula gives:

$$\theta_\mathrm{E} \approx 37'' \,. \tag{8.25}$$

Cluster-scale Einstein radii are of order *tens of arcseconds*, producing the spectacular giant arcs observed in massive galaxy clusters. Note that for an extended cluster only the mass *within* the Einstein radius enters the point-mass-equivalent estimate above; the total virial mass $M_{200} \sim 10^{15}\,M_\odot$ of a typical massive cluster is much larger, but most of it lies outside θ_E. An NFW treatment (Module 10) yields the same answer self-consistently.

These values are computed and verified in `lens_equation.wl`. Interactive exploration is available in `einstein_ring.nb`.

8.8 Image Positions for a Point Mass Lens

The two image positions $\theta_\pm$ from eq. (8.21) have several important properties. It is convenient to work with the dimensionless variable $u \equiv \beta/\theta_\mathrm{E}$ (source position in units of the Einstein radius). Then:

$$\theta_\pm = \frac{\theta_\mathrm{E}}{2}\left(u \pm \sqrt{u^2+4}\right). \tag{8.26}$$

8.8.1 Properties of the Images

1. θ_+ **is always positive** (on the same side of the lens as the source) and satisfies $\theta_+ > \theta_\mathrm{E}$. This is the **primary image**, which is always located *outside* the Einstein radius.

2. θ_- **is always negative** (on the opposite side of the lens from the source) and satisfies $|\theta_-| < \theta_\mathrm{E}$. This is the **secondary image**, located *inside* the Einstein radius. Its negative sign indicates opposite parity (the image is mirror-reflected).

3. **Sum and product:** From Vieta's formulas applied to the quadratic (8.20):

$$\theta_+ + \theta_- = \beta\,, \qquad \theta_+\,\theta_- = -\theta_\mathrm{E}^2\,. \tag{8.27}$$

   The product relation shows that the images always straddle the Einstein ring: one inside, one outside.

4. **Image separation:**

$$\Delta\theta = \theta_+ - \theta_- = \sqrt{\beta^2 + 4\,\theta_\mathrm{E}^2}\,. \tag{8.28}$$

   The minimum separation occurs at $\beta = 0$ (Einstein ring configuration), where $\Delta\theta = 2\,\theta_\mathrm{E}$.

8.8.2 Limiting Cases

Source on the optical axis ($\beta \to 0$, $u \to 0$): Both images approach the Einstein radius: $\theta_+ \to +\theta_{\rm E}$, $\theta_- \to -\theta_{\rm E}$. The two images merge into the **Einstein ring**.

Source far from the lens ($\beta \gg \theta_{\rm E}$, $u \gg 1$):

$$\theta_+ \approx \beta + \frac{\theta_{\rm E}^2}{\beta} \to \beta \quad \text{(approaches the unlensed position)}\,, \tag{8.29}$$

$$\theta_- \approx -\frac{\theta_{\rm E}^2}{\beta} \to 0 \quad \text{(approaches the lens, becomes infinitely demagnified)}\,. \tag{8.30}$$

In this limit the primary image is essentially at the unlensed source position (negligible lensing), and the secondary image is extremely close to the lens and too faint to observe.

8.9 Magnification of a Point Mass Lens

Gravitational lensing conserves surface brightness (a consequence of Liouville's theorem applied to the photon phase space), but it changes the *solid angle* subtended by the image compared to the source. The ratio of image area to source area is the **magnification** μ. We will derive the magnification formalism rigorously in Module 5; here we preview the key results for the point mass lens.

For a circularly symmetric lens, the magnification is (Narayan & Bartelmann eq. 26):

$$\mu = \frac{\theta}{\beta}\,\frac{d\theta}{d\beta}\,. \tag{8.31}$$

The first factor accounts for the tangential stretching of the image (the image is at a different angular radius than the source), and the second factor accounts for the radial stretching.

For the point mass lens equation $\beta = \theta - \theta_{\rm E}^2/\theta$, we have $d\beta/d\theta = 1 + \theta_{\rm E}^2/\theta^2$, and the magnification of each image is:

$$\mu_\pm = \frac{\theta_\pm}{\beta}\,\frac{1}{1 + \theta_{\rm E}^2/\theta_\pm^2} = \frac{\theta_\pm^4}{\theta_\pm^4 - \theta_{\rm E}^4}\,. \tag{8.32}$$

Using the dimensionless variable $u = \beta/\theta_{\rm E}$, the magnifications simplify to (Narayan & Bartelmann eq. 27):

$$\boxed{\mu_\pm = \frac{u^2+2}{2u\sqrt{u^2+4}} \pm \frac{1}{2}} \tag{8.33}$$

Several important properties follow:

- $\mu_+ > 0$ always (same parity as the source).
- $\mu_- < 0$ always (opposite parity — the image is mirror-reflected), because $|\theta_-| < \theta_{\rm E}$ makes the denominator of eq. (8.32) negative. As in Narayan & Bartelmann eq. 27, the lower sign in eq. (8.33) gives the *absolute* magnification $|\mu_-|$; the signed value is $\mu_- = \frac{1}{2} - \frac{u^2+2}{2u\sqrt{u^2+4}}$.

- The total magnification (sum of absolute values) is:

$$|\mu_+| + |\mu_-| = \frac{u^2+2}{u\sqrt{u^2+4}} > 1 \quad \text{for all } u > 0\,. \tag{8.34}$$

  Gravitational lensing always *increases* the total observed flux (the total magnification exceeds unity).

- As $u \to 0$ (source on the optical axis), both $|\mu_\pm| \to \infty$: the Einstein ring is *infinitely* magnified (in the geometric optics approximation).

- As $u \to \infty$, $\mu_+ \to 1$ (unlensed) and $|\mu_-| \to 0$ (secondary image disappears).

8.10 Summary

- The **thin-screen approximation** projects the three-dimensional mass distribution onto a single lens plane, justified by the small ratio of lens extent to cosmological distance ($\sim 10^{-4}$).

- The **lens equation** $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ maps image positions to source positions. The **reduced deflection angle** $\boldsymbol{\alpha} = (D_{\rm ds}/D_{\rm s})\,\hat{\boldsymbol{\alpha}}$ absorbs the distance factors.

- The **surface mass density** $\Sigma(\boldsymbol{\xi})$ and the deflection integral (8.10) allow the treatment of arbitrary continuous mass distributions.

- The **critical surface mass density** $\Sigma_{\rm cr} = (c^2/4\pi G)\,(D_{\rm s}/D_{\rm d}\,D_{\rm ds})$ separates weak lensing ($\kappa < 1$) from strong lensing ($\kappa \geq 1$).

- The **Einstein radius** $\theta_{\rm E} = \sqrt{4GM/c^2 \cdot D_{\rm ds}/(D_{\rm d}\,D_{\rm s})}$ is the characteristic angular scale of lensing. Typical values: ~ 1 mas for stellar microlensing, $\sim 1''$–$3''$ for galaxy lensing ($M \sim 10^{11}$–$10^{12}\,M_\odot$), $\sim 30''$–$40''$ for cluster lensing.

- A **point mass lens** always produces two images: $\theta_\pm = (\beta \pm \sqrt{\beta^2 + 4\theta_{\rm E}^2})/2$. The primary image ($\theta_+$) is outside $\theta_{\rm E}$ with the same parity; the secondary (θ_-) is inside $\theta_{\rm E}$ with opposite parity.

- When the source is exactly behind the lens ($\beta = 0$), the images merge into an **Einstein ring** at angular radius $\theta_{\rm E}$.

- The **magnification** of the point mass lens is $\mu_\pm = (u^2+2)/(2u\sqrt{u^2+4}) \pm 1/2$, where $u = \beta/\theta_{\rm E}$. The total magnification always exceeds unity.

8.11 Problems

Exercise 8.11.1. (Derivation of the lens equation.) Starting from the lensing geometry (Figure 8.1), derive the lens equation (8.7).

(a) Using the small-angle approximation, show that the physical position in the source plane is $\boldsymbol{\eta} = (D_\mathrm{s}/D_\mathrm{d})\,\boldsymbol{\xi} - D_\mathrm{ds}\,\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi})$.

(b) Substitute $\boldsymbol{\xi} = D_\mathrm{d}\,\boldsymbol{\theta}$ and $\boldsymbol{\eta} = D_\mathrm{s}\,\boldsymbol{\beta}$ to obtain the lens equation in angular coordinates.

(c) Define the reduced deflection angle and write the lens equation in the compact form $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$.

Exercise 8.11.2. (Point mass lens equation.) For a point mass lens with Einstein radius θ_E:

(a) Derive the lens equation $\beta = \theta - \theta_\mathrm{E}^2/\theta$ from the deflection angle $\hat{\alpha} = 4GM/(c^2\xi)$.

(b) Solve the resulting quadratic to obtain the image positions $\theta_\pm$.

(c) Show that two images always form (i.e., the discriminant is always positive).

(d) Verify the Vieta relations: $\theta_+ + \theta_- = \beta$ and $\theta_+\,\theta_- = -\theta_\mathrm{E}^2$.

(e) Show that $\theta_+ > \theta_\mathrm{E} > 0 > \theta_- > -\theta_\mathrm{E}$ for $\beta > 0$.

Verify with `problems_04.wl`.

Exercise 8.11.3. (Einstein radius for various systems.) Compute the Einstein radius θ_E for:

(a) **Stellar microlensing:** $M = 1\,M_\odot$, lens at distance $D_\mathrm{d} = 5$ kpc, source at $D_\mathrm{s} = 10$ kpc (so $D_\mathrm{ds} = 5$ kpc). Express the result in milliarcseconds.

(b) **Galaxy lensing:** $M = 10^{12}\,M_\odot$, $z_d = 0.3$, $z_s = 1$. Use the angular diameter distances from Module 3 (or the Mathematica script): $D_\mathrm{d} \approx 919$ Mpc, $D_\mathrm{s} \approx 1652$ Mpc, $D_\mathrm{ds} \approx 1055$ Mpc. Express the result in arcseconds.

(c) **Cluster lensing:** $M = 10^{15}\,M_\odot$, $z_d = 0.3$, $z_s = 2$. Use $D_\mathrm{d} \approx 919$ Mpc, $D_\mathrm{s} \approx 1727$ Mpc, $D_\mathrm{ds} \approx 1329$ Mpc. Express the result in arcseconds.

For each case, also compute the physical Einstein radius $R_\mathrm{E} = D_\mathrm{d}\,\theta_\mathrm{E}$ in kpc. Verify with `problems_04.wl`.

Exercise 8.11.4. (Critical surface mass density.)

(a) Starting from the definition (8.12), verify that Σ_cr has units of mass per area by performing a dimensional analysis.

(b) Compute Σ_cr (in $\mathrm{g\,cm^{-2}}$ and $M_\odot\,\mathrm{pc^{-2}}$) for the galaxy lens configuration of Exercise 8.11.3(b).

(c) Compute Σ_cr for the cluster lens configuration of Exercise 8.11.3(c).

(d) A typical elliptical galaxy has a central surface mass density of $\Sigma_0 \approx 5000\,M_\odot\,\mathrm{pc^{-2}}$. Is it a strong lens? Compare Σ_0 with your result from (b).

Verify with `problems_04.wl`.

Exercise 8.11.5. (Image positions for a point mass lens.) Consider a point mass lens with $\theta_{\rm E} = 1''$.

(a) For source positions $\beta = 0.1'', 0.5'', 1'', 2'', 5''$, compute both image positions θ_+ and θ_-. Tabulate your results.

(b) Plot θ_+ and θ_- as functions of β for $0 \leq \beta \leq 5''$. On your plot, also show the Einstein radius and the line $\theta = \beta$ (unlensed position).

(c) Compute the image separation $\Delta\theta = \theta_+ - \theta_-$ for each value of β and plot it. What is the minimum image separation?

(d) Compute the magnification of each image using eq. (8.33) and plot the total magnification $|\mu_+| + |\mu_-|$ as a function of β.

Use Mathematica to generate the plots and verify the numerical values. See `lens_equation.wl` and `problems_04.wl`.

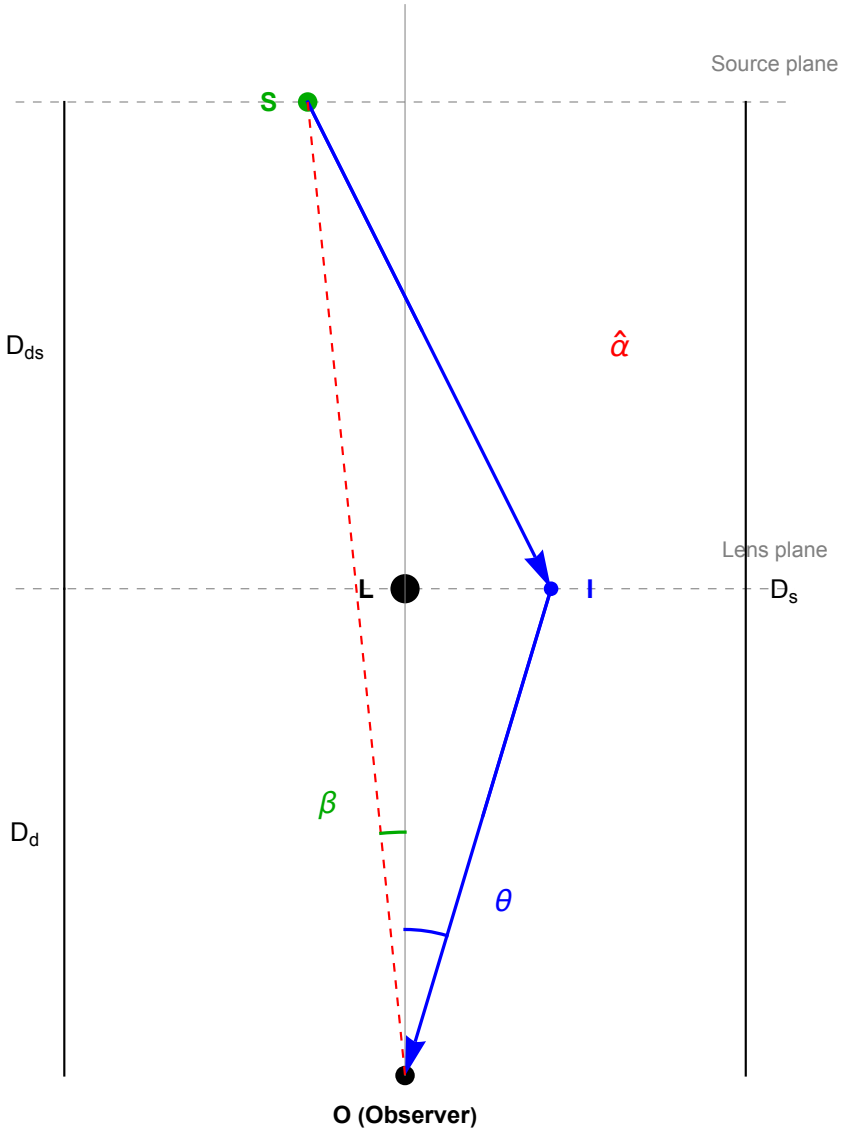


Figure 8.1: Geometry of gravitational lensing. A light ray from a source at angular position $\boldsymbol{\beta}$ in the source plane is deflected by the angle $\hat{\alpha}$ as it passes through the lens plane, and is observed at angular position $\boldsymbol{\theta}$. The angular diameter distances $D_{\rm d}$, $D_{\rm s}$, and $D_{\rm ds}$ connect angular separations to physical separations. The reduced deflection angle $\boldsymbol{\alpha} = (D_{\rm ds}/D_{\rm s})\,\hat{\boldsymbol{\alpha}}$ gives the apparent angular shift. Generated by `lens_equation.wl`.

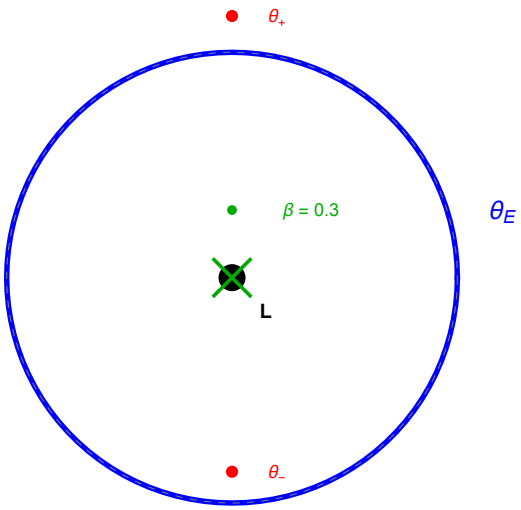


Figure 8.2: The Einstein ring. When a source (star symbol at center) lies exactly behind a point mass lens (filled circle at center), the image is a complete ring of angular radius $\theta_{\rm E}$. For a slightly offset source ($\beta \neq 0$), the ring breaks into two images at positions θ_+ and θ_-. Generated by `lens_equation.wl`.

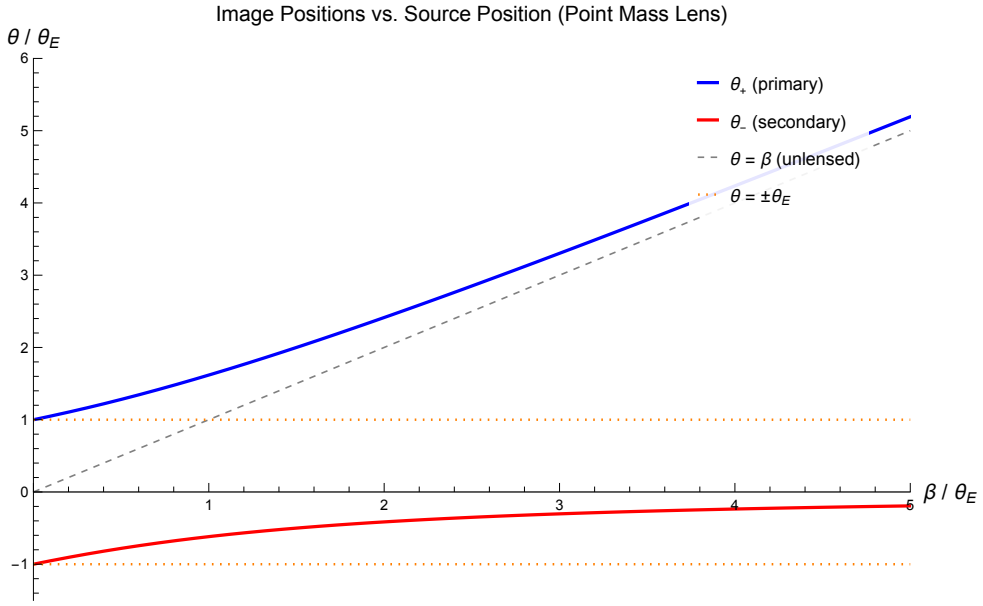


Figure 8.3: Image positions θ_+ and θ_- as functions of the source position β for a point mass lens, in units of the Einstein radius $\theta_{\rm E}$. The primary image θ_+ (blue) is always outside $\theta_{\rm E}$ and approaches the unlensed position (gray dashed line) for $\beta \gg \theta_{\rm E}$. The secondary image θ_- (red) is always inside $\theta_{\rm E}$ (in absolute value) and approaches the lens for $\beta \gg \theta_{\rm E}$. At $\beta = 0$, both images are at $\pm\theta_{\rm E}$ (Einstein ring). Generated by `lens_equation.wl`.

Chapter 9

Magnification, Convergence, and Shear

> *Convergence acting alone causes an isotropic focusing of light rays, leading to an isotropic magnification of a source. The source is mapped onto an image with the same shape but larger size. Shear introduces anisotropy (or astigmatism) into the lens mapping [. . .]* — Narayan & Bartelmann (1997), Sect. 3.2

In Module 8 we derived the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ and used it to find image positions for a point mass lens. We also previewed the magnification formula for the point mass case. Now we develop the *general* formalism that describes how lensing distorts images: their size, shape, and brightness relative to the unlensed source.

The key players are:

- The **lensing potential** $\psi(\boldsymbol{\theta})$, from which the deflection angle derives as a gradient.
- The **convergence** $\kappa(\boldsymbol{\theta})$, the dimensionless surface mass density, which causes isotropic focusing.
- The **shear** $\gamma(\boldsymbol{\theta})$, a spin-2 field that causes anisotropic stretching.
- The **Jacobian matrix** $\mathcal{A}$, which encodes all local lensing distortions.
- The **magnification** μ, the ratio of image area to source area.

Companion Mathematica files:

- `lensing_potential.wl` — symbolic derivation of ψ, convergence, shear, Jacobian, and magnification for the point mass lens; verification of the 2D Poisson equation; publication-quality figure generation
- `convergence_shear_demo.nb` — interactive notebook: Manipulate for convergence and shear effects on circular sources, magnification map for point mass lens

9.1 Introduction: From Image Positions to Image Properties

The lens equation tells us *where* images form. But observations reveal much more than positions: lensed images are *stretched*, *magnified*, and sometimes *mirror-reflected* relative to the unlensed source. Giant arcs in galaxy clusters are dramatic examples — background galaxies stretched into elongated curves by the cluster's gravity.

To describe these distortions quantitatively, we need to understand the *local* properties of the lens mapping $\boldsymbol{\beta}(\boldsymbol{\theta})$. Since the lens mapping is a smooth function (except at isolated singularities), its local behavior is captured by its **Jacobian matrix** — the matrix of first partial derivatives. This matrix encodes all information about magnification, stretching, and parity of each image.

We begin by introducing the **effective lensing potential** $\psi(\boldsymbol{\theta})$, a scalar field from which the deflection angle can be derived as a gradient. This potential provides a unified framework: its first derivatives give the deflection, and its second derivatives give the convergence and shear.

9.2 The Effective Lensing Potential

In Module 8 (eq. 8.11) we expressed the reduced deflection angle as an integral over the convergence. Remarkably, this deflection can be written as the gradient of a scalar potential.

Definition 9.2.1. The **effective lensing potential** (or simply the lensing potential) is defined as (Narayan & Bartelmann eq. 51; Congdon & Keeton eq. 4.14):

$$\boxed{\psi(\boldsymbol{\theta}) = \frac{1}{\pi}\int \kappa(\boldsymbol{\theta}')\,\ln|\boldsymbol{\theta}-\boldsymbol{\theta}'|\,d^2\theta'} \tag{9.1}$$

where $\kappa(\boldsymbol{\theta}') = \Sigma(D_{\rm d}\,\boldsymbol{\theta}')/\Sigma_{\rm cr}$ is the convergence (eq. 8.13) and the integral extends over the entire lens plane.

This is the two-dimensional analog of the Newtonian gravitational potential: just as the three-dimensional gravitational potential $\Phi(\mathbf{r}) = -G\int \rho(\mathbf{r}')/|\mathbf{r}-\mathbf{r}'|\,d^3r'$ satisfies Poisson's equation $\nabla^2\Phi = 4\pi G\rho$, the lensing potential satisfies the two-dimensional Poisson equation.

Remark 9.2.1. The lensing potential ψ is dimensionless when $\boldsymbol{\theta}$ is measured in radians. It is related to the physical (Shapiro) time delay through the Fermat potential, which we will encounter in Module 6.

9.2.1 Key Properties of the Lensing Potential

The lensing potential has two crucial properties:

(1) Its gradient gives the deflection angle. Taking the gradient of eq. (9.1):

$$\boxed{\boldsymbol{\alpha}(\boldsymbol{\theta}) = \nabla_\theta\,\psi(\boldsymbol{\theta})} \tag{9.2}$$

where $\nabla_\theta = (\partial/\partial\theta_1,\,\partial/\partial\theta_2)$ is the two-dimensional gradient with respect to the angular coordinates. This can be verified by differentiating eq. (9.1) under the integral sign and using $\nabla_\theta \ln|\boldsymbol{\theta}-\boldsymbol{\theta}'| = (\boldsymbol{\theta}-\boldsymbol{\theta}')/|\boldsymbol{\theta}-\boldsymbol{\theta}'|^2$.

(2) Its Laplacian gives twice the convergence (2D Poisson equation).

$$\boxed{\nabla_\theta^2 \psi(\boldsymbol{\theta}) = 2\,\kappa(\boldsymbol{\theta})} \tag{9.3}$$

The identity rests on the Green's function of the two-dimensional Laplacian,

$$\nabla_\theta^2 \ln|\boldsymbol{\theta} - \boldsymbol{\theta}'| \;=\; 2\pi\,\delta^{(2)}(\boldsymbol{\theta} - \boldsymbol{\theta}')\,, \tag{9.4}$$

which we prove in two steps.

Step 1: $\nabla^2 \ln r = 0$ *for* $r \neq 0$. In polar coordinates centered at $\boldsymbol{\theta}'$, let $r = |\boldsymbol{\theta} - \boldsymbol{\theta}'|$ and write $\nabla^2 = (1/r)\partial_r(r\,\partial_r) + (1/r^2)\partial_\phi^2$. Then $\partial_r(\ln r) = 1/r$ and $\nabla^2(\ln r) = (1/r)\partial_r(r \cdot 1/r) = (1/r)\partial_r(1) = 0$ for $r > 0$.

Step 2: the integrated strength is 2π. For any disk D_a of radius a centered at $\boldsymbol{\theta}'$, the divergence theorem gives

$$\int_{D_a} \nabla^2 \ln r\, d^2\theta \;=\; \oint_{\partial D_a} \nabla(\ln r)\cdot \hat{n}\, d\ell \;=\; \oint_{r=a} \frac{1}{r}\bigg|_{r=a} a\, d\phi \;=\; 2\pi.$$

Combined with step 1, $\nabla^2(\ln r)$ is a distribution supported at the origin with total weight 2π, i.e. eq. (9.4).

Differentiating eq. (9.1) under the integral sign and substituting eq. (9.4):

$$\nabla_\theta^2 \psi \;=\; \frac{1}{\pi}\int \kappa(\boldsymbol{\theta}')\, 2\pi\,\delta^{(2)}(\boldsymbol{\theta} - \boldsymbol{\theta}')\, d^2\theta' \;=\; 2\,\kappa(\boldsymbol{\theta})\,. \tag{9.5}$$

Equivalently, one can bypass eq. (9.4) entirely using $\boldsymbol{\alpha} = \nabla\psi$ (eq. 9.2) and $\kappa = \frac{1}{2}\nabla\cdot\boldsymbol{\alpha}$ (Narayan & Bartelmann eq. 56; Congdon & Keeton eq. 4.22): $\frac{1}{2}\nabla^2\psi = \frac{1}{2}\nabla\cdot(\nabla\psi) = \frac{1}{2}\nabla\cdot\boldsymbol{\alpha} = \kappa$. Both routes are verified symbolically in `lensing_potential.wl` (direct convolution for the point-mass and uniform-disk convergence).

9.3 Convergence

The convergence was already introduced in Module 8 (eq. 8.13):

$$\kappa(\boldsymbol{\theta}) = \frac{\Sigma(D_\mathrm{d}\,\boldsymbol{\theta})}{\Sigma_\mathrm{cr}} \tag{9.6}$$

It is the **dimensionless surface mass density** — the ratio of the actual surface mass density to the critical value. We now explore its role in image distortion.

9.3.1 Physical Meaning

The convergence measures the *isotropic focusing* caused by the lens:

- A region with $\kappa > 0$ acts as a converging lens, focusing light rays and magnifying images isotropically (without changing their shape).

- $\kappa = 1$ means $\Sigma = \Sigma_{\rm cr}$: this is the threshold for strong lensing. Regions where $\kappa \geq 1$ are **supercritical** and can produce multiple images.
- $\kappa < 1$ everywhere means the lens is subcritical: it can still distort images (via shear), but a single mass concentration cannot produce multiple images of a generic source.

9.3.2 Convergence from the Lensing Potential

From the 2D Poisson equation (9.3):

$$\kappa(\boldsymbol{\theta}) = \frac{1}{2}\,\nabla_\theta^2\,\psi = \frac{1}{2}\left(\frac{\partial^2\psi}{\partial\theta_1^2} + \frac{\partial^2\psi}{\partial\theta_2^2}\right). \tag{9.7}$$

Thus the convergence is the *trace* of the Hessian of the lensing potential (up to a factor of 1/2).

9.4 Shear

The shear describes the *anisotropic distortion* of images — the stretching in one direction and compression in the perpendicular direction.

The specific combinations that define γ_1 and γ_2 are forced by a clean geometric argument. The Hessian of the lensing potential,

$$H_{ij} \;\equiv\; \frac{\partial^2\psi}{\partial\theta_i\,\partial\theta_j}\,,$$

is a 2×2 real symmetric matrix — three independent components. Every such matrix decomposes uniquely into a trace piece (proportional to the identity) plus a traceless symmetric piece:

$$H \;=\; \frac{\operatorname{tr} H}{2}\,\mathbb{I} + H^{\rm TF}, \qquad H^{\rm TF} = \begin{pmatrix} \frac{1}{2}(H_{11}-H_{22}) & H_{12} \\ H_{12} & -\frac{1}{2}(H_{11}-H_{22}) \end{pmatrix}. \tag{9.8}$$

The trace part is the isotropic convergence $\frac{1}{2}\operatorname{tr} H = \kappa$ (eq. 9.7); the traceless part is the shear. Its two independent entries are named

$$\boxed{\gamma_1(\boldsymbol{\theta}) = \frac{1}{2}\left(\frac{\partial^2\psi}{\partial\theta_1^2} - \frac{\partial^2\psi}{\partial\theta_2^2}\right), \qquad \gamma_2(\boldsymbol{\theta}) = \frac{\partial^2\psi}{\partial\theta_1\,\partial\theta_2}\,,} \tag{9.9}$$

so that the Hessian itself reads $H = \big((\kappa+\gamma_1)\;\;\gamma_2;\;\gamma_2\;\;(\kappa-\gamma_1)\big)$. Under a rotation of the $\boldsymbol{\theta}$-axes by angle ϕ, $H^{\rm TF}$ transforms as a rank-2 symmetric tensor whose traceless part carries *two* powers of $e^{\pm i\phi}$. Forming the complex shear $\gamma \equiv \gamma_1 + i\gamma_2$,

$$\gamma \;\longrightarrow\; e^{-2i\phi}\,\gamma\,, \tag{9.10}$$

which is the spin-2 transformation law — the reason the shear has a 180° (not 360°) rotational symmetry. The decomposition (9.8) and the transformation law (9.10) are verified symbolically in `lensing_potential.wl`.

The shear magnitude and the complex shear are:

$$|\gamma| = \sqrt{\gamma_1^2 + \gamma_2^2}\,, \qquad \gamma = \gamma_1 + i\,\gamma_2 = |\gamma|\, e^{2i\phi_\gamma} \tag{9.11}$$

where ϕ_γ is the orientation angle of the shear.

9.4.1 Physical Meaning

The shear is a **spin-2 field**: it is invariant under rotations by 180 (not 360), reflecting the fact that an ellipse looks the same after a half-turn. The two components have specific geometric meanings:

- $\gamma_1 > 0$: stretching along the θ_1-axis, compression along θ_2.
- $\gamma_1 < 0$: stretching along the θ_2-axis, compression along θ_1.
- $\gamma_2 > 0$: stretching along the direction at 45 to the θ_1-axis.
- $\gamma_2 < 0$: stretching along the direction at 135 to the θ_1-axis.

Remark 9.4.1. The convergence and shear together account for all four independent second derivatives of the lensing potential (the Hessian is a symmetric 2×2 matrix with three independent components; κ captures the trace, and γ_1, γ_2 capture the traceless part):

$$\frac{\partial^2 \psi}{\partial\theta_i\,\partial\theta_j} = \begin{pmatrix} \kappa + \gamma_1 & \gamma_2 \\ \gamma_2 & \kappa - \gamma_1 \end{pmatrix}. \tag{9.12}$$

9.5 The Jacobian Matrix (Inverse Magnification Tensor)

The lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ defines a mapping from the image plane to the source plane. The local properties of this mapping are encoded in its **Jacobian matrix**:

$$\mathcal{A}_{ij} = \frac{\partial \beta_i}{\partial \theta_j} = \delta_{ij} - \frac{\partial \alpha_i}{\partial \theta_j} = \delta_{ij} - \frac{\partial^2 \psi}{\partial\theta_i\,\partial\theta_j} \tag{9.13}$$

where in the last step we used $\alpha_i = \partial\psi/\partial\theta_i$ (eq. 9.2).

Using the decomposition of the Hessian into convergence and shear (eq. 9.12), the Jacobian takes the elegant form (Narayan & Bartelmann eq. 58; Congdon & Keeton eq. 4.51):

$$\boxed{\mathcal{A} = \begin{pmatrix} 1 - \kappa - \gamma_1 & -\gamma_2 \\ -\gamma_2 & 1 - \kappa + \gamma_1 \end{pmatrix}} \tag{9.14}$$

Remark 9.5.1. The matrix $\mathcal{A}$ is sometimes called the **inverse magnification tensor** because its inverse $\mathcal{A}^{-1}$ gives the magnification tensor $\mathcal{M}$:

$$\mathcal{M} = \mathcal{A}^{-1} = \frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{\beta}}\,. \tag{9.15}$$

$\mathcal{M}$ maps source-plane displacements to image-plane displacements.

9.5.1 Decomposition of the Jacobian

The Jacobian can be decomposed into a convergence (isotropic) part and a shear (anisotropic) part:

$$\mathcal{A} = (1-\kappa)\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} - \begin{pmatrix}\gamma_1 & \gamma_2\\ \gamma_2 & -\gamma_1\end{pmatrix} \tag{9.16}$$

The first term is proportional to the identity matrix and represents isotropic focusing (convergence). The second term is a traceless, symmetric matrix (the shear matrix) that represents anisotropic distortion.

9.5.2 Eigenvalues of the Jacobian

The eigenvalues of $\mathcal{A}$ are (Congdon & Keeton eq. 4.54):

$$\lambda_\pm = 1 - \kappa \pm |\gamma| = (1-\kappa) \pm \sqrt{\gamma_1^2 + \gamma_2^2}\,. \tag{9.17}$$

These eigenvalues give the inverse magnification along the two principal axes of the distortion. The corresponding eigenvectors define the directions of maximum and minimum stretching.

9.6 Magnification

Gravitational lensing conserves surface brightness (a consequence of Liouville's theorem — the photon phase space density is preserved along geodesics). Therefore, the magnification is simply the ratio of the *solid angle* of the image to that of the source:

$$\mu = \frac{(\text{image area})}{(\text{source area})} = \frac{1}{\det \mathcal{A}}\,. \tag{9.18}$$

Computing the determinant of eq. (9.14):

$$\det \mathcal{A} = (1-\kappa-\gamma_1)(1-\kappa+\gamma_1) - \gamma_2^2 = (1-\kappa)^2 - (\gamma_1^2+\gamma_2^2) \tag{9.19}$$

and therefore:

$$\boxed{\mu = \frac{1}{\det \mathcal{A}} = \frac{1}{(1-\kappa)^2 - |\gamma|^2}} \tag{9.20}$$

This is the **general magnification formula** for gravitational lensing. Note that μ depends on position through $\kappa(\boldsymbol{\theta})$ and $\gamma(\boldsymbol{\theta})$.

9.6.1 Properties of the Magnification

1. **Positive parity ($\mu > 0$):** The image has the same orientation (handedness) as the source. This occurs when $\det \mathcal{A} > 0$.

2. **Negative parity ($\mu < 0$):** The image is mirror-reflected relative to the source. This occurs when $\det \mathcal{A} < 0$. The *absolute* magnification is $|\mu|$.

3. **Magnified images ($|\mu| > 1$):** The image subtends a larger solid angle than the source. The condition is $(1-\kappa)^2 - |\gamma|^2 < 1$.

4. **Demagnified images ($|\mu| < 1$):** The image is smaller than the source.

5. **Critical curves ($\det \mathcal{A} = 0$, $\mu \to \infty$):** Points where the magnification formally diverges. In practice, images near critical curves are very highly magnified (factors of 10–100), but finite source sizes prevent true divergences. These curves are the subject of Module 8.

9.6.2 Magnification in Terms of Eigenvalues

Using the eigenvalues of $\mathcal{A}$ (eq. 9.17):

$$\mu = \frac{1}{\lambda_+ \lambda_-} = \frac{1}{(1-\kappa-|\gamma|)(1-\kappa+|\gamma|)} \,. \tag{9.21}$$

The magnification diverges when either eigenvalue vanishes, i.e., when $1-\kappa = \pm|\gamma|$. These conditions define the two families of critical curves:

- $\lambda_- = 0$: the **tangential critical curve** ($1-\kappa = |\gamma|$), where images are stretched tangentially.

- $\lambda_+ = 0$: the **radial critical curve** ($1-\kappa = -|\gamma|$), where images are stretched radially.

9.7 Convergence vs. Shear: Physical Effects

To build intuition for the roles of convergence and shear, consider how a small circular source is distorted in three limiting cases.

9.7.1 Pure Convergence ($\kappa > 0$, $\gamma = 0$)

The Jacobian reduces to $\mathcal{A} = (1-\kappa)\,\mathbf{I}$, a uniform scaling. A circular source remains circular, but its angular size is scaled by $1/(1-\kappa)$. The magnification is:

$$\mu = \frac{1}{(1-\kappa)^2} \,. \tag{9.22}$$

For $\kappa < 1$, the image is magnified and has positive parity. For $\kappa > 1$, the image is still magnified and *retains positive parity*: $\det \mathcal{A} = (1-\kappa)^2 > 0$ for all $\kappa \neq 1$. Although both eigenvalues $(1-\kappa)$ are negative when $\kappa > 1$, their product is positive — a uniform negative scaling is equivalent to a 180° rotation, which preserves handedness. A genuine parity flip ($\det \mathcal{A} < 0$, $\mu < 0$) requires shear, so that the eigenvalues $\lambda_\pm = (1-\kappa) \pm |\gamma|$ have opposite signs.

9.7.2 Pure Shear ($\kappa = 0$, $\gamma > 0$)

The Jacobian becomes:

$$\mathcal{A} = \begin{pmatrix} 1 - \gamma_1 & -\gamma_2 \\ -\gamma_2 & 1 + \gamma_1 \end{pmatrix}. \tag{9.23}$$

A circular source is distorted into an **ellipse**. The magnification is:

$$\mu = \frac{1}{1 - |\gamma|^2}. \tag{9.24}$$

The semi-axes of the elliptical image are proportional to $1/(1 - |\gamma|)$ and $1/(1 + |\gamma|)$, so the axis ratio is:

$$\frac{b}{a} = \frac{1 - |\gamma|}{1 + |\gamma|}. \tag{9.25}$$

The orientation of the ellipse is determined by the shear angle $\phi_\gamma = \frac{1}{2}\arctan(\gamma_2/\gamma_1)$.

9.7.3 Both Convergence and Shear

In general, both κ and γ are nonzero. A circular source is mapped to a *larger ellipse*:

- The convergence sets the overall magnification (size change).
- The shear sets the ellipticity (shape change).

The semi-axes of the image are $\delta/(1 - \kappa - |\gamma|)$ and $\delta/(1 - \kappa + |\gamma|)$ for a circular source of angular radius δ.

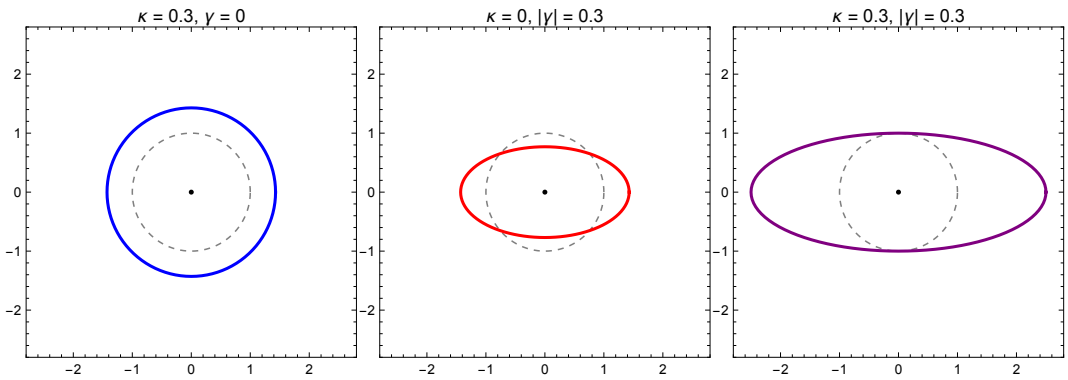


Figure 9.1: Effect of convergence and shear on a circular source. **Left:** Pure convergence ($\kappa = 0.3$, $\gamma = 0$) produces a larger circle (isotropic magnification). **Center:** Pure shear ($\kappa = 0$, $|\gamma| = 0.3$) produces an ellipse (anisotropic distortion; the magnification $\mu = 1/(1-|\gamma|^2) \approx 1.10$ differs from unity only at second order in $|\gamma|$). **Right:** Both ($\kappa = 0.3$, $|\gamma| = 0.3$) produces a larger ellipse. The dashed circle shows the original source in each panel. Generated by `lensing_potential.wl`.

9.8 Example: Point Mass Lens

We now compute κ, γ, and μ for the point mass lens, verifying consistency with the results of Module 8.

9.8.1 Lensing Potential for a Point Mass

For a point mass M at the origin, the convergence is a Dirac delta function:

$$\kappa(\boldsymbol{\theta}) = \frac{\Sigma(D_{\rm d}\boldsymbol{\theta})}{\Sigma_{\rm cr}} = \frac{M\,\delta^{(2)}(D_{\rm d}\boldsymbol{\theta})}{\Sigma_{\rm cr}} = \pi\theta_{\rm E}^2\,\delta^{(2)}(\boldsymbol{\theta})\,. \tag{9.26}$$

The mass is concentrated entirely at the origin; away from the origin, $\kappa = 0$.

Substituting into eq. (9.1), the lensing potential is:

$$\psi(\boldsymbol{\theta}) = \theta_{\rm E}^2 \ln|\boldsymbol{\theta}| \tag{9.27}$$

which satisfies $\nabla\psi = \theta_{\rm E}^2\,\boldsymbol{\theta}/|\boldsymbol{\theta}|^2$, giving the deflection $\boldsymbol{\alpha} = \theta_{\rm E}^2\,\boldsymbol{\theta}/|\boldsymbol{\theta}|^2$ as expected.

9.8.2 Shear for a Point Mass

Using polar coordinates on the lens plane, we can write $\theta_1 = \theta\cos\phi$ and $\theta_2 = \theta\sin\phi$, where $\theta = |\boldsymbol{\theta}|$. Computing the second derivatives of $\psi = \theta_{\rm E}^2\ln\theta$:

$$\begin{aligned}
\frac{\partial^2\psi}{\partial\theta_1^2} &= \theta_{\rm E}^2\,\frac{\theta_2^2-\theta_1^2}{\theta^4}\,,\\
\frac{\partial^2\psi}{\partial\theta_2^2} &= \theta_{\rm E}^2\,\frac{\theta_1^2-\theta_2^2}{\theta^4}\,,\\
\frac{\partial^2\psi}{\partial\theta_1\,\partial\theta_2} &= -\theta_{\rm E}^2\,\frac{2\theta_1\theta_2}{\theta^4}\,.
\end{aligned} \tag{9.28}$$

The shear components are (for $\theta \neq 0$):

$$\gamma_1 = \frac{1}{2}\left(\frac{\partial^2\psi}{\partial\theta_1^2} - \frac{\partial^2\psi}{\partial\theta_2^2}\right) = -\frac{\theta_{\rm E}^2(\theta_1^2-\theta_2^2)}{\theta^4} = -\frac{\theta_{\rm E}^2}{\theta^2}\cos 2\phi\,, \tag{9.29}$$

$$\gamma_2 = \frac{\partial^2\psi}{\partial\theta_1\,\partial\theta_2} = -\frac{2\theta_{\rm E}^2\,\theta_1\theta_2}{\theta^4} = -\frac{\theta_{\rm E}^2}{\theta^2}\sin 2\phi\,. \tag{9.30}$$

The shear magnitude is:

$$\boxed{|\gamma| = \sqrt{\gamma_1^2+\gamma_2^2} = \frac{\theta_{\rm E}^2}{\theta^2}} \tag{9.31}$$

The shear falls off as θ^{-2} and is aligned tangentially (the factor $e^{2i\phi}$ in the complex shear).

9.8.3 Convergence for a Point Mass (away from origin)

To compute the convergence correctly, we must be careful with the form of ψ. Writing $\psi = \frac{\theta_{\rm E}^2}{2}\ln(\theta_1^2+\theta_2^2)$ in Cartesian coordinates:

$$\frac{\partial\psi}{\partial\theta_1} = \frac{\theta_{\rm E}^2\,\theta_1}{\theta^2}\,, \qquad \frac{\partial^2\psi}{\partial\theta_1^2} = \theta_{\rm E}^2\,\frac{\theta^2-2\theta_1^2}{\theta^4}\,. \tag{9.32}$$

Similarly, $\partial^2\psi/\partial\theta_2^2 = \theta_{\rm E}^2\,(\theta^2-2\theta_2^2)/\theta^4$. Summing:

$$\nabla^2\psi = \theta_{\rm E}^2\,\frac{2\theta^2-2(\theta_1^2+\theta_2^2)}{\theta^4} = 0 \qquad (\theta\neq 0)\,. \tag{9.33}$$

Therefore $\kappa = 0$ for $\theta \neq 0$, confirming that the convergence of a point mass is indeed a delta function concentrated entirely at the origin. In the distributional sense, $\nabla^2\ln|\boldsymbol{\theta}| = 2\pi\,\delta^{(2)}(\boldsymbol{\theta})$, which recovers $\kappa = \pi\theta_{\rm E}^2\,\delta^{(2)}(\boldsymbol{\theta})$ via the Poisson equation.

9.8.4 Magnification for a Point Mass

With $\kappa = 0$ and $|\gamma| = \theta_{\rm E}^2/\theta^2$ (for $\theta \neq 0$), the magnification formula (9.20) gives:

$$\mu = \frac{1}{1-\theta_{\rm E}^4/\theta^4} = \frac{\theta^4}{\theta^4-\theta_{\rm E}^4}\,. \tag{9.34}$$

This is precisely eq. (8.32) from Module 8! Evaluating at the image positions $\theta_\pm$:

$$|\mu_\pm| = \frac{u^2+2}{2u\sqrt{u^2+4}} \pm \frac{1}{2}\,, \qquad \mu_+ > 0\,, \quad \mu_- < 0\,, \tag{9.35}$$

where $u = \beta/\theta_{\rm E}$, recovering eq. (8.33) exactly. (The inner image has $|\theta_-| < \theta_{\rm E}$, so $\theta_-^4 - \theta_{\rm E}^4 < 0$ and its signed magnification is negative.) The consistency is verified symbolically in `lensing_potential.wl`.

9.9 Example: Singular Isothermal Sphere (Preview)

The singular isothermal sphere (SIS) is the simplest realistic model for a galaxy lens. Its surface mass density profile is (Congdon & Keeton Sec. 2.3.2):

$$\Sigma(\xi) = \frac{\sigma_v^2}{2G\,\xi} \tag{9.36}$$

where σ_v is the one-dimensional velocity dispersion and $\xi = |\boldsymbol{\xi}|$ is the physical radius in the lens plane.

9.9.1 Convergence and Shear of the SIS

The convergence is:

$$\kappa(\theta) = \frac{\Sigma(D_{\rm d}\theta)}{\Sigma_{\rm cr}} = \frac{\theta_{\rm E}}{2\theta} \tag{9.37}$$

where $\theta_{\rm E} = 4\pi\sigma_v^2 D_{\rm ds}/(c^2 D_{\rm s})$ is the Einstein radius of the SIS. Note that κ diverges at $\theta = 0$ (the "singular" in the name) and that $\kappa(\theta_{\rm E}) = 1/2$.

The shear magnitude for the SIS is:

$$|\gamma(\theta)| = \frac{\theta_{\rm E}}{2\theta} = \kappa(\theta)\,. \tag{9.38}$$

Remarkably, $|\gamma| = \kappa$ at every point for the SIS. This equality does *not* hold for general lens models — it is a special property of the isothermal profile.

9.9.2 Magnification of the SIS

The magnification is:

$$\mu = \frac{1}{(1-\kappa)^2 - |\gamma|^2} = \frac{1}{1-2\kappa} = \frac{1}{1-\theta_{\rm E}/\theta} = \frac{\theta}{\theta - \theta_{\rm E}}\,. \tag{9.39}$$

The magnification diverges at $\theta = \theta_{\rm E}$ — this is the (tangential) critical curve of the SIS. We will treat the SIS model in full detail in Module 7.

9.10 Critical Curves and Caustics (Preview)

Definition 9.10.1. The **critical curves** are the loci in the image plane where the magnification diverges:

$$\det \mathcal{A}(\boldsymbol{\theta}) = 0 \qquad \Longleftrightarrow \qquad (1-\kappa)^2 = |\gamma|^2\,. \tag{9.40}$$

The **caustics** are the images of the critical curves under the lens mapping, i.e., the corresponding curves in the source plane:

$$\boldsymbol{\beta}_{\rm caustic} = \boldsymbol{\theta}_{\rm crit} - \boldsymbol{\alpha}(\boldsymbol{\theta}_{\rm crit})\,. \tag{9.41}$$

Critical curves and caustics play a central role in strong lensing:

- When a source crosses a caustic, the number of images changes by two (a pair of highly magnified images appears or disappears).
- Sources near caustics are very highly magnified — this is exploited in studies of distant, faint galaxies.
- The topology of critical curves and caustics determines the image configuration (doubly, quadruply, or multiply imaged sources).

Remark 9.10.1 (Physical intuition for critical curves). When $\det(\mathcal{A}) = 0$, the magnification is formally infinite. Physically, this means that images near a critical curve are highly stretched and magnified. A small source near a *caustic* (the mapping of the critical curve to the source plane) produces two images straddling the critical curve — one with positive parity (unreflected) and one with negative parity (mirror-reflected). These images appear/disappear in pairs as the source crosses a caustic, which is why the total number of images always changes by ± 2, preserving the odd-number theorem (Module 6). The full topological picture is developed in Module 12.

For the point mass lens, the critical curve is the Einstein ring ($\theta = \theta_{\rm E}$), and the caustic degenerates to a single point ($\beta = 0$). For more complex lens models, both critical curves and caustics have richer structures. We will develop the full theory in Module 8.

9.11 Summary

- The **lensing potential** $\psi(\boldsymbol{\theta}) = (1/\pi)\int \kappa(\boldsymbol{\theta}') \ln|\boldsymbol{\theta} - \boldsymbol{\theta}'|\, d^2\theta'$ is a scalar field whose gradient gives the deflection angle ($\boldsymbol{\alpha} = \nabla\psi$) and whose Laplacian gives twice the convergence ($\nabla^2\psi = 2\kappa$).
- The **convergence** $\kappa = \Sigma/\Sigma_{\rm cr}$ causes isotropic focusing: a circular source remains circular but changes size.
- The **shear** $\gamma = \gamma_1 + i\gamma_2$ causes anisotropic distortion: a circular source is stretched into an ellipse.
- The **Jacobian matrix** $\mathcal{A} = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}$ encodes all local lensing properties. Its eigenvalues $\lambda_\pm = 1 - \kappa \pm |\gamma|$ give the inverse magnifications along the principal axes.
- The **magnification** $\mu = 1/\det\mathcal{A} = 1/[(1-\kappa)^2 - |\gamma|^2]$. The sign of μ determines the image parity.
- For a **point mass lens**: $\kappa = 0$ (away from the origin), $|\gamma| = \theta_{\rm E}^2/\theta^2$, and the general magnification formula recovers the result $|\mu_\pm| = (u^2+2)/(2u\sqrt{u^2+4}) \pm 1/2$ from Module 4, with $\mu_- < 0$.
- **Critical curves** ($\det\mathcal{A} = 0$) and their mappings, **caustics**, govern the image multiplicity and are the sites of maximum magnification.

9.12 Problems

Exercise 9.12.1. (Derivation of the Jacobian matrix.) Starting from the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ with $\boldsymbol{\alpha} = \nabla\psi$:

(a) Compute $\mathcal{A}_{ij} = \partial\beta_i/\partial\theta_j$ and express it in terms of second derivatives of ψ.

(b) Decompose the Hessian matrix $\partial^2\psi/\partial\theta_i\partial\theta_j$ into a trace part (proportional to δ_{ij}) and a traceless part. Identify the trace part with κ and the traceless part with the shear.

(c) Write $\mathcal{A}$ in the form $\begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}$.

Verify with `problems_05.wl`.

Exercise 9.12.2. (The 2D Poisson equation.)

(a) Show that the two-dimensional Green's function of the Laplacian is $(1/2\pi)\ln|\boldsymbol{\theta}|$, i.e., $\nabla^2 \ln|\boldsymbol{\theta}| = 2\pi\,\delta^{(2)}(\boldsymbol{\theta})$. *Hint:* Use the fact that $\ln|\boldsymbol{\theta}|$ is harmonic for $\boldsymbol{\theta} \neq 0$ and apply Gauss's theorem to a small circle around the origin.

(b) Starting from the definition of ψ (eq. 9.1), apply the Laplacian to derive $\nabla^2\psi = 2\kappa$.

(c) Verify the 2D Poisson equation for the point mass lens: $\psi = \theta_{\rm E}^2 \ln|\boldsymbol{\theta}|$ and $\kappa = \pi\theta_{\rm E}^2\,\delta^{(2)}(\boldsymbol{\theta})$.

Verify with `problems_05.wl`.

Exercise 9.12.3. (Shear of a point mass lens.) For the point mass lensing potential $\psi = \theta_{\rm E}^2 \ln|\boldsymbol{\theta}|$:

(a) Compute γ_1 and γ_2 as functions of θ_1, θ_2.

(b) Show that $|\gamma| = \theta_{\rm E}^2/\theta^2$.

(c) Express γ_1 and γ_2 in polar coordinates (θ, ϕ) and show that the complex shear is $\gamma = -(\theta_{\rm E}^2/\theta^2)\,e^{2i\phi}$ (tangential shear).

(d) Compute the ratio $|\gamma|/\kappa$ at the image positions $\theta_\pm$ of the point mass lens. What happens for sources far from the Einstein radius?

Verify with `problems_05.wl`.

Exercise 9.12.4. (Magnification consistency check.)

(a) Using $\kappa = 0$ and $|\gamma| = \theta_{\rm E}^2/\theta^2$ for the point mass lens, show that $\mu = \theta^4/(\theta^4 - \theta_{\rm E}^4)$.

(b) Evaluate μ at the image positions $\theta_\pm = (\beta \pm \sqrt{\beta^2 + 4\theta_{\rm E}^2})/2$. Using $u = \beta/\theta_{\rm E}$, show that the result reduces to $\mu_+ = (u^2+2)/(2u\sqrt{u^2+4}) + 1/2$ and $\mu_- = 1/2 - (u^2+2)/(2u\sqrt{u^2+4})$, i.e., $|\mu_\pm| = (u^2+2)/(2u\sqrt{u^2+4}) \pm 1/2$.

(c) Verify that $\mu_+ > 0$ (positive parity) and $\mu_- < 0$ (negative parity) for all $u > 0$.

(d) Show that $|\mu_+| + |\mu_-| > 1$ for all $u > 0$ (gravitational lensing always increases the total flux).

Verify with `problems_05.wl`.

Exercise 9.12.5. (Elliptical image from constant κ and γ.) A small circular source of angular radius δ is lensed by a region with constant convergence κ and constant shear $\gamma = |\gamma|\,e^{2i\phi_\gamma}$ (with $\gamma_1 = |\gamma|\cos 2\phi_\gamma$ and $\gamma_2 = |\gamma|\sin 2\phi_\gamma$).

(a) Write the Jacobian matrix $\mathcal{A}$ for constant κ, γ_1, γ_2.

(b) Find the eigenvalues $\lambda_\pm$ of $\mathcal{A}$ and show that the image is an ellipse with semi-axes $a = \delta/|\lambda_-|$ and $b = \delta/|\lambda_+|$ (assuming $|\lambda_-| < |\lambda_+|$).

(c) Compute the axis ratio b/a and the magnification μ in terms of κ and $|\gamma|$.

(d) The orientation of the ellipse is along the eigenvector corresponding to λ_-. For $\gamma_2 = 0$ (shear aligned with the coordinate axes), what is the orientation of the major axis?

(e) Evaluate the axis ratio and magnification for $\kappa = 0.3$, $|\gamma| = 0.2$.

Verify with `problems_05.wl`.

Chapter 10

Fermat's Principle and Time Delays

> *The lensing properties of model gravitational lenses are especially easy to visualize by application of Fermat's principle of geometrical optics [...]* — Narayan & Bartelmann (1997), Sect. 3.3.1

In Module 8 we derived the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ as a geometric consequence of light deflection. In Module 6 we computed the Shapiro time delay experienced by photons passing through a gravitational potential, and in Module 5 we introduced the effective refractive index of a weak gravitational field.

In this module we show that the lens equation itself emerges from a variational principle: images form at stationary points of the **arrival-time surface**. This elegant reformulation — Fermat's principle for gravitational lensing — not only provides a deeper understanding of image formation, but also yields **observable time delays** between multiple images. These time delays are directly proportional to H_0^{-1}, making multiply-imaged variable sources (such as quasars) a powerful and independent probe of the Hubble constant.

Companion Mathematica files:

- `time_delay_function.wl` — symbolic computation of the Fermat potential, verification that $\nabla\tau = 0$ recovers the lens equation, time delay formulas for point mass and SIS lenses, numerical H_0 estimation, and publication-quality figure generation
- `arrival_time_surface.nb` — interactive notebook: 3D arrival-time surface with slider for source position, contour plots with critical curves, time delay visualization

10.1 Introduction: From Geometry to a Variational Principle

The lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ was derived in Module 8 by tracing light rays through the lensing geometry. While this approach is direct and intuitive, it does not explain *why* light "chooses" particular paths through the lens plane, nor does it immediately reveal the relative arrival times of multiple images.

A deeper perspective comes from **Fermat's principle**: among all possible paths from source to observer, light follows those for which the travel time is *stationary* (a minimum,

maximum, or saddle point of the arrival time). In classical optics, Fermat's principle explains Snell's law and the focusing properties of lenses. In gravitational lensing, it does far more:

1. It **derives** the lens equation as the stationarity condition $\nabla_{\boldsymbol{\theta}}\, t = 0$.
2. It **classifies** multiple images by the nature of the stationary point (minimum, saddle, or maximum), which determines their parity and arrival order.
3. It produces **observable time delays** between multiple images, which can be used to measure the Hubble constant.

The key quantity is the **time delay function** (or **arrival-time surface**) $t(\boldsymbol{\theta})$, which gives the arrival time of a photon traveling from a source at $\boldsymbol{\beta}$ via the lens plane at $\boldsymbol{\theta}$ to the observer. Images form at the stationary points of $t(\boldsymbol{\theta})$, and the differences in t between stationary points give the time delays.

10.2 The Time Delay Function

Consider a photon emitted by a source at angular position $\boldsymbol{\beta}$ in the source plane. The photon passes through the lens plane at angular position $\boldsymbol{\theta}$ and arrives at the observer. The total travel time relative to the unperturbed (unlensed, straight-line) path consists of two contributions:

- **Geometric delay** (extra path length): The deflected ray travels a longer geometric path than the straight-line ray. In the small-angle, thin-lens approximation, the extra path length corresponds to an angular excess of $\frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2$ (see Module 8, Section 8.3).
- **Shapiro (gravitational) delay**: The photon passes through the gravitational potential of the lens, experiencing the time delay derived in Module 6. In the thin-lens formalism, this delay is encoded in the **lensing potential** $\psi(\boldsymbol{\theta})$ (Module 9).

Combining these contributions, the total arrival time is (Narayan & Bartelmann eq. 63; Congdon & Keeton eqs. 4.23–4.25):

$$\boxed{t(\boldsymbol{\theta}) = \frac{(1+z_d)}{c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\left[\tfrac{1}{2}|\boldsymbol{\theta}-\boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})\right] + \text{const}} \tag{10.1}$$

where z_d is the redshift of the lens, the angular diameter distances $D_{\rm d}$, $D_{\rm s}$, $D_{\rm ds}$ are defined in Module 7, and the constant is an irrelevant additive term that depends on the unperturbed travel time.

The factor $(1 + z_d)$ arises from **cosmological time dilation**: the geometric and Shapiro delays are computed in the rest frame of the lens (at redshift z_d), but the time delay is *observed* in the observer's frame. A time interval $\Delta t_{\rm lens}$ at the lens is stretched to $\Delta t_{\rm obs} = (1+z_d)\,\Delta t_{\rm lens}$ in the observer's frame, just as photon wavelengths are redshifted by the same factor (eq. 7.37 from Module 7; cf. Schneider, Ehlers & Falco 1992, Sec. 4.6, eqs. 4.65–4.67). The combination $D_{\rm d}\,D_{\rm s}/D_{\rm ds}$ is the **effective lensing distance**, which encodes the geometry of the observer–lens–source system.

Definition 10.2.1 (Fermat potential)**.** The dimensionless quantity

$$\boxed{\tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = \tfrac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})} \tag{10.2}$$

is called the **Fermat potential** (or **time delay potential**). The arrival time is proportional to τ:

$$t(\boldsymbol{\theta}) = \frac{(1+z_d)}{c}\,\frac{D_\mathrm{d}\,D_\mathrm{s}}{D_\mathrm{ds}}\,\tau(\boldsymbol{\theta}, \boldsymbol{\beta}) + \text{const}\,. \tag{10.3}$$

Remark 10.2.1. The Fermat potential has a simple physical interpretation:

- The first term $\frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2$ is the **geometric delay**: it penalizes paths where the ray in the lens plane ($\boldsymbol{\theta}$) is far from the straight-line direction ($\boldsymbol{\beta}$).
- The second term $-\psi(\boldsymbol{\theta})$ is the **gravitational (Shapiro) delay**: for smooth extended profiles (such as the SIS), the lensing potential $\psi > 0$ everywhere in the vicinity of the lens, so this term *reduces* the arrival time — photons passing through a deeper potential arrive earlier due to the Shapiro effect (recall Module 6). For a point mass, however, $\psi = \theta_\mathrm{E}^2 \ln|\boldsymbol{\theta}| < 0$ inside the Einstein radius ($|\boldsymbol{\theta}| < \theta_\mathrm{E}$), so $-\psi > 0$ there and the Shapiro term *increases* the arrival time in that region.

The competition between geometric and gravitational delay determines the structure of the arrival-time surface and hence the number and types of images.

10.3 Fermat's Principle and the Lens Equation

Fermat's principle states that images form at stationary points of the arrival time $t(\boldsymbol{\theta})$, or equivalently, at stationary points of the Fermat potential $\tau(\boldsymbol{\theta}, \boldsymbol{\beta})$:

$$\nabla_{\boldsymbol{\theta}}\, t(\boldsymbol{\theta}) = 0 \quad \Longleftrightarrow \quad \nabla_{\boldsymbol{\theta}}\, \tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = 0\,. \tag{10.4}$$

Computing the gradient of τ:

$$\begin{aligned} \nabla_{\boldsymbol{\theta}}\, \tau &= \nabla_{\boldsymbol{\theta}} \left[\tfrac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})\right] \\ &= (\boldsymbol{\theta} - \boldsymbol{\beta}) - \nabla\psi(\boldsymbol{\theta})\,. \end{aligned} \tag{10.5}$$

Setting this to zero:

$$\boldsymbol{\theta} - \boldsymbol{\beta} - \nabla\psi(\boldsymbol{\theta}) = 0 \qquad \Longrightarrow \qquad \boldsymbol{\beta} = \boldsymbol{\theta} - \nabla\psi(\boldsymbol{\theta})\,. \tag{10.6}$$

Since the reduced deflection angle is $\boldsymbol{\alpha}(\boldsymbol{\theta}) = \nabla\psi(\boldsymbol{\theta})$ (from Module 9), this is precisely the **lens equation**:

$$\boxed{\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})} \tag{10.7}$$

Remark 10.3.1. This is a remarkable result: the lens equation, which we previously derived from ray-tracing geometry, follows directly from the **stationarity of the arrival time**. Fermat's principle provides a variational derivation of the fundamental equation of gravitational lensing. The connection between the gradient of the lensing potential and the deflection angle, $\boldsymbol{\alpha} = \nabla\psi$, was established in Module 9 from the Poisson equation $\nabla^2\psi = 2\kappa$.

10.4 Classification of Images: Morse Theory

At each stationary point $\boldsymbol{\theta}_i$ of the Fermat potential (i.e., at each image position), we can characterize the nature of the stationary point by the **Hessian matrix** of τ:

$$T_{ab} = \frac{\partial^2 \tau}{\partial\theta_a\,\partial\theta_b} = \delta_{ab} - \frac{\partial^2 \psi}{\partial\theta_a\,\partial\theta_b} = \mathcal{A}_{ab}\,, \tag{10.8}$$

where $\mathcal{A}_{ab}$ is the **Jacobian matrix** of the lens mapping (the inverse magnification matrix, from Module 9).

The eigenvalues of T_{ab} at a stationary point determine the *type* of the image. There are exactly three possibilities, classified by **Morse theory**:

1. **Type I image — Minimum of τ:** Both eigenvalues of T are positive. Equivalently, $\det(\mathcal{A}) > 0$ and $\mathrm{tr}(\mathcal{A}) > 0$. The magnification $\mu = 1/\det(\mathcal{A}) > 0$ (positive parity: the image is not mirror-reflected). This image corresponds to the *minimum* of the arrival-time surface and therefore **arrives first**.

2. **Type II image — Saddle point of τ:** One eigenvalue is positive and one is negative. Equivalently, $\det(\mathcal{A}) < 0$. The magnification $\mu = 1/\det(\mathcal{A}) < 0$ (negative parity: the image is mirror-reflected).

3. **Type III image — Maximum of τ:** Both eigenvalues of T are negative. Equivalently, $\det(\mathcal{A}) > 0$ and $\mathrm{tr}(\mathcal{A}) < 0$. The magnification $\mu = 1/\det(\mathcal{A}) > 0$ (positive parity). This image corresponds to the *maximum* of the arrival-time surface and therefore **arrives last**.

Table 10.1: Classification of images by Morse type. The Hessian of the Fermat potential at each image determines its type and parity. The global minimum of τ is a Type I image and arrives first; a saddle always arrives after it, but its order relative to other images is not fixed (for a point-mass or SIS lens, which has no Type III image, the saddle arrives last).

Type	**Stationary point**	**Eigenvalues**	$\det(\mathcal{A})$	**Parity**	**Arrival**
I	Minimum	$+,+$	>0	$+$ (same)	First
II	Saddle	$+,-$	<0	$-$ (flipped)	After I
III	Maximum	$-,-$	>0	$+$ (same)	Last

Remark 10.4.1 (The arrival-time surface determines image topology). The topology of the arrival-time surface $\tau(\boldsymbol{\theta})$ completely determines the image structure. Each minimum, saddle, and maximum of τ corresponds to an image. When a source crosses a caustic, two stationary points of τ merge and annihilate (or are created from nothing) — always one saddle and one extremum, preserving the constraint $N_{\min} + N_{\max} - N_{\text{saddle}} = 1$ (a consequence of Morse theory on the plane). This topological constraint is why images always appear and disappear in pairs with opposite parity, as explored in detail in Module 12.

Theorem 10.4.1 (Burke's Odd-Number Theorem)**.** *For a* ***transparent*** *lens (i.e., a lens with a smooth, bounded surface mass density $\Sigma(\boldsymbol{\theta})$ that falls off sufficiently rapidly at large $|\boldsymbol{\theta}|$), the total number of images is always* ***odd****. Specifically, if we denote by $n_{\min}$, $n_{\rm sad}$, and $n_{\max}$ the numbers of minima, saddle points, and maxima of τ, then:*

$$n_{\min} - n_{\rm sad} + n_{\max} = 1\,. \tag{10.9}$$

This is a topological result from Morse theory.

Remark 10.4.2. The smoothness hypothesis is essential. Burke's theorem does *not* apply to a point mass lens: Σ is a Dirac delta function and $\psi = \theta_{\rm E}^2 \ln|\boldsymbol{\theta}|$ diverges at the origin, so the origin is a singularity of τ rather than a true stationary point. A point mass therefore produces exactly two images ($\theta_\pm$ from Module 8), not three. For an extended lens with a smooth core (such as a non-singular isothermal sphere, Module 7), the central singularity is replaced by a genuine maximum of τ and a corresponding third image appears, typically highly demagnified. The singular isothermal sphere itself is intermediate: $\Sigma \propto 1/\theta$ is finite at every $\theta > 0$ but diverges at the origin, again removing the central image while keeping the two outer images.

10.5 The Arrival-Time Surface

The arrival-time surface is the three-dimensional plot of $\tau(\boldsymbol{\theta})$ as a function of the two-dimensional image-plane position $\boldsymbol{\theta} = (\theta_1, \theta_2)$ for a fixed source position $\boldsymbol{\beta}$. This is the iconic visualization of Fermat's principle in gravitational lensing.

10.5.1 Point Mass Lens

For a point mass lens with Einstein radius $\theta_{\rm E}$, the lensing potential is (from Module 9):

$$\psi(\boldsymbol{\theta}) = \theta_{\rm E}^2 \, \ln|\boldsymbol{\theta}|\,. \tag{10.10}$$

The Fermat potential becomes:

$$\tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = \tfrac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \theta_{\rm E}^2 \, \ln|\boldsymbol{\theta}|\,. \tag{10.11}$$

For a source at $\boldsymbol{\beta} = (\beta, 0)$ with $\beta > 0$ (placing the source along the θ_1-axis without loss of generality by the azimuthal symmetry of the lens), the arrival-time surface $\tau(\theta_1, \theta_2)$ has exactly **two** stationary points in the region $|\boldsymbol{\theta}| > 0$:

- A **minimum** (Type I) at $\boldsymbol{\theta} = (\theta_+, 0)$, where $\theta_+ = (\beta + \sqrt{\beta^2 + 4\theta_{\rm E}^2})/2$ is the primary image from eq. (8.21). This image arrives *first*.
- A **saddle point** (Type II) at $\boldsymbol{\theta} = (\theta_-, 0)$, where $\theta_- = (\beta - \sqrt{\beta^2 + 4\theta_{\rm E}^2})/2$ is the secondary image. This image arrives *second*.

The point $\boldsymbol{\theta} = \mathbf{0}$ is a *singularity* of ψ, not a true stationary point of τ, so the point-mass lens has only the two images $\theta_\pm$ above. A softened mass profile (Module 11) replaces this singularity with a smooth maximum and produces a corresponding (highly demagnified) third image.

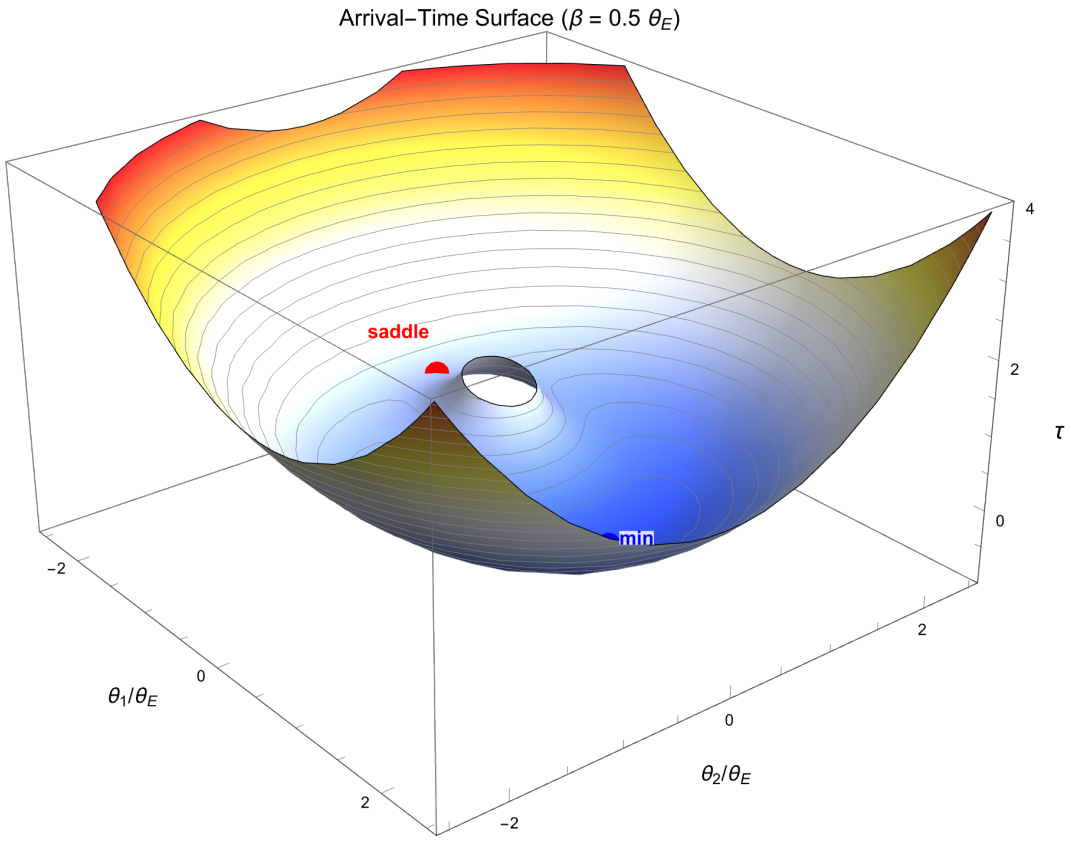


Figure 10.1: The arrival-time surface $\tau(\theta_1, \theta_2)$ for a point mass lens with source at $\boldsymbol{\beta} = (0.5, 0)\,\theta_{\rm E}$. The surface shows a minimum (Type I image, blue dot) and a saddle point (Type II image, red dot). The minimum has the smaller arrival time and corresponds to the image that is observed first. The difference in τ between the two stationary points gives the dimensionless time delay. Generated by `time_delay_function.wl`.

10.5.2 The Einstein Ring as a Degenerate Minimum

When the source is exactly behind the lens ($\boldsymbol{\beta} = 0$), the Fermat potential becomes circularly symmetric:

$$\tau(\boldsymbol{\theta}, \mathbf{0}) = \tfrac{1}{2}|\boldsymbol{\theta}|^2 - \theta_{\rm E}^2 \ln|\boldsymbol{\theta}|\,. \tag{10.12}$$

The stationarity condition $d\tau/d|\boldsymbol{\theta}| = 0$ gives $|\boldsymbol{\theta}| = \theta_{\rm E}$, which is the Einstein ring. The minimum of τ is no longer an isolated point but a **circle** of stationary points — the Einstein ring is a degenerate minimum of the arrival-time surface.

10.6 Observable Time Delays

The arrival time at each image $\boldsymbol{\theta}_i$ is given by eq. (10.1). The **time delay** between two images i and j is the difference in their arrival times:

$$\boxed{\Delta t_{ij} = t(\boldsymbol{\theta}_i) - t(\boldsymbol{\theta}_j) = \frac{(1+z_d)}{c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\left[\tau(\boldsymbol{\theta}_i, \boldsymbol{\beta}) - \tau(\boldsymbol{\theta}_j, \boldsymbol{\beta})\right]} \tag{10.13}$$

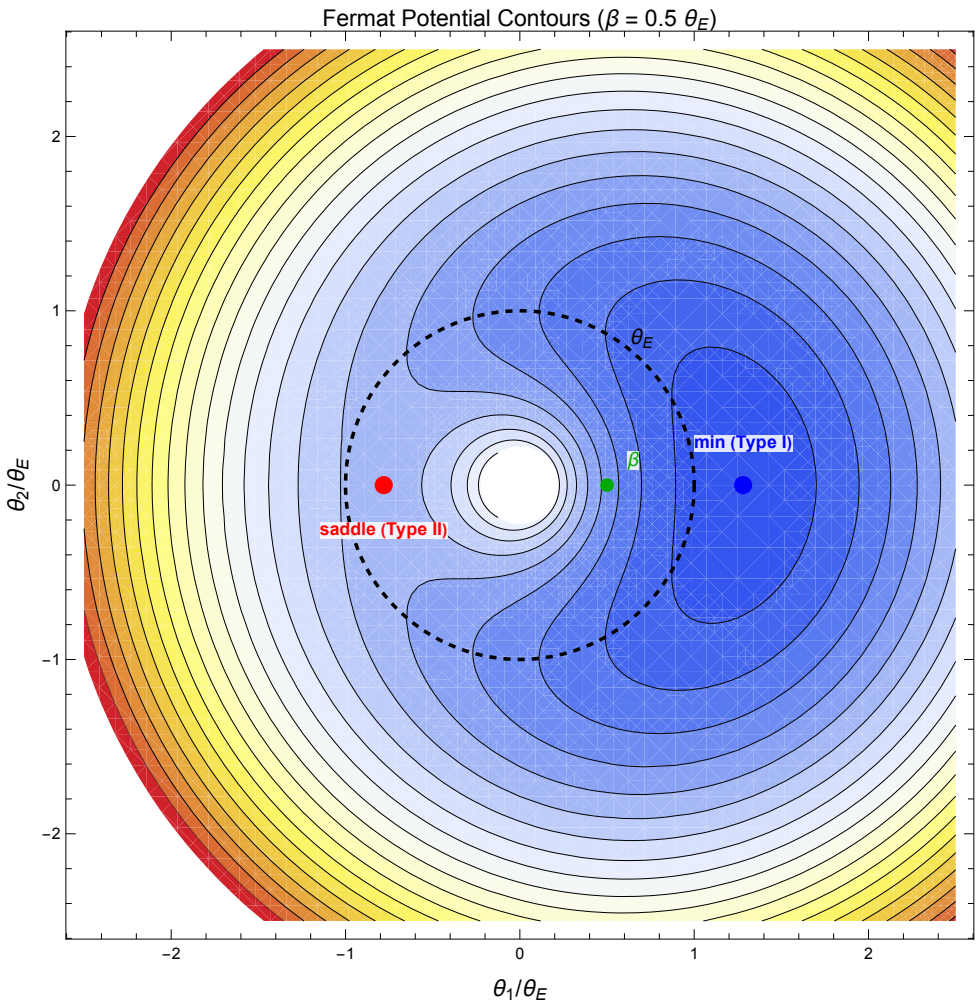


Figure 10.2: Contour plot of the Fermat potential $\tau(\theta_1, \theta_2)$ for a point mass lens with $\boldsymbol{\beta} = (0.5, 0)\,\theta_{\mathrm{E}}$. The two stationary points (minimum and saddle) are marked. The critical curve (Einstein ring, dashed circle) is overlaid. Generated by `time_delay_function.wl`.

This quantity is **directly observable**: if the source is variable (as quasars are), the light curves measured at the two image positions will show the same variability pattern, but shifted by Δt_{ij}. By cross-correlating the light curves, one can extract Δt_{ij} to high precision.

Remark 10.6.1. The time delay Δt_{ij} depends on three ingredients:

1. The **cosmological distances** $(1 + z_d)\, D_{\mathrm{d}}\, D_{\mathrm{s}}/D_{\mathrm{ds}}$, which encode the expansion history of the universe.
2. The **image positions** $\boldsymbol{\theta}_i$, $\boldsymbol{\theta}_j$, which are directly observed.
3. The **lensing potential** $\psi(\boldsymbol{\theta})$, which must be modeled from the observed lens properties (mass distribution, image positions, fluxes, etc.).

Given items (2) and (3), a measurement of Δt_{ij} constrains the distance combination in item (1) — and hence the cosmological parameters.

10.7 Time Delay and the Hubble Constant

The key cosmological application of time delays is the measurement of the Hubble constant H_0. The distance combination $D_{\rm d}\, D_{\rm s}/D_{\rm ds}$ is an angular diameter distance ratio that scales inversely with the Hubble constant:

$$\frac{D_{\rm d}\, D_{\rm s}}{D_{\rm ds}} \propto \frac{1}{H_0}\,. \tag{10.14}$$

This follows from the definition of angular diameter distances in Module 7: all comoving distances scale as c/H_0 times a dimensionless integral over the expansion history.

Therefore, the time delay between images scales as:

$$\boxed{\Delta t \propto H_0^{-1}} \tag{10.15}$$

This is the basis of **time-delay cosmography**: by measuring Δt from monitoring campaigns and modeling the lens potential ψ, one can determine H_0 *independently of the cosmic distance ladder*. The method is sometimes called "one-step" distance measurement, because it converts an observed time delay directly into a physical distance.

Remark 10.7.1. To extract H_0 from a time delay measurement, one needs:

1. A precise measurement of Δt (from light curve monitoring of a multiply-imaged variable source).
2. The lens and source redshifts z_d, z_s.
3. A model for the lens mass distribution $\psi(\boldsymbol{\theta})$, constrained by the observed image positions, fluxes, and extended image structure.
4. A cosmological model (usually flat ΛCDM with $\Omega_{\rm m}$ fixed) to convert the distance ratio into H_0.

The main systematic uncertainty is the lens model degeneracy (the "mass-sheet degeneracy"), which we will encounter in later modules.

10.7.1 Observational Milestones

Time-delay cosmography has been applied to a growing number of multiply-imaged quasars:

- **Q0957+561** (Walsh, Carswell & Weymann 1979): the first discovered gravitational lens. The time delay ($\Delta t \approx 417$ days) was measured after years of photometric monitoring.
- **B1608+656**: a four-image lens system with three independent time delays, providing strong constraints.
- **H0LiCOW** (2017–present): a systematic program combining precise time delays from COSMOGRAIL monitoring with detailed lens models and line-of-sight structure analysis. The H0LiCOW XIII result (Wong et al. 2020; six lenses) gives $H_0 = 73.3^{+1.7}_{-1.8}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, in agreement with local distance ladder measurements and in some tension with the CMB value ($H_0 \approx 67\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ from Planck). The successor **TDCOSMO** programme (Birrer et al. 2020) performed an independent analysis that explicitly marginalises over the mass-sheet degeneracy and obtained a less precise but consistent result ($H_0 = 74.5^{+5.6}_{-6.1}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$).

10.8 Example: Point Mass Time Delay

We now compute the time delay for a point mass lens explicitly. This illustrative calculation demonstrates the method and gives a sense of the physical scales involved.

10.8.1 Fermat Potential for the Point Mass

Using the lensing potential $\psi(\boldsymbol{\theta}) = \theta_{\rm E}^2 \ln|\boldsymbol{\theta}|$ and working in units of $\theta_{\rm E}$ (i.e., defining $x = \theta/\theta_{\rm E}$ and $y = \beta/\theta_{\rm E}$), the Fermat potential along the symmetry axis (where both images lie for a point mass) is:

$$\tau(x) = \tfrac{1}{2}(x - y)^2 - \ln|x|\,, \tag{10.16}$$

where we have set $\theta_{\rm E} = 1$ in the logarithmic term.

The two image positions are $x_\pm = (y \pm \sqrt{y^2+4})/2$.

10.8.2 Time Delay Computation

The Fermat potential at the two images is:

$$\tau(x_+) = \tfrac{1}{2}(x_+ - y)^2 - \ln x_+\,, \tag{10.17}$$
$$\tau(x_-) = \tfrac{1}{2}(x_- - y)^2 - \ln|x_-|\,. \tag{10.18}$$

The dimensionless time delay is $\Delta\tau = \tau(x_-) - \tau(x_+)$ (since $\tau(x_-) > \tau(x_+)$, the saddle image arrives later).

After algebraic simplification (verified in `time_delay_function.wl`), the result is:

$$\boxed{\Delta\tau = \frac{y\sqrt{y^2+4}}{2} + \ln\left(\frac{\sqrt{y^2+4}+y}{\sqrt{y^2+4}-y}\right)} \tag{10.19}$$

where $y = \beta/\theta_{\rm E}$ is the dimensionless source position.

The physical time delay is:

$$\Delta t = \frac{(1+z_d)}{c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\theta_{\rm E}^2\,\Delta\tau\,. \tag{10.20}$$

Using $\theta_{\rm E}^2 = 4GM/(c^2)\cdot D_{\rm ds}/(D_{\rm d}\,D_{\rm s})$:

$$\Delta t = \frac{4GM}{c^3}\,(1+z_d)\,\Delta\tau = \frac{R_{\rm S}}{c}\,(1+z_d)\,\Delta\tau\,, \tag{10.21}$$

where $R_{\rm S} = 2GM/c^2$ is the Schwarzschild radius.

Example 10.8.1. Numerical time delay for a galaxy-mass point lens.

Consider a lens mass $M = 10^{12}\,M_\odot$ at $z_d = 0.3$ with a source at $\beta = 0.5\,\theta_{\rm E}$ ($y = 0.5$). The Schwarzschild radius is:

$$R_{\rm S} = \frac{2\times 6.674\times10^{-11}\times10^{12}\times1.989\times10^{30}}{(2.998\times10^{8})^2} \approx 2.95\times10^{15}\ \text{m} \approx 0.096\ \text{pc}\,.$$

For $y = 0.5$: $\sqrt{y^2+4} = \sqrt{4.25} \approx 2.0616$.

$$\Delta\tau = \frac{0.5 \times 2.0616}{2} + \ln\left(\frac{2.0616 + 0.5}{2.0616 - 0.5}\right)$$
$$= 0.5154 + \ln(1.6401) = 0.5154 + 0.4947 = 1.0101\,.$$

The time delay is:

$$\Delta t = \frac{2.95 \times 10^{15}}{2.998 \times 10^{8}} \times 1.3 \times 1.0101 \approx 1.29 \times 10^{7}\ \text{s} \approx 149\ \text{days}\,.$$

This is a realistic order of magnitude — observed time delays in galaxy-scale lens systems range from days to months. Computed and verified in `time_delay_function.wl`.

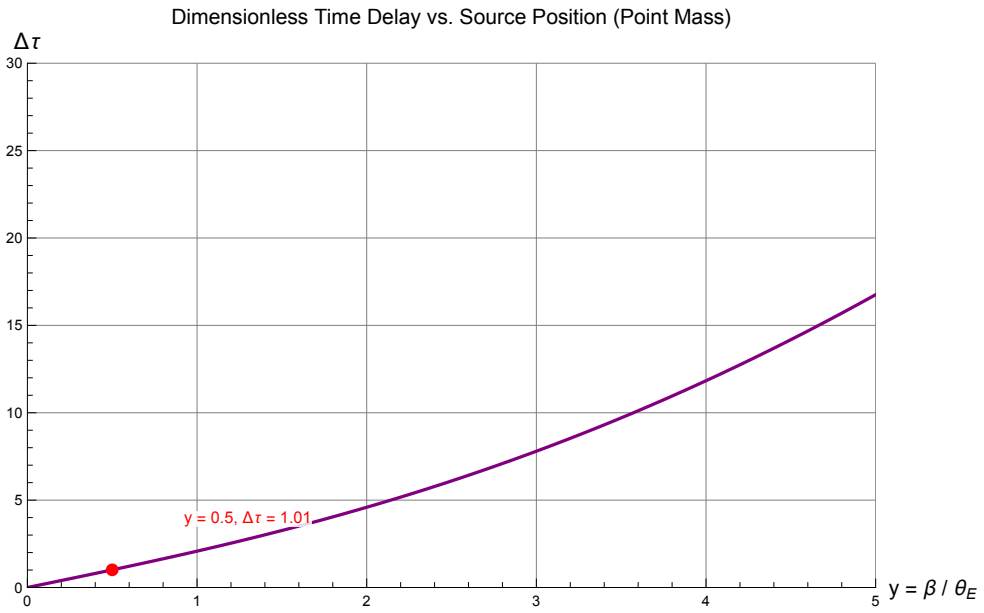


Figure 10.3: Dimensionless time delay $\Delta\tau$ between the two images of a point mass lens as a function of the source position $y = \beta/\theta_{\rm E}$. The time delay increases with the source offset. For $y \to 0$ (Einstein ring limit), $\Delta\tau \to 0$ because the two images become symmetric. Generated by `time_delay_function.wl`.

10.9 Example: SIS Time Delay

As a preview for Module 7 (axially symmetric lens models), we state the time delay for the **singular isothermal sphere** (SIS). The SIS is characterized by its velocity dispersion σ_v, and its Einstein radius is $\theta_{\rm E} = 4\pi\,(\sigma_v/c)^2\,D_{\rm ds}/D_{\rm s}$.

For a source inside the Einstein ring ($\beta < \theta_{\rm E}$), the SIS produces two images at positions $\theta_\pm = \beta \pm \theta_{\rm E}$ (both on the same side of the lens for θ_+, opposite for θ_-). The lensing potential

is $\psi(\boldsymbol{\theta}) = \theta_{\rm E}\,|\boldsymbol{\theta}|$, and the time delay between the two images is:

$$\boxed{\Delta t_{\rm SIS} = \frac{(1+z_d)}{c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\frac{\theta_{\rm E}^2}{2}\left(\frac{\theta_+^2-\theta_-^2}{\theta_{\rm E}^2}\right) = \frac{(1+z_d)}{2c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\left(\theta_+^2-\theta_-^2\right)} \tag{10.22}$$

Since $\theta_+^2 - \theta_-^2 = (\theta_+ + \theta_-)(\theta_+ - \theta_-) = 2\beta \cdot 2\theta_{\rm E} = 4\beta\,\theta_{\rm E}$, this simplifies to:

$$\Delta t_{\rm SIS} = \frac{2(1+z_d)}{c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\beta\,\theta_{\rm E}\,. \tag{10.23}$$

For the SIS, the time delay is proportional to the source position — the further the source from the optical axis, the larger the time delay. This simple relation makes the SIS a useful "first approximation" for time-delay estimates.

10.10 Summary

- The **time delay function** $t(\boldsymbol{\theta}) \propto \tau(\boldsymbol{\theta}, \boldsymbol{\beta})$ gives the arrival time of a photon passing through the lens plane at $\boldsymbol{\theta}$. The **Fermat potential** $\tau = \frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})$ combines the geometric delay (extra path length) and the Shapiro delay (gravitational time delay from the lensing potential).

- **Fermat's principle**: images form at stationary points of $\tau(\boldsymbol{\theta})$. The condition $\nabla_{\boldsymbol{\theta}}\tau = 0$ recovers the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \nabla\psi = \boldsymbol{\theta} - \boldsymbol{\alpha}$.

- Images are classified by **Morse theory**: **Type I** (minimum, positive parity, arrives first), **Type II** (saddle, negative parity), **Type III** (maximum, positive parity, arrives last). **Burke's theorem**: the number of images is always odd for a transparent lens.

- The **arrival-time surface** $\tau(\theta_1, \theta_2)$ visualizes the image-forming process. For a point mass lens, it has a minimum and a saddle point (plus a singular maximum at the lens center).

- The **time delay between images** $\Delta t_{ij} \propto (1+z_d)\,D_{\rm d}D_{\rm s}/D_{\rm ds} \times [\tau(\boldsymbol{\theta}_i) - \tau(\boldsymbol{\theta}_j)]$ is directly observable by monitoring variable sources.

- $\Delta t \propto H_0^{-1}$: measuring time delays and modeling the lens potential gives H_0 independently of the distance ladder. This is **time-delay cosmography**, implemented by programs such as H0LiCOW/TDCOSMO.

- For a point mass lens, the dimensionless time delay is $\Delta\tau = y\sqrt{y^2+4}/2 + \ln[(\sqrt{y^2+4}+y)/(\sqrt{y^2+4}-y)]$, where $y = \beta/\theta_{\rm E}$. The physical time delay for a galaxy-mass lens is of order days to months.

- For the SIS, $\Delta t_{\rm SIS} \propto \beta\,\theta_{\rm E}$, a simple linear dependence on the source position.

10.11 Problems

Exercise 10.11.1. (Deriving the lens equation from Fermat's principle.) Starting from the Fermat potential $\tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = \frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})$:

(a) Compute the gradient $\nabla_{\boldsymbol{\theta}}\, \tau$.

(b) Set $\nabla_{\boldsymbol{\theta}}\, \tau = 0$ and show that this gives the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \nabla\psi(\boldsymbol{\theta}) = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$.

(c) Explain physically why images form at stationary points of the arrival time.

Verify with `problems_06.wl`.

Exercise 10.11.2. (Arrival times for a point mass lens.) Consider a point mass lens with Einstein radius $\theta_{\rm E} = 1''$ and a source at $\beta = 0.5\,\theta_{\rm E}$.

(a) Find the two image positions θ_+ and θ_-.

(b) Compute the Fermat potential τ at each image position using $\tau(x) = \frac{1}{2}(x - y)^2 - \ln|x|$ with $y = 0.5$, where $x = \theta/\theta_{\rm E}$.

(c) Compute the dimensionless time delay $\Delta\tau = \tau(x_-) - \tau(x_+)$.

(d) For a lens mass $M = 10^{12}\,M_\odot$ at $z_d = 0.3$, compute the physical time delay Δt in days.

Verify with `problems_06.wl`.

Exercise 10.11.3. (Type I image arrives first.) For a point mass lens with source at $\beta > 0$:

(a) Show that $\tau(\theta_+) < \tau(\theta_-)$, i.e., the minimum-image (Type I) has a smaller value of τ than the saddle-image (Type II).

(b) Verify this by computing $\tau(\theta_\pm)$ for $\beta = 0.1, 0.5, 1.0, 2.0$ (in units of $\theta_{\rm E}$) and checking that $\tau(\theta_+) < \tau(\theta_-)$ in all cases.

(c) Explain why this implies the Type I image arrives *before* the Type II image.

Verify with `problems_06.wl`.

Exercise 10.11.4. (Measuring H_0 from time delays.) A multiply-imaged quasar has a measured time delay $\Delta t = 100$ days between two images. The lens model gives a dimensionless Fermat potential difference of $\tau_1 - \tau_2 = 2.0$ arcsec^2. The lens is at $z_d = 0.3$ and the source is at $z_s = 1.0$.

(a) Using the relation $\Delta t = (1 + z_d)/c \times D_{\rm d} D_{\rm s}/D_{\rm ds} \times (\tau_1 - \tau_2)$, solve for $D_{\rm d} D_{\rm s}/D_{\rm ds}$.

(b) For a flat ΛCDM cosmology with $\Omega_{\rm m} = 0.3$, show that $D_{\rm d} D_{\rm s}/D_{\rm ds} \propto 1/H_0$ and compute the proportionality constant (hint: use the dimensionless distance ratios from Module 3).

(c) Determine H_0 in km s^{-1} Mpc^{-1}. Compare with the H0LiCOW result and the Planck value.

This is a simplified version of the analysis performed in real time-delay cosmography. Verify with `problems_06.wl`.

Exercise 10.11.5. (Plotting the arrival-time surface.) Using Mathematica (or your preferred software):

(a) Plot the arrival-time surface $\tau(\theta_1, \theta_2)$ for a point mass lens with source at $\boldsymbol{\beta} = (0.3, 0)\,\theta_{\rm E}$. Use $\theta_{\rm E} = 1$.

(b) Identify the two stationary points on the surface. Label them as "minimum" and "saddle."

(c) Make a contour plot of τ and overlay the critical curve (the Einstein ring $|\boldsymbol{\theta}| = \theta_{\rm E}$).

(d) Verify that the stationary points agree with the analytical image positions $\theta_\pm$ from the lens equation.

See `arrival_time_surface.nb` and `problems_06.wl`.

Part III

Lens Models and Applications

Chapter 11

Axisymmetric Lens Models

> *In the special case of a circularly symmetric lens, we can shift the coordinate origin to the center of symmetry and reduce light deflection to a one-dimensional problem.* — Narayan & Bartelmann (1997), Sect. 2.1.2

In Modules 8 and 9 we developed the general framework of gravitational lensing: the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$, the convergence κ, the shear γ, and the magnification $\mu = 1/[(1-\kappa)^2 - |\gamma|^2]$. We applied these to the point mass lens and briefly previewed the singular isothermal sphere (SIS).

Now we move from formalism to *modeling*. Real gravitational lenses — galaxies and galaxy clusters — have extended mass distributions, and to interpret observations we need specific, physically motivated models. This module covers the most important **circularly symmetric** (axisymmetric) lens models:

- The **point mass** (recap from Module 4),
- The **singular isothermal sphere** (SIS),
- The **nonsingular isothermal sphere** (NIS),
- The **Navarro–Frenk–White** (NFW) profile.

These models form the building blocks for more complex, non-circular lens models (Module 8) and for practical lens modeling applications (Modules 9–10).

Companion Mathematica files:

- `axisymmetric_models.wl` — symbolic derivation of SIS, NIS, and NFW lensing quantities; numerical examples with realistic parameters; comparison figures exported to `Figures/07_Axisymmetric_Models/`
- `sis_nis_nfw.nb` — interactive notebook with `Manipulate` widgets for exploring SIS, NIS, and NFW models: vary velocity dispersion, core radius, concentration, and source position

11.1 Introduction: From Formalism to Models

The general formalism of Module 9 provides the machinery for computing lensing observables — but to apply it, one must specify a mass distribution. In practice, the choice of mass model is guided by the physics of the lens:

- **Stars and compact objects:** point mass (§11.3).
- **Elliptical galaxies:** the stellar and dark matter mass distribution is approximately described by an isothermal profile with $\rho \propto r^{-2}$ (§11.4–11.5).
- **Dark matter haloes:** N-body simulations show that the radial density profile of dark matter haloes is well-described by the universal NFW profile (§11.6).

In this module we restrict ourselves to **circularly symmetric** models, where all lensing quantities depend only on the angular radius $\theta = |\boldsymbol{\theta}|$. This simplification allows analytic progress and builds intuition before tackling the elliptical models of Module 8.

11.2 General Formalism for Circularly Symmetric Lenses

For a lens with circular symmetry about the optical axis, the surface mass density Σ, the convergence κ, the deflection angle $\boldsymbol{\alpha}$, and the shear γ all depend only on the angular radius $\theta = |\boldsymbol{\theta}|$. The vector deflection is purely radial:

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \alpha(\theta)\,\frac{\boldsymbol{\theta}}{\theta}\,, \tag{11.1}$$

and the lens equation reduces to a single dimension (Narayan & Bartelmann eq. 14):

$$\boxed{\beta = \theta - \alpha(\theta)} \tag{11.2}$$

where $\beta = |\boldsymbol{\beta}|$ (by symmetry, source and image are collinear with the lens center).

11.2.1 Deflection Angle from Enclosed Mass

For a circularly symmetric mass distribution, the deflection angle at radius θ depends only on the mass enclosed within that radius (just as in Newtonian gravity, by the circular analog of Newton's shell theorem). The projected mass within angular radius θ is:

$$M(\theta) = 2\pi \int_0^{D_\mathrm{d}\,\theta} \Sigma(\xi)\,\xi\,d\xi = 2\pi D_\mathrm{d}^2 \int_0^{\theta} \Sigma(D_\mathrm{d}\,\theta')\,\theta'\,d\theta'\,, \tag{11.3}$$

and the reduced deflection angle is (Congdon & Keeton Sec. 2.2.1):

$$\boxed{\alpha(\theta) = \frac{M(\theta)}{\pi\,\Sigma_\mathrm{cr}\,D_\mathrm{d}^2\,\theta}} \tag{11.4}$$

11.2.2 Mean Convergence

It is useful to define the **mean convergence** within radius θ:

$$\bar{\kappa}(\theta) = \frac{1}{\pi\theta^2}\int_0^{\theta} \kappa(\theta')\,2\pi\theta'\,d\theta' = \frac{2}{\theta^2}\int_0^{\theta} \kappa(\theta')\,\theta'\,d\theta'\,. \tag{11.5}$$

Since $M(\theta) = \pi D_\mathrm{d}^2 \Sigma_\mathrm{cr}\,\theta^2\,\bar{\kappa}(\theta)$, the deflection angle (11.4) can be written as:

$$\alpha(\theta) = \bar{\kappa}(\theta)\,\theta\,. \tag{11.6}$$

11.2.3 Shear for Circular Symmetry

For a circularly symmetric lens, the shear magnitude takes a remarkably simple form (cf. Narayan & Bartelmann Sec. 3.4):

$$\boxed{|\gamma(\theta)| = \bar{\kappa}(\theta) - \kappa(\theta)} \tag{11.7}$$

The shear is the difference between the mean convergence inside θ and the local convergence at θ. Physically, this makes sense: the shear arises from the *differential* focusing between interior and exterior mass.

11.2.4 Magnification for Circular Symmetry

From Module 8 (eq. 8.31), the magnification of a circularly symmetric lens is:

$$\mu = \frac{\theta}{\beta}\frac{d\theta}{d\beta} = \frac{1}{\left(1 - \frac{\alpha}{\theta}\right)\left(1 - \frac{d\alpha}{d\theta}\right)}\,. \tag{11.8}$$

The two factors correspond to the tangential and radial eigenvalues of the Jacobian:

$$\lambda_t = 1 - \frac{\alpha(\theta)}{\theta} = 1 - \bar{\kappa}(\theta)\,, \qquad \lambda_r = 1 - \frac{d\alpha}{d\theta} = 1 - 2\kappa(\theta) + \bar{\kappa}(\theta)\,. \tag{11.9}$$

The **tangential critical curve** occurs where $\lambda_t = 0$ (i.e., $\bar{\kappa} = 1$), and the **radial critical curve** occurs where $\lambda_r = 0$ (i.e., $2\kappa = 1 + \bar{\kappa}$).

11.3 Point Mass Lens (Recap)

We briefly collect the point mass results from Modules 8 and 9 for comparison with the extended models.

For a point mass M at the origin:

$$\Sigma(\theta) = M\,\delta^{(2)}(D_{\rm d}\,\boldsymbol{\theta})/D_{\rm d}^2\,, \qquad \kappa(\theta) = 0 \quad (\theta \neq 0)\,, \tag{11.10}$$

$$\alpha(\theta) = \frac{\theta_{\rm E}^2}{\theta}\,, \tag{11.11}$$

$$|\gamma(\theta)| = \frac{\theta_{\rm E}^2}{\theta^2} = \bar{\kappa}(\theta)\,, \tag{11.12}$$

where $\theta_{\rm E} = \sqrt{4GMD_{\rm ds}/(c^2 D_{\rm d}\,D_{\rm s})}$ is the Einstein radius (eq. 8.17).

The lens equation $\beta = \theta - \theta_{\rm E}^2/\theta$ yields two images at $\theta_\pm = (\beta \pm \sqrt{\beta^2 + 4\theta_{\rm E}^2})/2$, with magnifications

$$\mu_\pm = \frac{u^2 + 2}{2u\sqrt{u^2 + 4}} \pm \frac{1}{2}\,, \qquad u = \beta/\theta_{\rm E}\,. \tag{11.13}$$

Remark 11.3.1. The point mass has no radial critical curve (the convergence is a delta function, so $\lambda_r = 1 + \bar{\kappa} > 0$ everywhere away from the origin). It has only a tangential critical curve at $\theta = \theta_{\rm E}$ (the Einstein ring), whose corresponding caustic is the degenerate point $\beta = 0$.

11.4 Singular Isothermal Sphere (SIS)

The **singular isothermal sphere** is the simplest realistic model for a galaxy lens. It describes a self-gravitating system of particles with a Maxwellian velocity distribution and one-dimensional velocity dispersion σ_v. Despite its simplicity, the SIS captures many essential features of observed galaxy lenses.

11.4.1 Three-Dimensional Density Profile

The SIS density profile is (Congdon & Keeton eq. 2.42):

$$\boxed{\rho(r) = \frac{\sigma_v^2}{2\pi G\, r^2}} \tag{11.14}$$

where r is the three-dimensional (spherical) radius and σ_v is the one-dimensional velocity dispersion. This profile is "isothermal" because it corresponds to a self-gravitating gas at constant temperature $T \propto \sigma_v^2$, and "singular" because $\rho \to \infty$ as $r \to 0$.

Remark 11.4.1. The $\rho \propto r^{-2}$ profile gives a flat rotation curve: $v_c(r) = \sqrt{2}\,\sigma_v = \text{const}$. This is consistent with observations of spiral galaxies, which show approximately flat rotation curves out to large radii. For elliptical galaxies, the velocity dispersion σ_v plays the analogous role.

11.4.2 Surface Mass Density

Projecting along the line of sight (taking ξ as the impact parameter in the lens plane):

$$\Sigma(\xi) = \int_{-\infty}^{+\infty} \rho\Big(\sqrt{\xi^2 + z^2}\Big)\, dz = \frac{\sigma_v^2}{2G\,\xi}\,. \tag{11.15}$$

The surface mass density diverges as $\xi \to 0$ and falls off as ξ^{-1}.

11.4.3 Einstein Radius of the SIS

The physical deflection angle of the SIS is one of its most remarkable properties. The enclosed projected mass $M_{2\mathrm{D}}(\xi) = \pi\sigma_v^2\xi/G$ follows from the SIS surface density $\Sigma(\xi) = \sigma_v^2/(2G\xi)$ by a one-line integral,

$$M_{2\mathrm{D}}(\xi) \;=\; 2\pi\int_0^{\xi} \Sigma(\xi')\,\xi'\,d\xi' \;=\; 2\pi\cdot\frac{\sigma_v^2}{2G}\int_0^{\xi} d\xi' \;=\; \frac{\pi\sigma_v^2\xi}{G}\,,$$

and $\Sigma(\xi)$ itself follows from $\int_{-\infty}^{\infty}(\sigma_v^2/(2\pi G))/(\xi^2+z^2)\,dz = \sigma_v^2/(2G\xi)$. Substituting into the axisymmetric deflection formula (eq. 5.18 from Module 4):

$$\boxed{\hat{\alpha} = \frac{4\pi\sigma_v^2}{c^2}} \tag{11.16}$$

This is **constant** — independent of the impact parameter! Every light ray is deflected by the same angle, regardless of how close it passes to the lens center. This deeply counterintuitive result has a clean physical explanation: the SIS has $\rho \propto r^{-2}$, so the projected (surface) mass enclosed within a cylinder of radius ξ grows linearly, $M(\xi) = \pi\sigma_v^2\xi/G \propto \xi$ (eq. 11.3 and Exercise 11.9.1). Since the deflection angle is $\hat{\alpha} = 4GM(\xi)/(c^2\,\xi)$, the ξ in numerator and denominator cancel exactly, leaving $\hat{\alpha}$ independent of ξ. The $\rho \propto r^{-2}$ profile is the *unique* power law for which this cancellation occurs; any steeper or shallower profile gives a deflection that varies with impact parameter. This is also why the velocity dispersion σ_v — rather than a mass scale — is the natural parameter of the SIS (Congdon & Keeton Sec. 6.1; Narayan & Bartelmann Sec. 3.4; verified in `axisymmetric_models.wl`).

The reduced deflection angle is:

$$\alpha = \frac{D_{\rm ds}}{D_{\rm s}}\,\hat{\alpha} = \frac{4\pi\sigma_v^2}{c^2}\,\frac{D_{\rm ds}}{D_{\rm s}} = \theta_{\rm E}\,, \tag{11.17}$$

where the **Einstein radius** of the SIS is:

$$\boxed{\theta_{\rm E} = 4\pi\left(\frac{\sigma_v}{c}\right)^2\frac{D_{\rm ds}}{D_{\rm s}}} \tag{11.18}$$

Example 11.4.1. SIS Einstein radius for a typical galaxy lens. For $\sigma_v = 250$ km s^{-1}, $z_d = 0.3$, $z_s = 1$ ($D_{\rm ds}/D_{\rm s} \approx 1055/1652 \approx 0.638$):

$$\theta_{\rm E} = 4\pi\left(\frac{250\times10^3}{3\times10^8}\right)^2 \times 0.638 \approx 5.6\times10^{-6}\ \text{rad} \approx 1.15''\,. \tag{11.19}$$

This is typical of galaxy-scale strong lenses. The value is computed and verified in `axisymmetric_models.wl`.

11.4.4 Convergence and Shear of the SIS

The convergence is (previewed in Module 9, eq. 9.37):

$$\kappa(\theta) = \frac{\Sigma(D_{\rm d}\,\theta)}{\Sigma_{\rm cr}} = \frac{\theta_{\rm E}}{2\theta}\,. \tag{11.20}$$

The mean convergence within θ is:

$$\bar{\kappa}(\theta) = \frac{2}{\theta^2}\int_0^{\theta}\frac{\theta_{\rm E}}{2\theta'}\,\theta'\,d\theta' = \frac{\theta_{\rm E}}{\theta}\,. \tag{11.21}$$

The shear magnitude is (from eq. 11.7):

$$\boxed{|\gamma(\theta)| = \bar{\kappa} - \kappa = \frac{\theta_{\rm E}}{\theta} - \frac{\theta_{\rm E}}{2\theta} = \frac{\theta_{\rm E}}{2\theta} = \kappa(\theta)} \tag{11.22}$$

Remark 11.4.2. The equality $|\gamma| = \kappa$ at every point is a special property of the SIS. It does *not* hold for general lens models. For the point mass, $\kappa = 0$ (away from the origin) while $|\gamma| = \theta_{\rm E}^2/\theta^2 \neq 0$. For the NIS and NFW models below, $|\gamma| \neq \kappa$ in general.

11.4.5 SIS Lens Equation and Images

Take θ as a signed coordinate along the line through the lens and the source ($\beta \geq 0$; $\theta > 0$ on the source's side). The SIS deflection has constant magnitude $\theta_{\rm E}$ and points along $\theta/|\theta|$, so the 1D lens equation is

$$\beta = \theta - \theta_{\rm E} \frac{\theta}{|\theta|} \,. \tag{11.23}$$

This is *piecewise linear* in θ (unlike the point mass lens equation, which is quadratic). The analysis depends on whether the source lies inside or outside the Einstein radius:

Case 1: $\beta < \theta_{\rm E}$ **(source inside the caustic).** There are **two images**:

$$\theta_+ = \beta + \theta_{\rm E} \,, \qquad \theta_- = -(\theta_{\rm E} - \beta) = \beta - \theta_{\rm E} \,, \tag{11.24}$$

where $\theta_- < 0$ indicates the image is on the opposite side of the lens from the source. (For $\theta > 0$, eq. (11.23) reads $\beta = \theta - \theta_{\rm E}$, giving θ_+; for $\theta < 0$ it reads $\beta = \theta + \theta_{\rm E}$, giving θ_-, which is indeed negative when $\beta < \theta_{\rm E}$.)

Case 2: $\beta > \theta_{\rm E}$ **(source outside the caustic).** There is only **one image**:

$$\theta = \beta + \theta_{\rm E} \,. \tag{11.25}$$

The $\theta < 0$ branch would give $\theta = \beta - \theta_{\rm E} > 0$, which contradicts $\theta < 0$, so there is no second image.

Case 3: $\beta = 0$ **(source on the optical axis).** The image is a complete **Einstein ring** at $\theta = \theta_{\rm E}$.

11.4.6 SIS Magnification

The magnification of the SIS images is:

$$\mu_\pm = \frac{\theta_\pm}{\beta} \frac{d\theta}{d\beta} = \frac{\theta_\pm}{\beta} = 1 \pm \frac{\theta_{\rm E}}{\beta} \,, \tag{11.26}$$

where we used $d\theta/d\beta = 1$ (the SIS lens equation is linear). The total magnification for a doubly-imaged source ($\beta < \theta_{\rm E}$) is:

$$|\mu_+| + |\mu_-| = \left(1 + \frac{\theta_{\rm E}}{\beta}\right) + \left(\frac{\theta_{\rm E}}{\beta} - 1\right) = \frac{2\theta_{\rm E}}{\beta} \,. \tag{11.27}$$

11.4.7 Critical Curves and Caustics of the SIS

The SIS has a particularly simple critical curve structure:

- **Tangential critical curve:** $\lambda_t = 0$ requires $\bar{\kappa}(\theta) = 1$, i.e., $\theta_{\rm E}/\theta = 1$, giving $\theta_{\rm crit} = \theta_{\rm E}$. The critical curve is the Einstein ring.

- **Tangential caustic:** mapping the critical curve to the source plane: $\beta = \theta_{\rm E} - \theta_{\rm E} = 0$. The caustic degenerates to a single **point** at the origin.

- **Radial critical curve:** $\lambda_r = 0$ requires $2\kappa = 1 + \bar{\kappa}$. For the SIS, $2\kappa = \theta_{\rm E}/\theta$ and $1 + \bar{\kappa} = 1 + \theta_{\rm E}/\theta$, so the equation becomes $\theta_{\rm E}/\theta = 1 + \theta_{\rm E}/\theta$, i.e., $0 = 1$. There is **no radial critical curve** for the SIS — the singularity at the center prevents its formation.

Remark 11.4.3. The absence of a radial critical curve is a consequence of the central singularity. As we will see in Section 11.5, introducing a finite core radius (NIS) creates a radial critical curve and fundamentally changes the image configuration.

11.5 Nonsingular Isothermal Sphere (NIS)

The SIS has an unphysical singularity at $r = 0$: the density and surface mass density diverge. A more realistic model introduces a finite **core radius** θ_c that softens the central density cusp.

11.5.1 Surface Mass Density

The NIS surface mass density is (Congdon & Keeton Sec. 2.3.3):

$$\boxed{\Sigma(\theta) = \frac{\sigma_v^2}{2G\, D_{\rm d}\, \sqrt{\theta^2 + \theta_c^2}}} \tag{11.28}$$

where θ_c is the angular core radius. For $\theta \gg \theta_c$, $\Sigma \to \sigma_v^2/(2G\, D_{\rm d}\, \theta)$, recovering the SIS. For $\theta \to 0$, $\Sigma(0) = \sigma_v^2/(2G\, D_{\rm d}\, \theta_c)$ is finite. This is the projection of the 3D density (Congdon & Keeton eq. 2.53)

$$\rho(r) = \frac{\sigma_v^2}{2\pi G\, (r^2 + r_c^2)}\,, \tag{11.29}$$

with physical core radius $r_c = D_{\rm d}\, \theta_c$: since $\int_{-\infty}^{\infty} dz/(\xi^2 + r_c^2 + z^2) = \pi/\sqrt{\xi^2 + r_c^2}$, integrating along the line of sight at $\xi = D_{\rm d}\theta$ gives eq. (11.28).

11.5.2 Convergence of the NIS

$$\kappa(\theta) = \frac{\Sigma(\theta)}{\Sigma_{\rm cr}} = \frac{\theta_{\rm E}}{2\sqrt{\theta^2 + \theta_c^2}} \tag{11.30}$$

where $\theta_{\rm E}$ is the SIS Einstein radius (eq. 11.18). The central convergence is:

$$\kappa(0) = \frac{\theta_{\rm E}}{2\theta_c}\,. \tag{11.31}$$

For strong lensing to occur, we need $\kappa(0) > 1$, requiring $\theta_c < \theta_{\rm E}/2$.

11.5.3 Deflection Angle of the NIS

The axisymmetric formula $\alpha(\theta) = (2/\theta)\int_0^\theta \kappa(\theta')\,\theta'\,d\theta'$ reduces to an elementary integral once we substitute eq. (11.30):

$$\alpha(\theta) \;=\; \frac{2}{\theta}\int_0^\theta \frac{\theta_\mathrm{E}}{2\sqrt{\theta'^2+\theta_c^2}}\,\theta'\,d\theta' \;=\; \frac{\theta_\mathrm{E}}{\theta}\Big[\sqrt{\theta'^2+\theta_c^2}\Big]_0^\theta,$$

where the antiderivative follows from $d/d\theta'\sqrt{\theta'^2+\theta_c^2} = \theta'/\sqrt{\theta'^2+\theta_c^2}$. Evaluating the limits,

$$\alpha(\theta) = \theta_\mathrm{E}\,\frac{\sqrt{\theta^2+\theta_c^2}-\theta_c}{\theta}\,. \tag{11.32}$$

In the limit $\theta_c \to 0$:

$$\alpha(\theta) \;\to\; \theta_\mathrm{E}\,\frac{\theta-0}{\theta} \;=\; \theta_\mathrm{E}\,, \tag{11.33}$$

recovering the constant SIS deflection angle. The integral and the SIS limit are checked symbolically in `axisymmetric_models.wl`.

11.5.4 Effect of the Core Radius on Image Multiplicity

The NIS lens equation is:

$$\beta = \theta - \theta_\mathrm{E}\,\frac{\sqrt{\theta^2+\theta_c^2}-\theta_c}{\theta}\,. \tag{11.34}$$

The core radius has a dramatic effect on the critical curve structure:

- **For** $\theta_c < \theta_\mathrm{E}/2$ (supercritical core): the central convergence exceeds unity, and there exist both a tangential and a **radial** critical curve. A source near the center can produce **three images** (compared to two for the SIS).
- **For** $\theta_c = \theta_\mathrm{E}/2$: the central convergence is exactly unity. The radial critical curve contracts to a point at the center.
- **For** $\theta_c > \theta_\mathrm{E}/2$ (subcritical core): $\kappa < 1$ everywhere, and the lens cannot produce multiple images. There is only a single image for any source position.

The radial critical curve of the NIS creates a **radial caustic** in the source plane: a circle of radius $\beta_r > 0$. Sources inside this radial caustic (and inside the tangential caustic) produce three images; sources between the radial and tangential caustics produce one image.

Remark 11.5.1. The transition from SIS to NIS illustrates a general principle: the *central density profile* of a lens model determines whether a radial critical curve (and hence a third image) can form. Observed lens systems rarely show a detectable central (third) image, which constrains the core radius to be very small or zero — supporting cuspy density profiles.

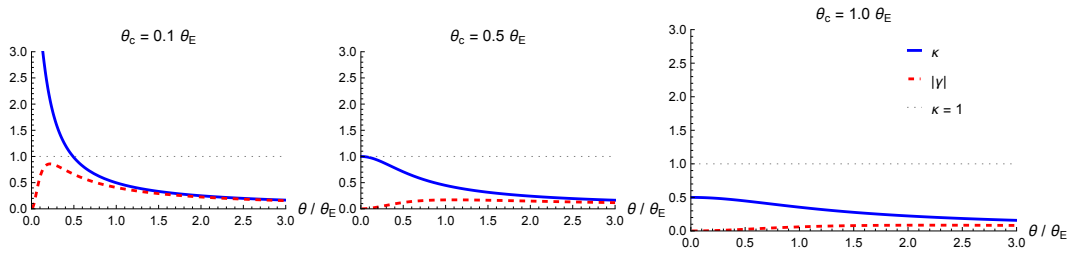


Figure 11.1: Effect of the core radius on the NIS convergence and shear, shown for three regimes that control the critical-curve topology. **Left:** for $\theta_c = 0.1\,\theta_{\rm E} < \theta_{\rm E}/2$, the central convergence $\kappa(0) > 1$, so both tangential and radial critical curves exist and three images can form. **Center:** at the critical value $\theta_c = 0.5\,\theta_{\rm E} = \theta_{\rm E}/2$, $\kappa(0) = 1$ exactly and the radial critical curve collapses to a point. **Right:** for $\theta_c = 1.0\,\theta_{\rm E} > \theta_{\rm E}/2$, $\kappa(0) < 1$ everywhere; no critical curves exist and only one image forms. Generated by `axisymmetric_models.wl`.

11.6 NFW Profile

The **Navarro–Frenk–White** (NFW) profile is the "universal" dark matter halo density profile found in cosmological N-body simulations (Navarro, Frenk & White 1996, 1997). It is the standard model for the mass distribution of galaxy clusters and is essential for cluster lensing (Module 10).

11.6.1 Three-Dimensional Density Profile

$$\boxed{\rho(r) = \frac{\rho_s}{\left(\dfrac{r}{r_s}\right)\left(1 + \dfrac{r}{r_s}\right)^2}} \tag{11.35}$$

where r_s is the **scale radius** and ρ_s is a characteristic density. The profile has two regimes:

- $r \ll r_s$: $\rho \propto r^{-1}$ (cusp, shallower than isothermal),
- $r \gg r_s$: $\rho \propto r^{-3}$ (steeper than isothermal, finite total mass within a virial radius).

11.6.2 NFW Parameters: Concentration and Virial Mass

The NFW profile is usually parametrized by two quantities:

- The **virial mass** M_{200}: the mass enclosed within the virial radius r_{200}, defined as the radius within which the mean density is 200 times the critical density of the universe.
- The **concentration parameter** $c = r_{200}/r_s$: the ratio of virial to scale radius. Typical values are $c \sim 5$ for massive clusters ($M_{200} \sim 10^{15}\,M_\odot$) and $c \sim 10$ for galaxy-mass haloes ($M_{200} \sim 10^{12}\,M_\odot$) [33, Part 2, p. 128].

The characteristic density is related to c by:

$$\rho_s = \frac{200}{3}\,\rho_{\rm crit}\,\frac{c^3}{\ln(1+c) - c/(1+c)}\,. \tag{11.36}$$

11.6.3 Surface Mass Density of the NFW

The line-of-sight projection of the NFW density gives the surface mass density

$$\Sigma(\xi) \;=\; \int_{-\infty}^{\infty} \rho\Big(\sqrt{\xi^2+z^2}\Big)\, dz\,, \tag{11.37}$$

where ξ is the physical impact parameter in the lens plane and z is the line-of-sight coordinate. We now evaluate this integral explicitly; the computation separates cleanly into three steps and is the first place where the distinctive NFW branch structure (arccosh vs. arctan) appears. The full derivation is verified symbolically in `nfw_projection.wl`.

Step 1: dimensionless variables. Let $x = \xi/r_s$ and $u = z/r_s$, so that $r/r_s = \sqrt{x^2+u^2}$ and $dz = r_s\, du$. The NFW density (eq. 11.35) becomes

$$\rho = \frac{\rho_s}{\sqrt{x^2+u^2}\,\big(1+\sqrt{x^2+u^2}\big)^2}\,.$$

Using the symmetry $z \leftrightarrow -z$,

$$\Sigma(\xi) \;=\; 2\,\rho_s\, r_s\, I(x)\,, \qquad I(x) = \int_0^{\infty} \frac{du}{\sqrt{x^2+u^2}\,\big(1+\sqrt{x^2+u^2}\big)^2}\,. \tag{11.38}$$

Step 2: change of variable $w = \sqrt{x^2+u^2}$. Then $u\,du = w\,dw$ and $du = w\,dw/\sqrt{w^2-x^2}$. The limits $u\!:0\to\infty$ map to $w\!:x\to\infty$, and

$$I(x) \;=\; \int_x^{\infty} \frac{dw}{(1+w)^2\,\sqrt{w^2-x^2}}\,. \tag{11.39}$$

Step 3: reduce to a simpler integral. A short calculation gives the identity

$$\frac{d}{dw}\left[\frac{\sqrt{w^2-x^2}}{1+w}\right] \;=\; \frac{w+x^2}{(1+w)^2\sqrt{w^2-x^2}}\,. \tag{11.40}$$

Writing $w + x^2 = (1+w) + (x^2-1)$ splits the right side into a piece involving one power of $(1+w)$ in the denominator and the target integrand $\big[(1+w)^2\sqrt{w^2-x^2}\big]^{-1}$ multiplied by (x^2-1). Integrating eq. (11.40) from $w = x$ to $w = \infty$ and using $\sqrt{w^2-x^2}/(1+w) \to 1$ at infinity and 0 at $w = x$:

$$1 \;=\; K(x) \;+\; (x^2-1)\, I(x)\,, \qquad K(x) \equiv \int_x^{\infty} \frac{dw}{(1+w)\sqrt{w^2-x^2}}\,, \tag{11.41}$$

so

$$I(x) \;=\; \frac{1-K(x)}{x^2-1}\,. \tag{11.42}$$

Step 4: evaluate $K(x)$. The substitution $w = x\cosh\tau$ (with $\sqrt{w^2-x^2} = x\sinh\tau$, $dw = x\sinh\tau\,d\tau$) turns K into

$$K(x) \;=\; \int_0^\infty \frac{d\tau}{1+x\cosh\tau}\,, \tag{11.43}$$

which is elementary. Using the Weierstrass half-angle substitution $t = \tanh(\tau/2)$ and the standard result $\int_0^1 2\,dt/(a-bt^2) = (2/\sqrt{ab})\,\operatorname{arctanh}\sqrt{b/a}$ (and its arctan analog when the signs flip), one finds

$$\boxed{K(x) = \begin{cases} \dfrac{\operatorname{arccosh}(1/x)}{\sqrt{1-x^2}}, & 0<x<1, \\ 1, & x=1, \\ \dfrac{\arctan\sqrt{x^2-1}}{\sqrt{x^2-1}}, & x>1. \end{cases}} \tag{11.44}$$

The two branches are related by the identity $\operatorname{arccosh}(1/x) \leftrightarrow \arccos(1/x)$ (analytic continuation through $x=1$), and the $x=1$ value follows from a Taylor expansion of either branch.

Collected result. Substituting eq. (11.44) into eq. (11.42), and then eq. (11.38),

$$\boxed{\Sigma(x) = 2\,\rho_s\,r_s\,f(x)\,, \qquad f(x) = \frac{1-K(x)}{x^2-1}\,,} \tag{11.45}$$

where $x = \xi/r_s = \theta/\theta_s$ and $\theta_s = r_s/D_\mathrm{d}$. Writing out the three branches explicitly,

$$f(x) = \begin{cases} \dfrac{1}{x^2-1}\left(1 - \dfrac{\operatorname{arccosh}(1/x)}{\sqrt{1-x^2}}\right) & 0<x<1, \\ \dfrac{1}{3} & x=1, \\ \dfrac{1}{x^2-1}\left(1 - \dfrac{\arctan\sqrt{x^2-1}}{\sqrt{x^2-1}}\right) & x>1. \end{cases} \tag{11.46}$$

This recovers the standard result of Bartelmann (1996) and Wright & Brainerd (2000).

Remark 11.6.1. The apparent singularity at $x=1$ in eq. (11.46) cancels between numerator and denominator. A Taylor expansion of $1-K(x)$ about $x=1$ (for either branch) gives $1-K(x) = \frac{1}{3}(x^2-1) + \mathcal{O}((x^2-1)^2)$, leaving $f(1) = 1/3$ — a smooth function of x.

11.6.4 Convergence of the NFW

Dividing eq. (11.45) by the critical surface density,

$$\boxed{\kappa(x) \;=\; \frac{\Sigma(x)}{\Sigma_\mathrm{cr}} \;=\; \frac{2\,\rho_s\,r_s}{\Sigma_\mathrm{cr}}\,f(x) \;\equiv\; \kappa_s\,f(x)\,, \qquad \kappa_s \equiv \frac{2\rho_s r_s}{\Sigma_\mathrm{cr}}.} \tag{11.47}$$

The characteristic convergence κ_s collects every dimensionful quantity from the density profile and the distance configuration, leaving $f(x)$ purely dimensionless.

Remark 11.6.2. Unlike the SIS (where $\kappa \propto \theta^{-1}$), the NFW convergence has a *logarithmic* divergence as $x \to 0$. This follows from the small-x expansion of eq. (11.44), $\operatorname{arccosh}(1/x) = \ln(2/x) + \mathcal{O}(x^2)$, which gives $K(x) \to \ln(2/x)$ and hence $f(x) \sim \ln(2/x)$ as $x \to 0$. The NFW cusp ($\rho \propto r^{-1}$) is shallower than the isothermal cusp ($\rho \propto r^{-2}$), which has important consequences for central image formation.

11.6.5 Deflection Angle and Shear of the NFW

For any axisymmetric profile, the reduced deflection angle satisfies eqs. (11.5) and (11.6),

$$\alpha(\theta) \;=\; \theta\,\bar{\kappa}(\theta)\,, \qquad \bar{\kappa}(\theta) = \frac{2}{\theta^2}\int_0^{\theta} \kappa(\theta')\,\theta'\,d\theta'. \tag{11.48}$$

Switching to $x = \theta/\theta_s$ and using $\kappa = \kappa_s f(x)$ (eq. 11.47) turns the integral into

$$\bar{\kappa}(x) = \frac{2\kappa_s}{x^2}\,h(x)\,, \qquad h(x) \equiv \int_0^{x} f(x')\,x'\,dx'. \tag{11.49}$$

Evaluation of $h(x)$. From eq. (11.42), $f(x) = [1 - K(x)]/(x^2-1)$, where $K(x)$ is given by eq. (11.44). A direct calculation shows

$$\frac{d}{dx}\left[\ln\tfrac{x}{2} + K(x)\right] \;=\; x\,f(x) \quad \text{for } 0 < x \neq 1, \tag{11.50}$$

and, using $\operatorname{arccosh}(1/x) = \ln(2/x) + \mathcal{O}(x^2)$ as $x \to 0$,

$$\lim_{x\to 0^+}\left[\ln(x/2) + K(x)\right] = 0\,.$$

Hence

$$h(x) \;=\; \ln\frac{x}{2} + g(x)\,, \qquad g(x) \equiv K(x), \tag{11.51}$$

with

$$g(x) = \begin{cases} \dfrac{\operatorname{arccosh}(1/x)}{\sqrt{1-x^2}}, & 0 < x < 1, \\[2mm] 1, & x = 1, \\[2mm] \dfrac{\arctan\sqrt{x^2-1}}{\sqrt{x^2-1}}, & x > 1. \end{cases} \tag{11.52}$$

Deflection angle and shear. Substituting eq. (11.51) into eqs. (11.48)–(11.49) and using $\theta = x\,\theta_s$:

$$\boxed{\alpha(x) = \frac{2\kappa_s\,\theta_s}{x}\left[\ln\frac{x}{2} + g(x)\right], \qquad \bar{\kappa}(x) = \frac{2\kappa_s}{x^2}\left[\ln\frac{x}{2} + g(x)\right].} \tag{11.53}$$

The tangential shear of any axisymmetric profile is $|\gamma(x)| = \bar{\kappa}(x) - \kappa(x)$ (cf. eq. 11.7):

$$|\gamma(x)| \;=\; \frac{2\kappa_s}{x^2}\left[\ln\frac{x}{2} + g(x)\right] \;-\; \kappa_s\,f(x)\,. \tag{11.54}$$

Every identity in this subsection — the antiderivative eq. (11.50), the $x \to 0$ limit, and the numerical consistency of eq. (11.53) with direct integration of κ — is verified symbolically and numerically in `nfw_projection.wl`.

11.6.6 Role of Concentration in NFW Lensing

The concentration parameter c has a strong effect on the lensing strength of an NFW halo:

- Higher c means more centrally concentrated mass, increasing the central convergence $\kappa(0)$ and making strong lensing more likely.
- For a fixed virial mass, increasing c increases κ_s and decreases θ_s, concentrating the lensing signal in a smaller angular region.
- Typical galaxy clusters ($c \sim 4$–5) are strong lenses for background sources at $z_s \gtrsim 1$, producing giant arcs with radii of $\sim 10''$–$30''$.

The NFW model will be applied extensively to cluster lensing in Module 10.

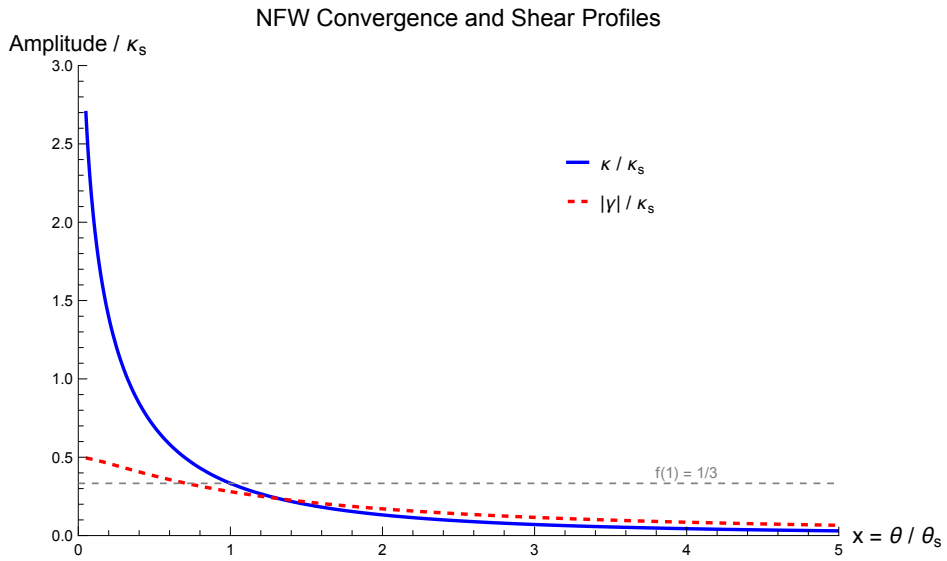


Figure 11.2: NFW convergence $\kappa(x)/\kappa_s = f(x)$ (solid) and shear $|\gamma(x)|/\kappa_s = \bar{\kappa}(x)/\kappa_s - f(x)$ (dashed) as functions of the dimensionless radius $x = \theta/\theta_s$. Both curves are normalized to κ_s and are universal functions of x (independent of concentration); the horizontal grey line marks $f(1) = 1/3$. The concentration c enters only through the overall scale κ_s and the angular scale θ_s, both of which absorb the dependence on c and on the cosmological distances. Generated by `axisymmetric_models.wl`.

11.7 Comparison of Models

Table 11.1 summarizes the key properties of the four axisymmetric lens models covered in this module.

The key qualitative differences between the models are:

1. **Central density:** The point mass and SIS have singular central densities; the NIS and NFW have finite (or logarithmically divergent) central densities. The central density determines whether a radial critical curve can form.

Table 11.1: Comparison of axisymmetric lens models. Here θ is the angular radius from the lens center in units of the appropriate scale ($\theta_{\rm E}$ for point mass and SIS, θ_s for NFW).

Property	Point Mass	SIS	NIS	NFW
$\rho(r)$	$\delta^{(3)}$	$\propto r^{-2}$	$\propto (r^2+r_c^2)^{-1}$	eq. (11.35)
$\kappa(\theta)$	$0\ (\theta \neq 0)$	$\theta_{\rm E}/2\theta$	$\theta_{\rm E}/2\sqrt{\theta^2+\theta_c^2}$	$\kappa_s\, f(x)$
$\alpha(\theta)$	$\theta_{\rm E}^2/\theta$	$\theta_{\rm E}$	eq. (11.32)	eq. (11.53)
$\kappa = \lvert\gamma\rvert$?	No ($\kappa = 0$)	Yes	No	No
Tangential CC	$\theta = \theta_{\rm E}$	$\theta = \theta_{\rm E}$	Yes	Yes
Radial CC	No	No	Yes ($\theta_c < \theta_{\rm E}/2$)	Yes
Max images	2	2	3	3
Caustic	Point	Point	Point + circle	Point + circle

2. **Deflection angle behavior:** The point mass deflection $\alpha \propto 1/\theta$ diverges at the center and falls off with distance. The SIS deflection is constant. The NIS and NFW deflections are intermediate.

3. **Image multiplicity:** Models with a radial critical curve (NIS, NFW) can produce a third (central) image. The SIS and point mass produce at most two images.

4. **Asymptotic behavior:** The SIS has $\Sigma \propto \theta^{-1}$ and infinite total mass. The NFW has $\Sigma \propto \theta^{-2}$ (for $\theta \gg \theta_s$) and a well-defined virial mass.

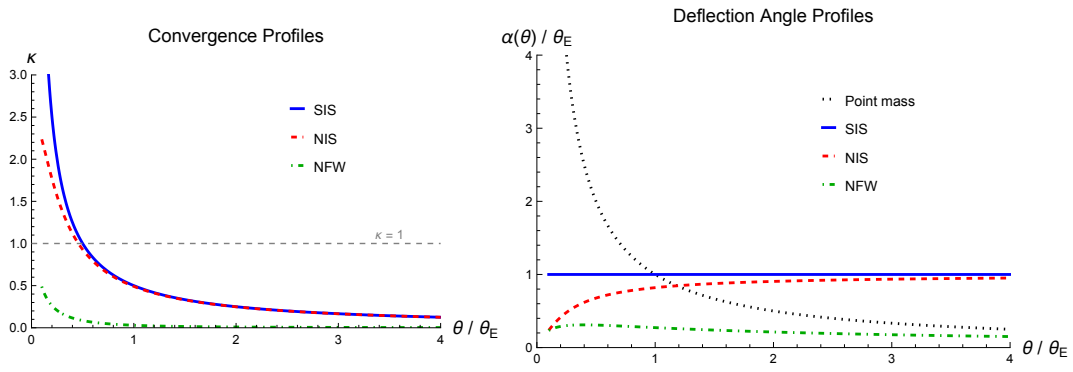


Figure 11.3: Comparison of convergence profiles (left) and deflection angles (right) for the four axisymmetric models. All curves use the SIS Einstein radius $\theta_{\rm E}$ as the unit of length on the horizontal axis; the SIS, NIS and NFW amplitudes are set so that the profiles can be compared on the same panel (specifically NIS with $\theta_c = 0.2\,\theta_{\rm E}$ and NFW with $\theta_s = 0.3\,\theta_{\rm E}$, $\kappa_s = 0.5$). Only the SIS and NIS are normalized to reproduce $\bar{\kappa}(\theta_{\rm E}) = 1$; the NFW deflection therefore does not pass through $\alpha = \theta_{\rm E}$ at $\theta = \theta_{\rm E}$ on this plot. The point mass convergence is a delta function and is not shown. Generated by `axisymmetric_models.wl`.

11.8 Summary

- For a **circularly symmetric** lens, all quantities depend only on $\theta = |\boldsymbol{\theta}|$. The deflection angle is $\alpha(\theta) = \bar{\kappa}(\theta)\,\theta$, and the shear is $|\gamma| = \bar{\kappa} - \kappa$.
- The **point mass** lens has $\alpha = \theta_{\mathrm{E}}^2/\theta$, $\kappa = 0$ (away from origin), and always produces two images.
- The **SIS** has $\rho \propto r^{-2}$, a constant physical deflection angle $\hat{\alpha} = 4\pi\sigma_v^2/c^2$, Einstein radius $\theta_{\mathrm{E}} = 4\pi(\sigma_v/c)^2\, D_{\mathrm{ds}}/D_{\mathrm{s}}$, and $\kappa = |\gamma| = \theta_{\mathrm{E}}/(2\theta)$ everywhere.
- The SIS produces two images for $\beta < \theta_{\mathrm{E}}$ and one image for $\beta > \theta_{\mathrm{E}}$, with magnifications $\mu_\pm = 1 \pm \theta_{\mathrm{E}}/\beta$. It has a tangential critical curve (Einstein ring) but no radial critical curve.
- The **NIS** introduces a core radius θ_c that removes the central singularity. For $\theta_c < \theta_{\mathrm{E}}/2$, a radial critical curve appears, enabling a third (central) image. The NIS reduces to the SIS for $\theta_c \to 0$.
- The **NFW** profile $\rho = \rho_s/[(r/r_s)(1 + r/r_s)^2]$ is the universal dark matter halo profile. It is parametrized by the virial mass M_{200} and concentration c. Higher concentration increases the lensing strength.
- Models with finite central density (NIS, NFW) possess a radial critical curve and can produce three images. The SIS and point mass cannot.

11.9 Problems

Exercise 11.9.1. (Derivation of the SIS deflection angle.) Starting from the SIS density profile $\rho(r) = \sigma_v^2/(2\pi G r^2)$:

(a) Compute the surface mass density $\Sigma(\xi) = \int_{-\infty}^{+\infty} \rho(\sqrt{\xi^2 + z^2})\, dz$ and show that $\Sigma(\xi) = \sigma_v^2/(2G\,\xi)$. *Hint:* Use the substitution $z = \xi \tan\phi$.

(b) Compute the projected enclosed mass $M(\xi) = 2\pi \int_0^\xi \Sigma(\xi')\,\xi'\, d\xi'$.

(c) Using $\hat{\alpha}(\xi) = 4GM(\xi)/(c^2\,\xi)$, show that $\hat{\alpha} = 4\pi\sigma_v^2/c^2$, independent of ξ.

(d) Derive the SIS Einstein radius θ_{E}.

Verify with `problems_07.wl`.

Exercise 11.9.2. (SIS: $\kappa = |\gamma|$ everywhere.)

(a) For the SIS with $\kappa(\theta) = \theta_{\mathrm{E}}/(2\theta)$, compute the mean convergence $\bar{\kappa}(\theta)$ using eq. (11.5).

(b) Show that $|\gamma(\theta)| = \bar{\kappa}(\theta) - \kappa(\theta) = \theta_{\mathrm{E}}/(2\theta) = \kappa(\theta)$.

(c) Explain physically why $\kappa = |\gamma|$ for the SIS but $\kappa = 0 \neq |\gamma|$ for the point mass. *Hint:* Consider the mass distribution — where is the mass located for each model?

(d) Show that for *any* power-law profile $\kappa(\theta) \propto \theta^{-n}$ with $n > 0$, the ratio $|\gamma|/\kappa$ is constant, and find the value of n for which $|\gamma| = \kappa$.

Verify with `problems_07.wl`.

Exercise 11.9.3. (SIS numerical example.) Consider an SIS lens with $\sigma_v = 250\ \mathrm{km\,s^{-1}}$ at $z_d = 0.3$ lensing a source at $z_s = 1$. Use $D_\mathrm{d} \approx 920$ Mpc, $D_\mathrm{s} \approx 1650$ Mpc, $D_\mathrm{ds} \approx 1055$ Mpc.

(a) Compute the Einstein radius θ_E in arcseconds.

(b) For a source at $\beta = 0.5''$:

(i) Is the source inside or outside the caustic?

(ii) Find the image positions θ_+ and θ_-.

(iii) Compute the magnifications μ_+ and μ_-.

(iv) Compute the total magnification.

(c) For what source position β does the SIS produce images with equal absolute magnification? Is this physically realizable?

(d) Compare the SIS Einstein radius with the point mass Einstein radius for $M = 10^{12}\,M_\odot$ at the same redshifts. Comment on the difference.

Verify with `problems_07.wl`.

Exercise 11.9.4. (NIS convergence and SIS limit.)

(a) Starting from the NIS surface mass density $\Sigma(\theta) = \sigma_v^2/[2G\,D_\mathrm{d}\,\sqrt{\theta^2 + \theta_c^2}]$, derive the NIS convergence $\kappa(\theta) = \theta_\mathrm{E}/[2\sqrt{\theta^2 + \theta_c^2}]$.

(b) Show that the NIS convergence reduces to the SIS convergence $\kappa = \theta_\mathrm{E}/(2\theta)$ in the limit $\theta_c \to 0$.

(c) Find the critical value of θ_c (in terms of θ_E) above which the NIS cannot produce multiple images. *Hint:* multiple images require $\kappa(0) \geq 1$.

(d) For the NIS with $\theta_c = 0.1\,\theta_\mathrm{E}$, compute $\kappa(0)$, $\kappa(\theta_\mathrm{E})$, and $\kappa(2\,\theta_\mathrm{E})$.

(e) Sketch (or plot) the NIS convergence profile for $\theta_c = 0$ (SIS), $\theta_c = 0.1\,\theta_\mathrm{E}$, $\theta_c = 0.3\,\theta_\mathrm{E}$, and $\theta_c = 0.5\,\theta_\mathrm{E}$ on the same axes.

Verify with `problems_07.wl`.

Exercise 11.9.5. (NFW concentration and lensing strength.)

(a) Using the NFW convergence formula $\kappa(x) = \kappa_s\,f(x)$ (eq. 11.47), plot $\kappa(x)/\kappa_s$ as a function of $x = \theta/\theta_s$ for $0.01 \leq x \leq 10$.

(b) For a cluster at $z_d = 0.3$ with $M_{200} = 10^{15}\,M_\odot$ and source at $z_s = 1$, compute κ_s and θ_s for $c = 5$, $c = 10$, and $c = 20$.

(c) Plot the convergence profiles $\kappa(\theta)$ (in arcsec) for the three concentrations on the same axes. At what angular radius does $\kappa = 1$ for each case?

(d) Discuss how the concentration parameter affects: (i) the Einstein radius, (ii) the size of the region where $\kappa > 1$, and (iii) the lensing cross-section for producing giant arcs.

Use Mathematica to generate the plots. See `axisymmetric_models.wl` and `problems_07.wl`.

Chapter 12

Non-Axisymmetric Models and Critical Curves

> *A circularly symmetric lens model is much too idealized and is unlikely to describe real galaxies. Therefore, considerable work has gone into developing non-circularly symmetric models. The breaking of the symmetry leads to qualitatively new image configurations [. . .]* — Narayan & Bartelmann (1997), Sect. 3.5

In Modules 9 and 11 we developed the Jacobian formalism for magnification and applied it to axisymmetric lens models: the point mass and the singular isothermal sphere (SIS). These models have a continuous rotational symmetry that makes the mathematics tractable but the phenomenology limited. The SIS, for instance, can produce at most two images of a point source, and its critical curve is a perfect circle.

Real galaxies, however, are *elliptical*. Their mass distributions have axis ratios $q \sim 0.5$–0.9, and they live in environments that contribute external tidal shear. Breaking circular symmetry introduces qualitatively new phenomena:

- Critical curves deform from circles into ellipses and ovals.
- Caustics develop a **diamond (astroid)** shape with cusps and folds.
- Sources can be **quadruply imaged** (not just doubly), producing Einstein crosses and arcs.
- The **topology** of image configurations — how many images exist, their parities, and their magnifications — is entirely governed by the source's position relative to the caustic structure.

This module introduces the two workhorse non-axisymmetric models (SIS + external shear and the singular isothermal ellipsoid) and then develops the theory of critical curves, caustics, and image topology in full detail. The image topology section is the conceptual heart of strong lensing theory.

Companion Mathematica files:

- `critical_curves_caustics.wl` — symbolic and numerical computation of critical curves and caustics for SIS+shear and SIE models; image position solver; figure generation for all module figures
- `image_configurations.nb` — interactive notebook: place a source on the source plane and watch images appear, colored by parity; animate caustic crossings; sliders for

ellipticity and shear

12.1 Introduction: Why Non-Axisymmetric Models Matter

The singular isothermal sphere (Module 11) is an excellent starting point for galaxy-scale lensing. Its velocity dispersion σ_v maps directly onto observable galaxy kinematics, and its Einstein radius $\theta_{\rm E} = 4\pi\sigma_v^2 D_{\rm ds}/(c^2 D_{\rm s})$ sets the angular scale of lensing. But the SIS has a fundamental limitation: perfect circular symmetry.

Observations of strong lens systems reveal:

1. **Quadruple images** are common. About 20–30% of known galaxy-scale lens systems are quads (four bright images plus a faint central image), which an SIS cannot produce.

2. **Giant luminous arcs** in galaxy clusters are tangentially stretched images that trace out the critical curves of the cluster potential. These arcs are not perfect circles, implying the mass distribution is not axisymmetric.

3. **Astrometric anomalies** in lens models require ellipticity and/or external shear to achieve acceptable fits.

The simplest way to break axial symmetry is to add a constant **external shear** to the SIS potential. The resulting SIS + shear model already captures the essential features: diamond caustics, quad images, and fold/cusp configurations. The more physically motivated **singular isothermal ellipsoid** (SIE) achieves the same through an intrinsically elliptical mass distribution and is the model used in modern lens-modeling codes such as `lenstronomy` and `PyAutoLens`.

12.2 SIS + External Shear

The simplest non-axisymmetric lens model augments the SIS potential with a constant external tidal shear $\gamma_{\rm ext}$. Physically, this shear arises from the tidal gravitational field of neighboring galaxies or the cluster environment in which the lens galaxy resides.

12.2.1 Lensing Potential

The lensing potential is (Congdon & Keeton eqs. 6.1–6.2):

$$\boxed{\psi(\boldsymbol{\theta}) = \theta_{\rm E}\,|\boldsymbol{\theta}| + \frac{1}{2}\,\gamma_{\rm ext}\,(\theta_1^2 - \theta_2^2)} \tag{12.1}$$

where $|\boldsymbol{\theta}| = \sqrt{\theta_1^2 + \theta_2^2}$ and we have chosen the shear to be aligned with the θ_1-axis (the general case involves a rotation by the shear angle ϕ_γ). The first term is the SIS potential; the second is the external shear contribution.

12.2.2 Deflection Angle

The deflection angle components are:

$$\alpha_1 = \frac{\partial \psi}{\partial \theta_1} = \theta_{\rm E}\,\frac{\theta_1}{|\boldsymbol{\theta}|} + \gamma_{\rm ext}\,\theta_1\,, \qquad \alpha_2 = \frac{\partial \psi}{\partial \theta_2} = \theta_{\rm E}\,\frac{\theta_2}{|\boldsymbol{\theta}|} - \gamma_{\rm ext}\,\theta_2\,. \tag{12.2}$$

The SIS contribution deflects radially inward; the shear contribution stretches along θ_1 and compresses along θ_2 (for $\gamma_{\rm ext} > 0$).

12.2.3 Convergence and Shear

The convergence of the SIS + shear model is the same as the pure SIS (the external shear does not add mass):

$$\kappa(\boldsymbol{\theta}) = \frac{\theta_{\rm E}}{2\,|\boldsymbol{\theta}|}\,. \tag{12.3}$$

The total shear now has contributions from both the SIS and the external shear. For the SIS alone, $|\gamma_{\rm SIS}| = \theta_{\rm E}/(2|\boldsymbol{\theta}|) = \kappa$. The external shear adds $\gamma_{\rm ext}$ aligned with the coordinate axes, so the total shear field is anisotropic and position-dependent.

12.2.4 Critical Curves

The critical curves are defined by $\det \mathcal{A} = 0$. Start from the Hessian of the SIS+shear potential $\psi = \theta_{\rm E}|\boldsymbol{\theta}| + \frac{1}{2}\gamma_{\rm ext}(\theta_1^2 - \theta_2^2)$ and let $u = \theta_{\rm E}/|\boldsymbol{\theta}|$, $g = \gamma_{\rm ext}$. In polar coordinates $(\theta_1, \theta_2) = (|\boldsymbol{\theta}|\cos\phi, |\boldsymbol{\theta}|\sin\phi)$ the Jacobian entries are

$$\mathcal{A}_{11} = 1 - g - u\sin^2\phi, \quad \mathcal{A}_{22} = 1 + g - u\cos^2\phi, \quad \mathcal{A}_{12} = \mathcal{A}_{21} = u\sin\phi\cos\phi,$$

and a short expansion gives

$$\det \mathcal{A} \;=\; (1 - g^2) - u\,(1 - g\cos 2\phi).$$

Setting $\det \mathcal{A} = 0$ yields the critical curve:

$$\boxed{\theta_{\rm crit}(\phi) \;=\; \theta_{\rm E}\,\frac{1 - \gamma_{\rm ext}\cos 2\phi}{1 - \gamma_{\rm ext}^2}\,.} \tag{12.4}$$

The derivation is verified symbolically in `critical_curves_caustics.wl`. For $\gamma_{\rm ext} = 0$ this reduces to $\theta = \theta_{\rm E}$ (the SIS critical circle). For $\gamma_{\rm ext} > 0$, the critical curve is an **oval**: elongated along the θ_2-axis (perpendicular to the shear direction) and compressed along θ_1 (approximately elliptical for small $\gamma_{\rm ext}$; the deviation from a true ellipse reaches $\sim$1.5% at $\gamma_{\rm ext} = 0.1$).

Remark 12.2.1. The critical curve is elongated *perpendicular* to the shear direction. This is physically intuitive: the shear stretches images along θ_1, which means the magnification diverges at smaller θ in that direction, pushing the critical curve inward along θ_1 and outward along θ_2.

12.2.5 Caustics: The Diamond (Astroid)

The caustics are obtained by mapping the critical curves through the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta}_{\rm crit} - \boldsymbol{\alpha}(\boldsymbol{\theta}_{\rm crit})$. For the SIS + shear model, the tangential caustic takes the shape of a **diamond** or **astroid** — a four-cusped curve.

In parametric form, the caustic is:

$$\beta_1(\phi) = \theta_{\rm crit}(\phi)\cos\phi - \theta_{\rm E}\cos\phi - \gamma_{\rm ext}\,\theta_{\rm crit}(\phi)\cos\phi\,, \tag{12.5}$$

$$\beta_2(\phi) = \theta_{\rm crit}(\phi)\sin\phi - \theta_{\rm E}\sin\phi + \gamma_{\rm ext}\,\theta_{\rm crit}(\phi)\sin\phi\,. \tag{12.6}$$

This curve has four cusps (at $\phi = 0, \pi/2, \pi, 3\pi/2$) and four fold segments connecting them. The size of the diamond is proportional to $\gamma_{\rm ext}$: for $\gamma_{\rm ext} = 0$, the caustic degenerates to a point (the SIS point caustic), and for larger shear values, the diamond grows.

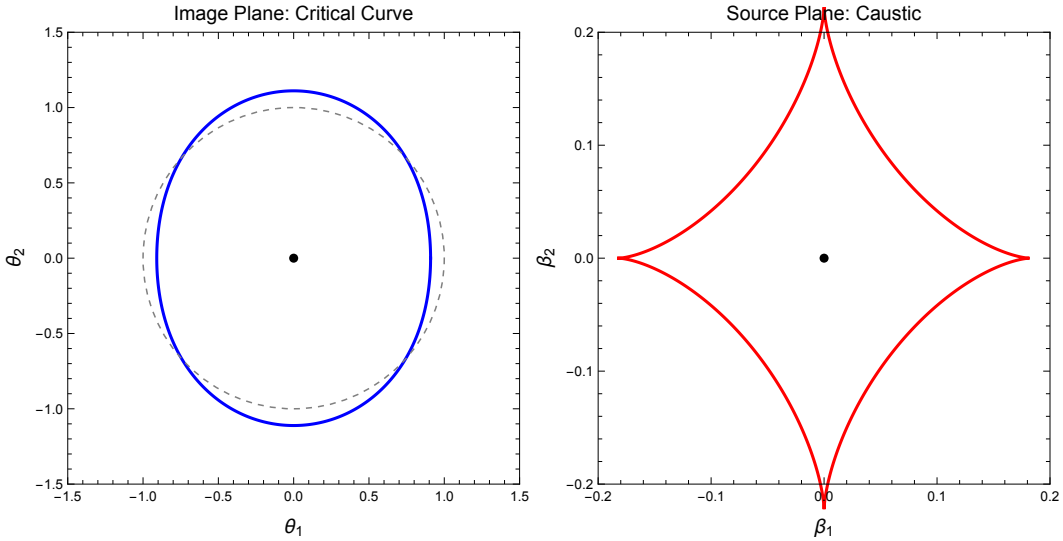


Figure 12.1: Critical curves (left, image plane) and caustics (right, source plane) for the SIS + external shear model with $\theta_{\rm E} = 1''$ and $\gamma_{\rm ext} = 0.1$. The critical curve is an oval (approximately elliptical for small shear); the caustic is a diamond (astroid) with four cusps. The shear axis is horizontal. Generated by `critical_curves_caustics.wl`.

12.2.6 Image Configurations

The image configurations for the SIS + shear model depend on the source position relative to the diamond caustic:

- **Source outside the diamond:** Two images (a double). One image is outside the critical curve (Type I, positive parity); the other is inside (Type II, negative parity, demagnified).
- **Source inside the diamond:** Four images (a quad). Two images outside the critical curve (positive parity); two inside (negative parity).

- **Source on a fold:** Three images, with two of them merging on the critical curve (highly magnified).
- **Source on a cusp:** Three images, with the merging triplet producing a characteristic cusp configuration.

Remark 12.2.2. The SIS is singular, so there is no separate radial critical curve or radial caustic. The SIS + shear model therefore has only a tangential caustic (the diamond). A non-singular model (e.g., a softened isothermal sphere + shear) would also have a radial caustic enclosing the tangential one.

12.3 The Singular Isothermal Ellipsoid (SIE)

The **singular isothermal ellipsoid** (SIE) is the workhorse model for galaxy-scale strong lensing. Rather than adding shear externally, the SIE builds ellipticity directly into the mass distribution.

12.3.1 Surface Mass Density

The SIE surface mass density is (Kormann, Schneider & Bartelmann 1994; Congdon & Keeton eq. 6.13):

$$\boxed{\Sigma(\xi_1, \xi_2) = \frac{\sigma_v^2}{2G} \frac{\sqrt{q}}{\sqrt{q^2\,\xi_1^2 + \xi_2^2}}} \tag{12.7}$$

where $q \leq 1$ is the axis ratio ($q = b/a$, with a the semi-major axis aligned along ξ_1), σ_v is the velocity dispersion, and (ξ_1, ξ_2) are physical coordinates in the lens plane. The iso-density contours $q^2\xi_1^2 + \xi_2^2 = \text{const}$ are ellipses of axis ratio q with the major axis along ξ_1. For $q = 1$ this reduces to the SIS surface mass density $\Sigma = \sigma_v^2/(2G\xi)$.

The factor $\sqrt{q}$ ensures that the total mass enclosed within a given elliptical radius is consistent with the isothermal profile.

Converting to angular coordinates via $\boldsymbol{\xi} = D_\mathrm{d}\boldsymbol{\theta}$ and dividing by the critical surface density gives the dimensionless convergence

$$\kappa(\theta_1, \theta_2) \;=\; \frac{\Sigma}{\Sigma_\mathrm{cr}} \;=\; \frac{\theta_\mathrm{E}\sqrt{q}}{2\sqrt{q^2\theta_1^2 + \theta_2^2}}\,, \tag{12.8}$$

where $\theta_\mathrm{E} = 4\pi\sigma_v^2 D_\mathrm{ds}/(c^2 D_\mathrm{s})$ is the SIS-equivalent Einstein radius (the Einstein radius of the circular isothermal sphere with the same σ_v).

12.3.2 Deflection Angle

The deflection angle for the SIE, first given in closed form by Kormann, Schneider & Bartelmann (1994, hereafter KSB94), is

$$\boxed{\alpha_1 = \frac{\theta_\mathrm{E}\,\sqrt{q}}{\sqrt{1-q^2}}\,\arctan\!\left(\frac{\theta_1\sqrt{1-q^2}}{\sqrt{q^2\theta_1^2 + \theta_2^2}}\right), \qquad \alpha_2 = \frac{\theta_\mathrm{E}\,\sqrt{q}}{\sqrt{1-q^2}}\,\operatorname{arctanh}\!\left(\frac{\theta_2\sqrt{1-q^2}}{\sqrt{q^2\theta_1^2 + \theta_2^2}}\right).} \tag{12.9}$$

The asymmetry — arctan along the major axis, arctanh along the minor — is intrinsic to the elliptical geometry. Rather than reproducing the long convolution integral that leads to these forms (a contour-integral computation spanning several pages of KSB94), we give a shorter derivation that shows eq. (12.9) is consistent with the SIE convergence (12.8) by using the scale-invariance of the lens. Every step is verified symbolically in `sie_deflection.wl`.

Step 1: scale invariance and the lensing potential. The SIE convergence (12.8) is homogeneous of degree -1 in $\boldsymbol{\theta}$,

$$\kappa(\lambda\boldsymbol{\theta}) = \lambda^{-1}\kappa(\boldsymbol{\theta}) \qquad (\lambda > 0).$$

Since $\alpha_i = \partial\psi/\partial\theta_i$ and $\boldsymbol{\alpha} = \kappa * \boldsymbol{\theta}/\pi$ is a convolution with a homogeneous-of-degree--1 kernel, the deflection is homogeneous of degree 0 and the potential ψ is homogeneous of degree 1. Euler's theorem for homogeneous functions then gives

$$\psi(\theta_1, \theta_2) \;=\; \theta_1\,\alpha_1(\theta_1, \theta_2) + \theta_2\,\alpha_2(\theta_1, \theta_2)\,. \tag{12.10}$$

Step 2: ansatz and its structure. For the SIE, any deflection consistent with the elliptical symmetry must involve only θ_1, θ_2, and the elliptical radius $F(\boldsymbol{\theta}) \equiv \sqrt{q^2\theta_1^2 + \theta_2^2}$. Dimensional and homogeneity requirements force the argument of the inverse trig/hyperbolic function to be dimensionless and of degree 0. Natural choices are $\theta_1\sqrt{1-q^2}/F$ and $\theta_2\sqrt{1-q^2}/F$ — the coefficients $\sqrt{1-q^2}$ are fixed uniquely (up to sign) by the requirement that $\alpha_i \to \theta_{\rm E}\theta_i/|\boldsymbol{\theta}|$ in the SIS limit $q \to 1$. This motivates the KSB94 ansatz (12.9).

Step 3: verify the gradient condition $\alpha_i = \partial_i\psi$. Combining (12.10) and (12.9) and differentiating gives

$$\frac{\partial\psi}{\partial\theta_i} \;=\; \alpha_i + \theta_j\,\frac{\partial\alpha_j}{\partial\theta_i}\,,$$

so the gradient condition is equivalent to $\theta_j(\partial\alpha_j/\partial\theta_i) = 0$. Direct differentiation of (12.9) shows that all cross terms cancel after using the identity $\partial_i(1/F) = -q^{2(1-\delta_{i2})}\theta_i/F^3$ and the derivative of arctan/arctanh. The detailed cancellation is carried out symbolically in `sie_deflection.wl`; the result is

$$\frac{\partial\psi}{\partial\theta_1} = \alpha_1, \qquad \frac{\partial\psi}{\partial\theta_2} = \alpha_2. \tag{12.11}$$

Step 4: verify $\nabla^2\psi = 2\kappa$. Computing $\nabla^2\psi = \partial_1\alpha_1 + \partial_2\alpha_2$ from (12.9):

$$\begin{aligned}
\partial_1\alpha_1 &= \frac{\theta_{\rm E}\sqrt{q}}{\sqrt{1-q^2}}\,\frac{\partial}{\partial\theta_1}\arctan\!\left(\frac{\theta_1\sqrt{1-q^2}}{F}\right) = \frac{\theta_{\rm E}\sqrt{q}}{\sqrt{1-q^2}}\cdot\frac{\sqrt{1-q^2}}{F}\cdot\frac{\theta_2^2}{q^2\theta_1^2+\theta_2^2} \\
&= \frac{\theta_{\rm E}\sqrt{q}\,\theta_2^2}{F^3}, \\
\partial_2\alpha_2 &= \frac{\theta_{\rm E}\sqrt{q}\,q^2\theta_1^2}{F^3},
\end{aligned}$$

after applying the chain rule to $\partial_i F = q^{2(1-\delta_{i2})}\theta_i/F$ and simplifying. Adding:

$$\nabla^2\psi \;=\; \frac{\theta_{\rm E}\sqrt{q}\,(q^2\theta_1^2+\theta_2^2)}{F^3} \;=\; \frac{\theta_{\rm E}\sqrt{q}}{F} \;=\; 2\,\kappa(\boldsymbol{\theta}), \tag{12.12}$$

matching (12.8). This completes the verification.

Step 5: the SIS limit $q \to 1$. As $q \to 1^-$, $\sqrt{1-q^2} \to 0$ and the arguments of the inverse functions vanish; using $\arctan x/x \to 1$ and $\operatorname{arctanh} x/x \to 1$,

$$\alpha_i \;\to\; \theta_{\rm E}\sqrt{q}\,\frac{\theta_i}{\sqrt{q^2\theta_1^2+\theta_2^2}} \;\to\; \theta_{\rm E}\,\frac{\theta_i}{|\boldsymbol{\theta}|},$$

which is the SIS deflection. The limit is taken symbolically in `sie_deflection.wl`.

12.3.3 Critical Curves and Caustics of the SIE

The critical curves of the SIE are found by solving $\det\mathcal{A}(\boldsymbol{\theta}) = 0$. Unlike the SIS (where the critical curve is a circle), the SIE critical curves are smooth oval curves whose shape depends on q.

For axis ratios $q \lesssim 0.9$, the SIE has:

- A **tangential critical curve**: a smooth oval close to the Einstein radius, elongated *along* the mass distribution's major axis. Concretely, with $q < 1$ and mass elongated along θ_1, the critical curve crosses the θ_1-axis at $\pm\theta_{\rm E}/\sqrt{q}$ and the θ_2-axis at $\pm\theta_{\rm E}\sqrt{q}$, so it stretches farther along θ_1 than along θ_2.
- A **tangential caustic**: a diamond (astroid) shape, similar to the SIS + shear case, with four cusps; for $q < 1$ the diamond is also wider along the mass major axis β_1.

The SIE, being singular at the origin, does not have a radial critical curve (just as the SIS does not). A non-singular isothermal ellipsoid would also have a radial critical curve mapping to a radial caustic.

The cross-section for quad lensing (the area enclosed by the tangential caustic) increases with ellipticity. For $q = 0.7$, the caustic area is roughly $0.066\,\theta_{\rm E}^2$, while for $q = 0.5$ it can be several times larger. This is why more elliptical galaxies are more likely to produce quad lens systems.

12.4 The Angular Structure of Lenses: Multipole Expansion

The SIE and the SIS + shear model are two *specific* ways of breaking circular symmetry. A more systematic view treats any lens potential as a circularly symmetric **monopole** plus a series of **angular multipoles**. This is the natural language for asking *which* deviations from circular symmetry a set of images can actually constrain — a question that, as Schneider et al. [33] emphasize (Part 2, §4.4–4.5, Kochanek), gravitational lensing answers better than any other probe of galaxy potentials.

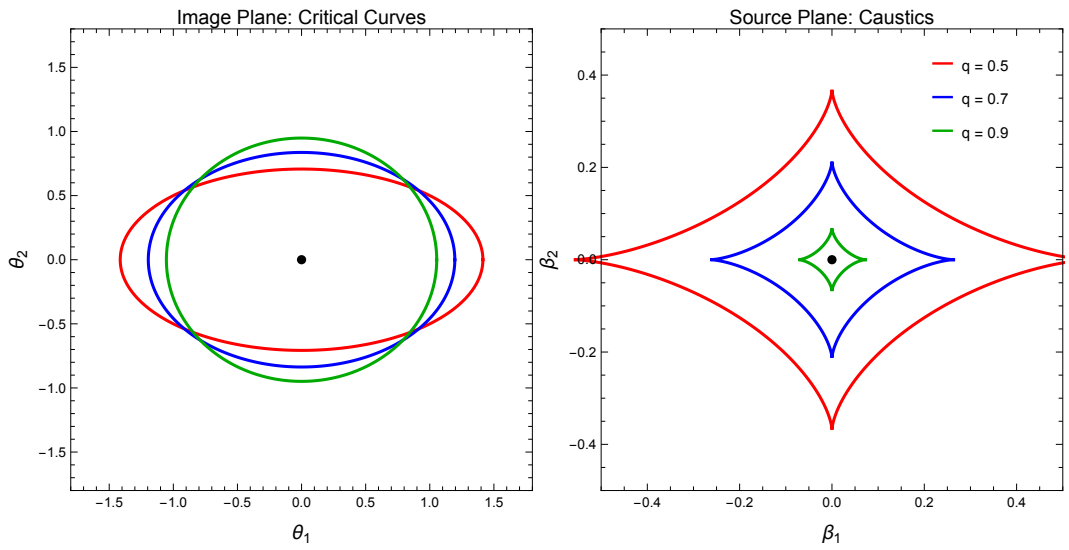


Figure 12.2: Critical curves (left) and caustics (right) for the SIE model with $\theta_{\rm E} = 1''$ and axis ratios $q = 0.5$, 0.7, and 0.9. As q decreases (more elliptical), the diamond caustic grows larger, increasing the cross-section for quad imaging. Generated by `critical_curves_caustics.wl`.

12.4.1 The Multipole Series

Expand the lensing potential in azimuthal harmonics of the polar angle φ (with $\boldsymbol{\theta} = \theta(\cos\varphi, \sin\varphi)$):

$$\psi(\theta, \varphi) = \psi_0(\theta) + \sum_{m=1}^{\infty} \psi_m(\theta)\, \cos\big[m(\varphi - \varphi_m)\big], \tag{12.13}$$

where ψ_0 is the monopole (the axisymmetric part of Module 11) and each $m \geq 1$ term has its own orientation φ_m. The convergence of a single multipole follows from the two-dimensional Poisson equation $\nabla^2\psi = 2\kappa$ (Module 9), with the Laplacian in polar coordinates $\nabla^2 = \partial_\theta^2 + \theta^{-1}\partial_\theta + \theta^{-2}\partial_\varphi^2$.

For the **isothermal-scaling** multipole $\psi_m = a_m\,\theta\,\cos[m(\varphi - \varphi_m)]$ (radial dependence $\propto \theta$, matching the flat rotation curve of the SIS), a direct computation gives

$$\boxed{\kappa_m(\theta, \varphi) = \frac{a_m\,(1 - m^2)}{2\,\theta}\, \cos\big[m(\varphi - \varphi_m)\big]} \tag{12.14}$$

(verified symbolically in the companion script; cf. the equivalent relation in Schneider et al. [33], Part 2, eq. 77). Two limits anchor this result:

- $m = 0$: $\kappa_0 = a_0/(2\theta)$ — the SIS profile $\kappa = \theta_{\rm E}/(2\theta)$ with $a_0 = \theta_{\rm E}$.
- $m = 2$: $\kappa_2 = -(3a_2/2\theta)\cos[2(\varphi - \varphi_2)]$ — the leading **quadrupole**, the elliptical distortion of an isothermal galaxy.

12.4.2 Internal vs. External Multipoles

Each angular harmonic m admits two radial "flavours" that are solutions of Laplace's equation away from the mass: an **internal** part sourced by mass *interior* to θ (decaying outward) and

an **external** part sourced by mass *exterior* to θ (growing outward). The quadrupole makes this concrete:

- The **external shear** of §12.2 is the massless external quadrupole, $\psi = \frac{1}{2}\gamma\,\theta^2\cos[2(\varphi - \varphi_\gamma)]$, with $\kappa = 0$ but nonzero γ: a pure tidal field from the environment.
- The isothermal quadrupole $\psi_2 \propto \theta$ of Eq. (12.14) is the internal quadrupole carried by the lens's own elliptical mass.

For a singular isothermal ellipsoid the internal and external quadrupole amplitudes are locked in the ratio $\psi_{2,\mathrm{int}}/\psi_{2,\mathrm{ext}} = (m-1)/(m+1) = 1/3$, so the internal fraction of the total quadrupole is $f_{\mathrm{int}} = 1/4$: *three-quarters of an SIE's quadrupole is generated by mass outside any given radius* [33, Part 2, eq. 74]. For lenses embedded in flat-rotation-curve dark haloes, most of the quadrupole therefore comes from the halo beyond the Einstein ring — which is precisely why an *external* shear term is an adequate stand-in for the environment in most models.

12.4.3 Which Multipoles Matter

The amplitude of successive poles falls steeply with ellipticity, $\psi_m \propto \epsilon^{m/2}$ (with $q = 1-\epsilon$), so for a typical lens with $\theta_{\mathrm{E}} \sim 1''$:

- the **quadrupole** ($m = 2$) produces angular deflections $\sim 0.46''\,\epsilon$ — important for essentially any ellipticity, given milliarcsecond astrometry;
- the $m = 4$ **pole** (boxy/discy isophotes) contributes $\sim 0.09''\,\epsilon^2$ — important only for $q \lesssim 0.75$;
- the **dipole** ($m = 1$, lopsidedness) is nearly degenerate with the source position and rarely matters.

The practical lesson — and a recurring theme of the applications in Module 13 — is that a monopole plus an external shear (a single quadrupole) captures most of the constrainable angular structure, with the $m = 4$ term becoming relevant only for markedly flattened lenses.

12.5 Critical Curves

We now develop the general theory of critical curves, building on the preview in Module 9 (Section 9.10).

Definition 12.5.1. The **critical curves** are the loci in the image plane where the Jacobian determinant vanishes:

$$\boxed{\det \mathcal{A}(\boldsymbol{\theta}) = 0 \qquad \Longleftrightarrow \qquad (1-\kappa)^2 - |\gamma|^2 = 0} \tag{12.15}$$

Equivalently, $\mu(\boldsymbol{\theta}) \to \infty$ on critical curves.

12.5.1 Two Families of Critical Curves

Using the eigenvalues $\lambda_\pm = 1 - \kappa \pm |\gamma|$ of the Jacobian (Module 9, eq. 9.17), the condition $\det \mathcal{A} = \lambda_+\,\lambda_- = 0$ is satisfied when either eigenvalue vanishes:

1. **Tangential critical curve** ($\lambda_- = 0$):

$$1 - \kappa(\boldsymbol{\theta}) = |\gamma(\boldsymbol{\theta})| \tag{12.16}$$

   Images near this curve are stretched *tangentially* (along the curve), because the eigenvalue corresponding to the tangential direction vanishes. Giant arcs in clusters trace the tangential critical curve.

2. **Radial critical curve** ($\lambda_+ = 0$):

$$1 - \kappa(\boldsymbol{\theta}) = -|\gamma(\boldsymbol{\theta})| \tag{12.17}$$

   Images near this curve are stretched *radially* (perpendicular to the curve). Radial arcs (pointing toward the lens center) form near radial critical curves, though they are rarer and fainter than tangential arcs.

12.5.2 Critical Curves for Axisymmetric vs. Elliptical Lenses

- **Axisymmetric lenses:** The critical curves are circles centered on the lens. For the SIS, there is only a tangential critical curve at $\theta = \theta_{\rm E}$. For the point mass, the critical curve is the Einstein ring.

- **Elliptical lenses:** The critical curves become ellipses or ovals. For the SIE (mass elongated along θ_1, $q < 1$) the tangential critical curve is elongated *along* the mass major axis, crossing θ_1 at $\pm\theta_{\rm E}/\sqrt{q}$ and θ_2 at $\pm\theta_{\rm E}\sqrt{q}$. The shapes are computed numerically by solving $\det \mathcal{A} = 0$ on a grid or parametrically.

12.5.3 Physical Meaning

Critical curves are the sites of maximum magnification. Images that form near a critical curve are:

1. **Highly magnified:** $|\mu| \gg 1$, because $\det \mathcal{A} \approx 0$.

2. **Highly stretched:** Along one direction (tangential or radial), the magnification diverges while it remains finite in the orthogonal direction. This produces the elongated arc morphology.

3. **On parity boundaries:** As we cross a critical curve, $\det \mathcal{A}$ changes sign, so the image parity flips. One side has positive parity (Type I); the other has negative parity (Type II).

The practical significance is enormous: critical curves are where we look for the most highly magnified background sources. Cluster lensing surveys map out critical curves to find magnified galaxies at very high redshift.

12.6 Caustics

Definition 12.6.1. The **caustics** are the images of the critical curves under the lens mapping. That is, the caustics are the curves in the *source plane* obtained by applying the lens equation to each point on a critical curve:

$$\boxed{\boldsymbol{\beta}_{\text{caustic}} = \boldsymbol{\theta}_{\text{crit}} - \boldsymbol{\alpha}(\boldsymbol{\theta}_{\text{crit}})} \tag{12.18}$$

12.6.1 Types of Caustics

Corresponding to the two families of critical curves, there are two types of caustics:

1. **Tangential caustic:** The mapping of the tangential critical curve. For elliptical lenses, this is typically a diamond (astroid) shape with four cusps. Sources inside the tangential caustic are quadruply imaged.

2. **Radial caustic:** The mapping of the radial critical curve. This is typically a smooth, roughly elliptical curve that encloses the tangential caustic. Sources inside the radial caustic but outside the tangential caustic are doubly imaged; sources outside both caustics have a single image.

12.6.2 Caustics for Specific Models

- **Point mass:** The tangential critical curve (Einstein ring $\theta = \theta_{\text{E}}$) maps to a single point ($\beta = 0$) — a **degenerate point caustic**. A source at $\beta = 0$ produces the Einstein ring; any source at $\beta > 0$ produces exactly two images.

- **SIS:** Same as the point mass — the critical curve $\theta = \theta_{\text{E}}$ maps to $\beta = 0$. The SIS has no radial caustic. Sources with $\beta < \theta_{\text{E}}$ produce two images; sources with $\beta > \theta_{\text{E}}$ produce one image.

- **SIS + shear:** The tangential caustic opens up into a diamond (astroid) with four cusps. The diamond size is proportional to γ_{ext}.

- **SIE:** Same diamond structure as SIS + shear, with the diamond size controlled by the ellipticity $(1 - q)$.

- **Non-singular models** (NIS, NFW): Both tangential and radial caustics are present. The radial caustic encloses the tangential one, creating regions with 1, 3, and 5 images.

12.7 Image Topology

This section is the conceptual heart of strong gravitational lensing. It explains *why* images appear and disappear as a source moves, why they always come in pairs, and how the full image configuration (positions, parities, magnifications) is determined entirely by the source's location relative to the caustic structure.

The key insight is that **caustics are the boundaries between regions of different image multiplicity**, and **critical curves are the boundaries between regions of different image parity**. Together, they provide a complete topological description of the lens mapping.

12.7.1 Critical Curves as Parity Boundaries

Recall from Module 9 that the magnification is $\mu = 1/\det \mathcal{A}$ and that the sign of μ determines the image parity:

- $\det \mathcal{A} > 0 \Rightarrow \mu > 0$: **positive parity** (Type I image). The image has the same orientation as the source.
- $\det \mathcal{A} < 0 \Rightarrow \mu < 0$: **negative parity** (Type II image). The image is mirror-reflected relative to the source.

Since $\det \mathcal{A}$ is a continuous function of position that passes through zero on critical curves, the sign of $\det \mathcal{A}$ must *change* across a critical curve. Therefore:

Proposition 12.7.1. ***Critical curves are parity boundaries.*** *Images on opposite sides of a critical curve have opposite parities. If a critical curve separates region R_+ ($\det \mathcal{A} > 0$) from region R_- ($\det \mathcal{A} < 0$), then:*

- *Images in R_+ have positive parity (Type I, minima of the arrival time surface).*
- *Images in R_- have negative parity (Type II, saddle points of the arrival time surface).*

For a typical galaxy lens with an elliptical mass distribution:

- The region **outside** the tangential critical curve has $\det \mathcal{A} > 0$: images here are Type I (positive parity, time-delay minima).
- The region **between** the tangential and radial critical curves has $\det \mathcal{A} < 0$: images here are Type II (negative parity, saddle points).
- The region **inside** the radial critical curve has $\det \mathcal{A} > 0$ again: images here are Type I or Type III (positive parity, but corresponding to time-delay maxima for Type III, as discussed in Module 10).

Remark 12.7.1. The connection to Morse theory (Module 10) is direct. The three image types correspond to the three types of stationary points of the Fermat potential: **Type I** = minimum, **Type II** = saddle, **Type III** = maximum. The magnification parity determines the Morse index.

12.7.2 Caustic Crossings and Image Creation/Destruction

The most important property of caustics is:

Proposition 12.7.2. ***Image creation/destruction at caustics.*** *When a source crosses a caustic, the number of images changes by exactly two. A pair of images either* ***appears*** *(as the source enters a higher-multiplicity region) or* ***disappears*** *(as it exits).*

This happens because, as a source approaches a caustic from the side with fewer images, two of the "potential" image positions on opposite sides of the corresponding critical curve converge toward the critical curve. At the moment the source is exactly on the caustic, the two images merge *on* the critical curve (where $\det \mathcal{A} = 0$ and the magnification diverges). As the source crosses the caustic, the two images separate and become distinct.

Crucially, the two images that appear or disappear:

1. Are located on **opposite sides** of the critical curve.
2. Have **opposite parities** (one Type I, one Type II).
3. Are both **highly magnified** near the caustic (their magnifications diverge as they approach the critical curve).
4. Have **nearly equal magnifications** (in absolute value) near the caustic.

Remark 12.7.2. The fact that images always appear or disappear in **pairs** with opposite parities is a topological necessity. It preserves **Burke's odd-number theorem** (Module 10): the total number of images is always odd (for a non-singular, transparent lens with a source inside the outermost caustic). Since images are created and destroyed in pairs, the parity of the count (odd or even) never changes.

12.7.3 Walking a Source Across Caustics: A Worked Example

To make the image creation/destruction process concrete, we walk through a complete example using an SIE lens with axis ratio $q = 0.7$ and Einstein radius $\theta_{\mathrm{E}} = 1''$. The SIE has a tangential caustic (diamond shape) in the source plane. For a non-singular model there would also be a radial caustic enclosing the diamond, but for the SIE (which is singular at the center) we focus on the tangential caustic.

Consider a source that starts far from the lens and moves inward, crossing the caustics:

Stage 1: Source far outside all caustics. The source position $\boldsymbol{\beta}$ is far from the lens center, well outside any caustic. For a singular model like the SIE there are **two images** (one Type I outside the critical curve, near the true source position, and one Type II inside the critical curve on the opposite side of the lens). In a non-singular model an additional third image would appear at the maximum of τ near the lens center, but for the SIE that central image is suppressed because Σ is singular at the origin. The two observable images are:

- **Type I** (positive parity, minimum of the arrival time), outside the tangential critical curve near the true source position, weakly magnified ($|\mu| \approx 1$).

- **Type II** (negative parity, saddle of the arrival time), inside the critical curve on the far side of the lens, also weakly magnified.

Stage 2: Source crosses the tangential caustic (outside → inside). As the source reaches the tangential (diamond) caustic, two new images appear simultaneously, straddling the tangential critical curve. The total count goes from **2 to 4 images**:

1. The **original Type I image** (positive parity, outside the tangential critical curve) persists, moving slightly.
2. A **new Type II image** (negative parity) appears just inside the tangential critical curve. This is a *saddle point* of the arrival time surface. It is highly magnified and tangentially stretched.
3. A **new Type I image** (positive parity) appears just outside the tangential critical curve, close to the Type II image. This is a *minimum* of the arrival time surface. It is also highly magnified.

The two new images are born as a highly magnified, closely spaced pair straddling the critical curve. They have opposite parities (one positive, one negative) and nearly equal $|\mu|$. As the source moves further inside the caustic, they separate and their magnifications decrease.

Stage 3: Source inside the tangential caustic (quad configuration). With the source well inside the diamond caustic, four bright images surround the lens (plus a faint, highly demagnified central image in non-singular models). The typical configuration is an **Einstein cross**:

- Two Type I images (positive parity, outside the tangential critical curve).
- Two Type II images (negative parity, inside the tangential critical curve).

The four images form a roughly cross-shaped pattern around the lens. The magnification depends on the source's proximity to the caustic folds and cusps.

Stage 4: Source crosses the tangential caustic again (inside → outside). As the source exits the diamond caustic, two images converge on the critical curve, merge, and disappear. The count drops from **4 back to 2**. The two images that disappear are a Type I/Type II pair, just as in the creation event.

For a non-singular lens model (e.g., a softened isothermal ellipsoid), there is also a **radial caustic** enclosing the tangential one. As the source crosses the radial caustic inward, two additional images appear near the radial critical curve (near the lens center), bringing the total from 3 to 5. This gives the complete sequence:

Region	Image count	Types
Outside radial caustic	1	I
Between radial & tangential	3	I, II, III
Inside tangential caustic	5	I, I, II, II, III

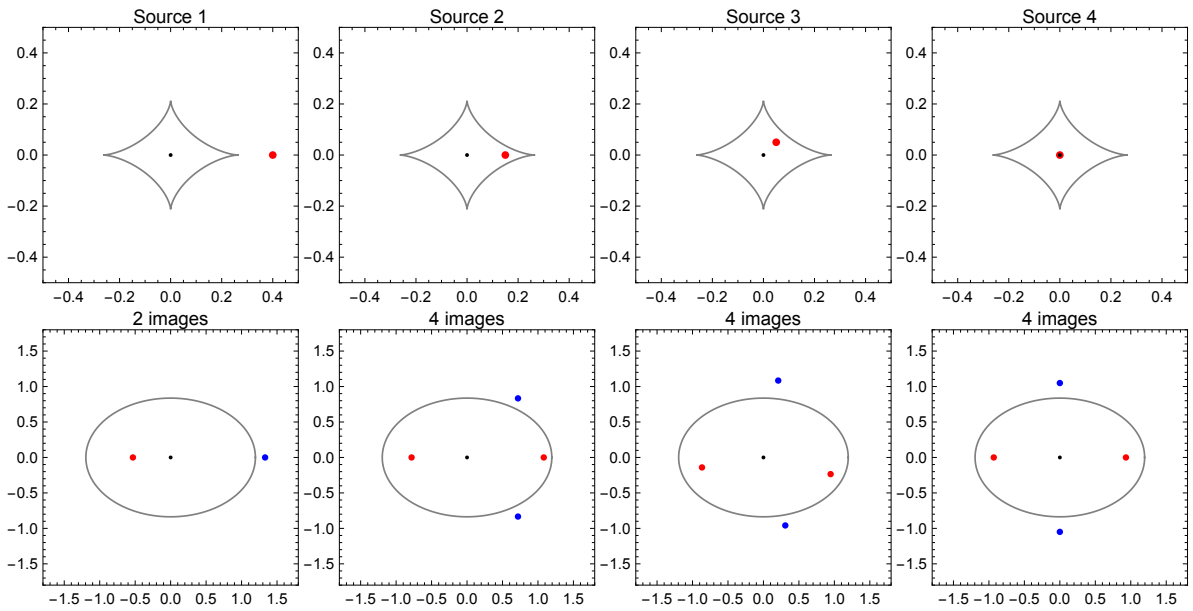


Figure 12.3: A source walks across the caustic of an SIE lens ($q = 0.7$). **Top row:** Source positions (red star) in the source plane, with the diamond caustic shown. **Bottom row:** Corresponding image configurations in the image plane, with the critical curve shown. Blue dots = Type I images (positive parity); red dots = Type II images (negative parity). As the source crosses the caustic, images appear/disappear in pairs. Generated by `critical_curves_caustics.wl`.

Here Type III is a maximum of the arrival time (the highly demagnified central image).

12.7.4 The Fold and Cusp Catastrophes

The caustic curves of a generic smooth lens consist of two types of points: **folds** (smooth segments) and **cusps** (pointed corners). These are the only two generic singularities of smooth two-dimensional mappings, a result from **catastrophe theory** (or singularity theory; see Petters, Levine & Wambsganss 2001 for a thorough treatment).

Fold Caustics

Near a **fold** (a smooth segment of the caustic), the following properties hold:

1. **Two images merge.** As a source crosses a fold caustic, two images appear or disappear. The images straddle the critical curve, one on each side.

2. **Magnification scaling.** Near a fold, the magnification of each of the two merging images scales as:

$$\boxed{|\mu| \propto \frac{1}{\sqrt{d_\perp}}} \tag{12.19}$$

where $d_\perp$ is the perpendicular distance of the source from the caustic. The magnification diverges as $d_\perp \to 0$ (source on the caustic), but only as an inverse square root — a relatively mild divergence.

3. **Equal and opposite magnifications.** The two merging images have magnifications that approach $+|\mu_0|$ and $-|\mu_0|$ (equal absolute values, opposite signs), so their *total signed magnification* approaches zero. The unsigned total $2|\mu_0| \propto 1/\sqrt{d_\perp}$ diverges.

4. **Tangential stretching.** The merging images are highly elongated tangentially (along the critical curve) and compressed radially. This is the origin of the "arc" morphology.

Cusp Caustics

Near a **cusp** (a pointed corner of the caustic), the behavior is richer:

1. **Three images merge.** As a source approaches a cusp from inside the caustic, three images converge toward a single point on the critical curve. These form the characteristic **cusp triplet**.

2. **Stronger magnification divergence.** Near a cusp, the sum of the (unsigned) image magnifications diverges more strongly than for a fold:

$$|\mu_A| + |\mu_B| + |\mu_C| \propto d_\perp^{-1} \tag{12.20}$$

   (compared to $d_\perp^{-1/2}$ for a fold).

3. **Cusp relation.** The *signed* magnifications of the three cusp images satisfy a (different) relation,

$$\mu_A + \mu_B + \mu_C \;\to\; 0 \quad (\text{in the limit } d_\perp \to 0)\,, \tag{12.21}$$

   where A, B, C are the three images of the cusp triplet (with signed magnifications). That is, the sum of the signed magnifications of the three brightest images in a cusp configuration should vanish. Violations of this relation indicate substructure (e.g., dark matter subhalos) perturbing the images.

4. **Configuration.** The three cusp images are closely spaced along the critical curve. One is on the opposite side of the critical curve from the other two, ensuring opposite parities.

Remark 12.7.3. The fold and cusp are the only **stable** (generic) singularities of smooth two-dimensional maps. Higher-order singularities (swallowtails, lips, etc.) require special symmetries or fine-tuning and are not generically present in gravitational lens models. This is a deep result of Whitney's theorem in singularity theory (Petters et al. 2001, Ch. 7, Sec. 7.5).

12.7.5 Image Configuration Taxonomy

The caustic structure of a lens determines all possible image configurations. For an elliptical galaxy lens (SIE or similar model), the following taxonomy applies:

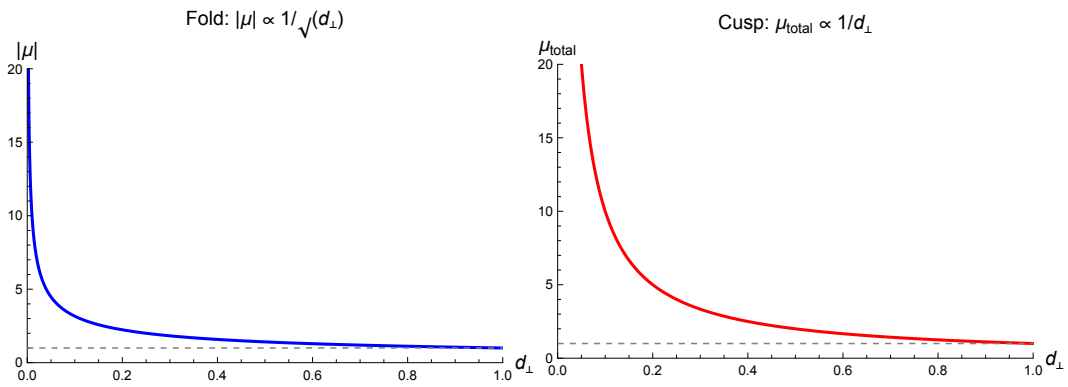


Figure 12.4: Magnification behavior near fold and cusp caustics. **Left:** Near a fold, two images straddle the critical curve with magnifications scaling as $|\mu| \propto 1/\sqrt{d_\perp}$. **Right:** Near a cusp, three images merge, with stronger divergence. The cusp relation $\mu_A + \mu_B + \mu_C \approx 0$ holds for the signed magnifications. Generated by `critical_curves_caustics.wl`.

Doubles

Source outside the tangential caustic. Two images form (or effectively one bright image plus one highly demagnified image near the lens center for singular models):

- One Type I image (positive parity, outside the critical curve): the dominant, brighter image.
- One Type II image (negative parity, inside the critical curve): typically fainter, on the opposite side of the lens from the source.

Doubles account for ∼70–80% of known galaxy-scale lens systems.

Quads / Einstein Crosses

Source inside the tangential caustic. Four bright images form (plus a faint central image):

- Two Type I images (positive parity), roughly on the side of the lens closer to the source.
- Two Type II images (negative parity), on the opposite side.

The four images form a roughly cross-shaped (**Einstein cross**) or diamond-shaped pattern around the lens. Quads account for ∼20–30% of galaxy-scale systems and are extremely valuable for cosmography because they provide three independent time delays.

Cusp Configurations

Source near a cusp of the tangential caustic. Three of the four images in a quad merge into a closely spaced triplet near one point on the critical curve. This produces the **naked cusp** or **cusp triplet** configuration:

- Three closely spaced images near the cusp of the critical curve, with the cusp relation $\mu_A + \mu_B + \mu_C \approx 0$.
- One isolated image on the opposite side of the lens.

Why Most Observed Quads Have a Diamond Caustic

The tangential caustic of an elliptical galaxy lens is generically a diamond (astroid) shape. This is because:

1. The dominant perturbation from circular symmetry is quadrupolar (ellipticity or shear), which produces exactly four cusps.
2. Higher-order multipoles (hexapole, octupole) are typically subdominant and do not change the topology of the caustic.
3. The diamond caustic produces Einstein crosses (quads with roughly equal magnifications) when the source is near the center, and fold/cusp configurations when the source is near a fold or cusp of the diamond.

12.7.6 Why Giant Arcs Form

One of the most spectacular phenomena in strong lensing is the formation of **giant luminous arcs** in galaxy clusters. The theory of critical curves and caustics provides a complete explanation.

Step 1: Sources near the tangential caustic. Giant arcs form when a background galaxy lies very close to (but just inside or outside) the tangential caustic. At this location, one or two of its images lie very close to the tangential critical curve.

Step 2: Tangential stretching near the critical curve. The tangential critical curve is where $\lambda_- = 0$ (the eigenvalue corresponding to the tangential direction vanishes). Near this curve, images are stretched enormously in the *tangential* direction (along the critical curve) while remaining relatively unstretched in the radial direction.

Quantitatively, if the source is at perpendicular distance $d_\perp$ from the tangential caustic, the tangential magnification scales as:

$$\mu_{\rm tan} \propto \frac{1}{\sqrt{d_\perp}} \tag{12.22}$$

while the radial magnification remains of order unity. The result is an image that is stretched by a factor of $\mu_{\rm tan}$ along the critical curve but not radially — an **arc**.

Step 3: Arc length and curvature. Since the arc follows the tangential critical curve, its curvature matches that of the critical curve. For a galaxy cluster, the tangential critical curve has a radius of ∼10–30 arcsec, so giant arcs appear as curved features with large opening angles. The length-to-width ratio of the arc is approximately $\mu_{\rm tan} \sim 1/\sqrt{d_\perp}$, which can easily reach 10:1 or more for sources very close to the caustic.

Step 4: Multiple arcs. A source just inside the tangential caustic produces two highly magnified images on opposite sides of the tangential critical curve (a fold pair). Both images are arcs. If the source is also near a cusp, three images merge into a very long arc (a "cusp arc").

Remark 12.7.4. The largest known arcs in galaxy clusters have length-to-width ratios exceeding 10:1 and total magnifications of $\mu \sim 20$–50. These extreme magnifications are used to study intrinsically faint galaxies at high redshift ("cosmic telescopes").

12.8 Image Configurations: Doubles, Quads, and Einstein Crosses

We now illustrate the different image configurations with explicit examples for an SIE lens.

12.8.1 Double Configuration

For a source at $\boldsymbol{\beta} = (0.5, 0.0)''$ lensed by an SIE with $\theta_{\mathrm{E}} = 1''$, $q = 0.7$: The source is outside the tangential caustic. Two images form:

- Image A at $\boldsymbol{\theta}_A \approx (1.43, 0.0)''$: Type I, positive parity, $\mu_A \approx 6.1$.
- Image B at $\boldsymbol{\theta}_B \approx (-0.43, 0.0)''$: Type II, negative parity, $\mu_B \approx -0.6$.

The brighter image (A) is on the same side as the source; the fainter image (B) is on the opposite side.

12.8.2 Quad / Einstein Cross Configuration

For a source at $\boldsymbol{\beta} = (0.05, 0.05)''$ (inside the tangential caustic): Four bright images form, arranged in a cross pattern. The two Type I images tend to be on the side closer to the source; the two Type II images are on the opposite side. The magnifications of the four images are comparable, with the images nearer the critical curve being brighter.

12.8.3 Cusp Configuration

For a source near a cusp of the tangential caustic (e.g., near $\boldsymbol{\beta} \approx (0.25, 0.0)''$ for an SIE with $q = 0.7$, where the cusp on the β_1-axis sits at $|\beta_1| \approx 0.263\,\theta_{\mathrm{E}}$): Three images cluster together near the corresponding point on the critical curve, forming a cusp triplet. The fourth image is isolated on the opposite side.

12.9 Summary

- **SIS + external shear** is the simplest non-axisymmetric model. The external shear γ_{ext} deforms the critical curve into an oval and opens the point caustic into a **diamond (astroid)** with four cusps.

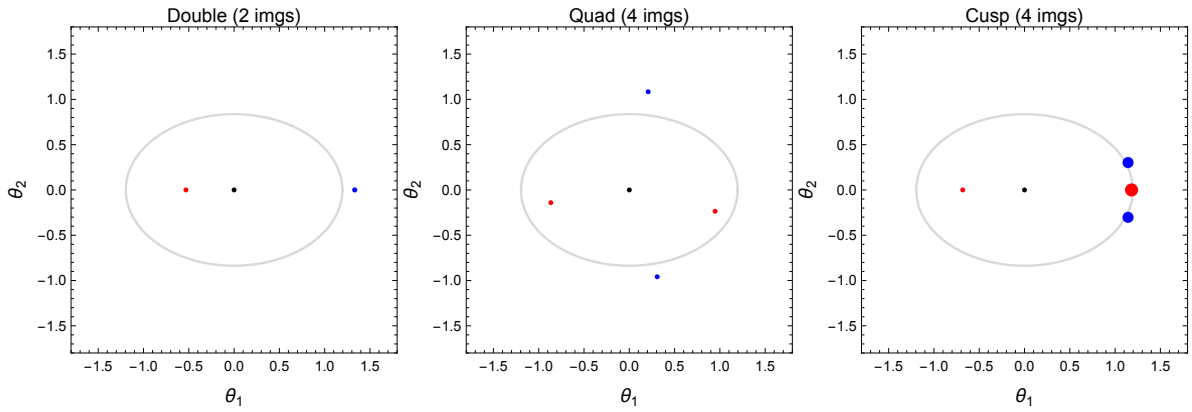


Figure 12.5: Image configurations for an SIE lens with $\theta_{\rm E} = 1''$ and $q = 0.7$. **Left:** Double (source outside caustic). **Center:** Quad / Einstein cross (source inside caustic). **Right:** Cusp configuration (source near a cusp). Blue = Type I (positive parity); red = Type II (negative parity). The critical curve and caustic are shown as gray curves. Generated by `critical_curves_caustics.wl`.

- The **singular isothermal ellipsoid** (SIE) builds ellipticity directly into the mass distribution. Its deflection angle involves arctan and arctanh functions (Kormann et al. 1994). It is the standard model for galaxy-scale lens modeling.

- **Critical curves** ($\det \mathcal{A} = 0$) are parity boundaries in the image plane. The tangential critical curve ($1-\kappa = |\gamma|$) is the site of giant arcs; the radial critical curve ($1-\kappa = -|\gamma|$) is the site of radial arcs.

- **Caustics** are the mappings of critical curves to the source plane. They divide the source plane into regions of different image multiplicity.

- **Image topology:** As a source crosses a caustic, two images appear or disappear on opposite sides of the critical curve, with opposite parities. This preserves Burke's odd-number theorem.

- **Fold caustics:** Two images merge; magnification $|\mu| \propto 1/\sqrt{d_\perp}$. This produces giant arcs.

- **Cusp caustics:** Three images merge; the cusp relation $\mu_A + \mu_B + \mu_C \approx 0$ holds.

- **Image configurations:** Doubles (source outside caustic), quads/Einstein crosses (source inside caustic), and cusp triplets (source near a cusp) account for the observed morphologies of strong lens systems.

- **Giant arcs** form when extended sources lie near the tangential caustic: images are tangentially stretched by factors of $\mu_{\rm tan} \propto 1/\sqrt{d_\perp}$ along the tangential critical curve.

12.10 Problems

Exercise 12.10.1. (SIS + shear: critical curves and caustics.) Consider an SIS + external shear model with $\theta_{\rm E} = 1''$ and $\gamma_{\rm ext} = 0.1$.

(a) Starting from the lensing potential (eq. 12.1), compute the Jacobian matrix $\mathcal{A}$ and the determinant $\det \mathcal{A}$ as functions of (θ_1, θ_2).

(b) Solve $\det \mathcal{A} = 0$ to find the critical curve. Express the result in polar coordinates $\theta_{\rm crit}(\phi)$. Plot the critical curve.

(c) Map the critical curve through the lens equation to obtain the caustic. Plot the caustic in the source plane. Verify that it has four cusps and the shape of an astroid.

(d) Estimate the area enclosed by the caustic (the quad cross-section).

Verify with `problems_08.wl`.

Exercise 12.10.2. (SIE tangential caustic.) For an SIE with $\theta_{\rm E} = 1''$ and axis ratio $q = 0.7$:

(a) Write down the deflection angle components α_1 and α_2 from eq. (12.9).

(b) Compute the tangential caustic numerically by: (i) finding the critical curve from $\det \mathcal{A} = 0$, and (ii) mapping it through the lens equation. Plot the caustic.

(c) Compute the area of the tangential caustic (the quad cross-section) numerically. How does it compare to the SIS + shear model with the same effective ellipticity?

Verify with `problems_08.wl`.

Exercise 12.10.3. (Fold magnification scaling.) Show that images appear or disappear in pairs at caustic crossings by analyzing $\det \mathcal{A}$ near a fold:

(a) Near a fold caustic, the lens mapping in the direction perpendicular to the fold can be approximated as $\beta_\perp \approx a\,\theta_\perp^2$, where $\theta_\perp$ is the signed distance from the critical curve and a is a constant. Show that for $\beta_\perp > 0$, there are two solutions $\theta_\perp = \pm\sqrt{\beta_\perp / a}$ (one on each side of the critical curve), while for $\beta_\perp < 0$ there are no solutions.

(b) Compute the Jacobian of the simplified mapping and show that the magnification scales as $|\mu| \propto 1/\sqrt{\beta_\perp}$ for each image.

(c) Show that the two images have opposite parities (opposite signs of $\det \mathcal{A}$).

Verify with `problems_08.wl`.

Exercise 12.10.4. (SIE image positions.) For a source at $\boldsymbol{\beta} = (0.1, 0.2)''$ lensed by an SIE with $\theta_{\rm E} = 1''$ and $q = 0.8$:

(a) Determine whether the source is inside or outside the tangential caustic. How many images do you expect?

(b) Find all image positions $\boldsymbol{\theta}_i$ numerically by solving the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$.

(c) For each image, compute the magnification μ_i and determine the parity (Type I or Type II).

(d) Compute the signed magnification sum $\sum_i \mu_i$ and discuss whether it is consistent with the expected value (considering the missing central image for the singular SIE).

Verify with `problems_08.wl`.

Exercise 12.10.5. (Burke's odd-number theorem for the SIE.) Verify Burke's odd-number theorem for an SIE with $\theta_{\mathrm{E}} = 1''$ and $q = 0.7$:

(a) For a source outside the tangential caustic (e.g., $\boldsymbol{\beta} = (0.5, 0.0)''$), find all image positions. Count the images (including any faint central image) and verify the count is odd.

(b) For a source inside the tangential caustic (e.g., $\boldsymbol{\beta} = (0.05, 0.05)''$), find all image positions. Verify the count is odd.

(c) For each case, classify each image by parity and verify that the number of positive-parity images exceeds the number of negative-parity images by one.

(d) Discuss why the SIE (being singular at the origin) requires care in applying Burke's theorem. What role does the central image play?

Verify with `problems_08.wl`.

Chapter 13

Strong Lensing by Galaxies: Applications

> *The Einstein radius is a most useful quantity to express the lensing strength of an object, and it is usually very robustly determined via strong lens models [. . .]. As a consequence, the mass enclosed in the cylinder of radius equal to the Einstein radius can be measured to within 1–2%, including all random and systematic uncertainties.* — T. Treu, *Ann. Rev. Astron. Astrophys.* **48**, 87 (2010)

In Modules 11 and 12 we developed the theory and the lens models: the SIS, SIE, critical curves, caustics, and image configurations. These are the *tools*. Now we apply them.

Galaxy-scale strong lensing is one of the most powerful techniques in modern astrophysics. It addresses fundamental questions:

- **What is the total mass of a galaxy?** Lensing measures the projected mass within the Einstein radius, including dark matter.
- **How is mass distributed?** Combining lensing with stellar kinematics constrains the density profile slope.
- **What is the expansion rate of the universe?** Time delays between multiple images, combined with a lens model, yield H_0 — independent of the cosmic distance ladder.
- **What are the limits of what lensing can tell us?** Fundamental degeneracies — most notably the *mass-sheet degeneracy* — limit the information that lensing alone can provide.

This module connects the lens modeling formalism to these real astrophysical applications. Understanding the degeneracies is as important as understanding the models themselves.

Companion Mathematica files:

- `galaxy_lensing.wl` — mass from Einstein radius, mass-sheet degeneracy demonstration, power-law profile analysis, H_0 from time delays; figure generation for all module figures
- `mass_sheet_demo.nb` — interactive notebook: MSD visualizer with slider for λ, power-law profile explorer, H_0 estimator from time delays

13.1 Lens Modeling Overview

Before applying lensing to astrophysics, we must understand what goes into building a lens model and what data constrain it.

13.1.1 Parametric Models: SIE + External Shear

The standard approach to modeling galaxy-scale strong lenses uses a **parametric** mass model. The workhorse is the singular isothermal ellipsoid (SIE, Module 12) plus external shear:

$$\kappa(\boldsymbol{\theta}) = \frac{\theta_{\rm E}\sqrt{q}}{2\sqrt{q^2\theta_1^2 + \theta_2^2}} \tag{13.1}$$

with an external shear $\gamma_{\rm ext}$ at position angle ϕ_γ. The model parameters are:

Parameter	Physical meaning
$\theta_{\rm E}$	Einstein radius (sets the mass scale)
q	Axis ratio (b/a, sets the ellipticity)
$\theta_{1,c}$, $\theta_{2,c}$	Lens center position
ϕ_q	Position angle of the mass distribution
$\gamma_{\rm ext}$	External shear magnitude
ϕ_γ	External shear position angle

This gives 7 lens parameters (or 5 if the center is fixed to the observed galaxy light). The source position $\boldsymbol{\beta} = (\beta_1, \beta_2)$ adds 2 more unknowns, for a total of 9 (or 7) free parameters.

13.1.2 What Data Constrain the Model

The observational constraints for a strong lens system are:

- **Image positions:** Each image gives 2 constraints ($\theta_{1,i}$, $\theta_{2,i}$). A quad system with 4 images gives 8 constraints.
- **Flux ratios:** Each flux ratio gives 1 constraint (the ratio of magnifications). Three independent flux ratios for a quad.
- **Time delays:** Each time delay gives 1 constraint. Three independent delays for a quad.
- **Extended arcs:** If the source is extended (a galaxy rather than a quasar), the arc morphology provides many more constraints (effectively constraining the lens potential on a grid).

13.1.3 Degrees of Freedom

For a quad lens modeled with SIE + shear:

- Constraints from image positions alone: $4 \times 2 = 8$.
- Free parameters (with center fixed): 7.
- Degrees of freedom: $8 - 7 = 1$ (marginally constrained).

This is why quad lenses are so valuable: they provide just enough constraints to determine the SIE + shear model from astrometry alone. Double lenses ($2 \times 2 = 4$ constraints) are under-constrained for SIE + shear. Extended arcs provide many more constraints and can support more complex models.

13.1.4 Free-Form Models

An alternative to parametric models is **free-form** or **pixelated** modeling, where the lens potential is reconstructed on a grid without assuming a specific functional form. Programs like `PixeLens` (Saha & Williams 2004) and `GLASS` (Coles et al. 2014) take this approach. Free-form models are less biased by the assumed mass profile but have more degrees of freedom and thus require regularization or priors.

13.2 The Einstein Radius as a Mass Estimator

13.2.1 The Model-Independent Mass Measurement

The single most important result in galaxy-scale strong lensing is that the Einstein radius directly gives the enclosed projected mass. For *any* circularly symmetric lens, the projected mass enclosed within the Einstein radius is (Congdon & Keeton eq. 7.9):

$$\boxed{M(\leq \theta_{\rm E}) = \pi\, \theta_{\rm E}^2\, D_{\rm d}^2\, \Sigma_{\rm cr}} \tag{13.2}$$

This follows from the definition of the Einstein radius as the angular radius where $\kappa_{\rm mean} = 1$, i.e., where the mean convergence within that radius equals unity. Recall that $\kappa = \Sigma/\Sigma_{\rm cr}$, so $\kappa_{\rm mean} = 1$ means the mean surface mass density within $\theta_{\rm E}$ equals the critical surface mass density. The enclosed mass is then the critical density times the area:

$$M_{\rm E} = \Sigma_{\rm cr} \times \pi (D_{\rm d}\, \theta_{\rm E})^2 = \pi\, \theta_{\rm E}^2\, D_{\rm d}^2\, \Sigma_{\rm cr}\,. \tag{13.3}$$

The critical surface mass density is (Module 8):

$$\Sigma_{\rm cr} = \frac{c^2}{4\pi G}\, \frac{D_{\rm s}}{D_{\rm d}\, D_{\rm ds}}\,. \tag{13.4}$$

Substituting:

$$\boxed{M_{\rm E} = \frac{c^2}{4G}\, \theta_{\rm E}^2\, \frac{D_{\rm d}\, D_{\rm s}}{D_{\rm ds}}} \tag{13.5}$$

Remark 13.2.1. Equation (13.5) is **model-independent** for any circularly symmetric lens. It depends only on the observed Einstein radius $\theta_{\rm E}$ and the angular diameter distances (which require knowing the redshifts z_d and z_s and adopting a cosmology). It does not depend on the radial mass profile. This is the great power of strong lensing as a mass measurement tool.

13.2.2 The SIE Einstein Radius and Velocity Dispersion

For the SIE model, the Einstein radius is related to the velocity dispersion by (Module 11):

$$\boxed{\theta_{\rm E} = 4\pi \, \frac{\sigma_v^2}{c^2} \, \frac{D_{\rm ds}}{D_{\rm s}}} \tag{13.6}$$

For a typical massive elliptical galaxy with $\sigma_v = 250$ km/s at $z_d = 0.3$ lensing a source at $z_s = 1.0$:

$$\theta_{\rm E} \approx 4\pi \left(\frac{250}{3 \times 10^5}\right)^2 \frac{D_{\rm ds}}{D_{\rm s}} \approx 4\pi \left(\frac{250}{3 \times 10^5}\right)^2 \times 0.64 \approx 1.15'' \, . \tag{13.7}$$

The enclosed mass is then $M_{\rm E} \sim 2$–$4 \times 10^{11} \, M_\odot$, comparable to the total stellar mass of the galaxy. The fact that $M_{\rm E}$ typically exceeds the stellar mass is direct evidence for dark matter within the Einstein radius.

13.2.3 Numerical Examples with Real Systems

Figure 13.1 shows the enclosed mass as a function of Einstein radius for different lens–source redshift configurations.

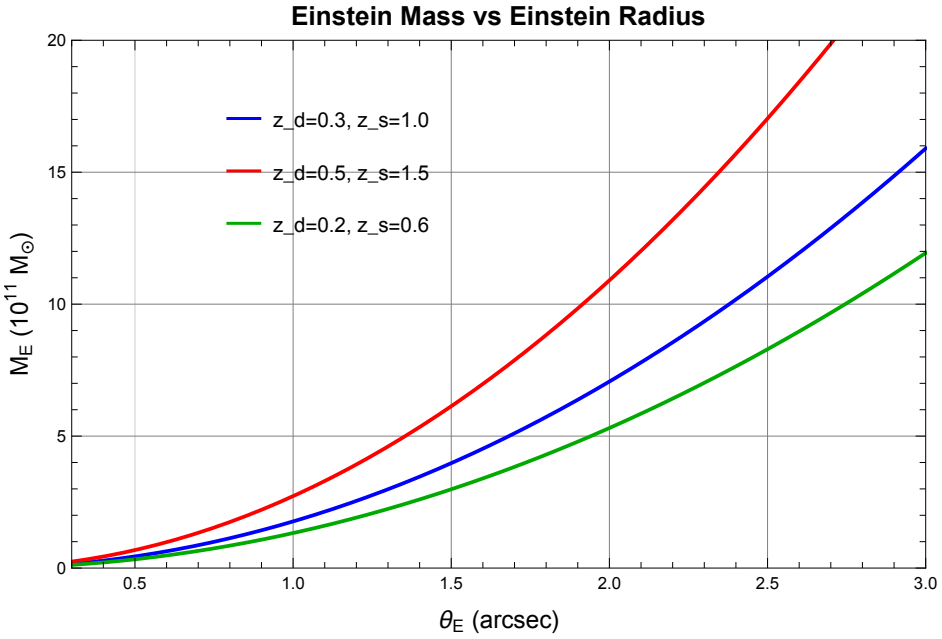


Figure 13.1: Enclosed Einstein mass $M_{\rm E}$ as a function of Einstein radius $\theta_{\rm E}$ for three lens–source redshift configurations (flat ΛCDM, $H_0 = 70$ km/s/Mpc, $\Omega_{\rm m} = 0.3$). Higher redshift lenses enclose more mass for the same angular Einstein radius. Generated by `galaxy_lensing.wl`.

Table 13.1 lists the Einstein radii and enclosed masses for several well-known lens systems:

Remark 13.2.2. Q2237+0305 (the Einstein Cross) has an unusually small enclosed mass because the lens galaxy is very nearby ($z_d = 0.04$), so the physical Einstein radius is small even though the angular Einstein radius is nearly $1''$.

System	z_d	z_s	$\theta_{\rm E}$ (arcsec)	$M_{\rm E}$ ($10^{11}\,M_\odot$)	Type
Q0957+561	0.36	1.41	1.0	~ 1.9	Double
Q2237+0305	0.04	1.69	0.9	~ 0.2	Quad
B1608+656	0.63	1.39	1.0	~ 3.9	Quad

Table 13.1: Einstein radii and enclosed masses for three well-known lens systems. Masses computed using eq. (13.5) with a flat ΛCDM cosmology ($H_0 = 70$ km/s/Mpc, $\Omega_{\rm m} = 0.3$). Computed by `galaxy_lensing.wl`.

13.3 The Mass-Sheet Degeneracy

The **mass-sheet degeneracy** (MSD) is the most important systematic limitation of strong gravitational lensing. It is a fundamental degeneracy that cannot be broken by lensing data alone (image positions, flux ratios, or arc morphology).

13.3.1 The Mass-Sheet Transform

Consider a lens with convergence $\kappa(\boldsymbol{\theta})$. The **mass-sheet transform** (MST) replaces this convergence with:

$$\boxed{\kappa_\lambda(\boldsymbol{\theta}) = \lambda\,\kappa(\boldsymbol{\theta}) + (1-\lambda)} \tag{13.8}$$

and simultaneously rescales the source plane:

$$\boxed{\boldsymbol{\beta}_\lambda = \lambda\,\boldsymbol{\beta}} \tag{13.9}$$

where λ is a free parameter.

Physically, the MST adds a **uniform sheet of convergence** $(1-\lambda)$ to the lens (hence the name "mass sheet") and rescales both the original convergence and the source plane by λ.

13.3.2 Image Positions Are Invariant

The key property of the MST is that it **preserves all image positions**. To see this, recall the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$. For the original lens, the deflection angle is related to the convergence via $\nabla \cdot \boldsymbol{\alpha} = 2\kappa$. Under the MST:

$$\boldsymbol{\alpha}_\lambda(\boldsymbol{\theta}) = \lambda\,\boldsymbol{\alpha}(\boldsymbol{\theta}) + (1-\lambda)\,\boldsymbol{\theta}\,. \tag{13.10}$$

The new lens equation becomes:

$$\begin{aligned}
\boldsymbol{\beta}_\lambda &= \boldsymbol{\theta} - \boldsymbol{\alpha}_\lambda(\boldsymbol{\theta}) \\
&= \boldsymbol{\theta} - \lambda\,\boldsymbol{\alpha}(\boldsymbol{\theta}) - (1-\lambda)\,\boldsymbol{\theta} \\
&= \lambda\,[\boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})] \\
&= \lambda\,\boldsymbol{\beta}\,.
\end{aligned} \tag{13.11}$$

So if $\boldsymbol{\theta}$ satisfies the original lens equation for source position $\boldsymbol{\beta}$, it also satisfies the transformed lens equation for source position $\boldsymbol{\beta}_\lambda = \lambda\boldsymbol{\beta}$. **The image positions are unchanged.**

13.3.3 What the MST Changes

Although image positions are preserved, the MST changes other observables:

Magnifications. Under the MST the transformed convergence and deflection are

$$\kappa_\lambda(\boldsymbol{\theta}) = \lambda\,\kappa(\boldsymbol{\theta}) + (1-\lambda), \qquad \boldsymbol{\alpha}_\lambda(\boldsymbol{\theta}) = \lambda\,\boldsymbol{\alpha}(\boldsymbol{\theta}) + (1-\lambda)\,\boldsymbol{\theta},$$

the second of which follows from the first via $\boldsymbol{\alpha} = \nabla\psi$ and $\nabla^2\psi = 2\kappa$ (the $(1-\lambda)\boldsymbol{\theta}$ piece is the deflection of a uniform mass sheet, $\nabla\cdot(\boldsymbol{\theta}/2) = 1 = 1-\lambda$ up to the overall factor). The Jacobian then transforms componentwise:

$$(\mathcal{A}_\lambda)_{ij} \;=\; \delta_{ij} - \frac{\partial\alpha_{\lambda,i}}{\partial\theta_j} \;=\; \delta_{ij} - \lambda\,\frac{\partial\alpha_i}{\partial\theta_j} - (1-\lambda)\,\delta_{ij} \;=\; \lambda\left(\delta_{ij} - \frac{\partial\alpha_i}{\partial\theta_j}\right) \;=\; \lambda\,\mathcal{A}_{ij}\,.$$

That is, every entry of $\mathcal{A}$ scales by λ:

$$\mathcal{A}_\lambda = \lambda\,\mathcal{A}, \qquad \det\mathcal{A}_\lambda = \lambda^2\,\det\mathcal{A}, \tag{13.12}$$

so the magnification $\mu = 1/\det\mathcal{A}$ scales as

$$\boxed{\mu_\lambda = \frac{1}{\lambda^2}\,\mu\,.} \tag{13.13}$$

Individual magnifications change, but **magnification ratios** $\mu_i/\mu_j = \mu_{\lambda,i}/\mu_{\lambda,j}$ are preserved. This means flux *ratios* cannot distinguish between the original model and the MST-transformed model. Both steps are verified symbolically in `galaxy_lensing.wl`.

Time delays. The Fermat-potential difference between images, and therefore the model-predicted time delay for a given H_0, scales as:

$$\boxed{\Delta t_\lambda = \lambda\,\Delta t} \tag{13.14}$$

That is, an MST-transformed model with parameter λ produces a time delay λ times that of the original ($\lambda = 1$) model at the same H_0. But the observed Δt is fixed; to match that observation under the transformed model, the inferred H_0^{naive} that one computes from a $\lambda = 1$ (no-sheet) fit and the underlying "true" H_0^{true} that accounts for an actual sheet of parameter λ differ by:

$$\boxed{H_0^{\text{true}} \;=\; \lambda\,H_0^{\text{naive}} \quad\Longleftrightarrow\quad H_0^{\text{naive}} \;=\; H_0^{\text{true}}/\lambda\,.} \tag{13.15}$$

With an unmodeled positive mass sheet ($\lambda < 1$), the naive analysis therefore *overestimates* H_0, and including the sheet brings the inferred value *downward*. This is the direction in which TDCOSMO has explored [3]: relaxing the lens profile (equivalently, allowing $\lambda < 1$) shifts the time-delay H_0 from the H0LiCOW value $H_0 \approx 73.3\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ toward the Planck CMB value $H_0 \approx 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Remark 13.3.1. The MSD is devastating for time-delay cosmography. If we model a lens and infer H_0 from the time delay, the answer is degenerate with λ: any extra convergence (a true mass sheet, or unmodeled line-of-sight structure, or a profile that is steeper than assumed) reduces H_0^{true} below H_0^{naive}. Breaking this degeneracy is the central challenge of modern time-delay cosmography.

13.3.4 Physical Interpretation

The mass-sheet transform can be interpreted in two ways:

1. **Literal mass sheet:** A uniform sheet of matter between the observer and the source (or at the lens redshift) adds convergence without changing image positions. In practice, line-of-sight structure (voids, filaments, galaxy groups along the line of sight) acts as an approximate mass sheet.

2. **Profile degeneracy:** Even without an actual mass sheet, the MST reflects the fact that different radial mass profiles can produce the same image positions. Steeper profiles can mimic shallower profiles plus a mass sheet.

Figure 13.2 illustrates the mass-sheet degeneracy: the left panel shows how the convergence profile changes for different λ, and the right panel shows how absolute magnifications scale as $1/\lambda^2$.

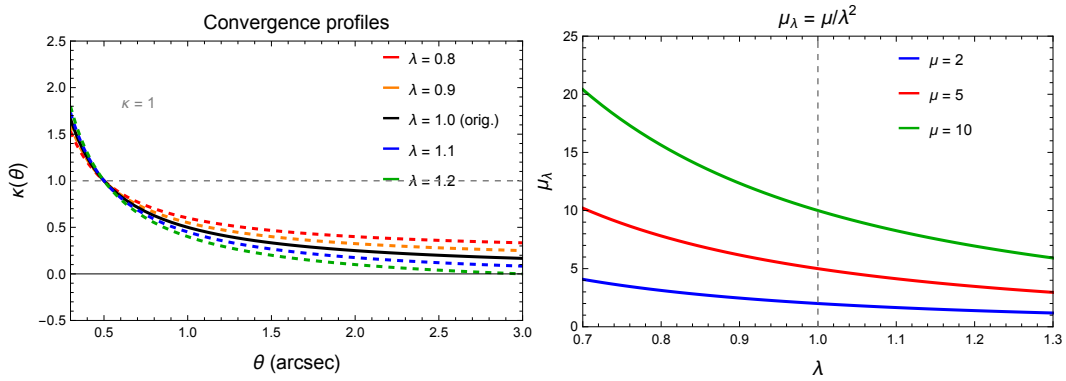


Figure 13.2: The mass-sheet degeneracy. **Left:** Convergence profiles for an SIS lens ($\theta_{\rm E} = 1''$) under the MST with different λ values. The solid curve is the original profile ($\lambda = 1$); dashed curves show the transformed profiles. Curves with $\lambda > 1$ correspond to a negative sheet, e.g. an underdense line of sight ($\kappa_{\rm ext} < 0$ in $\lambda = 1 - \kappa_{\rm ext}$). **Right:** Magnification scaling $\mu_\lambda = \mu/\lambda^2$ for three different original magnifications. Generated by `galaxy_lensing.wl`.

13.3.5 How to Break the MSD

Several methods can break or constrain the mass-sheet degeneracy:

1. **Stellar kinematics:** The velocity dispersion σ_v of the lens galaxy depends on the mass profile slope, not just the enclosed mass. A mass-sheet transform changes the predicted σ_v, so a kinematic measurement constrains λ (Section 13.5).

2. **Standardizable magnifications:** If the source has a known intrinsic luminosity (e.g., a Type Ia supernova), the absolute magnification can be measured, breaking the MSD.

3. **Multiple source planes:** If the same lens produces images of sources at different redshifts, the different $\Sigma_{\rm cr}$ values for each source plane break the degeneracy.

4. **Extended source reconstruction:** Detailed modeling of extended arcs constrains the lens potential more tightly, partially breaking the degeneracy (though not completely).

13.4 The Source Position Transformation

The mass-sheet degeneracy is a special case of a more general class of transformations that preserve image positions.

13.4.1 Definition

The **source position transformation** (SPT), introduced by Schneider & Sluse (2014), is any invertible mapping of the source plane:

$$\boldsymbol{\beta} \to \hat{\boldsymbol{\beta}}(\boldsymbol{\beta}) \tag{13.16}$$

that, when combined with a suitably modified lens equation, preserves all image positions. The modified lens equation is:

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\beta}) = \boldsymbol{\theta} - \hat{\boldsymbol{\alpha}}(\boldsymbol{\theta}) \tag{13.17}$$

where the new deflection field $\hat{\boldsymbol{\alpha}}$ is defined by the requirement that all observed image positions map to the transformed source position $\hat{\boldsymbol{\beta}}$.

The MST is the special case where $\hat{\boldsymbol{\beta}} = \lambda\boldsymbol{\beta}$ (a linear scaling). The SPT generalizes this to arbitrary nonlinear remappings of the source plane.

13.4.2 Implications

The SPT is an active area of research because it defines the *full* class of degeneracies in strong lensing. Key results include:

- Not all SPTs correspond to physically valid lens models (i.e., the transformed deflection field may not come from a non-negative mass distribution).
- The SPT preserves image positions and flux *ratios* (just as the MST does) but generically changes absolute magnifications and time delays.
- Breaking the SPT generally requires external information (kinematics, multiple source planes, or standardizable sources), just as for the MST.

Remark 13.4.1. The SPT shows that the mass-sheet degeneracy is not a peculiarity of a specific transform but rather the simplest member of a broad family of degeneracies inherent in the lensing inverse problem. Any attempt to measure H_0 from time delays must confront this fundamental limitation.

13.5 Strong Lensing and Stellar Kinematics

Combining strong lensing with stellar kinematics is one of the most powerful approaches to studying galaxy mass distributions. The two techniques are complementary in a fundamental way.

13.5.1 What Each Technique Measures

Strong lensing: Measures the **total projected mass** (dark matter + stars + gas) within the Einstein radius $\theta_{\rm E}$. The measurement is model-independent for a circularly symmetric lens (eq. 13.5). For an elliptical model, the enclosed mass depends weakly on the assumed axis ratio but is still robust.

Stellar kinematics: The stellar velocity dispersion σ_v of the lens galaxy, measured from absorption line broadening, probes the gravitational potential. Specifically, under the assumption of dynamical equilibrium (the Jeans equation), the luminosity-weighted line-of-sight velocity dispersion within an aperture $R_{\rm ap}$ depends on:

- The **mass profile slope** (how mass is distributed radially).
- The **total mass normalization**.
- The **orbital anisotropy** of the stellar orbits.

13.5.2 The Power-Law Mass Profile

The simplest model that captures the essential physics is a **spherical power-law** density profile:

$$\boxed{\rho(r) = \rho_0 \left(\frac{r}{r_0}\right)^{-\gamma'}} \tag{13.18}$$

where γ' is the power-law slope. The isothermal case is $\gamma' = 2$ (i.e., $\rho \propto r^{-2}$).

Figure 13.3 shows convergence profiles and enclosed mass scalings for power-law density profiles with different slopes.

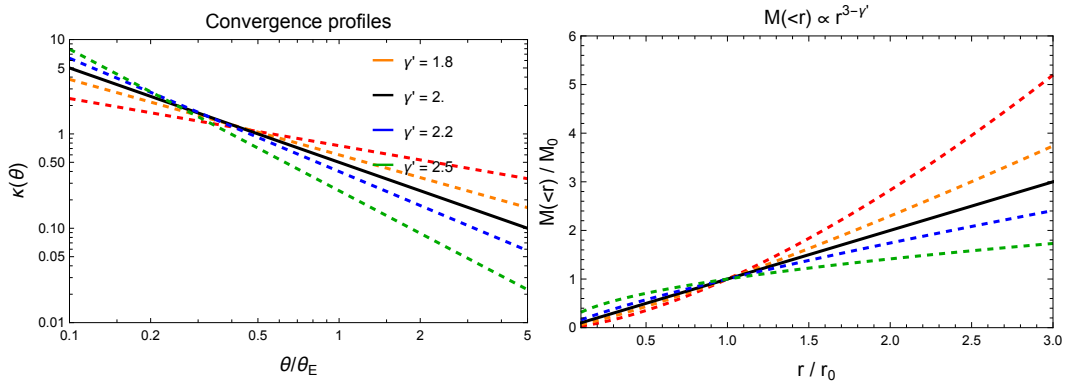


Figure 13.3: Power-law density profiles $\rho \propto r^{-\gamma'}$. **Left:** Convergence profiles for different slopes (the solid curve is the isothermal case $\gamma' = 2$). **Right:** Enclosed 3D mass $M(\leq r) \propto r^{3-\gamma'}$ for the same slopes. Generated by `galaxy_lensing.wl`.

The convergence follows from the line-of-sight projection. Writing $r = \sqrt{\xi^2 + z^2}$,

$$\Sigma(\xi) \;=\; \int_{-\infty}^{\infty} \rho_0 \left(\frac{\sqrt{\xi^2+z^2}}{r_0}\right)^{-\gamma'} dz \;=\; 2\rho_0\, r_0^{\gamma'}\, \xi^{1-\gamma'} \int_0^{\infty} (1+u^2)^{-\gamma'/2}\, du \qquad (u = z/\xi).$$

The remaining integral is a Beta function:

$$\int_0^\infty (1+u^2)^{-\gamma'/2}\,du \;=\; \frac{1}{2}\,B\!\left(\frac{1}{2},\frac{\gamma'-1}{2}\right) \;=\; \frac{\sqrt{\pi}}{2}\,\frac{\Gamma\!\left(\frac{\gamma'-1}{2}\right)}{\Gamma\!\left(\frac{\gamma'}{2}\right)},$$

finite for $\gamma' > 1$. Hence $\Sigma(\xi) \propto \xi^{1-\gamma'}$ and, using $\xi = D_{\rm d}\theta$ and dividing by $\Sigma_{\rm cr}$, the convergence has the same exponent. A short computation that fixes the overall normalization by matching to the isothermal case $\gamma' = 2$ (where $\theta_{\rm E}$ is the SIS Einstein radius) gives

$$\kappa(\theta) \;=\; \frac{3-\gamma'}{2}\,\left(\frac{\theta_{\rm E}}{\theta}\right)^{\gamma'-1}, \tag{13.19}$$

valid for $1 < \gamma' < 3$ (finite projection, finite enclosed mass). The projection integral and the normalization are verified in `galaxy_lensing.wl`. and the Einstein radius is:

$$\theta_{\rm E} = 4\pi\,\frac{\sigma_{\rm SIE}^2}{c^2}\,\frac{D_{\rm ds}}{D_{\rm s}}\,f(\gamma') \tag{13.20}$$

where $f(\gamma')$ is a function of the slope that equals unity for $\gamma' = 2$ (recovering the SIE result) and $\sigma_{\rm SIE}$ is the SIE-equivalent velocity dispersion.

13.5.3 Breaking the Degeneracy

The key insight is that lensing and kinematics respond differently to the mass profile slope γ':

- **Lensing** constrains the projected mass within $\theta_{\rm E}$. For a power-law profile, the enclosed mass $M_{\rm E}$ depends on the normalization ρ_0 and the slope γ' in a specific way.
- **Kinematics** constrains the mass within the spectroscopic aperture (typically a different physical radius than $R_{\rm E}$). The predicted σ_v^2 depends on γ' *differently* from how $M_{\rm E}$ does.

By measuring both $M_{\rm E}$ (from lensing) and σ_v (from spectroscopy), one can solve for *both* the mass normalization and the slope γ'. This breaks the mass-sheet degeneracy because the MST changes the effective γ': steepening or shallowing the profile to compensate for the added mass sheet.

13.5.4 The Bulge-Halo Conspiracy

One of the most remarkable results from combining lensing and kinematics is the **bulge-halo conspiracy**: the total (dark matter + stellar) density profiles of massive elliptical galaxies are remarkably close to isothermal over a wide range of radii:

$$\gamma' \approx 2.0 \pm 0.2 \tag{13.21}$$

This result, established by the SLACS survey (Auger et al. 2010) and confirmed by subsequent studies, means that the combined contribution of the stellar bulge (which has a steeper profile,

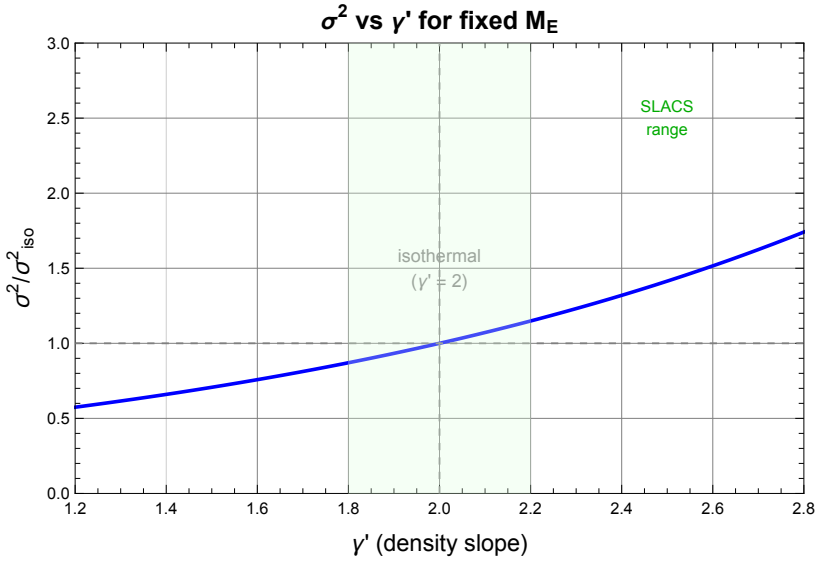


Figure 13.4: Predicted σ_v^2 (normalized to the isothermal value) as a function of the density slope γ', for fixed Einstein mass. The green band shows the range found by the SLACS survey ($\gamma' \approx 2.0 \pm 0.2$). Measuring σ_v constrains γ' and breaks the mass-sheet degeneracy. Generated by `galaxy_lensing.wl`.

$\gamma' \sim 3$) and the dark matter halo (which has a shallower inner profile, $\gamma' \sim 1$ for NFW) conspire to produce an approximately r^{-2} total density profile.

Figure 13.4 shows how the predicted velocity dispersion varies with γ' for a fixed Einstein mass, illustrating how kinematics breaks the degeneracy.

This "conspiracy" is not fully understood theoretically and remains an active area of research. It has important implications:

1. The SIE ($\gamma' = 2$) is a good approximation for most galaxy-scale lenses.
2. The small scatter in γ' suggests a universal mass assembly process for massive ellipticals.
3. It connects directly to the flat rotation curves observed in spiral galaxies (both are consequences of $\rho \propto r^{-2}$).

13.6 Time-Delay Cosmography

Strong lens time delays offer a one-step measurement of H_0, independent of the cosmic distance ladder.

13.6.1 Recap: Time Delays and H_0

From Module 10, the time delay between images A and B of a strongly lensed variable source is:

$$\Delta t_{AB} = \frac{1 + z_d}{c} \frac{D_\mathrm{d}\, D_\mathrm{s}}{D_\mathrm{ds}} \left[\Delta \tau_{AB}\right] \tag{13.22}$$

where $\Delta\tau_{AB}$ is the Fermat potential difference (computed from the lens model) and the prefactor $D_{\rm d} D_{\rm s}/D_{\rm ds}$ is called the **time-delay distance** $D_{\Delta t}$. Crucially:

$$D_{\Delta t} \propto \frac{1}{H_0} \tag{13.23}$$

so measuring Δt (from monitoring the variable source) and modeling $\Delta\tau$ (from the lens model) gives H_0.

13.6.2 The H0LiCOW / TDCOSMO Program

The most extensive application of this method is the **H0LiCOW** (H0 Lenses in COSMOGRAIL's Wellspring) program and its successor **TDCOSMO**, which have analyzed six well-studied time-delay lenses.

The headline result from Wong et al. (2020), assuming power-law mass profiles:

$$\boxed{H_0 = 73.3^{+1.7}_{-1.8}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}} \tag{13.24}$$

at 2.4% precision. This value is in 3.1σ tension with the Planck CMB measurement (Planck Collaboration 2020):

$$H_0^{\rm Planck} = 67.4 \pm 0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}} \tag{13.25}$$

and agrees well with the local distance ladder value from SH0ES (Riess et al. 2022):

$$H_0^{\rm SH0ES} = 73.0 \pm 1.0\ \mathrm{km\,s^{-1}\,Mpc^{-1}}\,. \tag{13.26}$$

Figure 13.5 illustrates the effect of the mass-sheet degeneracy on the inferred H_0.

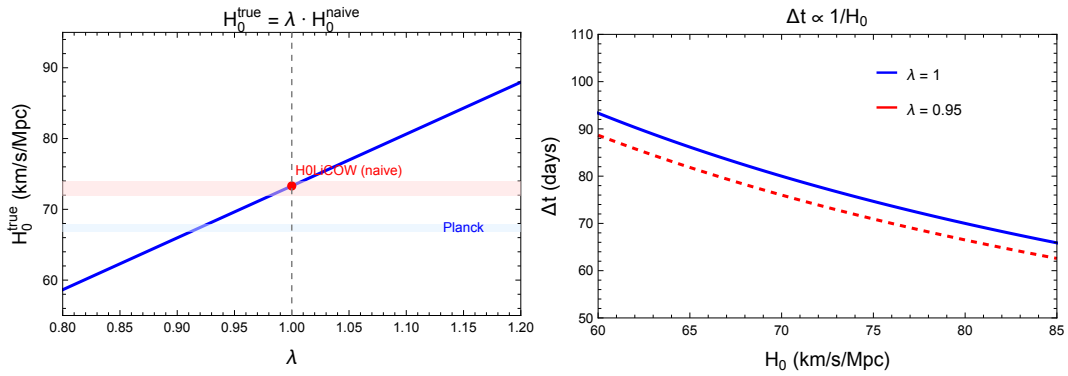


Figure 13.5: Time-delay cosmography and the mass-sheet degeneracy. **Left:** Inferred H_0 as a function of the MST parameter λ, showing the H0LiCOW measurement (red point) and the Planck and SH0ES bands. **Right:** Time delay vs. H_0 for $\lambda = 1$ (solid) and $\lambda = 0.95$ (dashed). Generated by `galaxy_lensing.wl`.

13.6.3 The Role of the Mass-Sheet Degeneracy

The mass-sheet degeneracy is the dominant systematic uncertainty in time-delay cosmography. From eq. (13.15):

$$H_0^{\rm true} = \lambda\, H_0^{\rm naive} \tag{13.27}$$

where $H_0^{\rm naive}$ is the H_0 inferred from a no-sheet ($\lambda = 1$) power-law fit and $H_0^{\rm true}$ is the actual cosmological value; since $\lambda < 1$ for an unmodeled positive mass sheet, $H_0^{\rm true} < H_0^{\rm naive}$. A 10% mass sheet ($\lambda = 0.9$, i.e., adding $\kappa_s = 0.1$ to the convergence) shifts $H_0^{\rm naive}$ downward by 10%.

The TDCOSMO collaboration (Birrer et al. 2020) showed that relaxing the assumption of a power-law mass profile (which partially encodes λ) broadens the uncertainty to:

$$H_0 = 74.5^{+5.6}_{-6.1}\ {\rm km\,s^{-1}\,Mpc^{-1}} \tag{13.28}$$

— still consistent with the local value but with much larger uncertainty. The increased uncertainty is almost entirely due to the mass-sheet degeneracy.

Constraining λ requires external information, primarily **stellar kinematics** of the lens galaxies (Section 13.5).

13.6.4 Current State and Future Prospects

The time-delay cosmography method is a key player in the Hubble tension debate. Upcoming surveys (Rubin Observatory/LSST, Euclid) will discover thousands of new lensed quasars, enabling a statistical approach that can average over individual modeling uncertainties. Combined with improved stellar kinematics from next-generation spectrographs, the method has the potential to reach $\sim$1% precision on H_0.

13.7 Real Lens Systems: Case Studies

13.7.1 Q0957+561: The First Gravitational Lens

Discovered by Walsh, Carswell & Weymann in 1979, Q0957+561 was the first confirmed strong gravitational lens. Key properties:

- **Configuration:** Double (two images of a $z_s = 1.41$ quasar).
- **Lens:** Giant elliptical galaxy at $z_d = 0.36$, in a galaxy cluster.
- **Einstein radius:** $\theta_{\rm E} \approx 1.0''$.
- **Time delay:** $\Delta t = 417 \pm 3$ days (optical monitoring; Kundić et al. 1997; see 33, Part 1, p. 12).
- **Historical significance:** Confirmation of Einstein's prediction of multiple imaging by galaxies. The long time delay was the first used for H_0 estimation (Refsdal's method).

13.7.2 Q2237+0305: The Einstein Cross

Q2237+0305 is one of the most photogenic gravitational lenses:

- **Configuration:** Quad — four images of a $z_s = 1.69$ quasar arranged in a near-perfect Einstein cross around the nucleus of a nearby barred spiral galaxy.

- **Lens:** A face-on barred spiral galaxy at $z_d = 0.04$ (Huchra's Lens). This is the nearest known strong lens galaxy.
- **Einstein radius:** $\theta_{\rm E} \approx 0.9''$.
- **Special feature:** Because the lens is so nearby, the Einstein radius corresponds to a very small physical radius (~ 0.7 kpc), probing deep into the bulge of the spiral galaxy. The system shows dramatic **microlensing** variability as individual stars in the lens galaxy cross the lines of sight to the quasar images.
- **Time delays:** Very short ($\lesssim 1$ day) due to the symmetric geometry, making this system unsuitable for H_0 measurement but ideal for microlensing studies.

13.7.3 B1608+656: Time-Delay Cosmography

B1608+656 is one of the best-studied time-delay lenses:

- **Configuration:** Quad — four images of a $z_s = 1.39$ radio source.
- **Lens:** Two interacting galaxies at $z_d = 0.63$.
- **Einstein radius:** $\theta_{\rm E} \approx 1.0''$.
- **Time delays:** Three independent delays measured with $\sim$1–2% precision from radio monitoring.
- **Cosmographic result:** One of the six TDCOSMO/H0LiCOW lenses used to constrain H_0. The lens modeling is more complex than usual due to the two-galaxy lens configuration.
- **Key insight:** Deep HST imaging resolves the lensed host galaxy of the radio source, providing extended arc constraints that significantly tighten the lens model.

13.8 Substructure and Flux-Ratio Anomalies

Everything so far has treated the lens as a *smooth* mass distribution. But ΛCDM predicts that galaxy haloes are lumpy: N-body simulations produce far more low-mass subhaloes than the number of luminous satellites we observe (the "missing satellites" problem), with of order a few percent of the projected mass near the images bound in substructure [33, Part 2, §8]. Strong lensing offers a way to weigh this *dark* substructure.

13.8.1 The Flux-Ratio Anomaly

Lens models that fit the image *positions* of a quad to milliarcsecond precision routinely fail to reproduce the image *flux ratios*. For a source straddling a fold or cusp of the caustic, smooth models make sharp predictions — e.g. the **cusp relation**, that the signed magnifications of the three merging images of a cusp image sum to zero — and real systems (the classic case is the radio quad B1422+231) violate them. Optical flux ratios can be corrupted by microlensing (Module 17) and dust; but *radio* flux ratios are largely immune to both, and they are anomalous just the same.

13.8.2 Why Flux Ratios Probe Low-Mass Substructure

The key is how many derivatives of the potential each observable requires. For virialized objects $M \propto \sigma_v^4$ and $R \propto \sigma_v^2$, so:

- **time delays** $\propto \psi \propto M \propto \sigma_v^4$ — unaffected by low-mass substructure;
- **deflections** $\propto \nabla\psi \propto \sigma_v^2$ — perturbed only by *massive* substructure;
- **magnifications** $\kappa, \gamma \propto \nabla^2\psi \propto \sigma_v^0$ — perturbed *equally by all mass scales.*

Flux ratios (magnifications) are therefore uniquely sensitive to the low-mass end of the substructure spectrum, exactly where CDM and warm dark matter differ. A further diagnostic is that substructure preferentially *demagnifies saddle-point* images relative to minima [33, Part 2, §8]. Applying a Bayesian analysis to a sample of radio quads, Dalal & Kochanek found a substructure mass fraction $f_{\rm sat} \approx 0.02$ (90% confidence 0.006–0.07), consistent with CDM and far above the luminous-satellite fraction. Modern work extends this with **gravitational imaging** — detecting individual dark subhaloes through their astrometric perturbations of extended arcs — turning flux-ratio and arc anomalies into a probe of the small-scale matter power spectrum.

13.9 Extended Sources and Einstein-Ring Reconstruction

Real sources are extended, and their lensed images — arcs and **Einstein rings** — carry far more information than a handful of point images. The foundational fact is that **gravitational lensing conserves surface brightness**:

$$f_I(\boldsymbol{\theta}) = f_S(\boldsymbol{\beta}(\boldsymbol{\theta})), \tag{13.29}$$

so what we call "magnification" is only the apparent brightening that follows from the image subtending a larger solid angle than the source; the intrinsic surface brightness is unchanged [33, Part 2, §10].

13.9.1 The Einstein Ring as a Pattern

An Einstein ring is not a set of discrete images but a *pattern*: the locus of peak surface brightness along radial spokes from the lens centre. For an isothermal lens with arbitrary angular structure, $\psi = b\,\theta\,F(\varphi)$ with $\langle F\rangle = 1$, the tangential critical curve sits at

$$\frac{r_{\rm crit}}{b} = F(\varphi) + F''(\varphi), \tag{13.30}$$

while, for a compact source offset by $\boldsymbol{\beta}_0$ from the lens centre, the ring curve is

$$\frac{r_E(\varphi)}{b} = F(\varphi) + \boldsymbol{\beta}_0 \cdot \hat{e}_\theta \tag{13.31}$$

[Kochanek, Keeton & McLeod 2001; see 33, Part 2, eq. 141]. The remarkable consequence is that the *angular shape* of the ring directly reads off the multipoles $F(\varphi)$ of the lens potential (§12.4 of Module 12): the even multipoles of the ring encode the lens, the odd multipoles the source. Rings thus break the angular degeneracies that a two-image lens cannot constrain — though they pin down the *radial* profile only weakly (the dependence scales as $|\Delta\beta|^4$).

13.9.2 Source Reconstruction

To use every pixel of an arc, one reconstructs the source. Because surface brightness is conserved, every image-plane pixel maps to a source position $\boldsymbol{\beta}(\boldsymbol{\theta}, \mathbf{p})$ for lens parameters $\mathbf{p}$; in a *correct* model all image pixels mapping to the same source pixel share a surface brightness, while an incorrect model scatters them. Minimizing that scatter — the "Ring Cycle" idea, now realized as regularized linear (semilinear) inversion onto a pixel or adaptive grid — underlies modern pipelines such as `PyAutoLens` and the SLaM framework used in surveys like AGEL. This is the practical engine behind galaxy-scale mass measurements and the source-plane reconstructions of high-redshift galaxies.

13.10 External Convergence and Lens Environments

No lens is isolated. Mass in the lens's group or cluster, and uncorrelated structure along the line of sight, contribute an **external convergence** $\kappa_{\rm ext}$ that is nearly uniform across the image region — i.e. a mass sheet. As we saw in §13.3, a uniform sheet leaves the image positions and flux ratios unchanged (it is the physical realization of the mass-sheet transform with $\lambda = 1 - \kappa_{\rm ext}$), but it rescales an inferred Hubble constant,

$$H_0^{\rm true} = (1 - \kappa_{\rm ext})\, H_0^{\rm naive}, \tag{13.32}$$

exactly as in Eq. (13.15). Estimating $\kappa_{\rm ext}$ — typically a few percent — from the photometric and spectroscopic weighting of the lens field, calibrated against numerical line-of-sight simulations, is an essential and irreducible step in time-delay cosmography [3, 36].

13.10.1 Dust and the Interstellar Medium of the Lens

A more mundane environmental effect is the **interstellar medium of the lens galaxy**. Differential extinction reddens and dims the optical and near-infrared images that pass through different lines of sight in the lens, corrupting optical flux ratios (and biasing colours) in a strongly wavelength-dependent way [33, Part 2, §9]. This is a principal reason the substructure studies of §13.8 rely on *radio* flux ratios, which are essentially dust-free, and a reminder that a clean lensing measurement must always account for the baryonic astrophysics of the deflector.

13.11 Summary

- The **Einstein radius** provides a model-independent measurement of the enclosed projected mass: $M_{\rm E} = \pi \theta_{\rm E}^2 D_{\rm d}^2 \Sigma_{\rm cr}$. For typical galaxy-scale lenses, this mass is 10^{11}–$10^{12}\, M_\odot$.

- The standard parametric lens model is **SIE + external shear**, with 7 parameters. Quad lenses provide 8 positional constraints, marginally determining the model.

- The **mass-sheet degeneracy** (MST: $\kappa \to \lambda\kappa + (1-\lambda)$, $\boldsymbol{\beta} \to \lambda\boldsymbol{\beta}$) preserves all image positions but changes magnifications by $1/\lambda^2$ and time delays by λ. It is the dominant systematic for time-delay cosmography.
- The **source position transformation** (Schneider & Sluse 2014) generalizes the MST to arbitrary remappings $\boldsymbol{\beta} \to \hat{\boldsymbol{\beta}}(\boldsymbol{\beta})$ of the source plane.
- **Lensing + kinematics** breaks the MSD by constraining the density profile slope γ'. The SLACS survey found $\gamma' \approx 2.0 \pm 0.2$ (the "bulge-halo conspiracy").
- **Time-delay cosmography** measures H_0 via $\Delta t \propto D_{\Delta t} \propto 1/H_0$. The H0LiCOW/TDCOSMO result is $H_0 = 73.3^{+1.7}_{-1.8}$ km/s/Mpc, in tension with Planck.
- **Case studies:** Q0957+561 (first lens, 1979), Q2237+0305 (Einstein Cross, microlensing laboratory), B1608+656 (time-delay cosmography).

13.12 Problems

Exercise 13.12.1. (Enclosed mass for an SIE lens.) Consider an SIE lens with $\theta_{\rm E} = 1.5''$, axis ratio $q = 0.8$, lens redshift $z_d = 0.3$, and source redshift $z_s = 1.0$. Assume a flat ΛCDM cosmology with $H_0 = 70$ km/s/Mpc and $\Omega_{\rm m} = 0.3$.

(a) Compute the angular diameter distances $D_{\rm d}$, $D_{\rm s}$, and $D_{\rm ds}$.

(b) Compute the critical surface mass density $\Sigma_{\rm cr}$.

(c) Compute the enclosed projected mass $M_{\rm E}$ using eq. (13.5). Express the result in solar masses.

(d) Estimate the velocity dispersion σ_v that would correspond to this Einstein radius using eq. (13.6).

Verify with `problems_09.wl`.

Exercise 13.12.2. (Mass-sheet degeneracy: invariance and observables.) Consider a general lens with convergence $\kappa(\boldsymbol{\theta})$ and a source at $\boldsymbol{\beta}$.

(a) Apply the mass-sheet transform $\kappa_\lambda = \lambda\kappa + (1-\lambda)$ and $\boldsymbol{\beta}_\lambda = \lambda\boldsymbol{\beta}$. Show explicitly that the lens equation $\boldsymbol{\beta}_\lambda = \boldsymbol{\theta} - \boldsymbol{\alpha}_\lambda(\boldsymbol{\theta})$ is satisfied at the same image positions $\boldsymbol{\theta}$.

(b) Show that the Jacobian matrix transforms as $\mathcal{A}_\lambda = \lambda\,\mathcal{A}$ and hence the magnification transforms as $\mu_\lambda = \mu/\lambda^2$.

(c) Show that magnification *ratios* μ_i/μ_j are invariant under the MST.

(d) Using the Fermat potential, show that time delays transform as $\Delta t_\lambda = \lambda\,\Delta t$.

Verify with `problems_09.wl`.

Exercise 13.12.3. (H_0 from a time-delay lens.) A quad lens at $z_d = 0.6$, $z_s = 1.5$ has a measured time delay $\Delta t = 80$ days between two images. The lens model predicts a Fermat potential difference $\Delta\tau = 0.64$ arcsec^2.

(a) Write the time-delay equation (eq. 13.22) and solve for H_0.

(b) Compute H_0 numerically, assuming a flat ΛCDM cosmology with $\Omega_{\rm m} = 0.3$ and using the angular diameter distances at the given redshifts. *(Hint: the time-delay distance $D_{\Delta t} = (1+z_d)D_{\rm d}D_{\rm s}/D_{\rm ds}$ depends on H_0, so you will need to solve self-consistently or use H_0-independent distance ratios.)*

(c) Discuss: if the true lens includes a mass sheet with $\kappa_s = 0.05$ (i.e., $\lambda = 0.95$) that was not accounted for in the model, how would the inferred H_0 change?

Verify with `problems_09.wl`.

Exercise 13.12.4. (Power-law profile: lensing and kinematics.) Consider a spherical power-law density profile $\rho(r) = \rho_0 (r/r_0)^{-\gamma'}$.

(a) Compute the projected surface mass density $\Sigma(\theta)$ by integrating ρ along the line of sight. Show that $\kappa(\theta) \propto \theta^{1-\gamma'}$.

(b) For $\gamma' = 2$ (isothermal), verify that you recover $\kappa \propto 1/\theta$ (the SIS).

(c) The luminosity-weighted line-of-sight velocity dispersion for an isotropic power-law profile is (Treu & Koopmans 2004):

$$\sigma_v^2 \propto M(\leq R_{\rm ap}) \cdot R_{\rm ap}^{-1} \cdot g(\gamma')$$

where $g(\gamma')$ is a known function of the slope. Argue that σ_v depends on γ' differently from $M_{\rm E}$, so combining the two measurements constrains γ'.

(d) For a lens with $\theta_{\rm E} = 1.2''$ and $\sigma_v = 230$ km/s, compute γ' assuming the relation holds. Discuss whether the result is consistent with the "bulge-halo conspiracy" ($\gamma' \approx 2.0$).

Verify with `problems_09.wl`.

Exercise 13.12.5. (Comparing real lens systems.) Consider the three lens systems in Table 13.1: Q0957+561 ($z_d = 0.36$, $z_s = 1.41$, $\theta_{\rm E} = 1.0''$), Q2237+0305 ($z_d = 0.04$, $z_s = 1.69$, $\theta_{\rm E} = 0.9''$), and B1608+656 ($z_d = 0.63$, $z_s = 1.39$, $\theta_{\rm E} = 1.0''$).

(a) For each system, compute the angular diameter distances $D_{\rm d}$, $D_{\rm s}$, $D_{\rm ds}$ and the critical surface mass density $\Sigma_{\rm cr}$.

(b) Compute the enclosed projected mass $M_{\rm E}$ for each system.

(c) Compute the physical Einstein radius $R_{\rm E} = D_{\rm d}\,\theta_{\rm E}$ in kpc for each system.

(d) Which system has the largest enclosed mass? Which has the largest physical Einstein radius? Discuss why the rankings are different.

Verify with `problems_09.wl`.

Chapter 14

Strong Lensing by Galaxy Clusters

> *In the hierarchical model of structure formation, galaxy clusters are the largest and the youngest structures in the universe. They form through the assembly of smaller-scale structures, and for this reason, they are often observed during mergers.* — M. Meneghetti, *Introduction to Gravitational Lensing* (2021), Sect. 5.7

This final module brings together the entire gravitational lensing framework developed in Modules 1–9 and applies it to the most spectacular setting: **galaxy clusters**. Where a single galaxy lens (Module 9) typically produces two or four images of a background source with an Einstein radius of $\sim 1''$, a massive galaxy cluster can produce dozens of multiple image systems, giant arcs with length-to-width ratios exceeding 10:1, and Einstein radii of $\sim 10''$–$50''$.

Galaxy clusters have total masses of 10^{14}–$10^{15}\,M_\odot$, with $\sim 85\%$ in dark matter, $\sim 13\%$ in hot intracluster gas, and only $\sim 2\%$ in stars. Their mass distributions are well-described by the NFW profile introduced in Module 11, now applied at scales 100–1000 times more massive than individual galaxy halos. The richness of cluster lensing — many image systems at multiple source redshifts — provides powerful constraints on the dark matter distribution and on cosmological parameters.

Companion Mathematica files:

- `cluster_lensing.wl` — NFW cluster lensing quantities; critical curves and caustics at cluster scale; ray-tracing through cluster potential to generate arcs; Einstein radius vs. cluster mass; figures exported to `Figures/10_Cluster_Lensing/`
- `nfw_cluster.nb` — interactive notebook with `Manipulate` widgets for NFW cluster parameters, giant arc formation, magnification maps, and weak lensing shear profiles

14.1 Introduction: Why Cluster Lensing Is Special

Galaxy clusters occupy a unique place in both astrophysics and gravitational lensing. They are the largest objects in the universe whose properties are determined by gravity alone — they sit at the top of the hierarchy of cosmic structure. Their enormous masses make them the most powerful gravitational lenses, producing phenomena that are qualitatively different from galaxy-scale lensing:

1. **Giant arcs.** Extended background galaxies are stretched into luminous arcs tens of arcseconds long, tracing out the tangential critical curve of the cluster potential. The first giant arcs were discovered independently by Lynds & Petrosian (1986) and Soucail et al. (1987), with the gravitational lensing interpretation confirmed spectroscopically by Soucail et al. (1988).

2. **Many multiple image systems.** A single massive cluster can produce 10–30 or more distinct sets of multiple images, each from a different background galaxy at a different redshift. Each set independently constrains the cluster mass model.

3. **Extreme magnification.** Magnification factors of $\mu \sim 10$–100 are common near the critical curves of clusters. This turns clusters into "gravitational telescopes" that reveal galaxies and even individual stars at redshifts $z > 6$ that would otherwise be undetectable.

4. **Sensitivity to dark matter.** Because clusters are dark-matter dominated, their lensing signal directly probes the dark matter distribution — its radial profile, substructure, and overall mass.

5. **Cosmological probe.** The abundance and properties of cluster lenses depend on cosmological parameters ($\Omega_{\rm m}$, σ_8, the dark energy equation of state), making cluster lensing a tool for precision cosmology.

14.2 NFW Model for Cluster Lensing

In Module 11 (Section 11.6) we introduced the NFW density profile as the universal dark matter halo profile. Here we apply it at **cluster scale**, where the relevant parameters are much larger than for galaxy halos.

14.2.1 Cluster-Scale NFW Parameters

The NFW profile (eq. 11.35) is parametrized by the virial mass M_{200} and concentration $c = r_{200}/r_s$. Typical values for galaxy clusters are:

Parameter	Galaxy halo	Cluster halo
M_{200}	$10^{12}\,M_\odot$	10^{14}–$10^{15}\,M_\odot$
c	10–20	3–8
r_s	15–30 kpc	200–500 kpc
r_{200}	200–300 kpc	1–2 Mpc

The key difference is that clusters are *less concentrated* ($c \sim 3$–8) than galaxy halos ($c \sim 10$–20), reflecting their later formation epoch in the hierarchical structure formation scenario. Despite the lower concentration, the vastly greater mass more than compensates, producing far stronger lensing.

14.2.2 Lensing Quantities at Cluster Scale

The NFW convergence, deflection angle, and shear from Module 11 (eqs. 11.47–11.54) apply directly. We collect the key formulae here for reference. Define $x = \theta/\theta_s$, where $\theta_s = r_s/D_{\rm d}$ is the angular scale radius. Then:

$$\kappa(x) = \kappa_s\, f(x)\,, \tag{14.1}$$

$$\alpha(x) = \frac{2\kappa_s\,\theta_s}{x}\left[\ln\frac{x}{2} + g(x)\right]\,, \tag{14.2}$$

$$\bar\kappa(x) = \frac{2\kappa_s}{x^2}\left[\ln\frac{x}{2} + g(x)\right]\,, \qquad |\gamma(x)| = \bar\kappa(x) - \kappa(x)\,, \tag{14.3}$$

with the functions $f(x)$ and $g(x)$ defined in eqs. (11.46)–(11.52), and the characteristic convergence:

$$\kappa_s = \frac{2\rho_s\, r_s}{\Sigma_{\rm cr}}\,. \tag{14.4}$$

Example 14.2.1. Characteristic convergence for a massive cluster. For a cluster with $M_{200} = 10^{15}\,M_\odot$, $c = 5$, at $z_d = 0.3$ lensing a source at $z_s = 2$ ($D_{\rm d} \approx 919$ Mpc, $D_{\rm s} \approx 1727$ Mpc, $D_{\rm ds} \approx 1328$ Mpc, from Module 7):

- $r_{200} \approx 1.86$ Mpc, $r_s = r_{200}/c \approx 372$ kpc,
- $\theta_s = r_s/D_{\rm d} \approx 84''$,
- $\kappa_s \approx 0.51$.

The local convergence $\kappa = \kappa_s f(x)$ exceeds unity for $x \lesssim 0.11$ (i.e., $\theta \lesssim 9''$), and the mean convergence $\bar\kappa$ reaches unity at $x \approx 0.18$ ($\theta \approx 15''$), so the cluster has a central region of super-critical convergence and is therefore a strong lens. These values are computed and verified in `cluster_lensing.wl`.

14.2.3 Critical Curves and Caustics for Cluster NFW

For the circularly symmetric NFW profile, the tangential and radial critical curves are circles. The tangential critical curve occurs where $\bar\kappa(x) = 1$, i.e.:

$$\frac{2\kappa_s}{x_t^2}\left[\ln\frac{x_t}{2} + g(x_t)\right] = 1\,, \tag{14.5}$$

and the radial critical curve where $2\kappa(x_r) - \bar\kappa(x_r) = 1$, i.e.:

$$2\kappa_s\, f(x_r) - \frac{2\kappa_s}{x_r^2}\left[\ln\frac{x_r}{2} + g(x_r)\right] = 1\,. \tag{14.6}$$

Both equations must be solved numerically. For the cluster parameters above ($M_{200} = 10^{15}\,M_\odot$, $c = 5$, $z_d = 0.3$, $z_s = 2$), the tangential critical curve lies at $x_t \approx 0.18$, giving $\theta_t \approx 15''$ — a physical radius of ~ 66 kpc at the cluster redshift.

Remark 14.2.1. The tangential critical curve radius is often called the **Einstein radius** of the cluster, $\theta_{\rm E}$. A *spherical* NFW halo with $M_{200} \sim 10^{15}\,M_\odot$ and $c \sim 5$ typically has $\theta_{\rm E} \sim 10''$–$20''$ at these redshifts. Observed clusters (Abell 1689, MACSJ 0717, etc.) frequently

display $\theta_{\rm E} \sim 30''$–$50''$; this additional lensing strength comes from ellipticity, substructure, merger boosts, and departures of the true mass distribution from a smooth spherical NFW. For a galaxy lens the Einstein radius is instead $\sim 1''$, and the enclosed projected mass within $\theta_{\rm E}$ ranges from $\sim 10^{11}\,M_\odot$ (galaxies) to $\sim 10^{13}$–$10^{14}\,M_\odot$ (clusters).

14.2.4 Einstein Radius Scaling with Cluster Mass

The Einstein radius of an NFW cluster depends on mass, concentration, and the lens/source redshift configuration. For a fixed concentration and redshift configuration, the scaling is approximately:

$$\theta_{\rm E} \propto M_{200}^{1/3}\,, \tag{14.7}$$

which arises because $\theta_{\rm E} \sim r_{200}/D_{\rm d}$ and $r_{200} \propto M_{200}^{1/3}$ (from the definition $M_{200} = (4\pi/3)\,200\,\rho_{\rm crit}\,r_{200}^3$). The actual relationship is more complex due to the dependence of κ_s on c and the nonlinear solution of eq. (14.5), but the cube-root scaling provides a useful approximation.

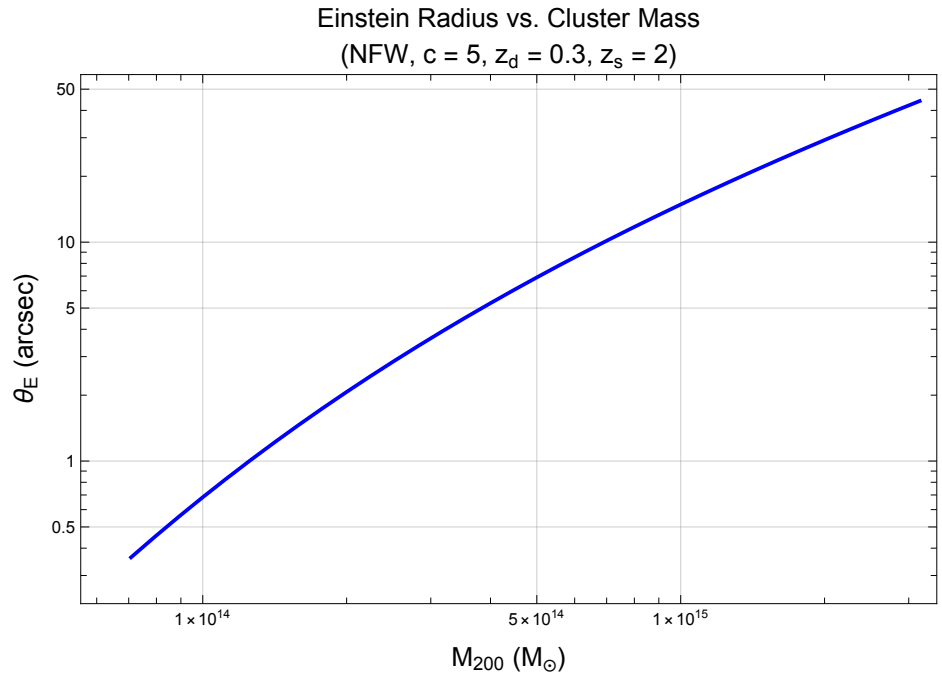


Figure 14.1: Einstein radius $\theta_{\rm E}$ as a function of cluster mass M_{200} for an NFW profile with $c = 5$, $z_d = 0.3$, and $z_s = 2$. The dashed line shows the approximate $M^{1/3}$ scaling. Clusters below $M_{200} \sim 10^{14}\,M_\odot$ have Einstein radii comparable to galaxy-scale lenses. Generated by `cluster_lensing.wl`.

14.3 Giant Arcs

Giant luminous arcs are the most visually striking signature of cluster strong lensing. They were first discovered by Lynds & Petrosian (1986) and Soucail et al. (1987) in the clusters

Abell 370 and Cl 2244-02, and their interpretation as gravitationally lensed images of background galaxies was confirmed spectroscopically by Soucail et al. (1988).

14.3.1 Formation Mechanism

Giant arcs form when an extended background galaxy lies close to the **tangential caustic** of the cluster. From the theory of critical curves (Module 12, Section 14.3):

1. The source galaxy is near (but not exactly on) the tangential caustic in the source plane.

2. Its images lie near the tangential critical curve in the image plane, where $\lambda_- \approx 0$.

3. The images are stretched *tangentially* (along the critical curve) by a large factor $\mu_{\rm tan} \propto 1/\sqrt{d_\perp}$, where $d_\perp$ is the perpendicular distance from the caustic.

4. The radial magnification remains of order unity, so the image is a thin, elongated **arc** that traces the critical curve.

14.3.2 Ray-Tracing Through a Cluster Potential

To visualize arc formation, we **ray-trace** an extended source through the cluster lens equation. The procedure is:

1. Define the source as a set of points in the source plane, e.g., a circular disk of angular radius β_0 centered at position $\boldsymbol{\beta}_0$.

2. For each source point $\boldsymbol{\beta}$, solve the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ for all image positions $\boldsymbol{\theta}$. For a cluster, this is done numerically on a fine grid.

3. Plot the union of all image positions to obtain the lensed image of the source.

In practice, it is more efficient to use the **inverse ray-tracing** approach: for each pixel $\boldsymbol{\theta}$ in the image plane, compute $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ and check whether $\boldsymbol{\beta}$ falls within the source. If it does, that pixel is part of a lensed image.

14.3.3 Length-to-Width Ratio

The **length-to-width ratio** (or axis ratio) of an arc is a direct measure of the tangential magnification:

$$\frac{L}{W} \approx \mu_{\rm tan} = \frac{1}{|1 - \bar{\kappa}(\theta)|} \propto \frac{1}{\sqrt{d_\perp}}\,, \tag{14.8}$$

where L and W are the length and width of the arc. Observationally, arcs with $L/W \geq 10$ are classified as "giant arcs", while those with $3 \leq L/W < 10$ are often called "arclets".

14.3.4 Arc Statistics

The probability that a cluster produces a giant arc depends on:

- **Cluster mass:** More massive clusters have larger Einstein radii and larger tangential caustics, increasing the cross-section for arc production. The arc cross-section scales steeply with mass, roughly as $\sigma_{\rm arc} \propto M_{200}^{2-3}$.
- **Source redshift:** Higher source redshifts increase $D_{\rm ds}/D_{\rm s}$, which increases κ_s and the Einstein radius.
- **Cluster ellipticity and substructure:** Non-axisymmetric mass distributions and merging clusters enhance the arc cross-section significantly, often by a factor of 2–5.
- **Source size and morphology:** Extended sources are more easily stretched into arcs; compact sources are more likely to appear as multiple point-like images.

Remark 14.3.1. Arc statistics have been a source of tension with cosmological models. Bartelmann et al. (1998) found that the observed number of giant arcs exceeded the prediction from ΛCDM by a factor of ∼10. This "arc statistics problem" has been partially resolved by accounting for cluster ellipticity, substructure, and the contribution of baryonic physics (cooling, AGN feedback) to the central mass concentration, but it remains an active area of research.

14.4 Multiple Image Systems in Clusters

One of the most powerful aspects of cluster lensing is that a single cluster can lens **many** background galaxies simultaneously, producing multiple sets of multiple images. Each set of images (called an "image family" or "image system") comes from a single background source at a particular redshift.

14.4.1 Image Families

A typical strong lensing cluster produces:

- **5–10 secure image families** with spectroscopic redshift confirmation in well-studied clusters.
- **20–50 candidate image families** identified by morphology, color, and photometric redshifts in deep imaging surveys.
- In extreme cases (e.g., Abell 1689), over 100 candidate images have been identified from more than 30 distinct background sources.

Each image family provides independent constraints on the cluster mass model:

1. The **image positions** constrain the deflection field $\boldsymbol{\alpha}(\boldsymbol{\theta})$ at several points.

2. The **magnification ratios** (flux ratios between images) constrain the convergence and shear fields.

3. The **redshift** of each family provides a distance ratio $D_{\rm ds}/D_{\rm s}$ that weights the lensing signal differently, breaking degeneracies in the mass model.

14.4.2 Spectroscopic Confirmation

Image families are confirmed by matching the redshifts of candidate images: if multiple images at different positions have the same spectroscopic redshift, they are identified as images of the same source. Modern multi-object spectrographs (VLT/MUSE, Keck/DEIMOS) have been transformative for cluster lensing, enabling redshift measurements of faint, highly magnified background galaxies.

14.4.3 Famous Cluster Lenses

Several galaxy clusters have become benchmarks for strong lensing studies:

- **Abell 370** ($z = 0.375$): Site of the first discovered giant arc (Soucail et al. 1987). Over 10 multiple image systems identified.

- **Abell 1689** ($z = 0.183$): One of the most powerful known lenses, with over 100 images from ∼30 background sources. Its Einstein radius of $\sim 47''$ (for $z_s \sim 2$) is among the largest known.

- **MACSJ0416.1-2403** ($z = 0.396$): A Hubble Frontier Field cluster with over 60 spectroscopically confirmed multiple images, used for detailed mass reconstruction.

- **Abell 2744** ($z = 0.308$): The "Pandora's Cluster", a complex merging system with extensive strong and weak lensing constraints.

14.5 Cluster Mass Estimation

Gravitational lensing provides a *direct* measurement of the projected mass distribution of a cluster, independent of assumptions about the dynamical state of the system. This makes it complementary to X-ray and kinematic mass estimates, which rely on assumptions of hydrostatic or virial equilibrium.

14.5.1 Strong Lensing Mass Estimates

Strong lensing constrains the mass distribution in the **inner region** of the cluster, typically within $\sim$ 200–300 kpc of the center. The basic approach is:

1. **Identify multiple image systems** and measure their positions and redshifts.

2. **Parametrize the mass model.** The cluster mass is described by one or more NFW halos (for the smooth dark matter component) plus additional components for cluster member galaxies (typically scaled isothermal profiles).

3. **Optimize the model parameters** by minimizing the difference between predicted and observed image positions. Modern codes (e.g., `lenstool`, `glafic`) use Bayesian methods (MCMC) to explore the parameter space.

4. **Derive the mass profile** from the best-fit model.

A quick estimate of the enclosed mass within the Einstein radius is provided by:

$$\boxed{M(\theta \leq \theta_{\rm E}) = \pi \left(D_{\rm d}\, \theta_{\rm E}\right)^2 \Sigma_{\rm cr}} \tag{14.9}$$

since $\bar{\kappa}(\theta_{\rm E}) = 1$ by definition. For a cluster with $\theta_{\rm E} = 30''$ at $z_d = 0.3$ with $z_s = 2$:

$$M(\theta_{\rm E}) \approx 1.3 \times 10^{14}\, M_\odot\,. \tag{14.10}$$

This is the projected mass within a cylinder of radius ~ 130 kpc — a substantial fraction of the total cluster mass. The exact value is computed in `cluster_lensing.wl`.

14.5.2 Weak Lensing Mass Estimates (Preview)

Strong lensing probes only the inner $\sim$10% of the cluster virial radius. To measure the total cluster mass, one must extend to larger radii using **weak lensing** (Section 14.7).

14.5.3 Combining Strong and Weak Lensing

The most accurate cluster mass profiles come from combining strong and weak lensing:

- **Strong lensing** provides high-resolution constraints on the inner mass profile ($r \lesssim$ 200 kpc), including the NFW concentration parameter.
- **Weak lensing** provides constraints at larger radii (200 kpc $\lesssim r \lesssim$ 3 Mpc), determining the virial mass M_{200}.
- The combined profile spans nearly two decades in radius and is well-fit by an NFW profile, supporting the universality of the dark matter halo profile.

14.5.4 Comparison with Other Mass Estimators

Cluster masses can also be estimated from:

- **X-ray observations:** The hot intracluster medium (ICM) emits thermal bremsstrahlung X-rays. Under the assumption of hydrostatic equilibrium, the gas temperature and density profiles yield the total mass profile. Hydrostatic masses are typically 10–20% lower than lensing masses (the "hydrostatic mass bias"), likely due to non-thermal pressure support from turbulence and bulk motions.

- **Galaxy velocity dispersion:** The velocities of cluster member galaxies, combined with the virial theorem, yield a mass estimate. This method is noisy (requiring many galaxy redshifts) and sensitive to interlopers.
- **Sunyaev–Zel'dovich (SZ) effect:** Inverse Compton scattering of CMB photons by the hot ICM produces a spectral distortion. The integrated SZ signal is proportional to the total thermal energy and correlates with total mass.

Remark 14.5.1. The complementarity of these methods is powerful. Comparing lensing masses (which are model-independent) with X-ray/SZ masses (which assume hydrostatic equilibrium) calibrates the hydrostatic mass bias, which is a critical systematic in using cluster counts for cosmology.

14.6 Magnification and the Cosmic Telescope

Perhaps the most exciting application of cluster lensing in contemporary astrophysics is the use of massive clusters as **gravitational telescopes**: natural magnifying glasses that amplify the light of distant background galaxies, allowing us to study objects that would otherwise be far too faint to detect.

14.6.1 Magnification by Clusters

Near the critical curves of a massive cluster, the magnification can reach $\mu \sim 10$–100 or more. Since the observed flux scales as μ (for sources smaller than the scale over which μ varies), a source at $z \sim 10$ behind a cluster lens can appear as bright as an unmagnified source at $z \sim 3$–4.

The trade-off is that magnification also reduces the effective area in the source plane. A magnification of μ enlarges the image area by μ but surveys a source-plane area that is μ times smaller. The net effect on the number of detected sources depends on the slope of the luminosity function — this is the **magnification bias**.

14.6.2 Magnification Bias

The magnification bias arises because lensing simultaneously:

1. **Amplifies flux:** Sources below the detection threshold are boosted above it, increasing the detected number.
2. **Dilutes area:** The effective survey area is reduced by $1/\mu$, decreasing the detected number.

For a cumulative luminosity function $N(>S) \propto S^{-\alpha}$ (where S is flux and α is the faint-end slope), the net number count behind a lens changes by a factor of $\mu^{\alpha-1}$. For $\alpha > 1$ (steep luminosity function), magnification bias *increases* the number of detected sources; for $\alpha < 1$, it *decreases* the count. At high redshift, the luminosity function is steep ($\alpha \gtrsim 2$), so there is a significant net increase in detected galaxies behind clusters.

14.6.3 The Hubble Frontier Fields

The Hubble Frontier Fields (HFF) program (2013–2017) was a major HST initiative that obtained ultra-deep imaging of six massive galaxy clusters chosen for their strong lensing power:

1. Abell 2744 ($z = 0.308$)
2. MACSJ0416.1-2403 ($z = 0.396$)
3. MACSJ0717.5+3745 ($z = 0.545$)
4. MACSJ1149.5+2223 ($z = 0.543$)
5. Abell S1063 ($z = 0.348$)
6. Abell 370 ($z = 0.375$)

These clusters were used as gravitational telescopes to detect galaxies at $z \sim 6$–10, probing the epoch of reionization. The HFF program discovered some of the most distant galaxies known at the time and provided detailed mass models of the six clusters from their many multiple image systems.

14.6.4 JWST and the New Era of Cluster Lensing

The James Webb Space Telescope (JWST, launched 2021) has revolutionized cluster lensing with its superior infrared sensitivity and angular resolution. Notable results include:

- **Earendel** (WHL0137-LS): The most distant individual star known, at $z = 6.2$, magnified by a factor of $\mu \sim 1000$–$40{,}000$ by the cluster WHL 0137-08 [35]. The extreme magnification occurs because the star lies almost exactly on the cluster's critical curve.
- **The Sunrise Arc**: A galaxy at $z = 6.2$ lensed into a long arc by the same cluster, in which Earendel was identified as a point source on the arc.
- **Early galaxy candidates at** $z > 10$: Lensed galaxies behind clusters observed with JWST/NIRCam have pushed the redshift frontier beyond $z \sim 13$, with magnifications of $\mu \sim 5$–30 making these detections possible.

Remark 14.6.1. The discovery of Earendel illustrates the extreme power of cluster lensing. Without magnification, Earendel would have an apparent magnitude of $m \approx 35$–39 — utterly undetectable. A magnification of $\mu \sim 1000$–$40{,}000$ corresponds to $\Delta m \approx -7.5$ to -11.5 magnitudes, bringing it to the observed $m \approx 27.2$, within reach of HST and JWST [35]. Such extreme magnification requires the image to lie within a few hundredths of an arcsecond of the critical curve ($\lesssim 0.036''$ in the Welch et al. [35] lens models), making such discoveries exceedingly rare.

14.7 Brief Introduction to Weak Lensing

Throughout Modules 1–9 and the preceding sections of this module, we have focused on **strong lensing**: the regime where $\kappa \gtrsim 1$ and multiple images or giant arcs are produced. But gravitational lensing does not stop at the strong lensing regime. At larger radii from the cluster center (and behind foreground mass distributions in general), the lensing effect is

weak — too small to produce multiple images, but still measurable statistically. This is the domain of **weak gravitational lensing**.

This section provides a brief introduction to weak lensing — just enough to show where the field goes beyond strong lensing and to connect to the rich literature on the subject. A full treatment of weak lensing is beyond the scope of these tutorials.

14.7.1 The Weak Lensing Regime

In the weak lensing regime, $\kappa \ll 1$ and $|\gamma| \ll 1$. There are no multiple images or giant arcs. Instead, the primary observable effect is a small, coherent **distortion** (shearing) of the shapes of background galaxies. An intrinsically circular galaxy is sheared into a slightly elliptical shape, with the orientation and magnitude of the induced ellipticity determined by the local shear field.

The key challenge is that individual galaxy shapes are intrinsically elliptical, with typical intrinsic ellipticities of $|\epsilon_{\rm int}| \sim 0.2$–$0.3$. The gravitational shear signal is $|\gamma| \sim 0.01$–0.05 in the weak lensing regime — much smaller than the intrinsic shape noise. Therefore, one must **average** over many background galaxies to extract the coherent shear signal.

14.7.2 The Reduced Shear

In the weak lensing regime, the observable quantity is the **reduced shear**:

$$\boxed{g = \frac{\gamma}{1-\kappa}} \tag{14.11}$$

This is because the lensing transformation preserves surface brightness, and the observed ellipticity of a lensed galaxy (for $|g| \leq 1$) is:

$$\epsilon^{\rm obs} = \frac{\epsilon^{\rm int} + g}{1 + g^* \, \epsilon^{\rm int}} \approx \epsilon^{\rm int} + g \,, \tag{14.12}$$

where the approximation holds for $|g| \ll 1$. Since intrinsic ellipticities are randomly oriented (in the absence of intrinsic alignments), their average vanishes: $\langle \epsilon^{\rm int} \rangle \approx 0$. Therefore:

$$\langle \epsilon^{\rm obs} \rangle \approx g \approx \gamma \quad (\text{for } \kappa \ll 1) \,. \tag{14.13}$$

The mean observed ellipticity of background galaxies is an unbiased estimator of the shear (in the weak limit).

14.7.3 Tangential Shear Profile Around Clusters

For a circularly symmetric mass distribution, the shear is purely tangential. The **tangential shear** at angular distance θ from the cluster center is:

$$\gamma_t(\theta) = \bar{\kappa}(\theta) - \kappa(\theta) \,, \tag{14.14}$$

which we recognize from eq. (11.7) (Module 11). The tangential shear profile $\gamma_t(\theta)$ can be measured by averaging the tangential component of the ellipticities of background galaxies in annular bins around the cluster center.

For an NFW profile, $\gamma_t(\theta)$ has a characteristic shape that rises toward the center (reaching the strong lensing regime) and falls off at large radii. Fitting the observed $\gamma_t(\theta)$ profile to the NFW prediction yields the cluster mass M_{200} and concentration c.

14.7.4 From Clusters to Cosmic Shear

The weak lensing framework extends far beyond individual cluster halos. The **cosmic shear** signal — the coherent shearing of galaxy shapes by the entire large-scale structure of the universe along the line of sight — is one of the primary probes of modern cosmology. Cosmic shear measurements constrain the matter power spectrum, the matter density $\Omega_{\rm m}$, and the amplitude of matter fluctuations σ_8.

Current and upcoming surveys measuring cosmic shear include:

- **Euclid** (ESA, launched 2023): space-based imaging of ∼1.5 billion galaxies over 15,000 deg^2.
- **Vera C. Rubin Observatory / LSST**: ground-based survey of ∼4 billion galaxies over 18,000 deg^2 (first light expected 2025).
- **Nancy Grace Roman Space Telescope** (NASA, expected launch 2027): space-based survey complementing Euclid.

These surveys will measure cosmic shear with percent-level precision, providing powerful constraints on dark energy, modified gravity, and the neutrino mass hierarchy.

14.8 The State of the Field and Future Directions

Strong lensing by galaxy clusters is a mature but rapidly evolving field. The combination of new observational facilities and sophisticated modeling techniques continues to push the boundaries of what cluster lensing can teach us.

14.8.1 Current Observational Facilities

- **HST:** The Hubble Space Telescope has been the workhorse for cluster strong lensing for three decades, with programs like the Hubble Frontier Fields and RELICS (Reionization Lensing Cluster Survey) producing the deepest cluster lensing data.

- **JWST:** With its infrared capabilities, JWST is discovering lensed galaxies at $z > 10$, measuring spectroscopic redshifts of multiple image systems, and resolving detailed structure in giant arcs.

- **Euclid:** The wide-field survey will discover thousands of new cluster lenses, dramatically increasing the statistical power of cluster lensing studies.

- **Rubin/LSST:** The deep, wide survey will combine strong and weak lensing for a large sample of clusters, with time-domain capability for detecting lensed transients.

14.8.2 Future Facilities

- **Nancy Grace Roman Space Telescope:** High-resolution, wide-field infrared imaging will complement JWST for cluster lensing studies.
- **Square Kilometre Array (SKA):** Radio weak lensing with the SKA will provide shape measurements free from many of the systematics that affect optical measurements (atmospheric seeing, PSF modeling).
- **Extremely Large Telescopes** (ELT, TMT, GMT): Ground-based 25–39m telescopes with adaptive optics will resolve sub-arcsecond structure in cluster arcs and obtain spectra of extremely faint lensed sources.

14.8.3 Open Questions

1. **Dark matter substructure:** CDM predicts a rich hierarchy of subhalos within clusters. The number and mass function of subhalos affect the lensing signal (flux ratio anomalies, perturbations to giant arcs). Comparing predictions to observations constrains alternative dark matter models (warm dark matter, self-interacting dark matter).
2. **The nature of dark matter:** The inner density profile of clusters (cusp vs. core) is sensitive to dark matter self-interactions. Strong lensing at the center of clusters provides the highest-resolution probe of the dark matter profile.
3. **The cluster mass function:** The abundance of massive clusters as a function of mass and redshift is a sensitive cosmological probe. Lensing-calibrated cluster masses are essential for turning cluster counts into cosmological constraints.
4. **Multi-plane lensing and line-of-sight effects:** Light rays from distant sources pass through multiple mass concentrations along the line of sight. Accounting for these "line-of-sight" effects is important for precision mass modeling.
5. **Time-delay cosmography with clusters:** Time delays between multiple images of time-varying sources (supernovae, quasars) in clusters constrain H_0. The multiply imaged supernova "Refsdal" in MACSJ1149 (Kelly et al. 2015) was the first such measurement.

14.8.4 Connection to the AGEL Project

The ASTRO 3D Galaxy Evolution with Lensing (AGEL) survey is a spectroscopic survey of galaxy-scale strong lens candidates selected from wide-field imaging surveys. While AGEL focuses on galaxy-scale lenses rather than clusters, many of the techniques and physical principles are the same:

- NFW profiles and isothermal models describe the lens mass distribution at both galaxy and cluster scales.
- Critical curves and caustics determine the image configurations at all mass scales.

- Strong lensing mass estimates and comparisons with dynamical masses apply to both galaxy and cluster lenses.

The transition from galaxy to cluster lensing is one of scale rather than fundamental physics — the same equations derived in Modules 1–9 govern both regimes.

14.9 Summary

- **Galaxy clusters** ($M_{200} \sim 10^{14}$–$10^{15}\,M_\odot$) are the most powerful gravitational lenses, producing giant arcs, dozens of multiple image systems, and magnification factors of $\mu \sim 10$–100.
- The **NFW profile** at cluster scale has $c \sim 3$–8 and $r_s \sim 200$–500 kpc. The Einstein radius is $\theta_{\rm E} \sim 10''$–$50''$, enclosing a projected mass of $\sim 10^{14}\,M_\odot$.
- **Giant arcs** form when extended sources lie near the tangential caustic. The length-to-width ratio $L/W \approx \mu_{\rm tan} \propto 1/\sqrt{d_\perp}$ can exceed 10:1 for sources very close to the caustic.
- **Multiple image systems** (10–30+ per cluster) each independently constrain the mass model. Spectroscopic confirmation via redshift matching identifies image families.
- **Cluster mass estimation** uses strong lensing (inner profile) and weak lensing (outer profile), complemented by X-ray and SZ measurements. Strong lensing gives $M(\theta_{\rm E}) = \pi(D_{\rm d}\,\theta_{\rm E})^2\,\Sigma_{\rm cr}$.
- Clusters serve as **gravitational telescopes**, magnifying high-redshift galaxies and enabling discoveries like Earendel ($z = 6.2$, $\mu \sim 10^3$–10^4) and galaxies at $z > 10$.
- **Weak lensing** measures the statistical shear $\langle\epsilon\rangle \approx g = \gamma/(1-\kappa)$ of background galaxies. The tangential shear profile $\gamma_t(\theta)$ yields the cluster mass profile at large radii.
- **Current and future facilities** (JWST, Euclid, Rubin/LSST, Roman, SKA) will transform cluster lensing with larger samples, deeper imaging, and new wavelength regimes.

Where to Go from Here

This module concludes the *Learning to Lens* tutorial suite. We have covered the foundations of general relativity (Modules 1a–1e), the theory of gravitational lensing (Modules 2–6), and the modeling and application of strong lensing (Modules 7–10). For readers who wish to continue deeper into the field, we suggest the following directions:

- **Advanced topics not covered in these tutorials:** dark matter substructure and flux ratio anomalies; line-of-sight effects and multi-plane lensing; free-form (non-parametric) mass reconstruction; lens modeling with extended source morphology; microlensing by stars in the lensing galaxy.

- **Software packages for lens modeling:**
  - `lenstronomy` (https://github.com/lenstronomy/lenstronomy): a comprehensive Python package for gravitational lens modeling and simulation.
  - `PyAutoLens` (https://github.com/Jammy2211/PyAutoLens): automated strong lens modeling with Bayesian inference.
  - `glafic` (https://www.slac.stanford.edu/~oguri/glafic/): a fast lens modeling code widely used for cluster lensing.
  - `GRALE` (https://research.edm.uhasselt.be/~jlieMDC/): free-form mass reconstruction using a genetic algorithm.

- **Key review articles for further reading:** Kneib & Natarajan (2011), "Cluster lenses", A&A Rev.; Treu (2010), "Strong Lensing by Galaxies", ARA&A; Bartelmann & Schneider (2001), "Weak Gravitational Lensing", Phys. Rep.; Kilbinger (2015), "Cosmology with Cosmic Shear", Rep. Prog. Phys.

- **Active research:** The AGEL project (Tran et al. 2022) exemplifies how the techniques developed in these tutorials are applied in practice. Finding, confirming, and modeling galaxy-scale strong lenses in wide-field surveys is an active area of research to which the reader is now well-prepared to contribute.

14.10 Problems

Exercise 14.10.1. (NFW cluster Einstein radius and enclosed mass.) Consider an NFW cluster with $M_{200} = 5 \times 10^{14}\, M_\odot$, $c = 5$, at $z_d = 0.3$ lensing a source at $z_s = 2$. Use $D_\mathrm{d} = 919$ Mpc, $D_\mathrm{s} = 1727$ Mpc, $D_\mathrm{ds} = 1328$ Mpc.

(a) Compute the NFW parameters: r_{200}, r_s, ρ_s, θ_s, and κ_s.

(b) Find the tangential critical curve radius θ_t (the Einstein radius θ_E) by solving $\bar{\kappa}(\theta_t) = 1$ numerically.

(c) Compute the projected mass enclosed within θ_E using eq. (14.9).

(d) Compare this enclosed mass to the total virial mass M_{200}. What fraction of the cluster mass is enclosed within the Einstein radius?

Verify with `problems_10.wl`.

Exercise 14.10.2. (Ray-tracing a source through a cluster potential.) Using the NFW cluster from Exercise 14.10.1:

(a) Set up inverse ray-tracing: for each pixel $\boldsymbol{\theta}$ in the image plane, compute $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ using the NFW deflection angle (eq. 14.2).

(b) Place a circular source of angular radius $0.5''$ at three positions: (i) $\beta = 0$ (on the optical axis), (ii) $\beta = 0.5\,\theta_\mathrm{E}$ (halfway to the critical curve), and (iii) $\beta = \theta_\mathrm{E}$ (on the tangential caustic).

(c) Plot the resulting lensed images for each source position. Identify the Einstein ring, giant arc, and counter-image morphologies.

(d) Estimate the length-to-width ratio of the arc in case (iii).

Verify with `problems_10.wl`.

Exercise 14.10.3. (Magnification of a high-redshift galaxy.) Consider a galaxy at $z_s = 9$ lensed by a cluster at $z_d = 0.3$ with $\theta_{\rm E} = 30''$. (For reference, $D_{\rm d} = 919$ Mpc, $D_{\rm s} = 921$ Mpc and $D_{\rm ds} = 802$ Mpc; at $z_s = 9$ the angular-diameter distance has turned over, so $D_{\rm s}$ is barely larger than $D_{\rm d}$. Parts (a)–(c) need only $\theta_{\rm E}$.)

(a) For an SIS model with the same Einstein radius, compute the magnification of a point source at $\beta = 3''$, $1''$, and $0.1''$.

(b) What is the apparent magnitude boost (Δm) for a source at $\beta = 0.1''$ from the lens center? ($\Delta m = -2.5\,\log_{10}\mu$.)

(c) At $z_s = 9$, an L_* galaxy has an apparent magnitude of $m \approx 31$ (AB) in the near-infrared. If the detection limit of JWST is $m \approx 29$, what minimum magnification is needed to detect such a galaxy? At what source position β does this magnification occur (for the SIS model)?

(d) Discuss the implications for surveys of high-redshift galaxies behind clusters.

Verify with `problems_10.wl`.

Exercise 14.10.4. (Einstein radii across mass scales.) Compare the Einstein radii of three lenses at $z_d = 0.3$ with a source at $z_s = 2$, using $D_{\rm d} = 919$ Mpc, $D_{\rm s} = 1727$ Mpc, $D_{\rm ds} = 1328$ Mpc:

(a) A galaxy lens modeled as an SIS with $\sigma_v = 250$ km s^{-1} ($M \approx 10^{12}\,M_\odot$).

(b) A group lens modeled as an SIS with $\sigma_v = 500$ km s^{-1} ($M \approx 10^{13}\,M_\odot$).

(c) A cluster lens modeled as an SIS with $\sigma_v = 1200$ km s^{-1} ($M \approx 10^{15}\,M_\odot$).

(d) Compute $\theta_{\rm E}$ for each case. By what factor does $\theta_{\rm E}$ increase from galaxy to cluster?

(e) Compute the enclosed mass within $\theta_{\rm E}$ for each case. Compare with the total mass of each object.

(f) Discuss why the scaling $\theta_{\rm E} \propto \sigma_v^2$ (from the SIS formula) leads to a qualitative difference in lensing phenomenology between galaxies and clusters.

Verify with `problems_10.wl`.

Exercise 14.10.5. (Weak lensing mass estimate.) A cluster at $z_d = 0.3$ has a measured tangential shear profile $\gamma_t(\theta) = 0.1 \times (\theta/1')^{-0.8}$.

(a) Using the relation $\gamma_t(\theta) = \bar{\kappa}(\theta) - \kappa(\theta)$, and assuming that $\kappa \ll \bar{\kappa}$ at large radii (so $\gamma_t \approx \bar{\kappa}$), estimate the mean convergence at $\theta = 1'$, $3'$, and $5'$.

(b) Convert $\bar{\kappa}(\theta)$ to the projected mass enclosed within θ: $M(\theta) = \pi\,(D_{\rm d}\,\theta)^2\,\Sigma_{\rm cr}\,\bar{\kappa}(\theta)$. Use $\Sigma_{\rm cr} \approx 2.8 \times 10^{15}\,M_\odot\,{\rm Mpc}^{-2}$ for $z_d = 0.3$, $z_s = 1$ (a typical background redshift for weak lensing), and $D_{\rm d} = 919$ Mpc.

(c) Estimate M within $\theta = 5'$ (corresponding to a physical radius of ~ 1.3 Mpc at $z_d = 0.3$).

(d) Compare your estimated mass with a typical cluster virial mass of $5 \times 10^{14}\,M_\odot$. Discuss whether the result is reasonable and what assumptions may affect the accuracy.

Verify with `problems_10.wl`.

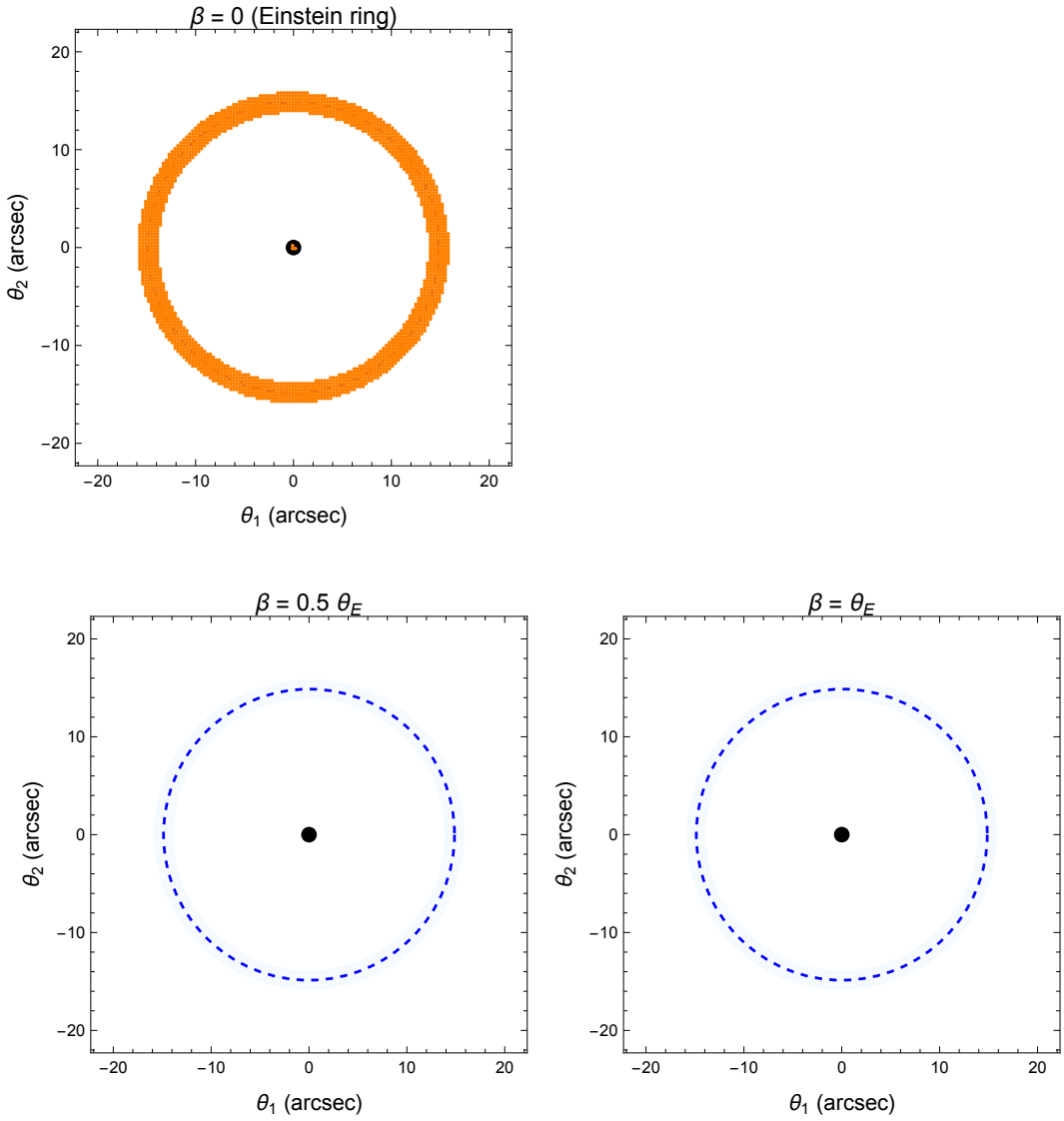


Figure 14.2: Giant arc formation by an NFW cluster ($M_{200} = 10^{15}\,M_\odot$, $c = 5$, $z_d = 0.3$, $z_s = 2$). A circular source of radius $0.5''$ is placed at three positions: $\beta = 0$ (Einstein ring, left), $\beta = 0.5\,\theta_{\rm E}$ (giant arc, center), and $\beta = \theta_{\rm E}$ (near the critical curve, right). The critical curve is shown as a dashed circle. Generated by `cluster_lensing.wl`.

Part IV

Statistical, Weak, and Micro Lensing

Chapter 15

Lensing Statistics and Cross-Sections

> *One is frequently interested in the probability that a specific gravitational lensing event occurs. [. . .] The results of such an investigation depend on the assumed distribution of lens masses and their individual density profiles. A comparison of these results with a statistically well defined sample of observed lens cases can in principle allow one to constrain the lens contents and/or the geometry of the Universe.* — P. Schneider, in Schneider, Kochanek & Wambsganss (2006), Part 1, §5

So far we have studied lenses one at a time. But a survey observes *populations*: many sources seen through many deflectors. The questions then become statistical — what fraction of distant quasars is multiply imaged? how many giant arcs should a cluster survey find? how does lensing distort the number counts of a flux-limited sample? These questions are answered with three ingredients built in this module:

- the **cross-section** of a single lens for a given imaging property (§15.1);
- the **optical depth**, obtained by summing cross-sections over the lens population along the line of sight (§17.4);
- the **magnification bias**, which reshapes the observed counts of a lensed source population (§15.3).

Together these make lens statistics a probe both of the lens population (masses, profiles, abundance) and of cosmology (through the volume element). We build on the point-mass and SIS imaging of Modules 8 and 11 and the caustic structure of Module 12.

Companion Mathematica files:

- `lensing_statistics.wl` — symbolic derivation and verification of the point-mass and SIS cross-sections (including the combined SIS threshold cross-section, Eq. (15.6)), the comoving optical-depth integrand and its Ω_Λ dependence, and the magnification-bias factor for power-law counts; figures exported to `Figures/11_Lensing_Statistics/`.

15.1 Lensing Cross-Sections

Fix a source and a lens at given distances. For some imaging property Q — "total magnification exceeds μ", "image separation exceeds $\Delta\theta$", "flux ratio below r" — there is a region of the

source plane in which the source must lie for the images to have property Q. The solid angle of that region is the **cross-section** σ_Q [33, Part 1, §5.1]. Everything in lens statistics is built from cross-sections.

15.1.1 The Point-Mass Lens

For a point mass there is a one-to-one relation between the scaled source position $y = \beta/\theta_{\rm E}$ and the total magnification $\mu_p(y) = (y^2+2)/\left(y\sqrt{y^2+4}\right)$ (Module 8). Inverting it, the source produces a total magnification greater than μ_p whenever it lies within

$$y^2(\mu_p) = 2\left(\frac{\mu_p}{\sqrt{\mu_p^2-1}} - 1\right), \tag{15.1}$$

so the cross-section for magnification $> \mu_p$ is

$$\sigma(\mu_p) = \pi\,\theta_{\rm E}^2\, y^2(\mu_p). \tag{15.2}$$

At high magnification $y^2 \to 1/\mu_p^2$, giving the universal $\sigma \propto \mu_p^{-2}$ (equivalently the differential $p(\mu)\,d\mu \propto -\frac{d\sigma}{d\mu}\,d\mu \propto \mu^{-3}\,d\mu$ that recurs in §15.3). Similarly, the cross-section for the image brightness ratio to be smaller than $r > 1$ is

$$\sigma(r) = \pi\,\theta_{\rm E}^2\left(r^{1/2} + r^{-1/2} - 2\right), \tag{15.3}$$

both verified symbolically in the companion script.

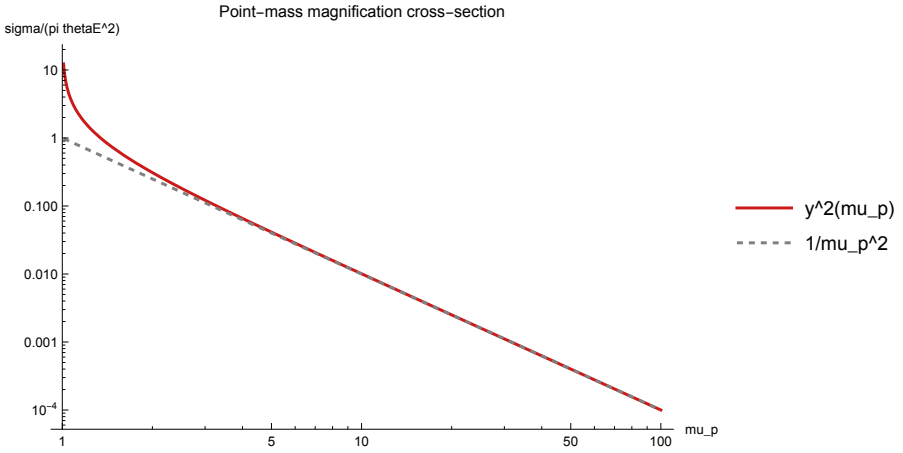


Figure 15.1: The point-mass magnification cross-section $\sigma(\mu_p)/(\pi\theta_{\rm E}^2) = y^2(\mu_p)$ of Eq. (15.1), as a function of the magnification threshold μ_p. At high magnification the cross-section falls as $\sigma \propto \mu_p^{-2}$ (dashed), the universal scaling that produces $p(\mu) \propto \mu^{-3}$ and drives the magnification bias. Generated by `lensing_statistics.wl`.

15.1.2 The Singular Isothermal Sphere

The SIS (Module 11) produces two images when $\beta < \theta_{\rm E}$, always separated by $2\theta_{\rm E}$. With $y = \beta/\theta_{\rm E}$, the total magnification and flux ratio of the pair are

$$\mu = \frac{2}{y}, \qquad r = \frac{1+y}{1-y}. \tag{15.4}$$

The basic multiple-imaging cross-section is therefore simply the area of the disk $\beta < \theta_{\rm E}$,

$$\sigma = \pi\,\theta_{\rm E}^2, \tag{15.5}$$

and with the resolution, magnification, and flux-ratio thresholds folded in [33, Part 1, eq. 101],

$$\sigma(\Delta\theta, r, \mu) = \pi\,\theta_{\rm E}^2 \left[\min\!\left(\frac{r-1}{r+1}, \frac{2}{\mu}\right)\right]^2 H(2\theta_{\rm E} - \Delta\theta), \tag{15.6}$$

where H is the Heaviside step function and r is finite, so that $(r-1)/(r+1) < 1$ keeps the source inside the multiple-imaging disk.[1] Because the SIS Einstein radius scales as $\theta_{\rm E} \propto \sigma_v^2$ (Eq. (11.18)), the multiple-imaging cross-section scales as $\sigma \propto \theta_{\rm E}^2 \propto \sigma_v^4$, so lens statistics are dominated by the most massive galaxies.

15.2 Optical Depth

The probability that a random source is lensed with property Q — the **optical depth** τ — is the fraction of the sky covered by the cross-sections of all intervening lenses. Summing the (non-overlapping) cross-sections of lenses with proper number density n along the line of sight,

$$\tau(Q) = \int_0^{z_s} dz_d\; D_{\rm ang}^2(z_d)\,\frac{dr_{\rm prop}}{dz} \int d\chi\; n(z_d,\chi)\,\sigma(Q; z_d, z_s, \chi), \tag{15.7}$$

where χ collects the lens parameters (mass, velocity dispersion, ellipticity) [33, Part 1, §5.2]. Written with the comoving number density $n = (1+z)^3 n_{\rm com}$ and comoving angular diameter distance f_K, this becomes the particularly clean

$$\tau(Q) = \int_0^{w_s} dw\; f_K^2(w) \int d\chi\; n_{\rm com}(w,\chi)\,\sigma(Q; w, w_s, \chi). \tag{15.8}$$

For a population of SIS galaxies with cross-section $\sigma = \pi\theta_{\rm E}^2 \propto \sigma_v^4$, integrating over the observed velocity-dispersion function yields the classic result that the multiple-imaging optical depth scales as $\tau \propto n_{\rm com}\,(\sigma_v/c)^4$ times a cosmological volume factor [34]. Because this volume factor is larger when $\Omega_\Lambda > 0$ (for fixed $n_{\rm com}$ and $z_s = 2$, about three times larger in a flat $\Omega_{\rm m} = 0.3$ universe than in an Einstein–de Sitter one), lens counts were an early probe of the cosmological constant [33, Part 2, §6]. The same calculation underpins forecasts of $\sim 10^5$ galaxy-scale lenses discoverable in *Euclid* and LSST imaging [10].

[1] The printed eq. 101 of Schneider et al. [33] carries an extra factor $H(\mu - 2)$. That factor would set $\sigma = 0$ for any magnification threshold $\mu < 2$, yet every SIS image pair has total magnification $2/y \geq 2$, so such a threshold is always met. Direct integration over the source plane confirms the form above for all μ (companion script, Test 6).

15.3 Magnification Bias

A flux-limited sample is not a fair sample of the sky: magnification both brightens sources and spreads them over a larger solid angle. If $p(\mu)$ is the magnification probability distribution and $N_0(>S)$ the unlensed counts, the observed counts are

$$N(>S) = \frac{1}{\langle\mu\rangle}\int d\mu\, p(\mu)\, N_0\!\left(>\frac{S}{\mu}\right), \tag{15.9}$$

with $\int p\, d\mu = 1$ and $\int \mu\, p\, d\mu = \langle\mu\rangle$ [33, Part 1, §5.3]. Two cases follow. For **power-law counts** $N_0(>S) = A\,S^{-\beta}$,

$$N(>S) = N_0(>S)\,\frac{1}{\langle\mu\rangle}\int d\mu\, p(\mu)\,\mu^{\beta}, \tag{15.10}$$

so the counts keep the same slope and are boosted by a factor that depends only on β and $p(\mu)$. Remarkably, for $\beta = 1$ the counts are *unchanged*: the solid-angle dilution exactly cancels the brightening. Counts steeper than $\beta = 1$ are boosted; flatter ones are depleted.

Because $p(\mu) \propto \mu^{-3}$ at high magnification [5] — a consequence of the universal fold magnification law, Eq. (12.19), met for the point mass in §15.1.1 — the bias integral grows sharply as $\beta \to 2$: a source population with steep counts, or with an exponential Schechter cutoff, is strongly biased toward including lensed objects. This is why several of the apparently most luminous objects in the sky — the IRAS galaxy F10214+4724 ($\mu \sim 50$), the Lyman-break galaxy cB58 ($\mu \sim 30$), the quasar APM 08279+5255 — turn out to be strongly magnified. It is also why lensed high-redshift galaxies are found preferentially behind massive clusters, the "cosmic telescope" effect of Module 14.

15.4 Summary

- A **cross-section** σ_Q is the source-plane solid angle producing a chosen imaging property Q. For a point mass, $\sigma(\mu_p) = \pi\theta_{\rm E}^2\, y^2(\mu_p)$ with $y^2 = 2(\mu_p/\sqrt{\mu_p^2-1} - 1)$; for an SIS the multiple-imaging cross-section is $\pi\theta_{\rm E}^2$.
- The **optical depth** sums cross-sections over the lens population along the line of sight, Eq. (15.8). For SIS galaxies $\tau \propto n_{\rm com}(\sigma_v/c)^4$ times a cosmological volume factor sensitive to Ω_Λ.
- **Magnification bias** reshapes flux-limited counts, Eq. (15.9); power-law counts are boosted by a β-dependent factor, unchanged at $\beta = 1$, and strongly biased as $\beta \to 2$ because $p(\mu) \propto \mu^{-3}$.

15.5 Problems

Exercise 15.5.1 (Point-mass magnification cross-section)**.** Starting from the total magnification of a point-mass lens, $\mu_p(y) = (y^2+2)/\big(y\sqrt{y^2+4}\big)$:

(a) Show that the source positions with $\mu_p(y) > \mu$ satisfy $y^2 < 2\big(\mu/\sqrt{\mu^2-1} - 1\big)$, giving the cross-section Eq. (15.2).

(b) Show that for $\mu \gg 1$ the cross-section scales as $\sigma \propto \mu^{-2}$, and explain how this yields the universal magnification distribution $p(\mu) \propto \mu^{-3}$.

(c) *Hint:* expand $y^2(\mu)$ for large μ; recall $p(\mu) \propto -\, d\sigma/d\mu$.

problems_11.wl

Exercise 15.5.2 (SIS flux ratio and magnification)**.** For the two images of an SIS with $y = \beta/\theta_{\rm E} < 1$, verify that the total magnification is $\mu = 2/y$ and the flux ratio is $r = (1+y)/(1-y)$, and hence that the cross-section for a flux ratio smaller than r is $\sigma \leq \pi\theta_{\rm E}^2$. problems_11.wl

Exercise 15.5.3 (The $\beta = 1$ null of magnification bias)**.** Using Eq. (15.10) and the normalization $\int \mu\, p(\mu)\, d\mu = \langle\mu\rangle$, show that flux-limited number counts with slope $\beta = 1$ are unchanged by lensing, for *any* magnification distribution $p(\mu)$. Explain physically why the brightening and the solid-angle dilution cancel. problems_11.wl

Chapter 16

Weak Gravitational Lensing and Cosmic Shear

> *One cannot 'see' the effect, nor can it be graphically displayed easily. Only by investigating many faint galaxy images can a signal be extracted from the data, and the human eye is not sufficient to perform this analysis.* — P. Schneider, *Weak Gravitational Lensing* (2006)

Modules 8–14 were concerned almost entirely with **strong** lensing: multiple images, giant arcs, Einstein rings — phenomena in which the Jacobian $\mathcal{A} = \partial\boldsymbol{\beta}/\partial\boldsymbol{\theta}$ differs so much from the identity that a single source produces strongly distorted images. This module treats the opposite, and far more common, regime. In **weak** gravitational lensing the Jacobian is very close to the unit matrix,

$$\mathcal{A} = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix} \approx \mathbb{1}\,, \qquad \kappa \ll 1\,, \quad |\gamma| \ll 1\,,$$

so the distortion and magnification of any *individual* background galaxy are tiny and cannot be recognized. The lensing signal is recovered only *statistically*, by averaging the shapes of many faint galaxies [33].

This is the natural complement to the convergence–shear formalism of Module 9. There we *defined* κ, γ, and the Jacobian; here we learn how to *measure* them from data, and what they tell us:

- how the shape of a faint galaxy provides a noisy but unbiased estimate of the (reduced) shear (§16.2);
- how the tangential shear around a mass concentration measures its projected mass profile (§16.3), extending the axisymmetric result $|\gamma| = \bar{\kappa} - \kappa$ of Module 11;
- how the shear field can be inverted into a parameter-free map of the projected mass — the Kaiser–Squires inversion (§16.5) — and why the *mass-sheet degeneracy* of Module 13 reappears here (§16.4);
- how the coherent distortion imprinted by the large-scale structure — *cosmic shear* — probes cosmology (§16.6); and
- how stacking foreground–background galaxy pairs measures the mean mass profile of galaxies (§16.7).

This chapter expands the brief weak-lensing sketch of §14.7 in Module 14 into a self-contained treatment of the core concepts.

Companion Mathematica files:

- `weak_lensing.wl` — symbolic verification of the Kaiser–Squires kernel and inversion, the tangential-shear relation for the SIS and point mass, reduced-shear invariance under the mass-sheet transform, and the image–ellipticity relations; tangential-shear profile and Kaiser–Squires schematic figures exported to `Figures/12_Weak_Lensing/`

16.1 Introduction: The Weak Regime

In strong lensing a single source tells the whole story: from the positions and fluxes of a handful of multiple images one constrains the deflector. In the weak regime, $\kappa \ll 1$ and $|\gamma| \ll 1$, no individual source carries a recognizable signal. Instead, one exploits the fact that lensing preserves surface brightness while distorting shapes: the observed ellipticity of each faint galaxy is a sum of its (unknown) intrinsic ellipticity and a small, coherent shear. Because intrinsic ellipticities are randomly oriented, they average to zero, and the mean observed ellipticity of a local ensemble of galaxies estimates the shear [33].

The accuracy of any weak-lensing measurement is therefore set by the number of usable background galaxies: the statistical error on the shear scales as $\sigma_\epsilon/\sqrt{N}$, where $\sigma_\epsilon \approx 0.3$ is the intrinsic ellipticity dispersion ("shape noise") and N is the number of galaxies averaged over. This is why deep imaging — reaching $n \sim 20$–30 galaxies per arcmin^2 in a few hours on a 4-m telescope — is the fundamental observational requirement, and why correcting the atmospheric and telescope point-spread function (which circularizes small images) is the central technical challenge of the field (33, Part 3, Sect. 3; 2).

Throughout this module we work in the thin-lens picture of Module 8 for individual galaxies and clusters, and generalize to a three-dimensional deflecting mass distribution — the large-scale structure — only in §16.6, where the lensing effect is described by an *effective* surface mass density.

16.2 Galaxy Shapes and the Shear Estimator

16.2.1 Complex Ellipticity

To turn shapes into shear we first need a quantitative measure of the shape of an image with arbitrary isophotes. Let $I(\boldsymbol{\theta})$ be the surface-brightness distribution of an isolated image. Its centroid is $\bar{\boldsymbol{\theta}} = \int d^2\theta\, I\, q_I[I]\, \boldsymbol{\theta} / \int d^2\theta\, I\, q_I[I]$ for a suitable weight q_I, and the tensor of second brightness moments is (33, Part 3, eq. 6)

$$Q_{ij} = \frac{\int d^2\theta\, I(\boldsymbol{\theta})\, q_I[I]\, (\theta_i - \bar{\theta}_i)(\theta_j - \bar{\theta}_j)}{\int d^2\theta\, I(\boldsymbol{\theta})\, q_I[I]}\,, \qquad i,j \in \{1,2\}\,. \tag{16.1}$$

The trace $Q_{11}+Q_{22}$ measures the size of the image, while the traceless part carries the ellipticity. One defines two complex ellipticities (33, Part 3, eq. 7)

$$\boxed{\chi \equiv \frac{Q_{11}-Q_{22}+2\mathrm{i}\,Q_{12}}{Q_{11}+Q_{22}}\,, \qquad \epsilon \equiv \frac{Q_{11}-Q_{22}+2\mathrm{i}\,Q_{12}}{Q_{11}+Q_{22}+2\left(Q_{11}Q_{22}-Q_{12}^{2}\right)^{1/2}}\,.} \tag{16.2}$$

Both share the same phase (that of the traceless part), so both encode the same orientation, but they differ in modulus. For an image with elliptical isophotes of axis ratio $r=b/a\le 1$ (33, Part 3, eq. 8),

$$|\chi| = \frac{1-r^{2}}{1+r^{2}}\,, \qquad |\epsilon| = \frac{1-r}{1+r}\,, \tag{16.3}$$

and the two are related by

$$\epsilon = \frac{\chi}{1+\sqrt{1-|\chi|^{2}}}\,, \qquad \chi = \frac{2\epsilon}{1+|\epsilon|^{2}}\,. \tag{16.4}$$

The phase of χ and ϵ is 2φ, twice the position angle φ of the major axis; the factor of two reflects the fact that an ellipse maps to itself under a rotation by 180°. It is this "spin-2" character that χ and ϵ share with the shear (Module 9, eq. 9.10).

16.2.2 From Source to Image, and the Estimator

Under the linearized lens mapping the source and image second-moment tensors are related by $Q^{(s)} = \mathcal{A}\,Q\,\mathcal{A}^{\mathsf{T}}$, which in terms of the complex ellipticity ϵ and the **reduced shear** $g=\gamma/(1-\kappa)$ becomes (33, Part 3, eq. 12)

$$\epsilon^{(s)} = \frac{\epsilon - g}{1-g^{*}\epsilon} \qquad (|g|\le 1)\,, \tag{16.5}$$

with the inverse obtained by $\epsilon \leftrightarrow \epsilon^{(s)}$ and $g\to -g$. Now comes the key statistical assumption: faint galaxies have no preferred orientation, so the intrinsic ellipticity averages to zero,

$$\mathrm{E}\left[\epsilon^{(s)}\right] = 0\,. \tag{16.6}$$

Averaging the transformation (16.5) over intrinsic orientations then gives the remarkable result (33, Part 3, eq. 14)

$$\boxed{\mathrm{E}[\epsilon] = g \qquad (|g|\le 1)\,.} \tag{16.7}$$

Each galaxy ellipticity is an unbiased — though very noisy — estimate of the local reduced shear. The noise is set by the intrinsic dispersion $\sigma_\epsilon = \sqrt{\langle \epsilon^{(s)}\epsilon^{(s)*}\rangle}$: averaging over N galaxies subject to the same g, the 1σ error on the mean is $\approx \sigma_\epsilon/\sqrt{N}$ (33, Part 3, eq. 15).

In the proper weak regime, $\kappa \ll 1$ and $|\gamma| \ll 1$, the reduced shear and shear coincide, and one has the useful chain (33, Part 3, eq. 16)

$$\gamma \approx g \approx \langle\epsilon\rangle \approx \frac{\langle\chi\rangle}{2}\,. \tag{16.8}$$

The factor $\tfrac{1}{2}$ between $\langle\chi\rangle$ and the shear follows directly from (16.3): for small distortions $|\chi|\approx 2|\epsilon|$.

Remark 16.2.1. Shape noise is irreducible: even with perfect measurements, the finite intrinsic ellipticity dispersion σ_ϵ sets a floor on the achievable accuracy. The only levers are the number density n of usable galaxies and the survey area. Increasing n means going deeper, which also pushes to higher-redshift (more strongly lensed) sources — a double benefit that motivates deep, wide imaging surveys.

16.3 Tangential and Cross Shear

16.3.1 Definitions

The shear is not a vector: under a rotation of the coordinate frame by ϕ its complex representation is multiplied by $e^{-2\mathrm{i}\phi}$, not $e^{-\mathrm{i}\phi}$. It is a **spin-2** (polar) quantity, like linear polarization. Relative to a chosen direction ϕ one defines the **tangential** and **cross** components (33, Part 3, eq. 17)

$$\boxed{\gamma_\mathrm{t} = -\mathrm{Re}\big(\gamma\, e^{-2\mathrm{i}\phi}\big)\ , \qquad \gamma_\times = -\mathrm{Im}\big(\gamma\, e^{-2\mathrm{i}\phi}\big)\ ,} \tag{16.9}$$

and identically for the ellipticity components $\epsilon_\mathrm{t}, \epsilon_\times$. Taking ϕ to be the polar angle of the point relative to a chosen center, $\gamma_\mathrm{t} > 0$ describes a shear oriented *tangentially* (an image stretched perpendicular to the line to the center), and $\gamma_\mathrm{t} < 0$ a radial stretch. The sign in (16.9) is fixed precisely so that a circular mass concentration — which stretches background images tangentially — produces a *positive* γ_t. For a circularly symmetric deflector the shear is purely tangential at every point, so $\gamma_\times = 0$; a non-zero cross component averaged over many pairs is a classic null test for systematics (its expectation vanishes by parity, §16.6).

16.3.2 Mean Tangential Shear and the Projected Mass

For a **circularly symmetric** lens we already know from Module 11 (eq. 11.7) that $\gamma_\mathrm{t}(\theta) = \bar\kappa(\theta) - \kappa(\theta)$, the mean convergence interior to θ minus the local convergence. Bartelmann [1] showed that a nearly identical relation holds for the *azimuthal average* of a completely general, non-symmetric mass distribution. Starting from Gauss's theorem applied to the deflection potential ψ (with $\nabla^2\psi = 2\kappa$) and differentiating the enclosed dimensionless mass $m(\theta) = \pi^{-1}\int_{|\vartheta|\le\theta} d^2\vartheta\,\kappa(\vartheta) = \theta^2\bar\kappa(\theta)$ with respect to θ, one obtains (33, Part 3, eq. 24)

$$\boxed{\langle\gamma_\mathrm{t}\rangle(\theta) = \bar\kappa(\theta) - \langle\kappa\rangle(\theta)\,,} \tag{16.10}$$

where $\langle\cdot\rangle(\theta)$ denotes the average over the circle of radius θ and $\bar\kappa(\theta)$ the mean convergence inside it. It follows that the tangential shear averaged on concentric circles gives the azimuthally averaged mass profile of any lens, symmetric or not.

It is customary to write this in physical units. Multiplying by the critical surface mass density Σ_cr (Module 8) and using $\kappa = \Sigma/\Sigma_\mathrm{cr}$,

$$\gamma_\mathrm{t}(\theta)\,\Sigma_\mathrm{cr} = \bar\Sigma(<\theta) - \Sigma(\theta) \equiv \Delta\Sigma(\theta)\,, \tag{16.11}$$

the **excess surface mass density** $\Delta\Sigma$: the mean surface density inside radius θ minus that at θ. Equation (16.11) is the basic relation of cluster and galaxy–galaxy weak lensing: the observable tangential shear translates directly into a projected mass profile.

Example 16.3.1. Tangential shear of an SIS. For the singular isothermal sphere (Module 11, eqs. 11.20–11.21) we have $\kappa = \theta_{\rm E}/(2\theta)$ and $\bar\kappa = \theta_{\rm E}/\theta$, so

$$\gamma_{\rm t}(\theta) = \bar\kappa - \kappa = \frac{\theta_{\rm E}}{\theta} - \frac{\theta_{\rm E}}{2\theta} = \frac{\theta_{\rm E}}{2\theta}\,. \tag{16.12}$$

The tangential shear falls off as θ^{-1}, exactly tracing the $\Sigma \propto \theta^{-1}$ isothermal profile. For a point mass, $\kappa = 0$ (away from the origin) and $\gamma_{\rm t} = \bar\kappa = \theta_{\rm E}^2/\theta^2$. Both are verified in `weak_lensing.wl`.

16.4 Reduced Shear and the Mass-Sheet Degeneracy

Galaxy ellipticities measure not the shear γ but the **reduced shear** $g = \gamma/(1-\kappa)$ (eq. 16.7). This has an important consequence. Consider the transformation, for an arbitrary constant λ, (33, Part 3, eq. 51)

$$\kappa(\boldsymbol{\theta}) \;\longrightarrow\; \kappa'(\boldsymbol{\theta}) = \lambda\,\kappa(\boldsymbol{\theta}) + (1-\lambda)\,, \qquad 1-\kappa' = \lambda\,(1-\kappa)\,. \tag{16.13}$$

Since the shear is the traceless part of the same Hessian that gives the convergence, scaling κ by λ scales the shear by λ, while the added *uniform sheet* $(1-\lambda)$ contributes no shear (its Hessian is isotropic). Hence $\gamma \to \gamma' = \lambda\gamma$, and therefore

$$\boxed{\,g' = \frac{\gamma'}{1-\kappa'} = \frac{\lambda\gamma}{\lambda(1-\kappa)} = \frac{\gamma}{1-\kappa} = g\,.\,} \tag{16.14}$$

The reduced shear — and hence every ellipticity measurement — is *invariant* under (16.13). This is the weak-lensing face of the **mass-sheet degeneracy** introduced in Module 13: from shear data alone the convergence is determined only up to the one-parameter family (16.13). In the Kaiser–Squires inversion below it appears as an undetermined additive constant κ_0.

The degeneracy also rescales the magnification. With $\mu^{-1} = (1-\kappa)^2 - |\gamma|^2$ (Module 9, eq. 9.20), the transform (16.13) gives

$$\mu \;\longrightarrow\; \mu' = \lambda^{-2}\,\mu\,, \tag{16.15}$$

which is why *magnification* information — from the lensing-induced change in background number counts, $n(>S)/n_0(>S) \approx 1 + 2(\alpha-1)\kappa$ in the weak limit for unlensed counts $n_0(>S) \propto S^{-\alpha}$ (here α is the count slope, not a deflection angle) (33, Part 3, eq. 27) — can in principle break the degeneracy that shear alone cannot. Both invariances are checked symbolically in `weak_lensing.wl`.

16.5 Cluster Mass Reconstruction

We now come to the method that made weak lensing quantitative: from a measured shear field one can reconstruct a **parameter-free map of the projected mass**, with no assumption of symmetry [16].

16.5.1 The Kaiser–Squires Inversion

The convergence and shear are both linear (second-derivative) functionals of the single potential ψ (Module 9), so they are related to each other by a non-local convolution. Explicitly (33, Part 3, eq. 41)

$$\gamma(\boldsymbol{\theta}) = \frac{1}{\pi}\int_{\mathbb{R}^2} d^2\theta'\, \mathcal{D}(\boldsymbol{\theta}-\boldsymbol{\theta}')\,\kappa(\boldsymbol{\theta}')\,, \qquad \mathcal{D}(\boldsymbol{\theta}) \equiv -\frac{\theta_1^2-\theta_2^2+2\mathrm{i}\,\theta_1\theta_2}{|\boldsymbol{\theta}|^4} = \frac{-1}{(\theta_1-\mathrm{i}\,\theta_2)^2}\,. \tag{16.16}$$

The complex kernel $\mathcal{D}$ is the shear produced by a point mass. In Fourier space the convolution becomes a product, $\hat\gamma(\boldsymbol{\ell}) = \pi^{-1}\hat{\mathcal{D}}(\boldsymbol{\ell})\,\hat\kappa(\boldsymbol{\ell})$, with (33, Part 3, eq. 43)

$$\hat{\mathcal{D}}(\boldsymbol{\ell}) = \pi\,\frac{\ell_1^2-\ell_2^2+2\mathrm{i}\,\ell_1\ell_2}{|\boldsymbol{\ell}|^2} = \pi\, e^{2\mathrm{i}\beta_\ell}\,, \qquad \hat{\mathcal{D}}\,\hat{\mathcal{D}}^* = \pi^2\,, \tag{16.17}$$

where β_ℓ is the polar angle of $\boldsymbol{\ell}$. Because $|\hat{\mathcal{D}}|^2 = \pi^2$ is a constant, the relation inverts trivially, $\hat\kappa = \pi^{-1}\hat\gamma\,\hat{\mathcal{D}}^*$ (for $\boldsymbol{\ell}\neq 0$), and transforming back to real space yields the **Kaiser–Squires inversion** (33, Part 3, eq. 44; 16)

$$\boxed{\kappa(\boldsymbol{\theta}) - \kappa_0 = \frac{1}{\pi}\int_{\mathbb{R}^2} d^2\theta'\,\mathrm{Re}[\mathcal{D}^*(\boldsymbol{\theta}-\boldsymbol{\theta}')\,\gamma(\boldsymbol{\theta}')]\,.} \tag{16.18}$$

Since the observed shear can be estimated at galaxy positions (via eq. 16.7), (16.18) converts a measured shear map into a mass map.

Remark 16.5.1. Three features of (16.18) deserve emphasis, all verified in `weak_lensing.wl`:

1. **The additive constant κ_0.** $\hat{\mathcal{D}}$ is undefined at $\boldsymbol{\ell}=0$, so the mean (DC) mode of κ is unconstrained. This is exactly the mass-sheet degeneracy (16.13): a uniform sheet produces no shear.
2. **Reduced shear, not shear.** What is measured is g, not γ; in "weak" weak lensing $\gamma \approx g$, but in the non-linear inner regions of clusters one must solve the integral equation
$$\kappa - \kappa_0 = \frac{1}{\pi}\int d^2\theta'\;[1-\kappa(\boldsymbol{\theta}')]\,\mathrm{Re}[\mathcal{D}^*(\boldsymbol{\theta}-\boldsymbol{\theta}')\,g(\boldsymbol{\theta}')]$$
iteratively (33, Part 3, eq. 50).
3. **Smoothing.** A naive discrete sum over galaxies has formally infinite variance (the θ^{-2} kernel); a smoothing kernel is required to obtain a finite-variance mass estimator.

16.5.2 Aperture Mass

The undetermined constant κ_0 motivates a statistic that is *immune* to the mass-sheet degeneracy: the **aperture mass** (33, Part 3, eq. 108)

$$M_{\mathrm{ap}}(\theta) = \int d^2\vartheta\; Q(|\vartheta|)\,\gamma_{\mathrm{t}}(\vartheta)\,, \tag{16.19}$$

a weighted integral of the tangential shear within an aperture of radius θ, with Q a weight function of compact support. If Q is derived from a *compensated* filter U (one with $\int d^2\vartheta\, U =$

0), then $M_{\rm ap}$ can equivalently be written as $\int d^2\vartheta\, U(|\vartheta|)\,\kappa(\vartheta)$: because U integrates to zero, adding a uniform sheet κ_0 contributes nothing, so $M_{\rm ap}$ is invariant under (16.13). Since it is built from the observable $\gamma_{\rm t}$ directly, it requires no inversion, and — as we see next — doubles as a clean cosmic-shear statistic and as a matched filter for detecting mass concentrations by their shear alone.

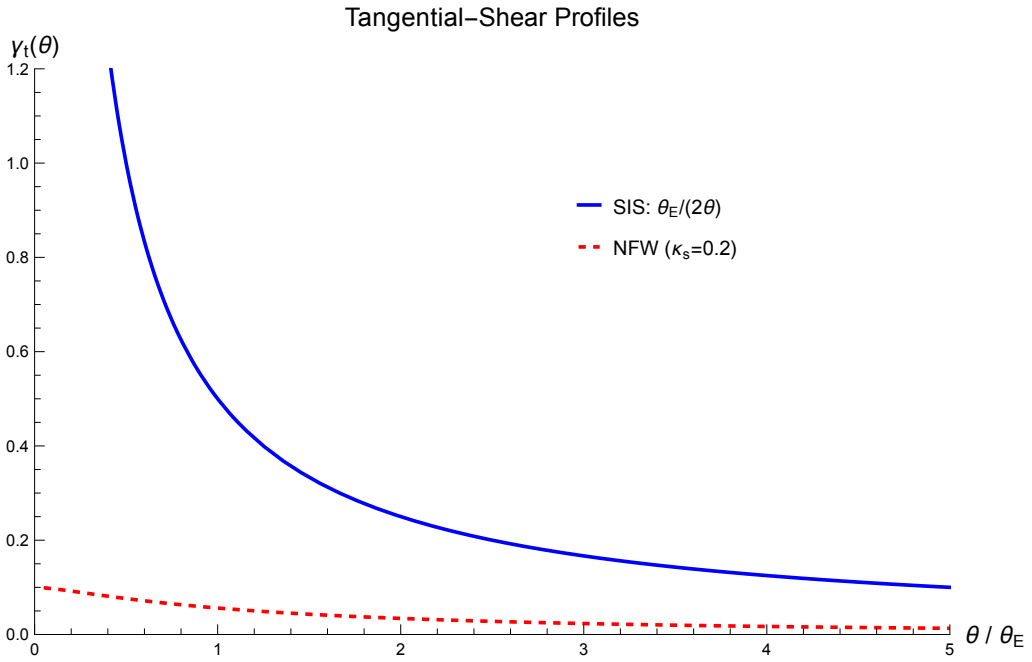


Figure 16.1: Tangential-shear profiles $\gamma_{\rm t}(\theta)$ for two axisymmetric lenses, illustrating eq. (16.10). **Solid:** the singular isothermal sphere, $\gamma_{\rm t} = \theta_{\rm E}/(2\theta)$ (eq. 16.12), with $\theta_{\rm E} = 1$. **Dashed:** an NFW halo (Module 11, eq. 11.54) with $\kappa_s = 0.2$ and scale $\theta_s = 1$. The NFW tangential shear decreases monotonically: it tends to the finite value $\kappa_s/2$ at the centre, is nearly flat inside θ_s, and steepens outside it, whereas the SIS is a single power law. The measurable $\gamma_{\rm t}$ maps directly to the excess surface mass density $\Delta\Sigma$ (eq. 16.11). Generated by `weak_lensing.wl`.

16.6 Cosmic Shear: Lensing by the Large-Scale Structure

So far the deflector has been a single, localized mass (galaxy or cluster) in a thin lens plane. But light bundles from distant galaxies are distorted continuously by *all* the inhomogeneous matter along the line of sight — the large-scale structure (LSS). This produces a weak, spatially coherent distortion of galaxy shapes across the sky called **cosmic shear**, whose statistics directly probe the matter distribution and hence cosmology [4, 15] (33, Part 3, Sect. 6).

16.6.1 The Effective Convergence

For a three-dimensional deflecting mass distribution, working to first order in the Newtonian potential Φ (the Born approximation, i.e. integrating along the unperturbed ray), the

distortion is still described by a Jacobian $\mathcal{A}_{ij} = \delta_{ij} - \psi_{,ij}$ with an *effective* deflection potential. Its convergence is a line-of-sight integral of the fractional density contrast $\delta = \Delta\rho/\bar{\rho}$, weighted by the geometric lensing efficiency (33, Part 3, eq. 93)

$$\kappa(\boldsymbol{\theta}) = \frac{3H_0^2\,\Omega_\mathrm{m}}{2c^2} \int_0^{w_h} dw\; g(w)\, f_K(w)\, \frac{\delta(f_K(w)\boldsymbol{\theta},\, w)}{a(w)}\,, \tag{16.20}$$

where w is comoving distance, a the scale factor, f_K the comoving angular-diameter distance, and $g(w) = \int_w^{w_h} dw'\, p_w(w')\, f_K(w'-w)/f_K(w')$ is the source-averaged efficiency factor. Two features are worth noting: $\kappa \propto \Omega_\mathrm{m}$, because lensing responds to the physical density contrast $\Delta\rho \propto \Omega_\mathrm{m}\,\delta$, not to δ alone; and κ can be negative, since δ can be (unlike the surface density of a real cluster).

16.6.2 The Convergence Power Spectrum

Because δ is a realization of a random field, the observables are statistical. Applying Limber's equation to the projection (16.20) relates the **convergence power spectrum** $P_\kappa(\ell)$ to the three-dimensional matter power spectrum $P_\delta(k)$ (33, Part 3, eq. 99)

$$\boxed{P_\kappa(\ell) = \frac{9H_0^4\,\Omega_\mathrm{m}^2}{4c^4} \int_0^{w_h} dw\; \frac{g^2(w)}{a^2(w)}\, P_\delta\!\left(\frac{\ell}{f_K(w)},\, w\right).} \tag{16.21}$$

The shear power spectrum equals the convergence power spectrum, since $\hat{\gamma}(\boldsymbol{\ell}) = e^{2\mathrm{i}\beta_\ell}\,\hat{\kappa}(\boldsymbol{\ell})$ from (16.17) and $|e^{2\mathrm{i}\beta_\ell}| = 1$. The strong prefactor makes the cosmic-shear amplitude scale roughly as $\sigma_8\,\Omega_\mathrm{m}^{0.5}$, much like cluster abundances, so cosmic shear is a leading probe of the clustering amplitude of matter [33, Part 3, Sect. 6.4].

16.6.3 Two-Point Statistics

P_κ is not measured directly; what one estimates from a shear catalogue are real-space two-point functions. Using the tangential and cross components (16.9) defined with respect to the line connecting each galaxy pair, one forms the two **shear correlation functions** (33, Part 3, eqs. 104–105)

$$\xi_\pm(\theta) = \langle\gamma_\mathrm{t}\gamma_\mathrm{t}\rangle(\theta) \pm \langle\gamma_\times\gamma_\times\rangle(\theta) = \int_0^\infty \frac{d\ell\,\ell}{2\pi}\, J_{0,4}(\ell\theta)\, P_\kappa(\ell)\,, \tag{16.22}$$

with J_0 for ξ_+ and J_4 for ξ_-. The mixed correlator $\xi_\times = \langle\gamma_\mathrm{t}\gamma_\times\rangle$ vanishes by parity: under a reflection $\gamma_\mathrm{t} \to \gamma_\mathrm{t}$ but $\gamma_\times \to -\gamma_\times$, so a non-zero $\xi_\times$ (or a "B-mode" signal) flags residual systematics. The correlation functions are the preferred observable because, unlike aperture statistics, they can be evaluated over gaps and masked regions — one simply uses every galaxy pair at a given separation. The aperture-mass dispersion $\langle M_\mathrm{ap}^2\rangle(\theta)$ built from the statistic (16.19) is a clean summary, because its filter is so narrow that it probes P_κ at essentially a single multipole $\ell \sim 5/\theta$ (33, Part 3, eq. 109 and Sect. 6.3).

Remark 16.6.1. **Tomography.** Splitting the source galaxies into photometric-redshift bins and measuring $\xi_\pm$ within and between bins gives the redshift evolution of the matter power

spectrum through the changing efficiency $g(w)$ in (16.21). This "weak-lensing tomography" turns a projected 2-D measurement into a partially 3-D probe, and is central to the dark-energy programs of Stage-IV surveys.

16.7 Galaxy–Galaxy Lensing

A single galaxy is far too weak a lens to detect individually — the signal-to-noise of an SIS detection scales as $S/N \propto \theta_{\rm E}\sqrt{n}\,/\sigma_\epsilon$, which for a galaxy ($\sigma_v \lesssim 200$ km s^{-1}) is hopeless (33, Part 3, eq. 19). But the signal of *many* galaxies can be stacked. In the mean, a background galaxy behind a foreground (lens) galaxy is stretched tangentially about it, so the mean tangential ellipticity of background sources, binned by the foreground–background separation θ, measures the mean tangential shear $\gamma_{\rm t}(\theta)$ — and hence, through eq. (16.11), the *average* projected mass profile of the lens galaxies (33, Part 3, Sect. 8.2; first detected by Brainerd et al. [6]).

Because a single galaxy's halo cannot be mapped, one parametrizes the lens population and fits. A standard choice is a **truncated isothermal sphere**: an isothermal $\rho \propto r^{-2}$ core turning to $\rho \propto r^{-4}$ beyond a truncation radius s, with the velocity dispersion and truncation scaled to luminosity via the Faber–Jackson / Tully–Fisher relations,

$$\sigma = \sigma_* \left(\frac{L}{L_*}\right)^{\beta/2}, \qquad s = s_* \left(\frac{L}{L_*}\right)^{\eta}, \qquad M \propto \sigma^2 s\,. \tag{16.23}$$

Fitting the stacked $\gamma_{\rm t}(\theta)$ constrains $(\sigma_*, s_*, \beta, \eta)$ and thereby the mass, extent, and luminosity scaling of galaxy dark-matter haloes out to radii far beyond the reach of stellar or gas dynamics. The first detection [6] yielded $\sigma_* \approx 160$ km s^{-1} for an L_* galaxy; modern surveys stack millions of lenses and split them by colour, stellar mass, and environment. On larger scales the same measurement transitions into a probe of the galaxy–mass cross-correlation (the "bias" of galaxies relative to the underlying matter).

16.8 Summary

- **Weak regime:** $\kappa \ll 1$, $|\gamma| \ll 1$, so $\mathcal{A} \approx \mathbb{1}$. No individual source shows a recognizable signal; the shear is recovered only statistically, by averaging galaxy shapes, with error $\sim \sigma_\epsilon/\sqrt{N}$ set by shape noise.
- **Shear estimator:** the complex ellipticities χ, ϵ (eq. 16.2) are spin-2 shape measures. With randomly oriented intrinsic ellipticities, $\mathrm{E}[\epsilon] = g$, the reduced shear — each galaxy is an unbiased, noisy shear estimate. In the weak limit $\gamma \approx g \approx \langle\epsilon\rangle \approx \langle\chi\rangle/2$.
- **Tangential shear:** $\gamma_{\rm t} = -\mathrm{Re}(\gamma e^{-2i\phi})$; averaged on circles, $\langle\gamma_{\rm t}\rangle = \bar\kappa - \langle\kappa\rangle$ for *any* mass distribution (eq. 16.10), so $\gamma_{\rm t}\Sigma_{\rm cr} = \Delta\Sigma$ measures the projected mass profile. For the SIS, $\gamma_{\rm t} = \theta_{\rm E}/(2\theta)$.
- **Mass-sheet degeneracy:** $g = \gamma/(1-\kappa)$ is invariant under $\kappa \to \lambda\kappa + (1-\lambda)$, $\gamma \to \lambda\gamma$ (with $\mu \to \lambda^{-2}\mu$), so shear determines κ only up to an additive constant. Magnification (number counts) can break it.

- **Kaiser–Squires:** γ and κ are related by convolution with $\mathcal{D}(\boldsymbol{\theta}) = -1/(\theta_1 - \mathrm{i}\theta_2)^2$; because $|\hat{\mathcal{D}}|^2 = \pi^2$, this inverts to $\kappa - \kappa_0 = \pi^{-1}\int \mathrm{Re}[\mathcal{D}^*\gamma]\, d^2\theta'$ (eq. 16.18), a parameter-free mass map. The aperture mass $M_{\rm ap}$ is a mass-sheet-invariant alternative.

- **Cosmic shear:** LSS distorts galaxy shapes coherently; the effective convergence (16.20) has power spectrum $P_\kappa \propto \Omega_{\rm m}^2 \int (g^2/a^2)\, P_\delta$ (eq. 16.21), measured via $\xi_\pm(\theta) = \int (d\ell\, \ell/2\pi) J_{0,4}(\ell\theta) P_\kappa$. Amplitude $\sim \sigma_8\, \Omega_{\rm m}^{0.5}$; tomography adds redshift information.

- **Galaxy–galaxy lensing:** stacking foreground–background pairs measures the mean $\gamma_{\rm t}$ and hence the average galaxy mass profile (truncated isothermal, eq. 16.23), extending to a probe of galaxy bias on large scales.

16.9 Problems

Exercise 16.9.1. (Reduced shear and the mass-sheet degeneracy.) The mass-sheet transform is $\kappa \to \kappa' = \lambda\kappa + (1-\lambda)$, $\gamma \to \gamma' = \lambda\gamma$, for constant λ.

(a) Show that the reduced shear $g = \gamma/(1-\kappa)$ is invariant under this transform.

(b) Using $\mu^{-1} = (1-\kappa)^2 - |\gamma|^2$, show that the magnification transforms as $\mu \to \lambda^{-2}\mu$.

(c) Explain why a purely local measurement of galaxy ellipticities cannot break the degeneracy, and name one observable that can.

(d) Why is the transform called a "mass sheet"? Compute the convergence of a uniform sheet $\kappa(\boldsymbol{\theta}) = 1 - \lambda$ and its shear.

Verify with `problems_12.wl`.

Exercise 16.9.2. (Tangential shear and the projected mass.)

(a) For the SIS with $\kappa(\theta) = \theta_{\rm E}/(2\theta)$, compute the mean convergence $\bar\kappa(\theta)$ and show that the tangential shear is $\gamma_{\rm t}(\theta) = \bar\kappa - \kappa = \theta_{\rm E}/(2\theta)$.

(b) For a point mass, $\kappa = 0$ away from the origin. What is $\gamma_{\rm t}(\theta)$? Contrast the radial fall-off of $\gamma_{\rm t}$ for the point mass and the SIS.

(c) Starting from $\langle\gamma_{\rm t}\rangle = \bar\kappa - \langle\kappa\rangle$ (eq. 16.10), show that in physical units the observable tangential shear equals the excess surface mass density, $\gamma_{\rm t}\Sigma_{\rm cr} = \bar\Sigma(<\theta) - \Sigma(\theta)$.

(d) For an NFW halo, sketch $\gamma_{\rm t}(\theta)$ using Module 11 (eq. 11.54). Show that it stays finite at the centre, and explain qualitatively why its logarithmic slope changes near θ_s, unlike the single power law of the SIS.

Verify with `problems_12.wl`.

Exercise 16.9.3. (The Kaiser–Squires kernel.) The forward relation is $\hat\gamma(\boldsymbol{\ell}) = \pi^{-1}\hat{\mathcal{D}}(\boldsymbol{\ell})\,\hat\kappa(\boldsymbol{\ell})$ with $\hat{\mathcal{D}} = \pi(\ell_1^2 - \ell_2^2 + 2\mathrm{i}\ell_1\ell_2)/|\boldsymbol{\ell}|^2$.

(a) Show that $\hat{\mathcal{D}}(\boldsymbol{\ell})\,\hat{\mathcal{D}}^{*}(\boldsymbol{\ell}) = \pi^{2}$ for all $\boldsymbol{\ell} \neq 0$.

(b) Use this to derive the inversion $\hat{\kappa} = \pi^{-1}\hat{\gamma}\,\hat{\mathcal{D}}^{*}$, and hence eq. (16.18).

(c) Explain why the inversion determines κ only up to an additive constant κ_0. Relate this to Exercise 16.9.1.

(d) Writing $\hat{\mathcal{D}} = \pi\, e^{2i\beta_\ell}$ (with β_ℓ the polar angle of $\boldsymbol{\ell}$), verify that the shear and convergence have identical power spectra, $P_\gamma(\ell) = P_\kappa(\ell)$.

Verify with `problems_12.wl`.

Exercise 16.9.4. (Complex ellipticity.) An image has elliptical isophotes with semi-axes $a \geq b$ and axis ratio $r = b/a$, aligned with the coordinate axes (so $Q_{12} = 0$, $Q_{11} = a^2/4$, $Q_{22} = b^2/4$ up to a common factor).

(a) Using the definitions (16.2), show $|\chi| = (1 - r^2)/(1 + r^2)$ and $|\epsilon| = (1 - r)/(1 + r)$.

(b) Verify the transformation $\epsilon = \chi/(1 + \sqrt{1 - |\chi|^2})$ and its inverse $\chi = 2\epsilon/(1 + |\epsilon|^2)$.

(c) In the weak-distortion limit ($r \to 1$), show $|\chi| \approx 2|\epsilon|$, and hence that the shear estimators satisfy $\langle\chi\rangle/2 \approx \langle\epsilon\rangle \approx \gamma$.

Verify with `problems_12.wl`.

Exercise 16.9.5. (Cosmic-shear correlation functions.)

(a) Starting from the definitions $\xi_{\pm} = \langle\gamma_{\rm t}\gamma_{\rm t}\rangle \pm \langle\gamma_{\times}\gamma_{\times}\rangle$ and the relation (16.22), explain why ξ_+ and ξ_- are *not* independent (both derive from the single field P_κ).

(b) Show, using the parity behaviour $\gamma_{\rm t} \to \gamma_{\rm t}$, $\gamma_{\times} \to -\gamma_{\times}$ under reflection, that the mixed correlator $\xi_{\times} = \langle\gamma_{\rm t}\gamma_{\times}\rangle$ vanishes. Why is this a useful systematics test?

(c) For a scale-free power spectrum $P_\kappa(\ell) = A\,\ell^{-1}$, use $\int_0^\infty J_n(\ell\theta)\,d\ell = 1/\theta$ (valid for $n = 0$ and $n = 4$) to compute $\xi_+(\theta)$ and $\xi_-(\theta)$. Why would the steeper $P_\kappa \propto \ell^{-2}$ not give a finite ξ_+?

(d) Explain why the aperture-mass dispersion $\langle M_{\rm ap}^2\rangle(\theta)$ gives more *localized* information about P_κ than the shear dispersion $\langle|\bar{\gamma}|^2\rangle(\theta)$.

Verify with `problems_12.wl`.

Chapter 17

Gravitational Microlensing

> *"Of course, there is no hope of observing this phenomenon directly."* —
> A. Einstein, *Science* **84**, 506 (1936), on star–star lensing

Einstein's pessimism [13] was well founded for the technology of his day, but wrong for ours. The images formed when one star lenses another are split by only $\sim 10^{-3}$ arcsec, far too small to resolve, so the individual images that dominated Modules 8 through 12 can never be seen. What *is* observable is the **total magnification**, and because the lens, source, and observer are all in relative motion, that magnification varies with time. Monitoring millions of stars turns this tiny, transient brightening into one of the most productive tools in modern astrophysics. This is the regime of **gravitational microlensing**, and it is the subject of this module, which follows Part 4 of Schneider et al. [33].

Microlensing recycles the physics we have already built — it is the point-mass lens of Module 8 and the magnification formalism of Module 9, read out in a new observational regime:

- the images are **unresolved**, so the observable is the time-variable **total magnification** $A(u)$, not image positions;
- the natural clock is the **Einstein-radius crossing time** $t_{\rm E}$, and the light curve has a universal, symmetric **Paczyński** shape;
- ensembles of unseen lenses are characterized statistically through the **optical depth** τ;
- small departures from the point-lens light curve reveal **binary companions and planets** (the caustics of Module 12 in miniature);
- the sub-milliarcsecond wobble of the light centroid (**astrometric microlensing**) and the micro-caustic networks that flicker across **multiply imaged quasars** (Modules 12–13) extend the method to lens masses and quasar sizes.

Companion Mathematica files:

- `microlensing.wl` — symbolic derivation of the point-lens magnification $A(u) = (u^2+2)/(u\sqrt{u^2+4})$ from the two images, its $u \to 0$ and $u \to \infty$ limits, and the symmetry of the Paczyński light curve about the peak; figures exported to `Figures/13_Microlensing/`
- `microlensing_explorer.nb` — interactive notebook with `Manipulate` widgets for the light curve (vary u_0, t_0, $t_{\rm E}$) and for the binary-lens caustic topology (vary mass ratio q and separation d)

17.1 The Microlensing Regime

Consider a foreground star of mass M at distance D_d lensing a background star at distance D_s. The angular Einstein radius is (eq. 8.17, specialized to a point mass)

$$\theta_\mathrm{E} = \sqrt{\frac{4GM}{c^2}\,\frac{D_\mathrm{ds}}{D_\mathrm{d}\,D_\mathrm{s}}}\,, \tag{17.1}$$

which for stellar masses and kiloparsec distances evaluates to $\theta_\mathrm{E} \sim 10^{-3}$ arcsec (a milliarcsecond, far below the arcsecond image splittings of galaxy lensing). The two images produced by a point lens straddle the lens and are separated by $\Delta\theta \approx 2\theta_\mathrm{E} \sim$ mas, far below the resolution of any conventional telescope.

As a result, **individual image positions, magnifications, and shapes are not measurable**. All the information carried by the image geometry in earlier modules is lost. What survives is the *sum* of the image fluxes — the total magnification — which, thanks to relative motion, sweeps out a light curve [33, Sect. 1]. The remainder of this module is largely the study of that light curve and its perturbations.

Remark 17.1.1. Microlensing is defined by the *observable*, not by the lens mass. The same point-mass physics describes a planet ($10^{-6}\,M_\odot$) perturbing a stellar event and a $30\,M_\odot$ black hole in the Galactic halo. What makes all of these "micro" is that the image splitting is unresolved, so the total magnification is the only handle.

17.2 Point-Lens Magnification

Scale all angles by θ_E and work with the dimensionless source position $u \equiv \beta/\theta_\mathrm{E}$ and image position $y \equiv \theta/\theta_\mathrm{E}$. The point-mass lens equation (Module 8) reduces to the one-dimensional form [33, eq. 1]

$$\boxed{u = y - \frac{1}{y}} \tag{17.2}$$

This is quadratic, $y^2 - u\,y - 1 = 0$, with the two roots

$$y_\pm = \frac{1}{2}\left(u \pm \sqrt{u^2+4}\right)\,. \tag{17.3}$$

The two images always straddle the lens: $y_+ > 0$ lies on the same side as the source (positive parity, brighter, always outside the Einstein ring), while $y_- < 0$ lies on the opposite side (negative parity, fainter, inside the ring). Their separation is $|y_+ - y_-| = \sqrt{u^2+4}$.

17.2.1 Magnification of Each Image

For a circularly symmetric lens the signed magnification of an image at y is $\mu = [\,1 - y^{-4}\,]^{-1}$ (the product of the tangential and radial eigenvalues, eq. 11.9 evaluated for $\alpha = 1/y$). Substituting eq. (17.3) [33, eq. 3],

$$\mu_\pm = \frac{1}{2}\left[\frac{u^2+2}{u\sqrt{u^2+4}} \pm 1\right]\,, \tag{17.4}$$

so that $\mu_+ > 0$, $\mu_- < 0$, and the images obey the sum rule $\mu_+ + \mu_- = 1$.

17.2.2 Total Magnification

Because only the combined flux is observable, the relevant quantity is the **total magnification** $A(u) = |\mu_+| + |\mu_-| = \mu_+ - \mu_-$:

$$\boxed{A(u) = \frac{u^2 + 2}{u\sqrt{u^2 + 4}}} \tag{17.5}$$

This is the central formula of single-lens microlensing [33, eq. 4]; the derivation by summing the two image magnifications is verified symbolically in `microlensing.wl`. Its two limits are physically important:

$$A(u) \longrightarrow \frac{1}{u} \qquad (u \to 0)\,, \tag{17.6}$$

$$A(u) \longrightarrow 1 \qquad (u \to \infty)\,. \tag{17.7}$$

Close alignment ($u \to 0$) gives arbitrarily high magnification (formally infinite for a point source at $u = 0$, where the images merge into an Einstein ring), while a distant source ($u \gg 1$) is essentially unlensed. At the fiducial separation $u = 1$ the magnification is $A(1) = 3/\sqrt{5} \approx 1.34$, corresponding to a brightening of $\Delta m = 2.5 \log_{10} A \approx 0.32$ mag; the positive-parity image then supplies 87% of the total flux [33, Sect. 1.2]. The value $u = 1$ is the conventional threshold defining a microlensing "event".

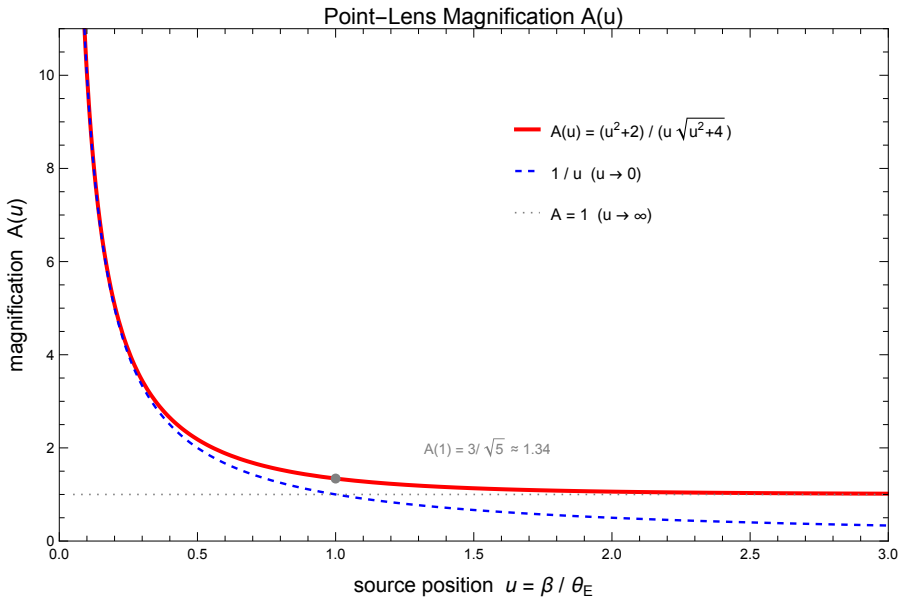


Figure 17.1: The point-lens total magnification $A(u)$ (eq. 17.5, red) as a function of the source–lens separation $u = \beta/\theta_{\mathrm{E}}$. The high-magnification asymptote $A \to 1/u$ (blue dashed) and the unlensed limit $A \to 1$ (grey dotted) are shown for reference; the marked point is $A(1) = 3/\sqrt{5} \approx 1.34$. Generated by `microlensing.wl`.

17.3 The Light Curve and the Einstein Timescale

A microlensing event is a *temporal* phenomenon. Over the days-to-months duration of an event the relative motion of lens and source is well approximated as rectilinear, so the separation traces a straight line past the lens. Writing t_0 for the time of closest approach and u_0 for the minimum (impact-parameter) separation, the source–lens separation is [33, eq. 8]

$$\boxed{u(t) = \sqrt{u_0^2 + \left(\frac{t - t_0}{t_{\rm E}}\right)^2}} \tag{17.8}$$

where the **Einstein (crossing) timescale**

$$t_{\rm E} \equiv \frac{D_{\rm d}\,\theta_{\rm E}}{v_\perp} = \frac{R_{\rm E}}{v_\perp} \tag{17.9}$$

is the time for the lens to traverse one (physical) Einstein radius $R_{\rm E} = D_{\rm d}\,\theta_{\rm E}$ at transverse speed $v_\perp$. Substituting eq. (17.8) into eq. (17.5) gives the observed light curve $F(t) = A(u(t))\,F_s$, where F_s is the unlensed source flux.

17.3.1 The Paczyński Curve

The resulting light curve — the **Paczyński curve** [23] — is smooth, achromatic, and *symmetric* about t_0. Symmetry follows immediately because $u(t)$ depends on time only through $(t - t_0)^2$: $u(t_0 + s) = u(t_0 - s)$, hence $A(u(t_0 + s)) = A(u(t_0 - s))$. Since A is a monotonically decreasing function of u (indeed $dA/du = -8/[u^2(u^2+4)^{3/2}] < 0$), the peak magnification occurs at closest approach:

$$A_{\rm max} = A(u_0) = \frac{u_0^2 + 2}{u_0\sqrt{u_0^2 + 4}} \;\approx\; \frac{1}{u_0} \quad (u_0 \ll 1)\,. \tag{17.10}$$

Both the symmetry and the identification $A_{\rm max} = A(u_0)$ are verified in `microlensing.wl`.

The point-source, point-lens light curve is described by just four parameters: the baseline flux F_0, the peak time t_0, the impact parameter u_0, and the timescale $t_{\rm E}$. Of these *only* $t_{\rm E}$ *carries physical information about the lens*, and it does so in the degenerate combination $t_{\rm E} = D_{\rm d}\theta_{\rm E}/v_\perp \propto \sqrt{M}$ that mixes the lens mass M, distance $D_{\rm d}$, and transverse velocity $v_\perp$ [33, Sect. 1.2]. Breaking this degeneracy — to weigh the lens — requires extra information: finite-source effects, parallax, or the astrometric shift of §17.6.

Example 17.3.1. A typical Galactic event. For a lens of $M = 0.3\,M_\odot$ halfway to a bulge source ($D_{\rm d} = 4$ kpc, $D_{\rm s} = 8$ kpc) moving at $v_\perp = 200\ {\rm km\,s^{-1}}$, eqs. (17.1) and (17.9) give $\theta_{\rm E} \approx 0.55$ mas, $R_{\rm E} \approx 2.2$ AU, and $t_{\rm E} \approx 19$ days — of order weeks, as observed. The numbers are computed in `problems_13.wl`.

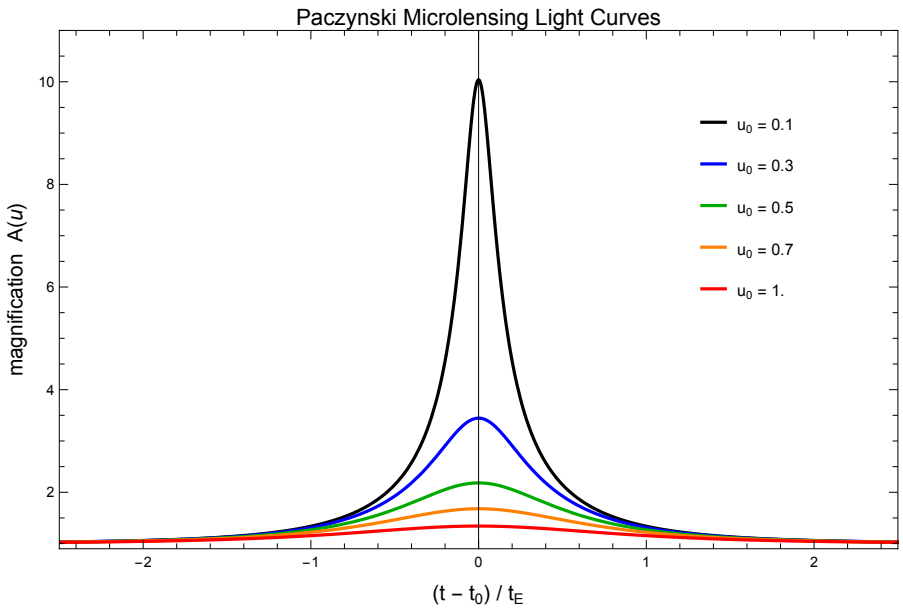


Figure 17.2: Paczyński microlensing light curves (eq. 17.8 substituted into eq. 17.5) for minimum impact parameters $u_0 = 0.1, 0.3, 0.5, 0.7, 1.0$ (top to bottom). Time is in units of the Einstein timescale $t_{\rm E}$; every curve is symmetric about the peak at $(t - t_0)/t_{\rm E} = 0$, and the smaller the impact parameter the higher and narrower the peak ($A_{\max} \approx 1/u_0$). Generated by `microlensing.wl`.

17.4 Optical Depth to Microlensing

A single event is rare, so microlensing is inherently a statistical enterprise: one monitors millions of stars to catch a handful of events. The relevant statistic is the **optical depth** τ, the probability that a random source lies within one Einstein radius of some lens at a given instant — equivalently, the fraction of the sky covered by the Einstein disks of all the lenses [33, Sect. 1.4].

For lenses distributed with mass density $\rho(D_{\rm d})$ along the line of sight, Paczynski [23] showed

$$\boxed{\tau = \int_0^{D_{\rm s}} \frac{4\pi G}{c^2}\,\frac{D_{\rm d}\,D_{\rm ds}}{D_{\rm s}}\,\rho(D_{\rm d})\,dD_{\rm d}} \tag{17.11}$$

The optical depth is *independent of the individual lens masses*: it depends only on the total mass density, because the Einstein area $\pi\theta_{\rm E}^2 \propto M$ while the number density $\propto 1/M$, and the two cancel. The equivalence of "fraction of sky in Einstein disks" and $\Sigma/\Sigma_{\rm cr}$ for a thin lens screen is verified in `problems_13.wl`.

17.4.1 The Dark-Matter Test Toward the Magellanic Clouds

Paczyński's motivation was this: if the dark halo of the Milky Way were made of compact objects (MACHOs — MAssive Compact Halo Objects), they would microlens background stars in the Large and Small Magellanic Clouds (LMC/SMC). For a standard isothermal halo

he predicted

$$\tau_{\rm halo} \sim 5 \times 10^{-7}\,, \tag{17.12}$$

so roughly one star in two million is measurably lensed at any time — a small but tractable number given large CCD surveys. The MACHO, EROS, and OGLE collaborations carried out the experiment through the 1990s and 2000s. Their verdict: the observed rate toward the LMC, $\tau_{\rm LMC} \approx 1.2^{+0.4}_{-0.3} \times 10^{-7}$ (MACHO, with a further $\sim$20% systematic uncertainty), is far too low for compact objects to constitute the entire halo. EROS limits objects of 2×10^{-7} to $1\,M_\odot$ to less than 25% of the halo, MACHO finds that objects of 0.3–30 $M_\odot$ cannot make up the whole halo, and the surviving events are consistent with ordinary stars in the disk or within the Clouds themselves [33, Sect. 3.5]. This important *null* result rules out compact objects of $\sim 10^{-7}$–30 $M_\odot$ as the dominant component of the halo dark matter.

17.4.2 Toward the Galactic Bulge

Monitoring the crowded Galactic bulge yields events with near certainty and a much higher optical depth, $\tau_{\rm bulge} \approx (0.9\text{–}4) \times 10^{-6}$ (EROS to MACHO/OGLE), tracing the stellar bar along the line of sight [33, Sect. 3.6]. The bulge is now the principal hunting ground for the planetary events of the next section.

17.5 Binary and Planetary Microlensing

A single point lens has only a degenerate point caustic at $u = 0$. As soon as a *second* mass is added, the lens becomes non-axisymmetric and develops **extended caustics** in the source plane — exactly the fold-and-cusp curves of Module 12, now realized at stellar scales. For a binary lens the deflection is the sum of two point-mass terms [33, eq. 14],

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \frac{4G}{c^2}\left[\frac{M_A(\boldsymbol{\theta}-\boldsymbol{\theta}_A)}{|\boldsymbol{\theta}-\boldsymbol{\theta}_A|^2} + \frac{M_B(\boldsymbol{\theta}-\boldsymbol{\theta}_B)}{|\boldsymbol{\theta}-\boldsymbol{\theta}_B|^2}\right], \tag{17.13}$$

and the system is described by three new parameters: the mass ratio $q = M_B/M_A$, the projected separation d (in units of $\theta_{\rm E}$ of the total mass), and the angle ϕ of the source trajectory relative to the binary axis. The caustic topology changes qualitatively with d. For equal masses: widely separated lenses ($d > 2$) have two small four-cusp (astroid-shaped) caustics; for $2^{-1/2} < d < 2$ these are merged into a single six-cusp caustic; and for $d < 2^{-1/2} \approx 0.707$ the caustic splits into a central four-cusp caustic plus two small triangular ones [25, 27] [33, Part 4, Sect. 2.1].[1]

17.5.1 Caustic Crossings

The observational signature is a sharp spike. When the source *crosses* a caustic, a pair of highly magnified images is created or destroyed, and because these images dominate the

[1] Schneider and Weiss [27], reproduced as Fig. 4 of Schneider et al. [33, Part 4], quote these transitions as 1 and $8^{-1/2}$ because they measure the *half*-separation of the two lenses. In the full-separation units used here the transitions are at $d = 2$ and $d = 2^{-1/2}$; both values are checked by counting critical curves in `microlensing.wl`.

(unresolved) total flux, the light curve shows a sharp spike. A point source would formally be infinitely magnified at the caustic; the finite spike height instead *measures the source size*, since the magnification is smeared over the time $t_{\rm cross} = R_{\rm source}/v_\perp$ it takes the source to move by its own radius. Caustic-crossing binaries thus break the point-source approximation in a useful way, occasionally even resolving the stellar surface brightness profile.

17.5.2 Detecting Planets

Mao and Paczynski [20] first showed that a planetary companion to the lens star produces the same kind of short, low-amplitude deviation on top of an otherwise ordinary stellar light curve, and estimated that of order 10% of events should show a detectable binary signature. A planet of mass ratio $q = M_{\rm planet}/M_\star$ generates a small caustic whose perturbation is strongest when the projected separation lies in the **lensing zone** $0.6 \lesssim d/\theta_{\rm E} \lesssim 1.6$, where the planetary caustic falls near the Einstein ring of the host star [33, Sect. 4.1]. By coincidence this angular range corresponds to physical separations of order 1 AU for typical Galactic lenses — overlapping the habitable zone. The planetary deviation is brief (hours to a day) and of small amplitude (a few percent), so dense, continuous monitoring of ongoing stellar events is required. Microlensing is especially sensitive to *cool, low-mass planets at* $\sim$*1–few AU* around distant ($\sim$ kpc) stars, a region of parameter space complementary to radial-velocity and transit surveys. Its cost is that each event is a unique, non-repeatable alignment, and only the mass *ratio* q (not the planet mass directly) is measured from the light curve alone.

17.6 Astrometric Microlensing

Although the two images cannot be resolved, their flux-weighted mean position — the **light centroid** — is displaced from the true source position, and that displacement can in principle be measured to sub-milliarcsecond precision. This is **astrometric microlensing** [33, Sect. 6]. Combining the image positions (17.3) with their magnifications (17.4), the centroid shift relative to the source is

$$\delta(u) = \frac{u}{u^2+2}\,\theta_{\rm E}\,. \tag{17.14}$$

Unlike the photometric magnification, the centroid shift *vanishes* both at perfect alignment ($u \to 0$) and at large separation ($u \to \infty$); it is maximized in between. Setting $d\delta/du = 0$ gives

$$\boxed{\delta_{\max} = \frac{1}{2\sqrt{2}}\,\theta_{\rm E} = 8^{-1/2}\,\theta_{\rm E} \approx 0.354\,\theta_{\rm E} \quad \text{at} \quad u = \sqrt{2}} \tag{17.15}$$

[33, eq. 22], a result due to Paczynski [24]; both the location $u = \sqrt{2}$ and the peak value are verified in `microlensing.wl`.

The astrometric signal is valuable because the centroid traces an ellipse whose *angular* size is set directly by $\theta_{\rm E} \propto \sqrt{M}$. Measuring it therefore yields $\theta_{\rm E}$, and combined with a parallax measurement it breaks the mass–distance–velocity degeneracy entirely, delivering the lens mass:

$$M = 0.123\,M_\odot\,\frac{\theta_{\rm E}^2}{\pi_{\rm ds}} \tag{17.16}$$

with $\theta_{\rm E}$ and the lens–source relative parallax $\pi_{\rm ds}$ in milliarcseconds [33, eq. 25]. The centroid shift is tens to hundreds of micro-arcseconds — beyond ground-based astrometry of the 2000s, but squarely within reach of space astrometry (the numerical coefficient is checked in `problems_13.wl`).

17.7 Quasar Microlensing

Microlensing is not confined to the Milky Way. When a distant quasar is multiply imaged by a foreground galaxy (Modules 12 and 13), the individual *stars* of that galaxy microlens each macro-image. This is **quasar microlensing** [33, Sect. 7]. The key contrast with Galactic microlensing is that at the position of a macro-image the stellar convergence is of order unity, $\kappa \sim 1$, so at any instant a *whole ensemble* of stars lenses the source coherently, producing an intricate **network of micro-caustics** in the source plane rather than a single isolated event.

The relevant scales, for typical lens and source redshifts $z_d \approx 0.5$, $z_s \approx 2$, are the Einstein radius projected into the quasar plane and its crossing time [33, Sect. 7.1]. For the concordance cosmology of Module 7 ($D_{\rm d} \approx 1260$, $D_{\rm s} \approx 1730$, $D_{\rm ds} \approx 1100$ Mpc),

$$r_{\rm E} = \sqrt{\frac{4GM}{c^2}\,\frac{D_{\rm s}\,D_{\rm ds}}{D_{\rm d}}} \approx 5\times10^{16}\left(\frac{M}{M_\odot}\right)^{1/2}\ {\rm cm}\,, \tag{17.17}$$

$$t_{\rm E} = \frac{r_{\rm E}}{v_\perp} \approx 28\left(\frac{M}{M_\odot}\right)^{1/2}\left(\frac{v_\perp}{600\ {\rm km\ s^{-1}}}\right)^{-1}\ {\rm yr}\,, \tag{17.18}$$

where $v_\perp$ is the relative transverse velocity projected into the source plane.[2] The full Einstein-crossing time is discouragingly long, but the observable fluctuations are governed instead by the much shorter *caustic*-crossing time $t_{\rm cross} = R_{\rm source}/v_\perp \approx 6\,R_{15}\,(v_\perp/600\ {\rm km\ s^{-1}})^{-1}$ months, where the quasar continuum radius R_{15} is in units of 10^{15} cm. The scale estimates and the $\propto\sqrt{M}$ mass scaling are checked in `problems_13.wl`.

The value of quasar microlensing is that $t_{\rm cross}$ — and hence the sharpness of the microlensing fluctuations — is set by the *source size*: a compact continuum-emitting region crosses caustics cleanly and flickers strongly, whereas a large emission region washes the peaks out. Monitoring the uncorrelated (non-time-delayed) variability between the macro-images therefore constrains the **quasar accretion-disk size**, independently of the time-delay cosmography of Module 13. The classic laboratory is the "Einstein Cross" Q2237+0305, whose four images vary independently by up to $\sim$1.3 mag; the fits imply a continuum region of order 10^{14} cm [33, Sect. 7.3]. Quasar microlensing also probes the fraction of the lens-galaxy mass in compact objects versus smooth dark matter, complementing the halo constraints of §17.4.

[2] Schneider et al. [33, Part 4, Sect. 7.1] quotes $r_{\rm E} \approx 4\times10^{16}$ cm, which corresponds to an older Einstein–de Sitter-type cosmology (EdS with $H_0 = 70$ gives 3.8×10^{16} cm). It also quotes $t_{\rm E} \approx 15$ yr and $t_{\rm cross} \approx 4$ months. Those two values are consistent with each other but are about 1.5× shorter than r/v for $v = 600\ {\rm km\,s^{-1}}$, which implies a differently defined effective velocity. Here we quote r/v directly; see `problems_13.wl`.

17.8 Summary

- In the **microlensing regime** the two point-lens images are split by only $\sim$ mas and are unresolved, so the sole observable is the time-variable **total magnification** $A(u) = (u^2+2)/(u\sqrt{u^2+4})$, with $u = \beta/\theta_{\rm E}$.
- $A(u) \to 1/u$ at high magnification ($u \to 0$) and $A(u) \to 1$ at large separation ($u \to \infty$); the event threshold is $u = 1$, where $A = 3/\sqrt{5} \approx 1.34$.
- Relative motion turns A into the symmetric **Paczyński light curve** $A(u(t))$ with $u(t) = \sqrt{u_0^2 + (t-t_0)^2/t_{\rm E}^2}$, peaking at $A_{\max} = A(u_0)$. Only the **Einstein timescale** $t_{\rm E} = D_{\rm d}\theta_{\rm E}/v_\perp \propto \sqrt{M}$ carries physical information, in a mass–distance–velocity degenerate combination.
- The **optical depth** $\tau = \int (4\pi G/c^2)(D_{\rm d}D_{\rm ds}/D_{\rm s})\rho\, dD_{\rm d}$ is mass-independent. Toward the LMC the measured $\tau \approx 1.2\times10^{-7}$ rules out compact objects of $\sim 10^{-7}$–$30\,M_\odot$ as the dominant halo dark matter (EROS: $< 25\%$ of the halo for 2×10^{-7}–$1\,M_\odot$); toward the bulge $\tau \sim 10^{-6}$.
- **Binary and planetary** lenses add extended caustics; caustic crossings give sharp, source-size-dependent spikes, and a planet in the lensing zone $0.6 \lesssim d/\theta_{\rm E} \lesssim 1.6$ produces a brief low-amplitude deviation revealing the mass ratio q.
- **Astrometric** microlensing measures the light-centroid shift, peaking at $\delta_{\max} = 8^{-1/2}\theta_{\rm E}$ for $u = \sqrt{2}$; with a parallax it yields the lens mass.
- **Quasar** microlensing ($\kappa \sim 1$, micro-caustic networks) constrains quasar accretion-disk sizes and the compact-object content of lens galaxies.

17.9 Problems

Exercise 17.9.1. (Total magnification and its limits.) Work in Einstein units, $u = \beta/\theta_{\rm E}$, $y = \theta/\theta_{\rm E}$.

(a) Solve the point-lens equation $u = y - 1/y$ for the two image positions $y_\pm$ and verify the separation $|y_+ - y_-| = \sqrt{u^2+4}$.

(b) Using $\mu = [1 - y^{-4}]^{-1}$, show that the total magnification is $A(u) = |\mu_+| + |\mu_-| = (u^2+2)/(u\sqrt{u^2+4})$.

(c) Verify the limits $A(u) \to 1/u$ as $u \to 0$ and $A(u) \to 1$ as $u \to \infty$.

(d) Compute the peak brightening Δm (in magnitudes) for an event with impact parameter $u_0 = 1$.

Verify with `problems_13.wl`.

Exercise 17.9.2. (Einstein radius and timescale.)

(a) For a lens of $M = 0.3\,M_\odot$ at $D_\mathrm{d} = 4$ kpc lensing a bulge source at $D_\mathrm{s} = 8$ kpc, compute the angular Einstein radius θ_E (in mas) and the physical Einstein radius R_E (in AU).

(b) For a transverse speed $v_\perp = 200\ \mathrm{km\,s^{-1}}$, compute the Einstein timescale t_E in days. Comment on why Galactic events last of order weeks.

(c) Show that at fixed geometry and velocity $t_\mathrm{E} \propto \sqrt{M}$, and hence that the light curve alone cannot separate M, D_d, and $v_\perp$. What additional measurements break this degeneracy?

Verify with `problems_13.wl`.

Exercise 17.9.3. (Optical depth and the MACHO test.)

(a) Explain physically why the optical depth τ (eq. 17.11) is independent of the individual lens masses. *Hint:* how do the Einstein area and the number density each scale with M?

(b) For a thin lens screen of surface mass density Σ, show that the fraction of sky covered by Einstein disks equals $\Sigma/\Sigma_\mathrm{cr}$.

(c) Given the Paczyński halo value $\tau \sim 5 \times 10^{-7}$, how many stars must be monitored to expect of order one lensed star at any instant?

(d) The measured LMC value is $\tau \approx 1.2 \times 10^{-7}$, well below the prediction for a fully compact halo. What does this imply about MACHOs as a dark-matter candidate?

Verify with `problems_13.wl`.

Exercise 17.9.4. (Peak magnification and event width.)

(a) An event reaches peak magnification $A_\mathrm{max} = 5$. Invert eq. (17.10) to find the impact parameter u_0.

(b) Show that the high-magnification approximation $A_\mathrm{max} \approx 1/u_0$ is accurate to better than 1% for $u_0 = 0.05$.

(c) Show that the source reaches the event threshold $u = 1$ (where $A = 3/\sqrt{5}$) at times $|t - t_0| = t_\mathrm{E}\sqrt{1 - u_0^2}$, for $u_0 < 1$.

Verify with `problems_13.wl`.

Exercise 17.9.5. (Astrometric centroid shift.) The light centroid of a point-lens event is displaced from the source by $\delta(u) = u\,\theta_\mathrm{E}/(u^2 + 2)$.

(a) Show that $\delta(u)$ is maximized at $u = \sqrt{2}$.

(b) Evaluate the maximum shift and show it equals $\delta_\mathrm{max} = 8^{-1/2}\,\theta_\mathrm{E} \approx 0.354\,\theta_\mathrm{E}$.

(c) Explain why the astrometric signal breaks the mass–distance degeneracy that the photometric light curve cannot, and identify the coefficient in the mass relation $M = 0.123\,M_\odot\,\theta_\mathrm{E}^2/\pi_\mathrm{ds}$ (with angles in mas).

Verify with `problems_13.wl`.

Exercise 17.9.6. (Quasar microlensing scales.)

(a) For a quasar at $z_s \approx 2$ microlensed by stars in a galaxy at $z_d \approx 0.5$, estimate the Einstein radius projected into the source plane for a solar-mass star, and show it is of order 10^{16} cm.

(b) Show $r_{\rm E} \propto \sqrt{M}$.

(c) Estimate the caustic-crossing time $t_{\rm cross} = R_{\rm source}/v_\perp$ for a continuum region of $R = 10^{15}$ cm and $v_\perp = 600\ \mathrm{km\,s^{-1}}$, and explain why this (not $t_{\rm E}$) sets the observable variability timescale.

(d) Why does the condition $\kappa \sim 1$ at a quasar macro-image mean that microlensing acts as a *network* of caustics rather than isolated events?

Verify with `problems_13.wl`.

Part V

Compound Lenses, Degeneracies, and Cosmography

Chapter 18

The Multi-Plane Lens Equation and Double-Source-Plane Lenses

[Chapter in preparation.]

Chapter 19

Power-Law Lenses with a Central Point Mass

[Chapter in preparation.]

Chapter 20

Odd Images, Central Images, and Their Suppression

[Chapter in preparation.]

Chapter 21

How an Unmodelled Central Mass Biases a Compound Lens

[Chapter in preparation.]

Chapter 22

The Mass-Sheet Degeneracy in Multi-Plane Lenses and Kinematic Breaking

[Chapter in preparation.]

Chapter 23

Azimuthal Multipoles and the Central-Mass Confusion

[Chapter in preparation.]

Chapter 24

Cosmological Sensitivity of Compound-Lens Ratios

[Chapter in preparation.]

Chapter 25

First-Order Estimators and Multi-Plane Coherence

[Chapter in preparation.]

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